

RRB MATHS UNIVERSAL FORMULAS

Dekhte Hi Answer – Dono Books ke Har Sawaal par Verified

28 Chapters • 185 Formulas (har chapter mein max 10)

8,463 book questions mapped (English book 5,213 + Hindi book 3,250)

1,235 solved examples computer-verified vs book answer key ✓

Har formula ke saath: PEHCHAN + FORMULA + TERMS + SOLVED (book Q. no.) + us formula ke saare book sawaal

Books: *RRB Mathematics – Chapterwise Solved Papers* (English) · *RRB Group D Maths – Chapterwise Solved Papers* (हिन्दी)

Kaise Use Karein (3 Steps)

1. **Question padho** → chapter ke formulas mein **PEHCHAN** box ke keywords milao ⇒ formula ID mil gaya (jaise ST-2).
2. **FORMULA** box mein values daalo. **TERMS** box har letter ka matlab batata hai.
3. **SOLVED** box mein book ka asli sawaal (Q. number ke saath) hal kiya hua hai — values badal ke waise hi karo. ✓ = book ki answer key se match.

Notation

- **EN 12** = English book ke *usi chapter* ka Q. 12 · **HI 30** = Hindi book ke *usi chapter* ka Q. 30.
- → **ST-3 (Ch 18)** = ye sawaal doosre chapter ke formula se hota hai (wahan dekho).
- Har formula ke neeche **BOOK KE SAWAAL** line mein us formula se hone wale *saare* sawaalon ke number hain — practice ke liye seedha book kholo.

Guarantee kya hai

- Dono books ke **8,463** sawaal (English 5,213 + Hindi 3,250) — har ek ko exactly ek formula ID se joda gaya; koi sawaal bina formula ke nahi bacha (har chapter ke end mein COVERAGE CHECK).
- **1,235** solved examples ko computer ne book ki answer key se match kiya — 1,235/1,235 ✓.
- Formulas pattern-based hain: naye (unseen) sawaal jo inhi types ke hon, unpar bhi seedhe lagte hain.
- Book ki galtiyan (galat answer key, chhapai ki galti, gayab number) aur jo sawaal page se wapas nikaale gaye — **Appendix A** mein.

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Chapter 1

Number System (Sankhya Paddhati) संख्या पद्धति

10 formulas · English book 345 sawaal · Hindi book 161 sawaal · 56 solved examples ✓

NS-1 — DIVISIBILITY TABLE (vibhajyata)

PEHCHAN – Question mein ye dikhe

“divisible by”, “which number is divisible”, “find digit x so that ... divisible”; विभाज्य, विभाजित, पूर्णतः विभाज्य

Divisor	Rule (sirf ye check karo)	Divisor	Rule
2 / 5 / 10	last digit even / 0,5 / 0	3 / 9	digit-sum $\div 3 / \div 9$
4 / 8	last 2 / last 3 digits $\div 4 / \div 8$	11	(odd-place sum) – (even-place sum) = 0, 11, 22...
7 / 13	last digit $\times 2$ ghatao (7) · ya 3-3 group alternate sum $\div 7/13$	17	seedha divide (ya last digit $\times 5$ ghatao)

Composite divisor = CO-PRIME tukde, dono pass: $6 = 2 \times 3$, $12 = 3 \times 4$, $18 = 2 \times 9$, $24 = 3 \times 8$, $33 = 3 \times 11$, $36 = 4 \times 9$, $44 = 4 \times 11$, $72 = 8 \times 9$, $88 = 8 \times 11$, $90 = 9 \times 10$.

Missing digit x: rule lagao $\rightarrow$ equation $\rightarrow x \in \{0, \dots, 9\}$ (max/min jo poocha ho).

TERMS – Har letter ka matlab

Odd place = right se 1st, 3rd, 5th... digit · even place = 2nd, 4th... · co-prime = jinka HCF 1 ho (4 aur 6 co-prime NAHI, isliye $24 \neq 4 \times 6$).

SOLVED – Asli book questions (verified)

Q (EN 1): 9 aur 11 dono se divisible?

10098: digit-sum 18 ✓(9); $(8 + 0 + 1) - (9 + 0) = 0$ ✓(11) ✓ **book ans (b)**

Q (EN 44): x4461 divisible by 11, x = ?

$(1 + 4 + x) - (6 + 4) = 0 \Rightarrow x = 5$ ✓ **book ans (d)**

Q (EN 5): 88p554085k6 divisible by 72 ($k \neq p$); $3k + 2p$?

$72 = 8 \times 9$: last 3 digits $5k6 \div 8 \Rightarrow k = 3$ ($536 = 8 \cdot 67$); digit-sum $52 + p + k \div 9 \Rightarrow p = 2 \Rightarrow 3k + 2p = 13$ ✓ **book ans (c)**

Q (HI 13): 28x232 को 88 से विभाज्य बनाने के लिए x?

$88 = 8 \times 11$: $232 \div 8$ ✓; 11-rule: $(2 + 2 + 8) - (3 + x + 2) = 0 \Rightarrow x = 7$ ✓ **book ans (d)**

BOOK KE SAWAAL (46) EN 1–5, 8–10, 16, 41, 42, 44–47, 51, 52, 67 · **HI** 1, 3, 5–7, 9–18, 20, 22–24, 26, 29, 32, 33, 37–40 | Ch 5 (L.S. – M.S.): EN 222

NS-2 — REMAINDER MACHINE (sheshfal)

PEHCHAN – Question mein ye dikhe

“remainder”, “leaves”, “divided by ... remainder”, quotient/divisor/dividend; शेषफल, भागफल, भाजक, भाज्य

$$N = D \times Q + R \quad (0 \leq R < D)$$

1. **Product / power:** har number ki jagah uska remainder likho (negative bhi chalega): $R(a \cdot b) = R(a) \cdot R(b)$, $R(a^n)$: agar $a \equiv \pm 1 \pmod{D}$ to $(\pm 1)^n$.
2. **Chain:** D_1 ko D_2 poori kaatta ho $\Rightarrow$ naya $R = R_1 \pmod{D_2}$.
3. **Multiple:** N ka remainder $R \Rightarrow kN$ ka remainder $= kR \pmod{D}$ (square: $R^2 \pmod{D}$).
4. **Ula divisor:** $N \pmod{R_1}$, $2N \pmod{R_2} \Rightarrow D = 2R_1 - R_2$.
5. **Galat divisor:** $N = \text{galat divisor} \times \text{mila answer} + \text{rem} \Rightarrow$ sahi divisor se divide.
6. **Polynomial:** $P(x) \div (x - a) \Rightarrow$ remainder $= P(a)$ (x ki jagah a daalo).

TERMS – Har letter ka matlab

N = dividend (bhajya) · D = divisor (bhajak) · Q = quotient (bhagfal) · R = remainder (sheshfal) · $\equiv$ = “same remainder”.

SOLVED – Asli book questions (verified)

Q (EN 43): $14528 \div d$: quotient 83, remainder 3; $d = ?$

$$d = \frac{14528 - 3}{83} = 175 \quad \checkmark \text{ book ans (d)}$$

Q (EN 6): $171 \times 172 \times 173 \div 17$ ka remainder

$$\text{rem: } 1 \times 2 \times 3 = 6 \quad \checkmark \text{ book ans (c)}$$

Q (EN 14): $19^{300} \div 20$

$$19 \equiv -1 \Rightarrow (-1)^{300} = 1 \quad \checkmark \text{ book ans (b)}$$

Q (EN 29): $N \div 280$ rem 73; $N \div 35 = ?$

$$280 = 8 \times 35 \Rightarrow 73 \pmod{35} = 3 \quad \checkmark \text{ book ans (c)}$$

Q (EN 7): $N \pmod{16}$, $2N \pmod{3}$; divisor?

$$D = 2(16) - 3 = 29 \quad \checkmark \text{ book ans (a)}$$

Q (EN 55): $5^{501} \div 126$

$$5^3 = 125 \equiv -1 \Rightarrow 5^{501} = (5^3)^{167} \equiv -1 \equiv 125 \quad \checkmark \text{ book ans (d)}$$

Q (HI 30): $ax^2 - 8x - 8 \div (x + 2)$, remainder 24; $a = ?$

$$P(-2) = 4a + 16 - 8 = 24 \Rightarrow a = 4 \quad \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (18) EN 6, 7, 11, 14, 20, 29, 34, 39, 43, 50, 53, 55, 331 · HI 8, 21, 27, 30, 57

NS-3 — NEAREST-MULTIPLE MACHINE (add / subtract / largest / smallest)

PEHCHAN – Question mein ye dikhe

“least number to be added/subtracted”, “greatest/smallest n-digit number divisible by”, “leaves remainder r”, “perfect square banane ke liye”; जोड़ा/घटाया जाए, सबसे बड़ी/छोटी संख्या, पूर्ण वर्ग

Pehle $R = N \pmod{d}$ nikalo:

$$\begin{array}{ll} \text{Add} = d - R & \text{Subtract} = R \\ \text{Largest n-digit} = \underbrace{99 \dots 9}_n - R & \text{Smallest n-digit} = 10 \dots 0 + (d - R) \end{array}$$

Remainder r bhi chahiye $\Rightarrow$ upar wale multiple mein + r . **Perfect square:** add = (agla square – N), subtract = (N – pichla square).

Sab se divide ho (1 se 10, ya 8 aur 12...) $\Rightarrow$ pehle LCM lo, phir yahi machine. Sabse bada n-digit = 99...9, sabse chhota = 10...0 (farak/jod seedha).

TERMS – Har letter ka matlab

d = divisor · R = remainder · 99...9 = sabse bada n-digit, 10...0 = sabse chhota n-digit · agar $R = 0$ to number khud answer.

SOLVED – Asli book questions (verified)**Q (EN 26):** Largest 4-digit divisible by 83

$$9999 \bmod 83 = 39 \Rightarrow 9999 - 39 = 9960 \checkmark \text{ book ans (c)}$$

Q (EN 37): Smallest 4-digit divisible by 47

$$1000 \bmod 47 = 13 \Rightarrow 1000 + (47 - 13) = 1034 \checkmark \text{ book ans (c)}$$

Q (EN 21): Greatest 4-digit + ? $\Rightarrow$ divisible by 307

$$9999 \bmod 307 = 175 \Rightarrow 307 - 175 = 132 \checkmark \text{ book ans (b)}$$

Q (EN 48): $1265 - ? \Rightarrow$ divisible by 29

$$R = 1265 \bmod 29 = 18 \checkmark \text{ book ans (c)}$$

Q (EN 33): Smallest 5-digit, rem 6 on $\div 7$

$$10000 \bmod 7 = 4 \Rightarrow 10000 + (6 - 4) = 10002 \checkmark \text{ book ans (b)}$$

Q (EN 265): $4042 + ? =$ perfect square

$$64^2 = 4096 \Rightarrow 4096 - 4042 = 54 \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (32) EN 15, 21, 24, 26, 30, 33, 35–38, 40, 48, 49, 68, 71–73, 152, 182, 253, 265, 268, 271 · **HI** 2, 19, 25, 35, 36, 119, 126, 158, 159**NS-4 — COUNT MACHINE (kitne numbers)****PEHCHAN – Question mein ye dikhe**

“how many numbers between A and B are divisible by”, “divisible by p or q”, “how many times digit k appears”, “how many perfect squares between”; कितनी संख्याएँ, के बीच

$$\# \text{multiples of } d \text{ in } [A, B] = \left\lfloor \frac{B}{d} \right\rfloor - \left\lfloor \frac{A-1}{d} \right\rfloor$$

- **p YA q:** $n(p) + n(q) - n(\text{LCM})$ · **p AUR q:** $n(\text{LCM})$ · **odd multiples:** $n(d) - n(2d)$
- **Perfect squares** between A, B = $\lfloor \sqrt{B} \rfloor - \lceil \sqrt{A} \rceil + 1$
- **Digit count** 1 se 99 tak: har non-zero digit 20 baar (units 10 + tens 10); 1–999: 300 baar.

TERMS – Har letter ka matlab

$\lfloor x \rfloor$ = sirf poora bhaag (decimal hatao) · “between” = dono ends chhod do · $n(d) = d$ ke multiples ki ginti.

SOLVED – Asli book questions (verified)**Q (EN 13):** 1–700 mein 17 se divisible

$$\lfloor 700/17 \rfloor = 41 \checkmark \text{ book ans (b)}$$

Q (EN 12): 1–100: 3 ya 4 se divisible

$$33 + 25 - 8 = 50 \checkmark \text{ book ans (a)}$$

Q (EN 23): 300–1000 mein 7 se divisible

$$142 - 42 = 100 \checkmark \text{ book ans (d)}$$

Q (EN 25): 250–1000: 5, 6, 7 teeno se

$$\text{LCM } 210 \Rightarrow 4 - 1 = 3 \checkmark \text{ book ans (d)}$$

Q (EN 264): 100 aur 200 ke beech perfect squares

$$11^2..14^2 \Rightarrow 4 \checkmark \text{ book ans (b)}$$

Q (EN 77): 1–100 mein digit 5 kitni baar

$$\text{units } 10 + \text{tens } 10 = 20 \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (15) EN 12, 13, 22, 23, 25, 65, 74, 76, 77, 264, 329, 332 · **HI** 4, 28, 34**NS-5 — FACTOR MACHINE (gunankhand)****PEHCHAN – Question mein ye dikhe**

“number of factors / divisors”, odd/even factors, “factors which are perfect squares”, “sum of prime factors”, “multiply/divide to make perfect square/cube”, “odd number of factors”; गुणनखंड, गुणक, भाजक, पूर्ण वर्ग

Prime powers mein todo: $N = 2^a \times p^b \times q^c$

$$\text{Total} = (a+1)(b+1)(c+1) \quad \text{Odd} = (b+1)(c+1) \quad \text{Even} = \text{Total} - \text{Odd}$$

- **Perfect-square factors** = $(\lfloor \frac{a}{2} \rfloor + 1)(\lfloor \frac{b}{2} \rfloor + 1) \dots$ (cube: $\lfloor \frac{a}{3} \rfloor + 1$)
- **Factors divisible by M** = factors of N/M · **Total prime factors** = $a + b + c$ · **sum of prime factors** = $2 + p + q$
- **Odd number of factors** $\Leftrightarrow$ **perfect square**. Square banana: jin primes ki power odd hai unse ek baar multiply/divide (cube: power 3 ka multiple).

TERMS – Har letter ka matlab

a, b, c = powers (ghatank) · p, q = odd primes · “multiples of $2^8 \times 3^2 \dots$ ” wale sawaal mein bhi factors hi poochhe gaye hain.

SOLVED – Asli book questions (verified)

Q (EN 281): Factors of 4200

$$2^3 \cdot 3 \cdot 5^2 \cdot 7 \Rightarrow 4 \cdot 2 \cdot 3 \cdot 2 = 48 \checkmark \text{ book ans (a)}$$

Q (EN 276): 120 ke odd, even divisors

$$2^3 \cdot 3 \cdot 5: \text{odd} = 2 \cdot 2 = 4, \text{even} = 16 - 4 = 12 \checkmark \text{ book ans (c)}$$

Q (EN 278): $2^7 3^4 5^3 7$ ke even factors

$$8 \cdot 5 \cdot 4 \cdot 2 - 5 \cdot 4 \cdot 2 = 320 - 40 = 280 \checkmark \text{ book ans (b)}$$

Q (EN 286): 256 ke perfect-square factors

$$2^8 \Rightarrow \lfloor 8/2 \rfloor + 1 = 5 \checkmark \text{ book ans (a)}$$

Q (EN 293): $216 \times ? \Rightarrow$ odd number of factors

$$216 = 2^3 3^3 \Rightarrow \text{dono power odd} \Rightarrow \times 2 \times 3 = 6 \checkmark \text{ book ans (b)}$$

Q (EN 296): $16^7 \times 27^6 \times 5^9$ ke total prime factors

$$2^{28} 3^{18} 5^9 \Rightarrow 28 + 18 + 9 = 55 \checkmark \text{ book ans (c)}$$

Q (EN 301): $2^{10} 3^6 5^3 7^5$ ke factors jo 2160 se divisible

$$2160 = 2^4 3^3 5 \Rightarrow N/M = 2^6 3^3 5^2 7^5 : 7 \cdot 4 \cdot 3 \cdot 6 = 504 \checkmark \text{ book ans (c)}$$

BOOK KESAWAAL (28) EN 266, 274, 276, 278, 281, 282, 286, 287, 289–291, 293, 295, 296, 299–301, 339 · **HI** 31, 143–146, 148, 157, 160, 161 | Ch 5 (L.S. – M.S.): EN 174

NS-6 — UNIT DIGIT MACHINE (ikai ank)

PEHCHAN – Question mein ye dikhe

“unit digit”, “last digit”, “unit’s place”, “ends with”; इकाई अंक, अंतिम अंक

Power a^n : $r = n \bmod 4$ ($r = 0$ ho to 4 lo) $\Rightarrow$ answer = unit digit of (last digit of a)^r

last digit	r=1	r=2	r=3	r=0(4)
2	2	4	8	6
3	3	9	7	1
7	7	9	3	1
8	8	4	2	6
4 / 9	4 / 9	6 / 1	odd power $\rightarrow$ 4/9, even $\rightarrow$ 6/1	
0,1,5,6	hamesha wahi digit			

Product/sum: sirf last digits se karo. Product mein **5** $\times$ **even** aaye $\Rightarrow$ unit digit **0**.

TERMS – Har letter ka matlab

a = base, n = power · $(\dots)^n$ ke liye sirf base ka aakhri digit dekho.

SOLVED – Asli book questions (verified)**Q (EN 292):** 213^6 ka last digit

$$6 \bmod 4 = 2 \Rightarrow 3^2 = 9 \checkmark \text{ book ans (d)}$$

Q (EN 285): $1373^{36} - 1442^{20}$

$$3^{4k} \rightarrow 1, 2^{4k} \rightarrow 6 \Rightarrow 1 - 6 \rightarrow 11 - 6 = 5 \checkmark \text{ book ans (c)}$$

Q (EN 284): $4 \times 38 \times 764 \times 1256$

$$4 \cdot 8 \cdot 4 \cdot 6 = 768 \Rightarrow 8 \checkmark \text{ book ans (b)}$$

Q (EN 280): $91 \times 92 \times \dots \times 99$

$$95 \times \text{even} \Rightarrow 0 \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (13) EN 262, 275, 279, 280, 283–285, 288, 292, 294, 297, 298 · **HI** 147**NS-7 — PRIME / CO-PRIME TOOLKIT****PEHCHAN – Question mein ye dikhe**

prime, composite, co-prime, twin prime, “how many primes between”, “sum of primes”; अभाज्य, भाज्य, सह-अभाज्य

Prime test: $\sqrt{N}$ tak ke primes se divide karo — koi na kaate $\Rightarrow$ prime.

1–10	11–20	21–30	31–40	41–50	51–60	61–70	71–80	81–90	91–100
2,3,5,7	11,13,17,19	23,29	31,37	41,43,47	53,59	61,67	71,73,79	83,89	97
4	8	10	12	15	17	19	22	24	25

(neeche wali row = **us tak kul primes**; 1–200 mein 46). **Co-prime** $\Leftrightarrow$ **HCF = 1** (consecutive numbers hamesha co-prime).Twin primes (diff 2): 1–100 mein **8 jode**. **Chain trick:** $p \cdot q = A, q \cdot r = B \Rightarrow q = \text{HCF}(A, B)$. **Sum trick:** 3 odd primes ka sum hamesha odd $\Rightarrow$ sum even ho to **2** zaroor shaamil.**TERMS – Har letter ka matlab**

1 na prime na composite · 2 akela even prime · sabse chhota composite 4 · prime = sirf 1 aur khud se divide.

SOLVED – Asli book questions (verified)**Q (EN 78):** 50 se kam primestable: 15 $\checkmark$ **book ans (b)****Q (EN 86):** 50 aur 60 ke beech primes ka sum

$$53 + 59 = 112 \checkmark \text{ book ans (d)}$$

Q (EN 91): 3 primes (desc): pehle do ka product 323, aakhri do ka 221; sabse bada?

$$q = \text{HCF}(323, 221) = 17 \Rightarrow 323/17 = 19 \checkmark \text{ book ans (b)}$$

Q (HI 54): 20 और 80 के बीच अभाज्य

$$22 - 8 = 14 \checkmark \text{ book ans (b)}$$

Q (EN 88): 1–100 twin prime pairs

$$(3,5), (5,7), (11,13), (17,19), (29,31), (41,43), (59,61), (71,73) = 8 \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (56) EN 31, 78–93, 95–109, 111, 257 · **HI** 52–56, 58–64, 66–74, 116**NS-8 — POWER-PATTERN DIVISIBILITY****PEHCHAN – Question mein ye dikhe** $a^n \pm b^n$ is divisible by”, $3^{71} + 3^{72} + \dots$, “always divisible by”, product of consecutive numbers, $n^3 - n$, $pq - qp$

- $a^n - b^n \Rightarrow$ hamesha $\div (a - b)$; n **even** ho to $\div (a + b)$ bhi.
- $a^n + b^n \Rightarrow$ n **odd** ho to $\div (a + b)$.
- $a^m + a^{m+1} + \dots + a^{m+k} = a^m(1 + a + \dots + a^k) \Rightarrow$ bracket ke factors dekho.
- k consecutive integers ka product $\Rightarrow \div k!$ (4 consecutive $\Rightarrow 24$) $\cdot n^3 - n = (n - 1)n(n + 1) \Rightarrow \div 6$.
- 2-digit: $\overline{pq} - \overline{qp} = 9(p - q)$, $\overline{pq} + \overline{qp} = 11(p + q) \cdot \underbrace{11 \dots 1}_n \div 9 \Leftrightarrow n = 9, 18, \dots$

TERMS – Har letter ka matlab

a, b = bases, n = power $\cdot$ “always divisible” = har value par sach.

SOLVED – Asli book questions (verified)

Q (EN 17): $41^{43} + 43^{43}$ divisible by?

odd power $\Rightarrow \div (41 + 43) = 84$ ✓ **book ans (d)**

Q (EN 32): $3^{71} + \dots + 3^{75}$ divisible by?

$3^{71}(1 + 3 + 9 + 27 + 81) = 3^{71} \cdot 121 \Rightarrow 11$ ✓ **book ans (c)**

Q (EN 54): 4 consecutive numbers ka product

$4! = 24$ ✓ **book ans (c)**

Q (EN 19): $n^3 - n$ always divisible by

$(n - 1)n(n + 1) \Rightarrow 6$ ✓ **book ans (b)**

Q (HI 142): $55^{100} + 55^{101} + 55^{102}$ ka factor NAHI

$55^{100}(1 + 55 + 3025) = 55^{100} \cdot 3081 = 5^{100} 11^{100} \cdot 3 \cdot 13 \cdot 79 \Rightarrow 71$ nahi ✓ **book ans (c)**

BOOK KE SAWAAL (7) EN 17, 19, 27, 28, 32, 54 $\cdot$ **HI 142**

NS-9 — TWO-NUMBER + EQUATION MACHINE**PEHCHAN – Question mein ye dikhe**

“sum of two numbers is . . . , difference/product . . .”, “2 tables + 3 chairs cost. . .”, “1/3 of a number is 6 more than 1/5. . .”, consecutive numbers, 2-digit number “digits interchanged/reversed”; दो संख्याओं का योग/अंतर/गुणनफल, मूल्य, क्रमागत, अंकों को आपस में बदलने

Sum S, Difference D, Product P:

$$\begin{aligned} \text{Bada} &= \frac{S+D}{2}, \text{ Chhota} = \frac{S-D}{2} & x^2 - y^2 &= S \cdot D & x^2 + y^2 &= S^2 - 2P \\ x - y &= \sqrt{S^2 - 4P} & x + y &= \sqrt{D^2 + 4P} & x^3 + y^3 &= S^3 - 3PS & \frac{1}{x} + \frac{1}{y} &= \frac{S}{P} \end{aligned}$$

Product P aur quotient/ratio k $\Rightarrow$ chhota $= \sqrt{P/k}$, bada $= k \times$ chhota.

Do equations $a_1x + b_1y = c_1$, $a_2x + b_2y = c_2$ (Cramer — seedha daalo):

$$x = \frac{c_1b_2 - c_2b_1}{a_1b_2 - a_2b_1} \quad y = \frac{a_1c_2 - a_2c_1}{a_1b_2 - a_2b_1}$$

- **Fraction of number:** $\frac{a}{b}x - \frac{c}{d}x = k \Rightarrow x = k \div (\frac{a}{b} - \frac{c}{d})$
- **n consecutive** ka sum S $\Rightarrow$ beech wala $= S/n$ (odd/even consecutive: gap 2) $\cdot$ **Sir-pair:** 4-pair wale $= \frac{\text{Legs} - 2 \cdot \text{Heads}}{2}$
- **2-digit** $10x + y$: reverse ka farak $= 9(x - y)$, jod $= 11(x + y) \Rightarrow$ digit-sum aur digit-diff mil gaye $\Rightarrow$ digits $= (\text{sum} \pm \text{diff})/2$. **3-digit** $\overline{abc}$: reverse ka farak $= 99(a - c)$.
- **Unitary:** 1 ka maan $= \text{total} \div \text{ginti}$, phir $\times$ chahiye wali ginti. **Square wali equation** ($x^2 \dots$): options daal kar check karo — sabse fast.

TERMS – Har letter ka matlab

S = sum, D = difference, P = product $\cdot$ x, y = dono numbers (2-digit mein x = tens, y = units) $\cdot$ a_1, b_1, c_1 = pehli equation ke coefficients.

SOLVED – Asli book questions (verified)**Q (EN 124):** Sum 45, difference 19

$$(45 + 19)/2 = 32, 13 \checkmark \text{ book ans (b)}$$

Q (EN 131): $S = 25, P = 136 \Rightarrow x^3 + y^3$

$$25^3 - 3(136)(25) = 15625 - 10200 = 5425 \checkmark \text{ book ans (a)}$$

Q (EN 142): $S = 20, P = 96 \Rightarrow$ difference

$$\sqrt{400 - 384} = 4 \checkmark \text{ book ans (a)}$$

Q (EN 123): $2T + 3C = 540, 2T + C = 470 \Rightarrow$ 5 chairs

$$C = \frac{2 \cdot 470 - 2 \cdot 540}{2 \cdot 1 - 2 \cdot 3} = 35 \Rightarrow 175 \checkmark \text{ book ans (c)}$$

Q (EN 125): $1/3$ of number is 6 more than $1/5$

$$x = 6 \div \left(\frac{1}{3} - \frac{1}{5}\right) = 45 \checkmark \text{ book ans (b)}$$

Q (EN 206): digit-sum 11, reverse 45 kam

$$\text{diff} = 45/9 = 5 \Rightarrow \text{digits } (11 + 5)/2 = 8, 3 \Rightarrow 83 \checkmark \text{ book ans (d)}$$

Q (EN 326): Heads 42, legs 138 $\Rightarrow$ chickens

$$\text{goats} = (138 - 84)/2 = 27 \Rightarrow \text{chickens } 15 \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (188) EN 18, 60, 66, 75, 94, 110, 112–118, 120–135, 137, 139, 142–148, 150, 153, 156–159, 161–172, 175–180, 184–192, 195, 197–215, 254–256, 258, 263, 267, 277, 323–327, 330, 333, 336, 338, 340, 343, 344 · **HI** 41–51, 65, 75–115, 117, 118, 120–122, 125, 149–152, 154, 156 | Ch 5 (L.S. – M.S.): EN 172 · **HI** 105 | Ch 7 (Labb aur Hani): EN 41, 108, 118, 154 | Ch 9 (Anupat aur Samanupat): EN 149 · **HI** 4 | Ch 10 (Saajhedaari): EN 20–22 · **HI** 5, 18

NS-10 — SERIES + PLACE-VALUE + RATIONAL BOX (ratta facts)**PEHCHAN – Question mein ye dikhe**

“sum of squares/cubes of 1 to n”, handshakes, “place value / face value”, rational / irrational, “which square root is rational”, perfect square test; स्थानीय मान, जातीय मान, परिमेय, अपरिमेय, वर्गों/घनों का योग

$$1 + \dots + n = \frac{n(n+1)}{2} \quad 1^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6} \quad 1^3 + \dots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

- **a se b tak** = $S(b) - S(a-1)$ · pehle n odd = n^2 · handshakes = $\frac{n(n-1)}{2} \cdot a + ar + ar^2 + \dots \infty = \frac{a}{1-r}$
- **Place value** = digit $\times$ (uski jagah ki value 1, 10, 100...) · **Face value** = digit khud.
- **Rational:** p/q, terminating/recurring decimal, perfect square/cube ka root. **Irrational:** $\sqrt{\text{(non-square)}}$, Π , non-recurring decimal. rational + irrational = irrational; do numbers ke beech rationals = ∞ .
- **Perfect square test:** last digit 2, 3, 7, 8 ho $\Rightarrow$ square NAHI; last zeros even hon; phir paas ke square se milao. Rows = columns $\Rightarrow$ rows = $\sqrt{N}$.
- **Digits se number:** sabse bada = digits descending; sabse chhota = sabse chhota non-zero digit pehle, baaki ascending. **Digit-operation** (odd digit +3, even -1...): har digit par wahi kaam karo, phir jo poocha.

TERMS – Har letter ka matlab

$S(n) = 1$ se n tak ka sum · p/q form = rational ($q \neq 0$).

SOLVED – Asli book questions (verified)**Q (EN 141):** $3^2 + \dots + 18^2$

$$S(18) - S(2) = 2109 - 5 = 2104 \checkmark \text{ book ans (d)}$$

Q (EN 138): $5^3 + \dots + 14^3$

$$105^2 - 10^2 = 11025 - 100 = 10925 \checkmark \text{ book ans (b)}$$

Q (EN 328): 40 log handshakes

$$40 \cdot 39/2 = 780 \checkmark \text{ book ans (b)}$$

Q (EN 306): 516372 mein 6 ka place – face value

$$6000 - 6 = 5994 \checkmark \text{ book ans (d)}$$

Q (EN 248): kiska square root irrational: 5184, 4465, 3025, 8836

$$4465 \text{ (paas ke squares } 66^2=4356, 67^2=4489) \Rightarrow \text{perfect square nahi } \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (102) EN 56–59, 61–64, 70, 119, 136, 138, 140, 141, 149, 151, 155, 160, 216–252, 259–261, 269, 270, 272, 273, 302–322, 328, 335 · **HI** 127–141 | Ch 3 (Ghatank – Karani): EN 123 | Ch 28 (Vividh): EN 42

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
NS-1	DIVISIBILITY TABLE (vibhajyata)	18	27
NS-2	REMAINDER MACHINE (sheshfal)	13	5
NS-3	NEAREST-MULTIPLE MACHINE (add / subtract / largest / smalles)	23	9
NS-4	COUNT MACHINE (kitne numbers)	12	3
NS-5	FACTOR MACHINE (gunankhand)	18	9
NS-6	UNIT DIGIT MACHINE (ikai ank)	12	1
NS-7	PRIME / CO-PRIME TOOLKIT	34	22
NS-8	POWER-PATTERN DIVISIBILITY	6	1
NS-9	TWO-NUMBER + EQUATION MACHINE	110	65
NS-10	SERIES + PLACE-VALUE + RATIONAL BOX (ratta facts)	85	15
→ DF-4	FRACTION OPERATION MACHINE (+ – × ÷, “kya jodein (Ch 2)	5	2
→ DF-6	DECIMAL SHIFT + ROUNDING MACHINE (Ch 2)	1	0
→ SIM-1	VBODMAS MACHINE (order of operations) (Ch 4)	1	0
→ SIM-2	DECIMAL COLUMN + “?” MACHINE (Ch 4)	2	0
→ LH-2	LCM + REMAINDER MACHINE (Ch 5)	0	1
→ LH-4	HCF APPLICATION (sabse bada naap / kam se kam hi (Ch 5)	1	1
→ PC-1	PERCENT BASICS (of · kitna % · poor number) (Ch 6)	2	0
→ PC-2	MULTIPLIER MACHINE (badha / ghata / ulta) (Ch 6)	1	0
→ PC-5	REVERSE + CONSUMPTION MACHINE (Ch 6)	1	0
Total (book ke sawaal)		345/345 ✓	161/161 ✓
→ DF-4: EN 173, 174, 181, 183, 341 HI 123, 124 · → DF-6: EN 69 · → LH-2: HI 153 · → LH-4: EN 154 HI 155 · → PC-1: EN 194, 196 · → PC-2: EN 334 · → PC-5: EN 193 · → SIM-1: EN 337 · → SIM-2: EN 342, 345			

Chapter 2

Decimal Fractions (Dashamlav Bhinn) दशमलव भिन्न

9 formulas · English book 232 sawaal · Hindi book 133 sawaal · 36 solved examples ✓

DF-1 — COMPARE + ORDER MACHINE

PEHCHAN – Question mein ye dikhe

largest/smallest fraction, ascending/descending order, “which is true $a/b < c/d$ ”, “between $7/12$ and $11/16$ ”, “as much greater than x as less than y ”; सबसे बड़ी/छोटी भिन्न, आरोही/अवरोही क्रम

$$\frac{a}{b} \text{ vs } \frac{c}{d} \Rightarrow a \times d \text{ vs } b \times c \quad (\text{bada product} \Rightarrow \text{wahi fraction bada})$$

- **Same numerator** $\Rightarrow$ chhota denominator = badi fraction. **Same denominator** $\Rightarrow$ bada numerator = badi.
- **(Neeche – upar) sab mein same** ($15/16, 24/25, 34/35$) $\Rightarrow$ proper: bade numbers wali badi; improper (upar $>$ neeche): chhote numbers wali badi.
- Bahut saari fractions $\Rightarrow$ har ek ko **3 decimal** tak bana lo ($\div$) ya sab ka denominator LCM kar do.
- **“x se utna bada jitna y se chhota”** $\Rightarrow \frac{x+y}{2}$. Do fractions ke beech: $\frac{a+c}{b+d}$ hamesha beech mein.

TERMS – Har letter ka matlab

a, c = numerators (ansh) · b, d = denominators (har) · negative fractions: jo zyada negative, wo chhoti.

SOLVED – Asli book questions (verified)

Q (EN 9): $15/16, 24/25, 34/35, 19/20$ mein greatest
neeche – upar = 1 sab mein $\Rightarrow$ sabse bade numbers: $34/35$ ✓ **book ans (c)**

Q (HI 2): $5/7, 9/14, 16/21, 29/42$ में सबसे बड़ी
LCM 42: $30, 27, 32, 29 \Rightarrow 16/21$ ✓ **book ans (d)**

Q (EN 8): $2/3, 3/4, 4/5, 5/6$: greatest – least
 $\frac{5}{6} - \frac{2}{3} = \frac{1}{6}$ ✓ **book ans (c)**

Q (EN 186): $4/7$ se utni badi jitni $5/6$ se chhoti
 $\frac{1}{2} \left(\frac{4}{7} + \frac{5}{6} \right) = \frac{59}{84}$ ✓ **book ans (a)**

BOOK KE SAWAAL (64) EN 1–22, 24–43, 125, 182, 183, 186, 188, 201, 219 · HI 1–13, 76, 77

DF-2 — TERMINATING / RECURRING TEST

PEHCHAN – Question mein ye dikhe

terminating / non-terminating / recurring decimal, “decimal will end after how many places”; सांत/शांत दशमलव, अवसान, आवर्ती दशमलव

Pehle fraction ko **lowest form** mein lao, phir sirf denominator dekho:

$$\text{den} = 2^m \times 5^n \text{ (aur kuch nahi)} \Rightarrow \text{Terminating, decimal places} = \max(m, n)$$

Denominator mein 3, 7, 11, 13... koi bhi aur prime bacha $\Rightarrow$ **Recurring** (non-terminating).

TERMS – Har letter ka matlab

m, n = 2 aur 5 ki powers · lowest form zaroori: $9/45 = 1/5$ terminating hai, jabki $45 = 3^2 \cdot 5$.

SOLVED – Asli book questions (verified)**Q (EN 52):** 57/120, 47/150, 61/110, 43/140: terminating?

$$\frac{57}{120} = \frac{19}{40}, 40 = 2^3 \cdot 5 \checkmark \text{ (baaki mein 3, 11, 7) } \checkmark \text{ book ans (a)}$$

Q (EN 50): 3/8 kitne decimal places ke baad khatam

$$8 = 2^3 \Rightarrow 3 \checkmark \text{ book ans (c)}$$

Q (EN 58): 6/144, 12/144, 3/144, 9/144: terminating?

$$\frac{9}{144} = \frac{1}{16} = \frac{1}{2^4} \checkmark \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (28) EN 44–46, 48, 50–61, 86, 87, 93 · **HI** 57–63, 66, 72**DF-3 — DECIMAL ↔ FRACTION (9-0 RULE)****PEHCHAN – Question mein ye dikhe**

“express 0.3 / 0.27 / 4.567 in p/q form”, recurring decimal ko fraction, “0.16 + 0.15 – ...”; p/q के रूप में, साधारण भिन्न, आवर्ती दशमलव

$$0.\overline{ab} = \frac{ab}{99} \quad 0.\overline{abc} = \frac{abc - a}{990} \quad 0.ab = \frac{ab}{100} \text{ (terminating)}$$

Rule: jitne digits **repeat** ⇒ utne **9**; jitne decimal digits repeat **nahi** ⇒ unke baad utne **0**. Numerator = (saare decimal digits) – (non-repeating digits). Whole number part alag se jodo: $4.\overline{5} = 4 + \frac{5}{9}$.**Jod/ghata/guna** wale sawaal: har ek ko fraction banao, phir karo (same bar-pattern ho to seedha numerators jodo). **Uta** (fraction → decimal): seedha divide; $\frac{n}{9} = 0.\overline{n}$, $\frac{n}{99} = 0.\overline{nn}$, $\frac{1}{11} = 0.\overline{09}$.**TERMS – Har letter ka matlab**bar () = repeat hone wala hissa · $0.\overline{3} = 0.333\ldots$ · p/q = fraction (q ≠ 0).**SOLVED – Asli book questions (verified)****Q (EN 73):** $0.\overline{12} = ?$

$$\frac{12}{99} = \frac{4}{33} \checkmark \text{ book ans (b)}$$

Q (EN 72): 0.27 = ?

$$\frac{27-2}{90} = \frac{25}{90} = \frac{5}{18} \checkmark \text{ book ans (c)}$$

Q (HI 14): $0.35\overline{9} = ?$

$$\frac{359-3}{990} = \frac{178}{495} \checkmark \text{ book ans (a)}$$

Q (EN 62): $0.1\overline{6} + 0.15 - 0.1\overline{3}$

$$\frac{15+14-12}{90} = \frac{17}{90} \checkmark \text{ book ans (b)}$$

Q (EN 80): $1.236576576\ldots = ?$

$$1 + \frac{236576-236}{99900} = \frac{123534}{9990} \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (91) EN 49, 62–85, 88–92, 94–96, 108, 110, 175 · **HI** 14–18, 20–56, 64, 65, 67–71, 80, 81, 108, 110, 111, 121**DF-4 — FRACTION OPERATION MACHINE (+ – × ÷, “kya jodein”)****PEHCHAN – Question mein ye dikhe**

sum/difference of fractions, “what should be added to a/b to get c/d”, “subtracted from”, remaining share of cake, mixed numbers, rod cut into pieces, “3/4 of weight is ...”; क्या जोड़ा जाए, घटाने पर, भिन्नों का योग

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd} \quad \frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd} \quad \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

- **A mein kya jodein ki B bane** = $B - A$ · **A se kya ghatayein ki B bache** = $A - B$ · **kis se A ghatayein ki B =** $A + B$.
- **Bacha hissa** (cake/kaam) = $1 -$ (sab diye hue hisse ka jod).
- **Mixed** $W\frac{n}{d} = \frac{W \cdot d + n}{d}$ · pieces = total ÷ ek piece · “ $\frac{3}{4} = 7/8$ kg” ⇒ $\text{poora} = \frac{7}{8} \div \frac{3}{4}$, phir × jo chahiye.

TERMS – Har letter ka matlab

a/b = pehli fraction, c/d = doosri · W = whole number part · A = di hui, B = chahiye wali.

SOLVED – Asli book questions (verified)

Q (EN 139): $10/11$ mein kya jodein ki $11/10$ bane

$$\frac{11}{10} - \frac{10}{11} = \frac{121-100}{110} = \frac{21}{110} \quad \checkmark \text{ book ans (a)}$$

Q (EN 216): Tapan $1/4$, Trisha $2/3$, Ravi = baaki

$$1 - \frac{1}{4} - \frac{2}{3} = \frac{1}{12} \quad \checkmark \text{ book ans (b)}$$

Q (EN 213): $208\frac{4}{5}$ rod ke $23\frac{1}{5}$ ke tukde

$$\frac{1044}{5} \div \frac{116}{5} = 9 \quad \checkmark \text{ book ans (d)}$$

Q (EN 215): $3/4$ weight = $7/8$ kg $\Rightarrow 5/7$ weight

$$\frac{7}{8} \div \frac{3}{4} \times \frac{5}{7} = \frac{5}{6} \text{ kg} \quad \checkmark \text{ book ans (b)}$$

Q (EN 113): $4/11 + 2/7 + 3/5$

$$\frac{140+110+231}{385} = \frac{481}{385} \quad \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (75) EN 98, 101, 104, 106, 112, 113, 121–124, 126, 128, 129, 132–136, 138–146, 148–156, 170, 192, 193, 196, 207, 210, 213–216, 228 · **HI** 82–85, 87–94, 96–100, 103, 126, 131 | Ch 1 (Sankhya Paddhati): EN 173, 174, 181, 183, 341 · **HI** 123, 124 | Ch 9 (Anupat aur Samanupat): EN 169

DF-5 — RECIPROCAL MACHINE**PEHCHAN – Question mein ye dikhe**

reciprocal / inverse, “fraction and its reciprocal ka sum/difference”, “reciprocal of sum of reciprocals”, inversely proportional; व्युत्क्रम, व्युत्क्रमानुपाती

$$\text{Reciprocal of } \frac{a}{b} = \frac{b}{a} \quad \frac{a}{b} + \frac{b}{a} = \frac{a^2 + b^2}{ab} \quad \frac{1}{\frac{1}{x} + \frac{1}{y}} = \frac{xy}{x+y}$$

- Sum $\frac{N}{D}$ diya $\Rightarrow D$ ko $a \times b$ mein todo jahan $a^2 + b^2 = N \Rightarrow \text{fraction} = \frac{a}{b}$ ya $\frac{b}{a}$.
- $x - \frac{1}{x} = D \Rightarrow x^3 - \frac{1}{x^3} = D^3 + 3D \cdot x + \frac{1}{x} = S \Rightarrow x^2 + \frac{1}{x^2} = S^2 - 2, x^3 + \frac{1}{x^3} = S^3 - 3S$

TERMS – Har letter ka matlab

x = fraction · S = sum, D = difference · “inversely proportional to a/b ” = b/a .

SOLVED – Asli book questions (verified)

Q (EN 117): $5/7$ aur $9/5$ ke reciprocals ke sum ka reciprocal

$$\frac{1}{\frac{1}{5} + \frac{1}{9}} = \frac{45}{88} \quad \checkmark \text{ book ans (c)}$$

Q (EN 120): $x - \frac{1}{x} = \frac{9}{11} \Rightarrow$ cubes ka difference

$$\frac{729}{1331} + \frac{27}{11} = \frac{3996}{1331} \quad \checkmark \text{ book ans (c)}$$

Q (EN 116): $2/11 \times$ reciprocal of $(-5/14)$

$$\frac{2}{11} \times \frac{-14}{5} = -\frac{28}{55} \quad \checkmark \text{ book ans (b)}$$

Q (EN 118): fraction + inverse = $2\frac{25}{66} = \frac{157}{66}$; badi?

$$66 = 6 \times 11, 36 + 121 = 157 \Rightarrow \frac{11}{6} = 1\frac{5}{6} \quad \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (10) EN 107, 116–120 · **HI** 104–107

DF-6 — DECIMAL SHIFT + ROUNDING MACHINE**PEHCHAN – Question mein ye dikhe**

decimal $\times \div$, “if $23 \times 19 = 437$ then $0.0437 \div 1.9$ ”, “correct to 3 decimal places”, “upper/lower limit of $x + y$ ”, interval of x , powers of 10; दशमलव स्थान, पूर्णांकित

- $\times$: decimal hatao, multiply karo, dono ke decimal places jod kar point lagao. $\div$: dono ko same decimal places tak 10^n se multiply karke normal divide.
- **Diya hua product use karo**: digits wahi rahenge, sirf point khisko — (upar ke decimal places) — (neeche ke decimal places).
- **Rounding** (n places): $(n+1)$ -th digit $\geq 5 \Rightarrow$ upar, warna waisa hi.
- **Bounds**: “v, n decimal tak sahi” $\Rightarrow v - \frac{1}{2} \cdot 10^{-n} \leq x < v + \frac{1}{2} \cdot 10^{-n}$. Sum/product ki upper limit = upper limits se; lower limit = lower limits se.

TERMS – Har letter ka matlab

n = kitne decimal places $\cdot 10^{-n} = 0.00\dots 1$ (n jagah) $\cdot$ shift = point kitni jagah khiskega.

SOLVED – Asli book questions (verified)

Q (EN 160): $0.1404 \div 0.06$

$$\frac{14.04}{6} = 2.34 \checkmark \text{ book ans (b)}$$

Q (EN 178): $23 \times 19 = 437 \Rightarrow 0.0437 \div 1.9$

0.0437 (4 places) $\div 1.9$ (1 place) $\Rightarrow 23$ ke saath 3 places: $0.023 \checkmark \text{ book ans (c)}$

Q (EN 163): $x = 4.51$, $y = 2.48$ (2 dp sahi); $x + y$ ki upper limit

$$4.515 + 2.485 = 7.000 \checkmark \text{ book ans (a)}$$

Q (EN 157): 0.0006697 ko 3 decimal tak

4th digit $6 \geq 5 \Rightarrow 0.001 \checkmark \text{ book ans (c)}$

Q (EN 173): $484 / 48.4 = 4.84 / x$

$$x = 4.84 \times \frac{48.4}{484} = 0.484 \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (44) EN 47, 105, 127, 137, 157, 159–165, 167, 168, 171–174, 176–181, 184, 185, 187, 218, 224, 231 • **HI** 19, 86, 109, 112–115, 117, 119, 120, 122, 124, 125 | Ch 1 (Sankhya Paddhati): EN 69

DF-7 — TELESCOPING SERIES (cancel wali series)**PEHCHAN – Question mein ye dikhe**

“ $1/2 + 1/6 + 1/12 + \dots + 1/n(n+1)$ ”, “ $1/(1 \cdot 4) + 1/(4 \cdot 7) + \dots$ ”, product/sum jisme har term ek hi gap wale do numbers se bani ho

$$\frac{1}{a(a+d)} + \frac{1}{(a+d)(a+2d)} + \dots + \frac{1}{l(l+d)} = \frac{1}{d} \left(\frac{1}{a} - \frac{1}{l+d} \right)$$

$$\text{Special: } \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1} \cdot \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \dots \left(1 - \frac{1}{n}\right) = \frac{1}{n}$$

TERMS – Har letter ka matlab

a = pehla factor, d = gap ($4-1=3$), $l+d$ = aakhri factor $\cdot$ beech ke sab terms cancel ho jaate hain.

SOLVED – Asli book questions (verified)

Q (EN 102): $1/(1 \cdot 4) + 1/(4 \cdot 7) + \dots + 1/(47 \cdot 50)$

$$\frac{1}{3} \left(1 - \frac{1}{50}\right) = \frac{49}{150} \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (2) EN 99, 102

DF-8 — FRACTION EQUATION MACHINE (ansh-har wale sawaal)**PEHCHAN – Question mein ye dikhe**

“numerator is 1 less than denominator...”, “if 2 is added to numerator...”, “same number subtracted from numerator and denominator”, “numerator +30%, denominator –35%”; अंश, हर, भिन्न ज्ञात कीजिए

Sabse fast: har option ko conditions mein daal kar check karo (5–10 second).

$$\frac{a-k}{b-k} = \frac{c}{d} \Rightarrow k = \frac{bc-ad}{c-d} \quad (\text{same } k \text{ dono se ghata/joda})$$

- **% change:** numerator $+x\%$, denominator $-y\% \Rightarrow \text{naya} = \text{purana} \times \frac{100+x}{100-y}$ (ulta chahiye to ulta multiply).
- **Sirf numerator $+k$** aur fraction Δ badhi $\Rightarrow \frac{k}{d} = \Delta \Rightarrow d = \frac{k}{\Delta}$.
- Do conditions $\Rightarrow$ do equations $\Rightarrow$ NS-9 ka Cramer formula. Chain ($A = \frac{2}{3}B$, $B = \frac{1}{2}C$) $\Rightarrow C = 6$, $B = 3$, $A = 2$ maan lo.

TERMS – Har letter ka matlab

a/b = asli fraction · k = joda/ghataya number · c/d = naya fraction · Δ = kitna badha.

SOLVED – Asli book questions (verified)

Q (EN 130): 15/19 ke dono se kya ghatayein ki $3/4$

$$k = \frac{19 \cdot 3 - 15 \cdot 4}{3 - 4} = 3 \quad \checkmark \text{ book ans (d)}$$

Q (EN 131): 42/45 se kya ghatayein ki $5/6$

$$k = \frac{45 \cdot 5 - 42 \cdot 6}{5 - 6} = 27 \quad \checkmark \text{ book ans (a)}$$

Q (EN 199): num $+30\%$, den $-35\% \Rightarrow 3/15$; asli?

$$\frac{3}{15} \times \frac{65}{130} = \frac{1}{10} \quad \checkmark \text{ book ans (b)}$$

Q (EN 222): num $+6 \Rightarrow$ fraction $3/4$ badhi; denominator?

$$d = 6 \div \frac{3}{4} = 8 \quad \checkmark \text{ book ans (a)}$$

Q (HI 78): अंश + हर = 13; दोनों में $+6 \Rightarrow 8/17$; |हर – अंश|

$$17(n+6) = 8(d+6), n+d=13 \Rightarrow n=2, d=11 \Rightarrow 9 \quad \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (37) EN 130, 131, 147, 166, 169, 189, 191, 194, 195, 197, 199, 202, 203, 205, 206, 208, 209, 211, 222, 226, 227 · **HI** 73–75, 78, 79, 95, 116, 118, 127–129 | Ch 6 (Pratishatata): EN 162 | Ch 11 (Samay aur Karya): EN 117 | Ch 16 (Aayu Sambandhi Prashn): EN 217, 225, 236

DF-9 — EQUIVALENT + CONVERSION BOX

PEHCHAN – Question mein ye dikhe

equivalent fraction, “which is not equal to $4/11$ ”, lowest form, reducible, %, $g \rightarrow kg$, “what fraction of a day”, “fraction of matches lost”; तुल्य/समतुल्य भिन्न, सरलतम रूप

$$\frac{a}{b} = \frac{c}{d} \iff a \times d = b \times c \quad \text{Lowest form: dono ko HCF se divide}$$

- **Reducible** $\iff$ HCF(upar, neeche) > 1 . **Part of whole** = hissa $\div$ poora (same unit mein).
- $x\% = \frac{x}{100} \cdot 1 \text{ kg} = 1000 \text{ g} \cdot 1 \text{ km} = 1000 \text{ m} \cdot 1 \text{ din} = 24 \text{ h} = 1440 \text{ min} = 86400 \text{ s} \cdot$ terminating decimal = digits/ 10^n phir cancel.
- Kisi bhi fraction ki equivalent fractions / do decimals ke beech numbers = **infinite**.

TERMS – Har letter ka matlab

HCF = sabse bada common factor · $a/b = c/d$ cross-multiply se check.

SOLVED – Asli book questions (verified)

Q (EN 232): $60/75 = 4/x$

$$x = \frac{4 \times 75}{60} = 5 \quad \checkmark \text{ book ans (d)}$$

Q (EN 111): 7 min 12 s = din ka kitna hissa

$$\frac{432}{86400} = \frac{1}{200} \quad \checkmark \text{ book ans (c)}$$

Q (EN 198): $4/5\% = ?$

$$\frac{4}{5 \times 100} = \frac{1}{125} \quad \checkmark \text{ book ans (b)}$$

Q (EN 109): $4/11$ ke equal NAHI

$$84 \times 11 = 924 \neq 209 \times 4 = 836 \Rightarrow 84/209 \quad \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (26) EN 23, 97, 100, 103, 109, 111, 114, 115, 158, 190, 198, 200, 204, 217, 220, 221, 223, 225, 229, 230, 232 · HI 101, 102, 123, 132, 133

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
DF-1	COMPARE + ORDER MACHINE	49	15
DF-2	TERMINATING / RECURRING TEST	19	9
DF-3	DECIMAL $\leftrightarrow$ FRACTION (9-0 RULE)	36	55
DF-4	FRACTION OPERATION MACHINE (+ $- \times \div$, “kya jodein”)	47	20
DF-5	RECIPROCAL MACHINE	6	4
DF-6	DECIMAL SHIFT + ROUNDING MACHINE	30	13
DF-7	TELESCOPING SERIES (cancel wali series)	2	0
DF-8	FRACTION EQUATION MACHINE (ansh-har wale sawaal)	21	11
DF-9	EQUIVALENT + CONVERSION BOX	21	5
$\rightarrow$ IS-4	CONJUGATE + RATIONALISE + $\sqrt{A \pm 2\sqrt{B}}$ (Ch 3)	1	1
Total (book ke sawaal)		232/232 ✓	133/133 ✓
$\rightarrow$ IS-4: EN 212 HI 130			

Chapter 3

Indices and Surds (Ghatank – Karani) घातांक तथा करणी

7 formulas · English book 139 sawaal · Hindi book 103 sawaal · 32 solved examples ✓

IS-1 — SQUARE ROOT IN 10 SECONDS (unit digit + range)

PEHCHAN – Question mein ye dikhe

“square root of 519841”, “value of $\sqrt{142884}$ ”, “how many digits in the square root”, “ $\sqrt{90}$ lies between”, “ $x^2 = 841$ ”; वर्गमूल, का मान ज्ञात कीजिए

Step 1 – last digit (N ke last digit se):

N ends in	1	4	5	6	9	0 (00)
$\sqrt{N}$ ends in	1 / 9	2 / 8	5	4 / 6	3 / 7	0

Step 2 – baaki digits: last 2 digits hata do; bache number se neeche wala sabse bada square $k^2 \Rightarrow \text{root} = k_$. **Step 3 – do mein se kaun:** bacha number $\geq k(k+1) \Rightarrow$ bada candidate, warna chhota.

N ke **d** digits $\Rightarrow \sqrt{N}$ ke $\lceil d/2 \rceil$ digits. $\sqrt{N}$ kin do integers ke beech: $k^2 < N < (k+1)^2$. N ka last digit 2, 3, 7, 8 $\Rightarrow$ perfect square nahi.

TERMS – Har letter ka matlab

N = diya number · k = neeche wala nearest square root · $k(k+1) = 72 \times 73$ jaisa product (sirf compare ke liye).

SOLVED – Asli book questions (verified)

Q (EN 13): $\sqrt{519841}$

last 1 $\Rightarrow 1/9$; 5198: $72^2 = 5184 \Rightarrow 72_$; $5198 < 72 \cdot 73 = 5256 \Rightarrow$ chhota $\Rightarrow$ **721 ✓ book ans (a)**

Q (EN 34): $\sqrt{243049}$

last 9 $\Rightarrow 3/7$; 2430: $49^2 = 2401$; $2430 < 49 \cdot 50 = 2450 \Rightarrow$ **493 ✓ book ans (c)**

Q (HI 25): $\sqrt{1234321}$

last 1; 12343: $111^2 = 12321$; $12343 < 111 \cdot 112 \Rightarrow$ **1111 ✓ book ans (d)**

Q (EN 10): $\sqrt{18769}$ mein kitne digits

5 digits $\Rightarrow \lceil 5/2 \rceil = 3$ **✓ book ans (b)**

Q (EN 6): $\sqrt{90}$ kin ke beech

$81 < 90 < 100 \Rightarrow 9$ aur 10 **✓ book ans (a)**

BOOK KESAWAAL (66) EN 4–7, 9, 10, 12–27, 29–39, 76 · HI 4, 7, 10, 23, 25, 26, 29–40, 42–52, 54, 65, 78

IS-2 — DECIMAL + FRACTION ROOT MACHINE

PEHCHAN – Question mein ye dikhe

$\sqrt{0.0324}$, $\sqrt{42.25}$, $\sqrt{0.5}$, $\sqrt[3]{0.000216}$, $\sqrt{(576/625)}$, “if $\sqrt{225} = 15$ then $\sqrt{2.25} + \sqrt{0.0225} \dots$ ”; दशमलव संख्या का वर्गमूल

• **Decimal:** point se **do-do** digits ke jode (cube root: teen-teen). Root mein decimal places = aadhe (cube: tihaai). $\sqrt{0.0324} = 0.18$

• **Non-square decimal:** $\sqrt{0.5} = \frac{\sqrt{50}}{10} = 0.707$, $\sqrt{0.6} = \frac{\sqrt{60}}{10} = 0.774$, $\sqrt{0.8} = 0.894$ ($\sqrt{2} = 1.414$, $\sqrt{3} = 1.732$, $\sqrt{5} = 2.236$).

• **Fraction:** $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$; mixed number pehle improper banao ($5\frac{4}{9} = \frac{49}{9}$).

• “ **$\sqrt{225} = 15$ diya**”: sirf decimal point khisko — har 2 jagah ke liye root mein 1 jagah.

TERMS – Har letter ka matlab

decimal places = point ke baad ke digits · jode (pairs) hamesha decimal point se banao.

SOLVED – Asli book questions (verified)

Q (EN 44): $\sqrt{0.0324}$

$\sqrt{324} = 18$, 4 places $\Rightarrow$ 2 places: 0.18 ✓ **book ans (a)**

Q (EN 43): $\sqrt{0.00069169}$

$\sqrt{69169} = 263$, 8 places $\Rightarrow$ 4: 0.0263 ✓ **book ans (c)**

Q (EN 46): $\sqrt{0.5}$

$\sqrt{50}/10 = 7.07/10 = 0.707$ ✓ **book ans (a)**

Q (EN 136): $\sqrt[3]{0.000216}$

$\sqrt[3]{216} = 6$, 6 places $\Rightarrow$ 2: 0.06 ✓ **book ans (a)**

Q (EN 93): $\sqrt{(576/625)}$

$\frac{24}{25} = 0.96$ ✓ **book ans (a)**

Q (EN 61): $\sqrt{225} = 15 \Rightarrow \sqrt{2.25} + \sqrt{0.0225} + \sqrt{0.000225}$

$1.5 + 0.15 + 0.015 = 1.665$ ✓ **book ans (c)**

BOOK KESAWAAL (57) EN 40–51, 53, 54, 57, 59, 61, 62, 67, 69, 86–88, 93–95, 97, 98, 100–102, 104, 105, 108, 109, 111, 136 · **HI** 3, 9, 11, 12, 18, 22, 27, 28, 53, 56–61, 68, 73, 101–103

IS-3 — SURD SIMPLIFY + LIKE-SURD MACHINE**PEHCHAN – Question mein ye dikhe**

“mixed surd form of $\sqrt{1350}$ ”, $7\sqrt{48} + 7\sqrt{147}$, “if $\sqrt{45} + \sqrt{20} = 11.180$ then $\sqrt{180} + 4\sqrt{5} = ?$ ”, “if $\sqrt{5} = 2.236\dots$ ”; करणी

$$\sqrt{a^2 b} = a\sqrt{b} \quad p\sqrt{b} + q\sqrt{b} = (p + q)\sqrt{b}$$

- Har surd ko **sabse chhote common surd** ($\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, $\sqrt{6}\dots$) mein todo: $\sqrt{48} = 4\sqrt{3}$, $\sqrt{147} = 7\sqrt{3}$.
- “**Ek expression = value**” **diya**: dono expressions ko same surd ke multiple banao $\Rightarrow$ answer = value $\times$ (naya multiple $\div$ purana multiple).

TERMS – Har letter ka matlab

a^2 = perfect square factor · b = square-free bacha hissa · like surds = jinke andar same number.

SOLVED – Asli book questions (verified)

Q (EN 3): $\sqrt{1350}$ mixed surd

$1350 = 15^2 \times 6 \Rightarrow 15\sqrt{6}$ ✓ **book ans (d)**

Q (EN 74): $7\sqrt{48} + 7\sqrt{147}$

$28\sqrt{3} + 49\sqrt{3} = 77\sqrt{3}$ ✓ **book ans (d)**

Q (EN 81): $4\sqrt{18} + 7\sqrt{32} - 2\sqrt{50}$

$12\sqrt{2} + 28\sqrt{2} - 10\sqrt{2} = 30\sqrt{2}$ ✓ **book ans (a)**

Q (EN 70): $\sqrt{45} + \sqrt{20} = 11.180 \Rightarrow \sqrt{180} + 4\sqrt{5}$

$5\sqrt{5} = 11.18 \Rightarrow 10\sqrt{5} = 22.36$ ✓ **book ans (a)**

Q (EN 79): $\sqrt{54} + \sqrt{150} = a \Rightarrow \sqrt{96} + \sqrt{216}$

$8\sqrt{6} = a \Rightarrow 10\sqrt{6} = 1.25a$ ✓ **book ans (d)**

BOOK KESAWAAL (20) EN 3, 56, 63, 64, 70–74, 79, 81, 107 · **HI** 2, 72, 85–88, 91 | Ch 22 (Beejganit): **HI** 71

IS-4 — CONJUGATE + RATIONALISE + $\sqrt{(A \pm 2\sqrt{B})}$ **PEHCHAN – Question mein ye dikhe**

$(a + \sqrt{b})(a - \sqrt{b})$, “rationalise / rational value of denominator”, “ $\sqrt{(8 + 2\sqrt{15})} \cdot \sqrt{(8 - 2\sqrt{15})}$ ”, “sum of square roots of two integers is $\sqrt{(14 + 8\sqrt{3})}$ ”; परिमेयीकरण, हर का परिमेय मान

$$(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b \quad \frac{1}{a + \sqrt{b}} = \frac{a - \sqrt{b}}{a^2 - b}$$

$$\sqrt{A \pm 2\sqrt{B}} = \sqrt{m} \pm \sqrt{n} \quad \text{jahan} \quad m + n = A, \quad m \times n = B$$

$8\sqrt{3} = 2\sqrt{48}$ — pehle bahar ka number **2** banao, phir m, n dhoondo.

TERMS – Har letter ka matlab

conjugate = beech ka sign ulta · A = bahar wala number · B = 2 hata kar andar wala number · $m > n$.

SOLVED – Asli book questions (verified)

Q (EN 78): $\sqrt{(8 + 2\sqrt{15})} \times \sqrt{(8 - 2\sqrt{15})}$

$\sqrt{64 - 60} = 2$ ✓ book ans (b)

Q (EN 65): $\sqrt{x} + \sqrt{y} = \sqrt{(14 + 8\sqrt{3})}$; $x^2 + y^2$?

$14 + 2\sqrt{48}$; $m + n = 14$, $mn = 48 \Rightarrow 8, 6 \Rightarrow 64 + 36 = 100$ ✓ book ans (c)

Q (EN 135): $\sqrt{(18 + 8\sqrt{5})} \Rightarrow x^2 + y^2$

$18 + 2\sqrt{80}$; $10, 8 \Rightarrow 164$ ✓ book ans (a)

Q (EN 131): $1/(5 + 3\sqrt{2})$ rationalise

$\frac{5-3\sqrt{2}}{25-18} = \frac{5-3\sqrt{2}}{7}$ ✓ book ans (d)

BOOK KE SAWAAL (18) EN 2, 65, 68, 78, 89, 110, 131, 133, 135 · **HI** 1, 6, 16, 84, 90, 93, 96 | Ch 2 (Dashamlav Bhinn): **EN** 212 · **HI** 130

IS-5 — NESTED ROOT + ($a^2 - b^2$) ROOT

PEHCHAN – Question mein ye dikhe

$\sqrt{(214 + \sqrt{(107 + \sqrt{196})})}$, $\sqrt{(34^2 - 16^2)}$, $\sqrt{(544^2 - 256^2)}$, “positive square root of ...”; धनात्मक वर्गमूल

- **Andar se bahar:** sabse andar wala perfect square pehle nikalo ($\sqrt{196} = 14$), phir agla. . .
- $\sqrt{a^2 - b^2} = \sqrt{(a - b)(a + b)} \Rightarrow$ product ko perfect square mein todo: $\sqrt{18 \times 50} = \sqrt{900} = 30$.
- $\sqrt{x} \times \sqrt{y} = \sqrt{xy}$ · $\frac{\sqrt{x}}{\sqrt{y}} = \sqrt{\frac{x}{y}}$ · ghata/jod ke andar root kabhi mat todo: $\sqrt{a + b} \neq \sqrt{a} + \sqrt{b}$.

TERMS – Har letter ka matlab

nested = root ke andar root · $a^2 - b^2 = (a - b)(a + b)$.

SOLVED – Asli book questions (verified)

Q (EN 77): $\sqrt{(214 + \sqrt{(107 + \sqrt{196})})}$

$\sqrt{214 + \sqrt{121}} = \sqrt{225} = 15$ ✓ book ans (b)

Q (EN 1): $\sqrt{(34^2 - 16^2)}$

$\sqrt{18 \cdot 50} = 30$ ✓ book ans (d)

Q (EN 8): $\sqrt{(544^2 - 256^2)}$

$\sqrt{288 \cdot 800} = \sqrt{230400} = 480$ ✓ book ans (b)

BOOK KE SAWAAL (23) EN 1, 8, 11, 28, 52, 77, 92, 96, 103, 106 · **HI** 5, 13–15, 17, 21, 24, 41, 67, 70, 79, 89, 92

IS-6 — POWER LAWS (ghatank ke niyam)

PEHCHAN – Question mein ye dikhe

$6^{x+4} = 6^7$, $2^x \cdot 4^{12} \cdot 8^3 = 16^{11}$, $8^{2/3}$, 3^{-5} , “ $(\sqrt{5})^n = 625$ ”, $\sqrt{(3^{38} + 3^{39})}$; घात, घातांक

$$a^m a^n = a^{m+n} \quad \frac{a^m}{a^n} = a^{m-n} \quad (a^m)^n = a^{mn} \quad a^{-n} = \frac{1}{a^n} \quad a^0 = 1 \quad a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

- **Equation:** dono taraf **same base** (2, 3, 5...) banao $\Rightarrow$ powers barabar rakh do.
- **Sum of powers:** $a^m + a^{m+1} = a^m(1 + a) \Rightarrow \sqrt{3^{38} + 3^{39}} = \sqrt{3^{38} \cdot 4} = 2 \cdot 3^{19}$.
- Yaad: $0.25 = 2^{-2}$, $0.125 = 2^{-3}$, $0.008 = 5^{-3}$, $625 = 5^4$, $729 = 3^6 = 9^3$, $1024 = 2^{10}$.

TERMS – Har letter ka matlab

a = base, m, n = powers · same base banana = sab ko 2 ya 3 ya 5 ki power mein likhna.

SOLVED – Asli book questions (verified)

Q (EN 115): $2^x \cdot 4^{12} \cdot 8^3 = 16^{11}$

$x + 24 + 9 = 44 \Rightarrow x = 11$ ✓ **book ans (b)**

Q (EN 112): $6^{x+4} = 6^7$

$x + 4 = 7 \Rightarrow 3$ ✓ **book ans (d)**

Q (EN 126): $8^{2/3}$

$(\sqrt[3]{8})^2 = 4$ ✓ **book ans (c)**

Q (EN 124): $(\sqrt{5})^n = 625$

$5^{n/2} = 5^4 \Rightarrow n = 8$ ✓ **book ans (c)**

Q (HI 81): $(-2)^4 \div (-2)^6$

$(-2)^{-2} = \frac{1}{4}$ ✓ **book ans (a)**

Q (EN 134): $\sqrt{(3^{38} + 3^{39})}$

$\sqrt{3^{38} \cdot 4} = 2 \cdot 3^{19}$ ✓ **book ans (c)**

BOOK KE SAWAAL (39) EN 112–120, 122, 124–129, 134, 137, 138 · **HI** 64, 66, 71, 75–77, 80–83, 95, 97 | Ch 22 (Beejganit): **EN** 254, 255, 258, 291 · **HI** 19, 41 | Ch 28 (Vividh): **EN** 12, 57

IS-7 — ROOT EQUATION + WORD MACHINE

PEHCHAN – Question mein ye dikhe

“ $\sqrt{x + x} = 90$ ”, “ $1849 \times x = 2451 (\sqrt{\quad})$ ”, “each member gives as much money as the number of members”, “ $1/3 - 1/4$ of a number = its square root”; x का मान

- **Options daalo:** $\sqrt{\quad}$ wali equation mein har option daal kar check karo (perfect squares se shuru).
- **Har aadmi utna de jitne log** $\Rightarrow \log = \sqrt{\text{total (paise mein)}} \cdot \text{rows} = \text{columns} \Rightarrow \sqrt{N}$.
- $k \cdot x = \sqrt{x} \Rightarrow x = \frac{1}{k^2} \cdot \sqrt{a} \times \sqrt{x} = \sqrt{b} \Rightarrow x = \frac{b}{a} \cdot \frac{\sqrt{x}}{c} = d \Rightarrow x = (cd)^2$.

TERMS – Har letter ka matlab

x = anjaan number · total ko paise mein badlo ($\text{₹}62.41 = 6241$ paise).

SOLVED – Asli book questions (verified)

Q (EN 139): har member utne ₹ de jitne members; total ₹62.41

$\sqrt{6241} = 79$ ✓ **book ans (d)**

Q (EN 130): $(1/3 - 1/4)$ of number = $\sqrt{\text{number}}$

$\frac{x}{12} = \sqrt{x} \Rightarrow x = 144$ ✓ **book ans (b)**

Q (EN 83): $\sqrt{x + x} = 90$

option 81: $9 + 81 = 90$ ✓ **book ans (a)**

BOOK KE SAWAAL (26) EN 55, 60, 66, 75, 80, 82–85, 90, 91, 99, 121, 130, 132, 139 · **HI** 8, 19, 55, 62, 63, 69, 94, 98–100

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
IS-1	SQUARE ROOT IN 10 SECONDS (unit digit + range)	34	32
IS-2	DECIMAL + FRACTION ROOT MACHINE	37	20
IS-3	SURD SIMPLIFY + LIKE-SURD MACHINE	12	7
IS-4	CONJUGATE + RATIONALISE + $\sqrt{(A \pm 2\sqrt{B})}$	9	7
IS-5	NESTED ROOT + $(a^2 - b^2)$ ROOT	10	13
IS-6	POWER LAWS (ghatank ke niyam)	19	12
IS-7	ROOT EQUATION + WORD MACHINE	16	10
→ NS-10	SERIES + PLACE-VALUE + RATIONAL BOX (ratta facts (Ch 1)	1	0
→ SIM-3	APPROXIMATION MACHINE (“closest approximate valu (Ch 4)	0	2
→ SIM-4	IDENTITY CANCEL TABLE (bada calculation kabhi na (Ch 4)	1	0
Total (book ke sawaal)		139/139 ✓	103/103 ✓
→ NS-10: EN 123 · → SIM-3: HI 20, 74 · → SIM-4: EN 58			

Chapter 4

Simplification (Saralikaran) सरलीकरण

8 formulas · English book 248 sawaal · Hindi book 243 sawaal · 40 solved examples ✓

SIM-1 — VBODMAS MACHINE (order of operations)

PEHCHAN – Question mein ye dikhe

lamba expression with brackets () { } [], “of”, $\div$, $\times$, negative signs; “which two signs interchanged”; सरल कीजिए, का मान ज्ञात कीजिए, “का”

Vinculum/Bracket $\rightarrow$ Of $\rightarrow$ Division $\rightarrow$ Multiplication $\rightarrow$ Addition $\rightarrow$ Subtraction

- Brackets **andar se bahar**: () $\rightarrow$ { } $\rightarrow$ []. “of” (का) = $\times$, par $\div$ se **pehle**.
- $\div$ aur $\times$ barabar mile $\Rightarrow$ **left se right**. Sign: $(-)(-) = +$, $(-)(+) = -$, $-(a - b) = -a + b$.
- Sign swap** wale sawaal: har option ke do sign badal kar seedha check karo. Variable wale (x, y, z) mein bhi yahi order.

TERMS – Har letter ka matlab

V = vinculum (line ke upar wala bracket) · of = ka (multiply) · left $\rightarrow$ right = jo pehle aaye wo pehle.

SOLVED – Asli book questions (verified)

Q (EN 46): $3 + [3 \times \{3 - (3 + 3) \div 6\}]$
 $(3 + 3) \div 6 = 1 \Rightarrow 3 \times 2 = 6 \Rightarrow 9$ ✓ **book ans (b)**

Q (EN 43): $72 \div 4 \times \{8 \times 4 - (14 - 19)\}$
 $\{32 + 5\} = 37$; $72 \div 4 = 18$; $18 \times 37 = 666$ ✓ **book ans (a)**

Q (HI 43): $[65 + \{31 - (45 \text{ का } 3 \div 5)\}] \times 4$
 $45 \times 3 = 135$, $\div 5 = 27$; $31 - 27 = 4$; $65 + 16 = 81$ ✓ **book ans (a)**

Q (EN 62): $2550 - [510 - \{270 - (90 - 80 + 70)\}]$
 $80 \rightarrow 190 \rightarrow 320 \Rightarrow 2230$ ✓ **book ans (b)**

BOOK KE SAWAAL (121) EN 10, 17, 29, 34, 35, 39–53, 55, 56, 58, 60–81, 129, 134, 236 · **HI** 4, 5, 8, 9, 17, 20–23, 25, 26, 30–32, 34, 37, 39–50, 52, 54–63, 65–76, 78–91, 93, 94, 100, 101, 186, 236 | Ch 1 (Sankhya Paddhati): EN 337 | Ch 28 (Vividh): EN 20

SIM-2 — DECIMAL COLUMN + “?” MACHINE

PEHCHAN – Question mein ye dikhe

decimals ka jod/ghata ($919 + 9.019 + \dots$), “?” = equation, “if $19 \times 23 = 437$ then 190×0.023 ”, kg–g, hours $\rightarrow$ h m s; ? के स्थान पर

- Jod/ghata**: decimal point ek line mein (khaali jagah 0) — column wise.
- “?” nikalna**: ? ko akela chhodo, baaki ko ulta operation laga kar doosri taraf bhejo: $a + ? = b \Rightarrow ? = b - a$; $a \times ? = b \Rightarrow ? = b \div a$; $\sqrt{?} = k \Rightarrow ? = k^2$.
- Diya hua product** $\Rightarrow$ digits same, sirf decimal points gino (DF-6).
- Units**: $1 \text{ kg} = 1000 \text{ g} \Rightarrow 6 \text{ kg } 5 \text{ g} = 6.005 \text{ kg}$; $0.84 \text{ h} = 0.84 \times 60 = 50.4 \text{ min} = 50 \text{ min } 24 \text{ s}$.

TERMS – Har letter ka matlab

? = anjaan number · ulta operation: $+$ $\leftrightarrow$ $-$, $\times$ $\leftrightarrow$ $\div$, $\sqrt{}$ $\leftrightarrow$ square.

SOLVED – Asli book questions (verified)**Q (EN 37):** $9876 + 34.567 - ? = 9908.221$ $? = 9910.567 - 9908.221 = 2.346$ ✓ **book ans (d)****Q (EN 140):** $19 \times 23 = 437 \Rightarrow 190 \times 0.023$ 1 zero aage + 3 decimal $\Rightarrow$ 2 decimal: 4.37 ✓ **book ans (d)****Q (EN 28):** 6 kg 5 g = ? kg $6 + \frac{5}{1000} = 6.005$ ✓ **book ans (d)****Q (EN 239):** 8.17 hours = h m s $0.17 \times 60 = 10.2$ min, $0.2 \times 60 = 12$ s $\Rightarrow$ 8 h 10 min 12 s ✓ **book ans (b)****BOOK KE SAWAAL (100)** EN 1, 2, 5–9, 11, 13–16, 18, 19, 21, 24–28, 30, 32, 33, 37, 94, 96, 101, 102, 106, 116, 118, 119, 121, 124, 126–128, 130–133, 135–138, 140, 146, 149, 151, 160, 176, 177, 181, 183, 188, 189, 191, 192, 195, 199, 201, 210, 237, 239 · **HI** 97–99, 102–106, 109, 124, 135–137, 144, 148, 159, 175–179, 182–185, 187–189, 191–196 | Ch 1 (*Sankhya Paddhati*): EN 342, 345**SIM-3 — APPROXIMATION MACHINE (“closest approximate value”)****PEHCHAN – Question mein ye dikhe**

“closest approximate value”, “लगभग कितना मान”, “निकटतम लगभग मान”, numbers like 19.99, 24.93, 15.01

Har number ko nearest whole (ya aasaan) number bana do $\Rightarrow$ VBODMAS $\Rightarrow$ sabse paas wala option

- 19.99 $\rightarrow$ 20, 10.01 $\rightarrow$ 10, 49.93 $\rightarrow$ 50, 5.1 $\rightarrow$ 5; % of wale: 20.01% of 320 $\rightarrow$ 20% of 320 = 64.
- Division mein numbers ko aise round karo ki poora kate ($325 \div 17.99 \rightarrow 324 \div 18 = 18$).

TERMS – Har letter ka matlab

approx = lagbhag · options door hote hain, isliye chhota error chalta hai.

SOLVED – Asli book questions (verified)**Q (HI 115):** $19.99 - 10.01 + 24.99 \times 10.01$ $20 - 10 + 25 \times 10 = 260$ ✓ **book ans (a)****Q (HI 118):** $29.99 + 15.01 + 44.99 \times 15.01$ $30 + 15 + 675 = 720$ ✓ **book ans (c)****Q (HI 33):** 20.01% of 320.002 + 121.99 $64 + 122 = 186$ ✓ **book ans (a)****Q (EN 148):** $18.96 + 12.96 + 15.16 - 17.89 \times 2.04 + 49.93 \div 5.1$ $19 + 13 + 15 - 18 \times 2 + 50 \div 5 = 21$ ✓ **book ans (d)****BOOK KE SAWAAL (59)** EN 3, 4, 95, 147, 148 · **HI** 33, 38, 108, 110–113, 115, 117–122, 125, 127–130, 132–134, 138, 139, 141, 143, 145–147, 149–158, 161–165, 167, 169, 170, 172–174, 180, 235 | Ch 3 (*Ghatank – Karani*): **HI** 20, 74**SIM-4 — IDENTITY CANCEL TABLE (bada calculation kabhi nahi)****PEHCHAN – Question mein ye dikhe**same numbers repeat: $a \times a - b \times b$, $(a+b)^2 - (a-b)^2$, $a^3 + b^3$, $x + y + z$ & $xy + yz + zx$, 9991×10009 , $(0.11^2 + \dots)/(0.011^2 + \dots)$; सरल कीजिए

Ye dikhe	Turant likho	Ye dikhe	Turant likho
$\frac{a^2 - b^2}{a - b}$	$a + b$	$a^2 + b^2 \pm 2ab$	$(a \pm b)^2$
$(a + b)^2 - (a - b)^2$	$4ab$	$(a + b)^2 + (a - b)^2$	$2(a^2 + b^2)$
$\frac{a^3 + b^3}{a^2 - ab + b^2}$	$a + b$	$\frac{a^3 - b^3}{a^2 + ab + b^2}$	$a - b$
$a^3 \pm 3a^2b + 3ab^2 \pm b^3$	$(a \pm b)^3$	$a + b + c = 0$	$a^3 + b^3 + c^3 = 3abc$
$x^2 + y^2 + z^2$	$(x + y + z)^2 - 2(xy + yz + zx)$	$(A - x)(A + x)$	$A^2 - x^2$

Scale trick: upar-neeche same digits, sirf decimal khiska $\Rightarrow$ answer = 10^{2k} (squares) / 10^{3k} (cubes).

TERMS – Har letter ka matlab

a, b, c = koi bhi numbers (ya decimals) · k = decimal kitna khiska.

SOLVED – Asli book questions (verified)

Q (EN 84): $(3.7)^3 - 3(3.7)^2(0.7) + 3(3.7)(0.7)^2 - (0.7)^3$
 $(3.7 - 0.7)^3 = 27$ ✓ **book ans (b)**

Q (EN 98): $73 \times 73 + 42 \times 42 - 2 \times 73 \times 42$
 $(73 - 42)^2 = 961$ ✓ **book ans (a)**

Q (HI 13): $(183^3 - 73^3)/(183^2 + 183 \cdot 73 + 73^2)$
 $183 - 73 = 110$ ✓ **book ans (b)**

Q (EN 86): $x + y + z = 11$, $xy + yz + zx = 42 \Rightarrow x^2 + y^2 + z^2$
 $121 - 84 = 37$ ✓ **book ans (b)**

Q (EN 20): $(-20)^3 + 13^3 + 7^3$
sum 0 $\Rightarrow 3(-20)(13)(7) = -5460$ ✓ **book ans (c)**

Q (HI 11): 9991×10009
 $10000^2 - 9^2 = 99999919$ ✓ **book ans (d)**

Q (EN 156): $(0.11^2 + 0.06^2 + 0.031^2)/(0.011^2 + 0.006^2 + 0.0031^2)$
decimal 1 jagah $\Rightarrow 10^2 = 100$ ✓ **book ans (c)**

BOOK KE SAWAAL (52) EN 20, 82–86, 88, 90, 92, 93, 98, 99, 112, 120, 122, 123, 125, 141, 142, 156, 190, 238, 241, 243, 244, 246–248 · **HI** 6, 11, 13, 14, 18, 29, 92, 107, 114, 123, 126, 131, 140, 142, 160, 166, 168, 190, 197, 233, 234, 243 | *Ch 3 (Ghatank – Karani):* **EN** 58 | *Ch 28 (Vividh):* **EN** 21

SIM-5 — NEAR-BASE MULTIPLY MACHINE

PEHCHAN – Question mein ye dikhe

97×97 , 1093^2 , 95×103 , 66666×9999 , 999^3 , “ $4.6 \times 13.5 + 5.4 \times 13.5$ ”

$$(B \pm d)^2 = B^2 \pm 2Bd + d^2 \quad (B + x)(B + y) = B^2 + B(x + y) + xy$$

- $N \times \underbrace{99 \dots 9}_k = N \times 10^k - N \cdot a \cdot c + b \cdot c = (a + b) \cdot c$ (common nikaalo)
- $(B - 1)^3 = B^3 - 3B^2 + 3B - 1 \Rightarrow 999^3 = 997002999$

TERMS – Har letter ka matlab

B = paas ka aasaan base (100, 1000, 1100) · d, x, y = base se farak (+ ya -).

SOLVED – Asli book questions (verified)**Q (EN 23):** 97×97

$$(100 - 3)^2 = 10000 - 600 + 9 = 9409 \checkmark \text{ book ans (b)}$$

Q (EN 38): 1093×1093

$$(1100 - 7)^2 = 1210000 - 15400 + 49 = 1194649 \checkmark \text{ book ans (a)}$$

Q (HI 10): 95×103

$$100^2 + 100(-5 + 3) - 15 = 9785 \checkmark \text{ book ans (d)}$$

Q (EN 36): 66666×9999

$$666660000 - 66666 = 666593334 \checkmark \text{ book ans (d)}$$

Q (EN 12): $4.6 \times 13.5 + 5.4 \times 13.5$

$$(4.6 + 5.4) \times 13.5 = 135 \checkmark \text{ book ans (b)}$$

Q (HI 217): 999^3

$$10^9 - 3 \cdot 10^6 + 3 \cdot 10^3 - 1 = 997002999 \checkmark \text{ book ans (d)}$$

BOOK KESAWAAL (6) EN 12, 23, 36, 38 · HI 10, 217**SIM-6 — FRACTION CHAIN + CONTINUED FRACTION + SERIES****PEHCHAN – Question mein ye dikhe**mixed fractions with $\div \times$ of, $1/(2 + 1/(3 + \dots))$, “ $1/(5 \times 8) + 1/(8 \times 11) + \dots$ ”, $1 + 2 + \dots + n + \dots + 1$, $176 + 17.6 + 1.76 \dots$; भिन्न

- Mixed $\rightarrow$ improper pehle; “of” = $\times$; $\div = \times$ ulta. Phir VBODMAS.
- Continued fraction** $(a + \frac{1}{b + \frac{1}{c}})$: sabse neeche se upar hal karo.
- Telescoping**: $\frac{1}{a(a+d)} + \dots + \frac{1}{l(l+d)} = \frac{1}{d} \left(\frac{1}{a} - \frac{1}{l+d} \right)$ (DF-7).
- Mountain sum**: $1 + 2 + \dots + n + \dots + 2 + 1 = n^2 \cdot a + (r-1)a + (r-1)ar + \dots$ (n terms) = $a r^{n-1}$.

TERMS – Har letter ka matlabcontinued fraction = fraction ke andar fraction $\cdot 1$ = aakhri term ka pehla factor.**SOLVED – Asli book questions (verified)****Q (EN 164):** $1/(2 + 1/(3 + 1/(1 + 1/4)))$

$$1 + \frac{1}{4} = \frac{5}{4} \rightarrow \frac{4}{5} \rightarrow \frac{19}{5} \rightarrow \frac{5}{19} \rightarrow \frac{43}{19} \rightarrow \frac{19}{43} \checkmark \text{ book ans (a)}$$

Q (EN 198): $(1 + 2/3) \div [(1 + 1/3) \div (2/3 + 1)]$

$$\frac{5}{3} \div \frac{4}{5} = \frac{25}{12} \checkmark \text{ book ans (d)}$$

Q (EN 168): $1/(5 \cdot 8) + 1/(8 \cdot 11) + 1/(11 \cdot 14) + 1/(14 \cdot 17)$

$$\frac{1}{3} \left(\frac{1}{5} - \frac{1}{17} \right) = \frac{4}{85} \checkmark \text{ book ans (b)}$$

Q (EN 22): $1 + 2 + \dots + 31 + 30 + \dots + 1$

$$31^2 = 961 \checkmark \text{ book ans (c)}$$

Q (EN 31): $4 + 3 \cdot 4 + 3 \cdot 4^2 + \dots + 3 \cdot 4^5$

$$a = 4, r = 4, 6 \text{ terms} \Rightarrow 4^6 \checkmark \text{ book ans (b)}$$

BOOK KESAWAAL (59) EN 22, 31, 54, 57, 139, 143–145, 150, 152–155, 161–175, 178–180, 182, 184–187, 193, 194, 196–198, 200 · HI 7, 12, 15, 16, 19, 24, 27, 28, 35, 36, 51, 64, 95, 96, 116, 171, 199**SIM-7 — RATIO SUBSTITUTION MACHINE****PEHCHAN – Question mein ye dikhe**“if $x : y = 2 : 3$ then $(5x + 3y) : (5x - 3y)$ ”, “ $x/6 = 2y/7$ find $\dots$ ”, “ $0.25a = 0.05b$ ”; अनुपात

$$\frac{x}{y} = \frac{a}{b} \Rightarrow \text{seedha } x = a, y = b \text{ daal do (k cancel ho jaata hai)}$$

- $px = qy \Rightarrow x : y = q : p$ ($0.25a = 0.05b \Rightarrow a : b = 1 : 5$)
- Expression mein har term same degree ho (sab linear ya sab square) tabhi ye chalta hai — RRB mein hamesha hota hai.

TERMS – Har letter ka matlab

x, y = variables · $a : b$ = diya hua ratio · k = common multiplier (cancel).

SOLVED – Asli book questions (verified)

Q (EN 234): $x : y = 2 : 3 \Rightarrow (5x + 3y) : (5x - 3y)$

$(10 + 9) : (10 - 9) = 19 : 1$ ✓ book ans (d)

Q (EN 235): $a : b = 3 : 2 \Rightarrow (7a + 9b) : (5a + 7b)$

$39 : 29$ ✓ book ans (c)

Q (HI 181): $0.25a = 0.05b \Rightarrow (b - a)/(b + a)$

$a = 1, b = 5 \Rightarrow \frac{4}{6} = 0.67$ ✓ book ans (a)

Q (EN 115): $x/6 = y/5 \Rightarrow (x^2 + y^2)/(x^2 - y^2)$

$\frac{36+25}{36-25} = \frac{61}{11}$ ✓ book ans (b)

BOOK KE SAWAAL (14) EN 87, 97, 113–115, 117, 157, 233–235 · **HI** 181 | Ch 22 (Beejganit): EN 264 · **HI** 43, 45

SIM-8 — POWER + ROOT SHORTCUTS (infinite root bhi)

PEHCHAN – Question mein ye dikhe

$5^{n+2} = 3125, 4^{2n+1} = 2^{3n+9}, 3^{x-2} + 3^x = 7290, \sqrt{(56 + \sqrt{(56 + \dots)})}, \sqrt{(30 - \sqrt{(30 - \dots)})}$, surds in simplification; घात, वर्गमूल

- **Power laws (IS-6):** same base banao $\Rightarrow$ powers barabar. $a^m + a^{m+k} = a^m(1 + a^k) \Rightarrow$ common nikaalo.
- **Infinite roots:** $\sqrt{N + \sqrt{N + \dots}} = k$ jahan $k(k-1) = N$ (bada factor) · $\sqrt{N - \sqrt{N - \dots}} = k$ jahan $k(k+1) = N$ (chhota factor) · $\sqrt{N} \sqrt{N} \sqrt{N} \dots = N$.
- **Surds:** IS-3 (like surds), IS-4 (rationalise) — phir VBODMAS.

TERMS – Har letter ka matlab

N = andar ka number · $k(k-1)$ = do consecutive numbers ka product ($56 = 8 \times 7$).

SOLVED – Asli book questions (verified)

Q (EN 215): $\sqrt{(56 + \sqrt{(56 + \dots)})}$

$56 = 8 \times 7 \Rightarrow 8$ ✓ book ans (d)

Q (EN 221): $\sqrt{(30 - \sqrt{(30 - \dots)})}$

$30 = 5 \times 6 \Rightarrow 5$ ✓ book ans (a)

Q (HI 238): $\sqrt{(30 + \sqrt{(30 + \dots)})}$

$30 = 6 \times 5 \Rightarrow 6$ ✓ book ans (c)

Q (HI 203): $5^{n+2} = 3125$

$3125 = 5^5 \Rightarrow n = 3$ ✓ book ans (d)

Q (HI 211): $4^{2n+1} = 2^{3n+9}$

$4n + 2 = 3n + 9 \Rightarrow n = 7$ ✓ book ans (b)

Q (HI 225): $3^{x-2} + 3^x = 7290$

$3^{x-2}(1 + 9) = 7290 \Rightarrow 3^{x-2} = 729 = 3^6 \Rightarrow x = 8$ ✓ book ans (a)

BOOK KE SAWAAL (91) EN 59, 89, 91, 100, 103–105, 107–111, 158, 159, 202–209, 211–232, 240, 242, 245 · **HI** 1–3, 53, 77, 198, 200–216, 218–232, 237–242

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
SIM-1	VBODMAS MACHINE (order of operations)	48	71
SIM-2	DECIMAL COLUMN + “?” MACHINE	64	34
SIM-3	APPROXIMATION MACHINE (“closest approximate value”)	5	52
SIM-4	IDENTITY CANCEL TABLE (bada calculation kabhi nahi)	28	22
SIM-5	NEAR-BASE MULTIPLY MACHINE	4	2
SIM-6	FRACTION CHAIN + CONTINUED FRACTION + SERIES	42	17
SIM-7	RATIO SUBSTITUTION MACHINE	10	1
SIM-8	POWER + ROOT SHORTCUTS (infinite root bhi)	47	44
Total (book ke sawaal)		248/248 ✓	243/243 ✓

Chapter 5

LCM & HCF (L.S. – M.S.) लघुत्तम समापवर्त्य एवं महत्तम समापवर्तक

8 formulas · English book 231 sawaal · Hindi book 126 sawaal · 34 solved examples ✓

LH-1 — POWER RULE (HCF/LCM nikaalna)

PEHCHAN – Question mein ye dikhe

“find the LCM/HCF of ...”, prime-power form ($2^4 \times 3^4 \dots$), “LCM of 4, 6 and x cannot be”, HCF of $(3^{45}-1)$ and $(3^{35}-1)$, consecutive/co-prime; लघुत्तम समापवर्त्य (ल.स.), महत्तम समापवर्तक (म.स.)

Prime factorize karo:

HCF = common primes ki **MIN** power LCM = saare primes ki **MAX** power

- Bade do numbers ka HCF: **bade ko chhote se divide, remainder se divisor ko...** jab tak 0 (Euclid).
- Consecutive / co-prime $\Rightarrow$ HCF = 1, LCM = product. $\text{HCF}(a^m - 1, a^n - 1) = a^{\text{HCF}(m,n)} - 1$.
- “**LCM cannot be**”: LCM hamesha diye numbers ke LCM ka **multiple** hota hai (4, 6 $\Rightarrow$ 12 ka multiple). Algebraic ($x^3 y^2 \dots$): same min/max power rule.
- Sabse chhota perfect square** jo sab se kate: LCM lo, jin primes ki power odd hai unse ek baar aur multiply (NS-5).

TERMS – Har letter ka matlab

HCF = sabse bada common factor (M.S./GCD) · LCM = sabse chhota common multiple (L.S.) · min/max = sabse chhoti/badi power.

SOLVED – Asli book questions (verified)

Q (EN 1): LCM of 98, 28, 112

$2 \cdot 7^2, 2^2 \cdot 7, 2^4 \cdot 7 \Rightarrow 2^4 \cdot 7^2 = 784$ ✓ **book ans (a)**

Q (EN 54): $x = 2^3 3^{25} \cdot 7^3, y = 2^2 3^3 5^2 7^2, z = 2^4 3 \cdot 5^3 7$ ka HCF

$2^2 \cdot 3 \cdot 5 \cdot 7 = 420$ ✓ **book ans (c)**

Q (EN 85): HCF of 2189 and 2587

$2587 - 2189 = 398, 2189 = 5 \cdot 398 + 199, 398 = 2 \cdot 199 \Rightarrow 199$ ✓ **book ans (c)**

Q (EN 11): LCM of 4, 6 and x CANNOT be

12 ka multiple nahi: 18 ✓ **book ans (b)**

Q (EN 71): HCF of $(3^{45} - 1)$ and $(3^{35} - 1)$

$3^{\text{HCF}(45,35)} - 1 = 3^5 - 1 = 242$ ✓ **book ans (b)**

BOOK KE SAWAAL (86) EN 1–4, 6–17, 19–23, 52, 54–57, 60, 61, 68, 70–81, 85, 88, 90, 128, 134, 140, 173, 176, 199, 201, 202, 215, 224, 225, 231 · **HI** 1–5, 7–11, 26, 29–45, 94, 126

LH-2 — LCM + REMAINDER MACHINE

PEHCHAN – Question mein ye dikhe

“least number which when divided by 9, 8, 10, 12 leaves remainder 3”, “remainders 9, 8, 7 when divided by 10, 9, 8”, “...and divisible by 13”, “largest 4-digit ...leaves remainder”, “when increased/reduced by 6 is divisible by”; छोटी से छोटी संख्या, शेषफल

Same remainder R : $N = k \cdot \text{LCM} + R$ **Same gap ($d_i - r_i = g$):** $N = k \cdot \text{LCM} - g$

- Sabse chhota: $k = 1$ (R ho to $\text{LCM} + R$). “x badhane par divisible” $\Rightarrow \text{LCM} - x$; “x ghatane par” $\Rightarrow \text{LCM} + x$.
- Upar se ek aur divisor m** (poora kate): $k = 1, 2, 3 \dots$ daalo jab tak $k \cdot \text{LCM} + R, m$ se kate.
- Largest/smallest n-digit:** 99...9 (ya 10...0) ke paas LCM ka multiple (NS-3) phir + R . Range (400–500) mein saare: LCM ke multiples + R .

TERMS – Har letter ka matlab

$R = \text{har baar bacha remainder} \cdot g = (\text{divisor} - \text{remainder}), \text{ sab mein same} \cdot k = 1, 2, 3 \dots$

SOLVED – Asli book questions (verified)

Q (EN 27): 9, 8, 10, 12 se rem 3

$\text{LCM} = 360 \Rightarrow 363 \checkmark \text{ book ans (c)}$

Q (EN 39): 10, 9, 8 se rem 9, 8, 7

$\text{gap } 1 \Rightarrow 360 - 1 = 359 \checkmark \text{ book ans (d)}$

Q (EN 36): 5, 6, 8, 9, 12 se rem 1, aur 13 se poor

$360k + 1: k = 10 \Rightarrow 3601 (\div 13 \checkmark) \checkmark \text{ book ans (c)}$

Q (EN 35): Largest 4-digit, 7, 9, 11 se rem 5

$\text{LCM } 693; 693 \times 14 = 9702 \Rightarrow 9707 \checkmark \text{ book ans (a)}$

Q (EN 40): 6 ghatane par 12, 15, 20, 27 se divisible

$540 + 6 = 546 \checkmark \text{ book ans (c)}$

BOOK KE SAWAAL (78) EN 5, 24–50, 197, 198, 200, 203–206, 209–214, 216–221, 223, 226–230 · **HI** 6, 12–17, 19–25, 110, 116, 119–123, 125 | Ch 1 (*Sankhya Paddhati*): **HI** 153 | Ch 28 (*Vividh*): **EN** 76 · **HI** 59

LH-3 — HCF + REMAINDER MACHINE**PEHCHAN – Question mein ye dikhe**

“greatest number which divides 2000 and 2200 leaving 22 and 38”, “largest number that divides ... leaving the same remainder”; बड़ी से बड़ी संख्या, समान शेषफल

Alag remainders: $\text{HCF}(a - r_1, b - r_2, c - r_3)$ Same remainder (anjaan): $\text{HCF}(|a - b|, |b - c|)$

Same remainder **diya** (R) $\Rightarrow \text{HCF}(a - R, b - R, c - R)$.

TERMS – Har letter ka matlab

$a, b, c = \text{diye numbers} \cdot r_1, r_2, r_3 = \text{unke remainders} \cdot |a - b| = \text{farak (hamesha positive)}$.

SOLVED – Asli book questions (verified)

Q (EN 91): 2000, 2200 ko rem 22, 38

$\text{HCF}(1978, 2162) = 46 \checkmark \text{ book ans (a)}$

Q (EN 97): 2270, 3739, 6677 same remainder

$\text{HCF}(1469, 2938) = 1469 \checkmark \text{ book ans (d)}$

Q (EN 94): 55, 72, 123 rem 3, 7, 6

$\text{HCF}(52, 65, 117) = 13 \checkmark \text{ book ans (a)}$

BOOK KE SAWAAL (30) EN 91–99, 101–103, 177, 207, 208 · **HI** 18, 47–60

LH-4 — HCF APPLICATION (sabse bada naap / kam se kam hisse)**PEHCHAN – Question mein ye dikhe**

largest measuring can / rod / tile / carton, “maximum number of students/packs”, “minimum number of rows/rooms/sections (one variety only)”, square tiles; अधिकतम लंबाई, कंटेनर, कमरों की न्यूनतम संख्या

Bada naap / har hisse ka size = $\text{HCF}(q_1, q_2, q_3)$ Kam se kam hisse (rows/rooms) = $\frac{q_1 + q_2 + q_3}{\text{HCF}}$

- **Square tiles** ($L \times B$): tile side = $\text{HCF}(L, B)$, tiles = $\frac{L \times B}{\text{HCF}^2}$ (pehle same unit — cm).
- “k cheezein bach gayi” $\Rightarrow$ pehle har quantity se k ghatao, phir HCF.

TERMS – Har letter ka matlab

q = har cheez ki quantity · HCF = har packet/row/can mein kitna · L, B = length, breadth.

SOLVED – Asli book questions (verified)

Q (EN 63): 72, 90, 144 L ke liye bada can

HCF = 18 L ✓ **book ans (b)**

Q (EN 83): 188, 282, 423 students: min sections

HCF $47 \Rightarrow 4 + 6 + 9 = 19$ ✓ **book ans (c)**

Q (EN 179): 18 m 72 cm $\times$ 13 m 20 cm: kam se kam square tiles

HCF(1872, 1320) = 24 $\Rightarrow \frac{1872 \cdot 1320}{24^2} = 4290$ ✓ **book ans (b)**

Q (EN 87): 50, 80, 65 mein se 5-5 bache; students?

HCF(45, 75, 60) = 15 ✓ **book ans (c)**

BOOK KE SAWAAL (29) EN 51, 53, 58, 62–66, 69, 82–84, 86, 87, 89, 100, 179–182 · **HI** 28, 46, 112, 113, 115, 117, 124 | Ch 1 (Sankhya Paddhati): **EN** 154 · **HI** 155

LH-5 — LCM APPLICATION (bells / lights / track — saath kab)**PEHCHAN – Question mein ye dikhe**

bells toll / lights change / clocks ring together, circular track start point par milna, “how many times together between 8 and 10”; घंटियाँ एक साथ, ट्रैफिक लाइट

Dobara saath = start time + LCM(t_1, t_2, t_3) T samay mein kitni baar = $\left\lfloor \frac{T}{\text{LCM}} \right\rfloor$ (+1 agar shuru wala bhi gina)

TERMS – Har letter ka matlab

t = har ek ka interval (same unit mein) · T = poora samay · +1 = starting wali ghanti.

SOLVED – Asli book questions (verified)

Q (EN 183): 3, 6, 12, 15, 18 s; 9:58:45 par saath $\Rightarrow$ agla?

LCM 180 s = 3 min $\Rightarrow$ 10:01:45 ✓ **book ans (b)**

Q (EN 193): 16, 24, 36, 42 min

LCM = 1008 min ✓ **book ans (d)**

Q (HI 106): 15, 24, 42 s; 8 से 10 बजे तक कितनी बार (8 बजे सहित)

LCM 840 s; $\lfloor 7200/840 \rfloor + 1 = 8 + 1 = 9$ ✓ **book ans (a)**

Q (EN 188): Track: 42, 63, 84 s

LCM = 252 s ✓ **book ans (b)**

BOOK KE SAWAAL (21) EN 183–196 · **HI** 106–109, 111, 114, 118

LH-6 — PRODUCT LINK (HCF $\times$ LCM = $a \times b$)**PEHCHAN – Question mein ye dikhe**

HCF & LCM diye + ek number $\Rightarrow$ doosra, product of numbers, sum of reciprocals, “LCM is 42 times HCF and LCM + HCF = 602”; गुणनफल

$$\text{HCF} \times \text{LCM} = a \times b \quad (\text{sirf DO numbers}) \quad b = \frac{\text{HCF} \times \text{LCM}}{a} \quad \frac{1}{a} + \frac{1}{b} = \frac{a+b}{\text{HCF} \times \text{LCM}}$$

- LCM = $k \times$ HCF aur LCM + HCF = S $\Rightarrow$ HCF = $\frac{S}{k+1}$ (LCM – HCF = D $\Rightarrow$ HCF = $\frac{D}{k-1}$).
- LCM + HCF aur LCM – HCF diye $\Rightarrow$ LCM = (sum+diff)/2, HCF = (sum–diff)/2 (NS-9).

TERMS – Har letter ka matlab

$a, b = \text{do numbers} \cdot k = \text{LCM kitne guna HCF}.$

SOLVED – Asli book questions (verified)

Q (EN 121): HCF 1080, LCM 30240, ek 4320

$$\frac{1080 \cdot 30240}{4320} = 7560 \checkmark \text{ book ans (c)}$$

Q (EN 119): sum 54, LCM 84, HCF $6 \Rightarrow 1/a + 1/b$

$$\frac{54}{504} = \frac{3}{28} \checkmark \text{ book ans (c)}$$

Q (EN 139): LCM = $42 \cdot \text{HCF}$, LCM + HCF = 602, ek 84

$$\text{HCF} = 602/43 = 14, \text{LCM } 588 \Rightarrow \frac{14 \cdot 588}{84} = 98 \checkmark \text{ book ans (a)}$$

Q (HI 71): गुणनफल 8410, HCF $29 \Rightarrow \text{LCM}$

$$8410/29 = 290 \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (56) EN 119–121, 123–127, 129–132, 137–139, 141, 143, 146, 148, 149, 152, 157–160, 162–165, 178 · **HI** 65, 67–71, 73–80, 82–87, 90, 91, 93, 97, 103, 104

LH-7 — RATIO LINK + PAIR COUNT**PEHCHAN – Question mein ye dikhe**

“ratio 5 : 7 and HCF $8 \Rightarrow \text{LCM}$ ”, “LCM 84, ratio 2 : 3 $\Rightarrow \text{sum}$ ”, “ratio 3:4:5, LCM 2400 $\Rightarrow \text{HCF}$ ”, “how many pairs with HCF 12 and LCM 720 / sum 288 and HCF 16”; अनुपात, युग्म

$$\text{Numbers} = h \cdot a, h \cdot b \text{ (} a, b \text{ co-prime)} \quad \text{LCM} = h \cdot \text{LCM}(a, b, \dots) \quad h = \frac{\text{LCM}}{\text{LCM}(a, b, \dots)}$$

- **Pairs with HCF h, LCM L:** L/h ke alag-alag primes = $k \Rightarrow \text{pairs} = 2^{k-1}$.
- **Pairs with sum S, HCF h:** S/h ko do co-prime numbers ke jod mein kitni tarah tod sakte ho — gino.

TERMS – Har letter ka matlab

$h = \text{HCF} \cdot a : b = \text{ratio (co-prime)} \cdot k = \text{distinct prime factors ki ginti}.$

SOLVED – Asli book questions (verified)

Q (EN 122): ratio 5:7, HCF $8 \Rightarrow \text{LCM}$

$$8 \cdot 5 \cdot 7 = 280 \checkmark \text{ book ans (d)}$$

Q (EN 145): LCM 84, ratio 2:3 $\Rightarrow \text{sum}$

$$h = 84/6 = 14 \Rightarrow 28 + 42 = 70 \checkmark \text{ book ans (b)}$$

Q (EN 168): ratio 3:4:5, LCM 2400 $\Rightarrow \text{HCF}$

$$2400/60 = 40 \checkmark \text{ book ans (b)}$$

Q (EN 135): HCF 12, LCM 720: kitne pairs

$$60 = 2^2 \cdot 3 \cdot 5, k = 3 \Rightarrow 2^2 = 4 \checkmark \text{ book ans (b)}$$

Q (EN 67): sum 288, HCF 16: pairs

$$18 = 1 + 17 = 5 + 13 = 7 + 11 \Rightarrow 3 \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (38) EN 59, 67, 118, 122, 133, 135, 142, 144, 145, 147, 150, 151, 153–156, 161, 166–171, 175 · **HI** 27, 66, 72, 81, 88, 89, 92, 95, 96, 98–102

LH-8 — FRACTION + DECIMAL HCF/LCM**PEHCHAN – Question mein ye dikhe**

LCM/HCF of fractions $(2/3, 4/9 \dots)$, of decimals $(1.75, 5.6, 7 / 0.24, 0.36)$, “LCM of a set of fractions is”; भिन्न / दशमलव का ल.स., म.स.

$$\text{HCF} \left(\frac{a}{b}, \frac{c}{d} \right) = \frac{\text{HCF}(a, c)}{\text{LCM}(b, d)} \quad \text{LCM} \left(\frac{a}{b}, \frac{c}{d} \right) = \frac{\text{LCM}(a, c)}{\text{HCF}(b, d)}$$

Decimals: sab ko same decimal places tak laao ($\times 100$ ya $\times 1000$), integers ka HCF/LCM nikaalo, phir wapas $\div 100$.

TERMS – Har letter ka matlab

a, c = numerators · b, d = denominators (fraction lowest form mein ho).

SOLVED – Asli book questions (verified)

Q (EN 106): LCM of $2/3$, $4/9$, $7/12$, $3/5$

$$\frac{\text{LCM}(2,4,7,3)}{\text{HCF}(3,9,12,5)} = \frac{84}{1} \quad \checkmark \text{ book ans (c)}$$

Q (EN 104): HCF of $2/3$, $4/5$, $3/2$

$$\frac{\text{HCF}(2,4,3)}{\text{LCM}(3,5,2)} = \frac{1}{30} \quad \checkmark \text{ book ans (b)}$$

Q (EN 112): LCM, HCF of 1.75, 5.6, 7

$$\times 100: 175, 560, 700 \Rightarrow \text{LCM } 2800, \text{HCF } 35 \Rightarrow 28, 0.35 \quad \checkmark \text{ book ans (c)}$$

Q (HI 63): 0.24 और 0.36 का HCF

$$\text{HCF}(24, 36)/100 = 0.12 \quad \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (20) EN 18, 104–117, 136 · **HI** 61–64

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
LH-1	POWER RULE (HCF/LCM nikaalna)		56 30
LH-2	LCM + REMAINDER MACHINE		53 22
LH-3	HCF + REMAINDER MACHINE		15 15
LH-4	HCF APPLICATION (sabse bada naap / kam se kam hisse)		20 7
LH-5	LCM APPLICATION (bells / lights / track — saath kab)		14 7
LH-6	PRODUCT LINK ($\text{HCF} \times \text{LCM} = a \times b$)		30 26
LH-7	RATIO LINK + PAIR COUNT		24 14
LH-8	FRACTION + DECIMAL HCF/LCM		16 4
→ NS-1	DIVISIBILITY TABLE (vibhajyata) (Ch 1)		1 0
→ NS-5	FACTOR MACHINE (gunankhand) (Ch 1)		1 0
→ NS-9	TWO-NUMBER + EQUATION MACHINE (Ch 1)		1 1
Total (book ke sawaal)		231/231 ✓	126/126 ✓
→ NS-1: EN 222 · → NS-5: EN 174 · → NS-9: EN 172 HI 105			

Chapter 6

Percentage (Pratishatata) प्रतिशतता

9 formulas · English book 247 sawaal · Hindi book 117 sawaal · 40 solved examples ✓

PC-1 — PERCENT BASICS (of · kitna % · poora number)

PEHCHAN – Question mein ye dikhe

“x% of y”, “27.5% of a number is 11”, “48 is what percent of 60”, “what % of a day is 6 h”, “35% of x = 40% of y”, fraction $\leftrightarrow$ %; प्रतिशत, का ...%

$$x\% \text{ of } y = \frac{x \cdot y}{100} \quad \text{Number} = \frac{\text{Value} \times 100}{x} \quad \text{A, B ka kitna \%} = \frac{A}{B} \times 100$$

- $a\%$ of $A = b\%$ of $B \Rightarrow A : B = b : a$ (ulta) · “a% of A = b% of B $\Rightarrow$ A ka kitna% B” = $\frac{a}{b} \times 100$.
- **Yaad table:** $\frac{1}{2} = 50\%$, $\frac{1}{3} = 33\frac{1}{3}\%$, $\frac{1}{4} = 25\%$, $\frac{1}{5} = 20\%$, $\frac{1}{6} = 16\frac{2}{3}\%$, $\frac{1}{8} = 12.5\%$, $\frac{1}{9} = 11\frac{1}{9}\%$, $\frac{1}{11} = 9\frac{1}{11}\%$, $\frac{1}{12} = 8\frac{1}{3}\%$
- Time: 1 din = 24 h = 1440 min; 1 h = 3600 s (same unit mein badal kar %).

TERMS – Har letter ka matlab

x = percent · y = jiska % lena hai · Value = x% ke barabar diya hua number.

SOLVED – Asli book questions (verified)

Q (EN 85): 27.5% of number = 11

$$\frac{11 \times 100}{27.5} = 40 \quad \checkmark \text{ book ans (c)}$$

Q (EN 87): k% of 280 = 50% of 350

$$k = \frac{175 \times 100}{280} = 62.5 \quad \checkmark \text{ book ans (c)}$$

Q (EN 90): 35% of x = 40% of y $\Rightarrow x : y$

$$40 : 35 = 8 : 7 \quad \checkmark \text{ book ans (d)}$$

Q (EN 129): 48 is what % of 60

$$\frac{48}{60} \times 100 = 80 \quad \checkmark \text{ book ans (d)}$$

Q (EN 117): 1 min 12 s = 1 h ka kitna %

$$\frac{72}{3600} \times 100 = 2\% \quad \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (110) EN 17, 25, 29, 31, 32, 39, 42, 43, 46, 54, 71, 73, 85–88, 90–93, 95–100, 102–110, 112–117, 119–124, 126–130, 132–134, 165, 167–170, 172, 173, 176, 178, 182–189, 200–203, 208, 209, 214, 217, 219, 221 · **HI** 15, 25, 32, 57, 59, 70, 71, 73, 74, 76, 78–82, 96, 99, 100, 102 | *Ch 1 (Sankhya Paddhati):* EN 194, 196 | *Ch 8 (Batta / Chhoot):* EN 49 · **HI** 74 | *Ch 12 (Mishran / Alligation):* EN 23 · **HI** 7 | *Ch 28 (Vividh):* EN 53, 64 · **HI** 50

PC-2 — MULTIPLIER MACHINE (badha / ghata / ulta)

PEHCHAN – Question mein ye dikhe

increased/decreased by x%, “becomes 725 when increased by 45%”, “salary after 30% increment is 910”, “% increase from 18000 to 22500”, “x% more than”; वृद्धि, कमी, बढ़कर

$$\text{New} = \text{Old} \times \frac{100 \pm x}{100} \quad \text{Old} = \text{New} \times \frac{100}{100 \pm x} \quad \% \text{ change} = \frac{\text{New} - \text{Old}}{\text{Old}} \times 100$$

Chain (A = 150% of B, C = 120% of A): B = 100 maan lo $\Rightarrow$ A = 150, C = 180 $\Rightarrow$ ratio se baanto. Saal-dar-saal average % (simple) = total % $\div$ saal.

TERMS – Har letter ka matlab

Old = pehle ki value · New = baad ki · + badhne par, – ghatne par · % change hamesha OLD par.

SOLVED – Asli book questions (verified)**Q (EN 131):** 45% badhne par 725

$$725 \times \frac{100}{145} = 500 \quad \checkmark \text{ book ans (a)}$$

Q (EN 83): 30% increment ke baad ₹910

$$910 \times \frac{100}{130} = 700 \quad \checkmark \text{ book ans (c)}$$

Q (EN 15): 18000 → 22500 ka % increase

$$\frac{4500}{18000} \times 100 = 25\% \quad \checkmark \text{ book ans (a)}$$

Q (EN 181): 1584, 20% zyada hai pichhle se

$$1584 \times \frac{100}{120} = 1320 \quad \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (30) EN 8, 15, 24, 64, 67, 70, 83, 89, 101, 118, 125, 131, 143, 151, 175, 179, 181 · **HI** 6, 13, 14, 40, 72, 75, 77, 83, 85, 101, 104 | Ch 1
(Sankhya Paddhati): EN 334 | Ch 7 (Labh aur Hani): HI 56

PC-3 — NET CHANGE MACHINE (do/teen change ek saath)**PEHCHAN – Question mein ye dikhe**

successive increase/decrease, “first ... then ...”, two discounts, price × sale, hours × wage, area when length & breadth change, square/circle side/radius change; पहले ... फिर, शुद्ध प्रतिशत, क्षेत्रफल

$$\text{Net}\% = a + b + \frac{a \times b}{100} \quad (\text{badha +, ghata -})$$

- **Teen changes:** $(1 \pm \frac{a}{100})(1 \pm \frac{b}{100})(1 \pm \frac{c}{100}) - 1, \times 100$.
- **Area:** rectangle L + a%, B + b% ⇒ upar wala; square/circle (side/radius + a%) ⇒ $2a + \frac{a^2}{100}$. Revenue = price × sale, earning = hours × rate ⇒ same formula.
- +x% phir -x% ⇒ hamesha **ghata** $\frac{x^2}{100}\%$.

TERMS – Har letter ka matlab

a, b = dono percent changes (sign ke saath) · answer – aaye to overall kami.

SOLVED – Asli book questions (verified)**Q (EN 137):** +10% phir +8%

$$10 + 8 + 0.8 = 18.8\% \quad \checkmark \text{ book ans (c)}$$

Q (EN 140): +40% phir -40%

$$-\frac{40^2}{100} = -16\% \quad \checkmark \text{ book ans (d)}$$

Q (EN 149): +10%, -20%, +50%

$$1.1 \times 0.8 \times 1.5 = 1.32 \Rightarrow 32\% \quad \checkmark \text{ book ans (c)}$$

Q (EN 224): square side +50% ⇒ area

$$100 + 25 = 125\% \quad \checkmark \text{ book ans (d)}$$

Q (EN 154): discounts 12% & 5% = single?

$$-12 - 5 + 0.6 = -16.4\% \quad \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (59) EN 9, 16, 111, 135–142, 144–150, 153, 154, 157, 159, 161, 196, 223–237 · **HI** 5, 7, 37, 84, 86–91, 93, 95, 111–115 | Ch 10
(Saajhedaari): EN 15 · **HI** 4 | Ch 28 (Vividh): EN 39

PC-4 — POPULATION / GROWTH MACHINE**PEHCHAN – Question mein ye dikhe**

population increases r% every year, “2 years ago”, different % in successive years, “died ... then left”, bacteria growth, “in how many years 1000 → 1331”; जनसंख्या, प्रति वर्ष

$$P_n = P \left(1 + \frac{r}{100}\right)^n \quad P_{\text{pehle}} = \frac{P}{\left(1 + \frac{r}{100}\right)^n} \quad \text{alag rates: } P\left(1 + \frac{a}{100}\right)\left(1 - \frac{b}{100}\right) \cdots$$

Mare/chale gaye (a% phir baaki ka b%) $\Rightarrow$ bache = $P\left(1 - \frac{a}{100}\right)\left(1 - \frac{b}{100}\right) \Rightarrow$ ulta: divide.

TERMS – Har letter ka matlab

P = abhi ki population $\cdot$ r = har saal ka % $\cdot$ n = saal $\cdot$ ghatne par $(1 - r/100)$.

SOLVED – Asli book questions (verified)

Q (EN 4): 2 saal pehle 10,000; 10% growth $\Rightarrow$ ab

$$10000 \times 1.1^2 = 12100 \quad \checkmark \text{ book ans (a)}$$

Q (EN 3): 15625; +8%, +12%, -22%

$$15625 \times 1.08 \times 1.12 \times 0.78 = 14742 \quad \checkmark \text{ book ans (c)}$$

Q (EN 19): abhi 1,85,220, 5% growth $\Rightarrow$ saal bhar pehle

$$185220 \div 1.05 = 176400 \quad \checkmark \text{ book ans (c)}$$

Q (EN 5): 20% mare, baaki ke 5% mare $\Rightarrow$ 15200

$$\frac{15200}{0.8 \times 0.95} = 20000 \quad \checkmark \text{ book ans (a)}$$

Q (EN 11): 1000 $\rightarrow$ 1331 at 10%

$$1.1^3 = 1.331 \Rightarrow 3 \text{ saal} \quad \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (30) EN 2-7, 11, 12, 14, 18, 19, 21, 22, 163 $\cdot$ HI 1-4, 8-11, 16, 17, 19, 20 | Ch 7 (Labh aur Hani): HI 57 | Ch 8 (Batta / Chhoot): EN 24, 27 | Ch 15 (Chakravardhi Byaj): EN 204

PC-5 — REVERSE + CONSUMPTION MACHINE

PEHCHAN – Question mein ye dikhe

“A is 20% less than B, B is how much % more than A”, “price increases 20%, reduce consumption to keep expenditure same”, “restore the original price”, “A, B less than C by 15%, 32% $\Rightarrow$ B less than A”; उपभोग/खपत कम

$$x\% \text{ zyada} \Rightarrow \text{ulta } \frac{x}{100+x} \times 100\% \text{ kam} \quad x\% \text{ kam} \Rightarrow \text{ulta } \frac{x}{100-x} \times 100\% \text{ zyada}$$

- **Price +r%, kharcha same** $\Rightarrow$ consumption $\frac{r}{100+r} \times 100\%$ kam (price -r% $\Rightarrow \frac{r}{100-r}$ zyada).
- **Kharcha bhi e% badhe** $\Rightarrow$ consumption change = $\left(\frac{100+e}{100+r} - 1\right) \times 100$.
- Kisi teesre se compare (C se A 15% kam, B 32% kam) $\Rightarrow$ C = 100 maan lo: A = 85, B = 68 $\Rightarrow$ PC-2.
- **Price x% ghati** $\Rightarrow$ same paise mein k kg zyada: naya rate = $\frac{x\% \text{ of paise}}{k}$, purana = naya $\times \frac{100}{100-x}$.

TERMS – Har letter ka matlab

x = diya % $\cdot$ r = price change $\cdot$ e = kharche ka change $\cdot$ denominator hamesha jis se compare ho raha.

SOLVED – Asli book questions (verified)

Q (EN 197): A, B se 20% kam $\Rightarrow$ B, A se kitna zyada

$$\frac{20}{80} \times 100 = 25\% \quad \checkmark \text{ book ans (b)}$$

Q (EN 210): tea price +20%, budget same $\Rightarrow$ consumption

$$\frac{20}{120} \times 100 = 16\frac{2}{3}\% \quad \checkmark \text{ book ans (b)}$$

Q (EN 204): petrol +52%, kharcha +14% $\Rightarrow$ consumption

$$\frac{114}{152} - 1 = -25\% \quad \checkmark \text{ book ans (b)}$$

Q (EN 207): price 20% ghati; wapas karne ke liye

$$\frac{20}{80} \times 100 = 25\% \quad \checkmark \text{ book ans (d)}$$

Q (EN 94): A, B, C se 15%, 32% kam $\Rightarrow$ B, A se kitna kam

$$\frac{85-68}{85} \times 100 = 20\% \quad \checkmark \text{ book ans (a)}$$

BOOK KESAWAAL (30) EN 94, 152, 155, 156, 158, 160, 164, 166, 197–199, 204–207, 210–213, 215, 216, 222 · **HI** 12, 51, 68, 92, 94, 110 | *Ch 1 (Sankhya Paddhati)*: EN 193 | *Ch 8 (Batta / Chhoot)*: EN 35

PC-6 — EXAM MARKS MACHINE (pass / fail)

PEHCHAN – Question mein ye dikhe

“obtained 45% and failed by 18 marks; 65% would get 6 more than pass”, “must score 40% to pass, gets 70 and fails by 20”, maximum marks; अंक, उत्तीर्ण, अनुत्तीर्ण

$$\text{Do students: Max} = \frac{(\text{fail by}) + (\text{extra})}{p_2\% - p_1\%} \times 100 \quad \text{Ek student: Max} = \frac{(\text{mila} + \text{fail by}) \times 100}{\text{pass}\%}$$

Pass marks = $p_1\%$ of Max + (fail by) · Pass % = $\frac{\text{pass marks}}{\text{Max}} \times 100$ · seedha: marks = $p\%$ of Max, Max = marks $\times 100 \div p$.

TERMS – Har letter ka matlab

p_1, p_2 = dono ke % · fail by = kitne se fail · extra = pass se kitne zyada.

SOLVED – Asli book questions (verified)

Q (EN 1): 45% fail by 18; 65% $\Rightarrow$ 6 zyada

$$\frac{18+6}{20} \times 100 = 120 \quad \checkmark \text{ book ans (c)}$$

Q (EN 35): 40% chahiye; 70 mile, 20 se fail

$$\frac{90 \times 100}{40} = 225 \quad \checkmark \text{ book ans (d)}$$

Q (EN 23): 46% fail by 10; 52% $\Rightarrow$ 8 zyada; pass marks

$$\text{Max} = \frac{18}{6} \times 100 = 300; \text{pass} = 138 + 10 = 148 \quad \checkmark \text{ book ans (a)}$$

Q (EN 47): 40% fail by 20; 45% $\Rightarrow$ 30 zyada; pass %

$$\text{Max } 1000, \text{pass } 420 \Rightarrow 42\% \quad \checkmark \text{ book ans (b)}$$

BOOK KESAWAAL (29) EN 1, 23, 30, 35, 38, 40, 41, 44, 45, 47–53, 55, 57, 58 · **HI** 23, 24, 26–31, 33 | *Ch 28 (Vividh)*: EN 43

PC-7 — INCOME – EXPENDITURE – SAVINGS MACHINE

PEHCHAN – Question mein ye dikhe

income/salary, spends $x\%$, saves ₹. . ., rent then $x\%$ of remaining, “income increases 25% and expenditure 15%, % change in savings”, food % after increment; आय, वेतन, व्यय, बचत

$$\text{Income} = \text{Expenditure} + \text{Savings} \quad \text{Income} = \frac{\text{Savings} \times 100}{\text{saving}\%}$$

- **Chain** ($a\%$ rent, phir baaki ka $b\%$. . .): bacha = $I(1 - \frac{a}{100})(1 - \frac{b}{100}) \cdots \Rightarrow I = \text{bacha} \div (\dots)$.
- **Change wale:** Income = 100 maan lo $\Rightarrow$ exp, savings likho $\Rightarrow$ naye nikaalo $\Rightarrow$ % change (PC-2).
- A aur B ki savings barabar $\Rightarrow s_A\% \cdot A = s_B\% \cdot B \Rightarrow A : B = s_B : s_A$.

TERMS – Har letter ka matlab

I = income · E = kharcha · S = bachat · $s\%$ = kitna % bachata hai.

SOLVED – Asli book questions (verified)**Q (EN 62):** 80% kharch, ₹5000 bachat

$$\frac{5000 \times 100}{20} = 25000 \quad \checkmark \text{ book ans (a)}$$

Q (EN 63): 15% rent, baaki ka 20%, phir 10% $\Rightarrow$ ₹42,840 bache

$$\frac{42840}{0.85 \times 0.8 \times 0.9} = 70000 \quad \checkmark \text{ book ans (d)}$$

Q (EN 65): 80% kharch; income +25%, exp +15% $\Rightarrow$ savings

$$I \ 100 \rightarrow 125, E \ 80 \rightarrow 92, S \ 20 \rightarrow 33 \Rightarrow +65\% \quad \checkmark \text{ book ans (d)}$$

Q (EN 66): A+B = 4300; A 95%, B 80% kharch; bachat same

$$5A = 20B \Rightarrow A : B = 4 : 1 \Rightarrow 3440 \quad \checkmark \text{ book ans (c)}$$

Q (EN 69): food 25%; ₹1000 increment ke baad 20%

$$0.25I = 0.2(I + 1000) \Rightarrow I = 4000 \quad \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (50) EN 59–63, 65, 66, 68, 69, 72, 74–82, 84 · **HI** 34–36, 38, 39, 41–50, 52–56, 58, 60–67, 97**PC-8 — ELECTION MACHINE (votes)****PEHCHAN – Question mein ye dikhe**

election, votes, winner got x%, lost by margin, invalid votes, “90% cast their vote, 80% valid”; चुनाव, मत, वैध/अवैध

Do candidates: Margin = $(W\% - L\%) \times \text{Valid}$ (sirf winner $x\% \Rightarrow (2x - 100)\%$)Chain: winner votes = Total $\times$ turnout% $\times$ valid% $\times$ winner% $\Rightarrow$ Total = votes $\div$ (sab fractions ka product).**TERMS – Har letter ka matlab**W, L = jeetne/haarne wale ka % $\cdot$ Valid = sahi votes $\cdot$ turnout = kitne logon ne vote diya.**SOLVED – Asli book questions (verified)****Q (EN 239):** 37% wala 5,20,000 se haara

$$(63 - 37)\% = 26\% \Rightarrow \frac{520000}{0.26} = 20,00,000 \quad \checkmark \text{ book ans (b)}$$

Q (EN 238): 90% voted, 80% valid, winner 60% = 64800

$$\frac{64800}{0.9 \times 0.8 \times 0.6} = 150000 \quad \checkmark \text{ book ans (a)}$$

Q (EN 247): 62% wala 35640 se jeeta

$$24\% = 35640 \Rightarrow 148500 \quad \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (12) EN 238–247 · **HI** 116, 117**PC-9 — GROUP % / MIXTURE % MACHINE****PEHCHAN – Question mein ye dikhe**

boys : girls with scholarship %, male/female population increase, average salary of men & women, salt/glucose solution %, weightage; लड़के-लड़कियाँ, पुरुष-महिला, विलयन

$$\text{Combined \%} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} \quad \frac{n_1}{n_2} = \frac{p_2 - p}{p - p_1} \quad (\text{alligation, MX-1})$$

• **Men +a%, women +b%, total +t%** $\Rightarrow$ women : men = $(t - a) : (b - t)$.• **Concentration** = $\frac{\text{solute}}{\text{solute} + \text{solvent}} \times 100$ (paani mein namak: dono jodo).• **Do subjects:** dono mein pass% = $P_1 + P_2 - (100 - \text{dono mein fail\%})$ · Weighted marks: $\frac{\sum (\% \times \text{weight})}{\sum \text{weight}}$.**TERMS – Har letter ka matlab**

n = groups ki ginti/ratio · p = unka % · p (bina subscript) = mila hua overall %.

SOLVED – Asli book questions (verified)

Q (EN 33): boys:girls 3:2; 20%, 25% scholarship $\Rightarrow$ NOT

$$\frac{60+50}{5} = 22\% \Rightarrow 78\% \checkmark \text{ book ans (a)}$$

Q (EN 10): 10,000; male +5%, female +10% $\Rightarrow$ 10,800; females?

$$\text{total} + 8\% \Rightarrow F : M = 3 : 2 \Rightarrow 6000 \checkmark \text{ book ans (b)}$$

Q (EN 174): men ₹5200, women ₹4200, all ₹5000 $\Rightarrow$ men %

$$\frac{5000-4200}{5200-4200} = 80\% \checkmark \text{ book ans (b)}$$

Q (EN 190): 33 g salt in 320 g water

$$\frac{33}{353} \times 100 = 9.35\% \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (27) EN 10, 13, 20, 26–28, 33, 34, 36, 37, 56, 174, 190–195, 220 · **HI** 18, 21, 22, 105–109

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
PC-1	PERCENT BASICS (of · kitna % · poora number)	82	19
PC-2	MULTIPLIER MACHINE (badha / ghata / ulta)	17	11
PC-3	NET CHANGE MACHINE (do/teen change ek saath)	39	17
PC-4	POPULATION / GROWTH MACHINE	14	12
PC-5	REVERSE + CONSUMPTION MACHINE	22	6
PC-6	EXAM MARKS MACHINE (pass / fail)	19	9
PC-7	INCOME – EXPENDITURE – SAVINGS MACHINE	20	30
PC-8	ELECTION MACHINE (votes)	10	2
PC-9	GROUP % / MIXTURE % MACHINE	19	8
$\rightarrow DF-8$	FRACTION EQUATION MACHINE (ansh–har wale sawaal) (Ch 2)	1	0
$\rightarrow PL-1$	BASIC P/L % (CP, SP, kharcha, fraction) (Ch 7)	1	0
$\rightarrow PL-6$	FALSE WEIGHT MACHINE (nakli baat) (Ch 7)	0	1
$\rightarrow PL-7$	SAME-SP MACHINE (do cheezein, ek hi SP) (Ch 7)	1	1
$\rightarrow PL-8$	WEIGHTED PARTS / TWO-ITEM MACHINE (Ch 7)	1	1
$\rightarrow SI-5$	SPLIT / MIX INVESTMENT MACHINE (Ch 14)	1	0
Total (book ke sawaal)		247/247 ✓	117/117 ✓

$\rightarrow$ **DF-8:** EN 162 · $\rightarrow$ **PL-1:** EN 218 · $\rightarrow$ **PL-6:** HI 69 · $\rightarrow$ **PL-7:** EN 177 HI 103 · $\rightarrow$ **PL-8:** EN 171 HI 98 · $\rightarrow$ **SI-5:** EN 180

Chapter 7

Profit & Loss (Labh aur Hani) लाभ और हानि

10 formulas · English book 168 sawaal · Hindi book 114 sawaal · 86 solved examples ✓

PL-1 — BASIC P/L % (CP, SP, kharcha, fraction)

PEHCHAN – Question mein ye dikhe

CP aur SP dono diye $\Rightarrow$ profit/loss %, “spends ₹400 on repairs / transport / tax”, “SP is 5/4 of CP”, “profit = 1/5 of SP”, “loses 1/7 of CP”, do logon ke profit % ka ratio; क्रय मूल्य, विक्रय मूल्य, मरम्मत, लाभ/हानि प्रतिशत

$$P/L\% = \frac{SP - CP}{CP} \times 100 \quad CP = \text{kharid} + \text{repair/transport/tax/rent} \quad (\text{sab kharche CP mein})$$

- $SP = \frac{k}{m} CP \Rightarrow P\% = \frac{k-m}{m} \times 100$ ($k < m \Rightarrow$ loss).
- Profit = $\frac{a}{b}$ of **SP** $\Rightarrow P\% = \frac{a}{b-a} \times 100$ · Profit = $\frac{a}{b}$ of **CP** $\Rightarrow P\% = \frac{a}{b} \times 100$ · Loss = $\frac{a}{b}$ of **CP** $\Rightarrow SP = (1 - \frac{a}{b})CP$.
- SP ₹S par loss ₹L $\Rightarrow CP = S + L$ · kai items $\Rightarrow$ total CP vs total SP.

TERMS – Har letter ka matlab

CP = cost price (kharid + saare kharche) · SP = selling price · + aaye to profit, – aaye to loss.

SOLVED – Asli book questions (verified)

Q (EN 1): CP ₹4500, SP ₹3500 $\Rightarrow$ loss %

$$\frac{1000}{4500} \times 100 = 22\frac{2}{9}\% \quad \checkmark \text{ book ans (c)}$$

Q (EN 2): ₹222 mein becha, ₹48 loss

$$CP = 270 \Rightarrow \frac{48}{270} \times 100 = 17\frac{7}{9}\% \quad \checkmark \text{ book ans (a)}$$

Q (EN 20): ₹30,000 + ₹5000 repair, becha ₹42,000

$$\frac{7000}{35000} \times 100 = 20\% \quad \checkmark \text{ book ans (a)}$$

Q (EN 18): SP = 5/4 of CP

$$\frac{5-4}{4} \times 100 = 25\% \quad \checkmark \text{ book ans (c)}$$

Q (HI 47): profit = SP ka 1/5 (₹15,000)

$$\frac{1}{5-1} \times 100 = 25\% \quad \checkmark \text{ book ans (b)}$$

Q (EN 27): ₹144 mein CP ka 1/7 loss; ₹189 pe?

$$CP = 144 \times \frac{7}{6} = 168 \Rightarrow \frac{21}{168} = 12.5\% \quad \checkmark \text{ book ans (b)}$$

Q (HI 111): A: 384 $\rightarrow$ 576, B: 1254 $\rightarrow$ 1672 $\Rightarrow$ profit % ka ratio

$$50\% : 33\frac{1}{3}\% = 3 : 2 \quad \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (24) EN 1, 2, 11, 18, 20, 25–27, 31, 32, 55 · **HI** 1, 3, 4, 6, 29, 38, 47, 50, 111 | Ch 6 (Pratishatata): EN 218 | Ch 8 (Batta / Chhoot): EN 44 · **HI** 71 | Ch 28 (Vividh): EN 65

PL-2 — CP $\leftrightarrow$ SP MULTIPLIER + SP TRANSFER + CHAIN

PEHCHAN – Question mein ye dikhe

“sold for ₹207 at 28% loss $\Rightarrow$ CP”, “CP ₹5000, 25% profit $\Rightarrow$ SP”, “₹4125 pe 10% gain, 18% ke liye kitne mein”, “₹211.20 pe 12% loss; ₹248.40 pe kya”, A $\rightarrow$ B $\rightarrow$ C sellers, sold back, profit amount, CP : SP ratio; क्रय मूल्य ज्ञात, किस मूल्य पर बेचे, क्रमिक विक्रेता

$$SP = CP \times \frac{100 \pm P}{100} \quad CP = SP \times \frac{100}{100 \pm P} \quad SP_2 = SP_1 \times \frac{100 \pm P_2}{100 \pm P_1}$$

- **Naya %** (SP_1 pe P_1 tha, SP_2 pe kya?): $\frac{SP_2(100 \pm P_1)}{SP_1} - 100$.
- Profit $\text{₹} = CP \times \frac{P}{100} = SP \times \frac{P}{100 + P}$.
- **Chain** $A \rightarrow B \rightarrow C$: aakhri price $= CP(1 \pm \frac{a}{100})(1 \pm \frac{b}{100}) \dots$ (ulta = divide); beech ka kharcha $\text{₹}k$ wahin jodo. Wapas khareeda (sold back) $\Rightarrow$ profit $= SP_1 - SP_2$.
- **Ratio**: $CP : SP = 100 : (100 \pm P) \cdot SP_1 : SP_2 = (100 + a) : (100 + b)$ (same CP).

TERMS – Har letter ka matlab

P = profit/loss % (profit +, loss –) · SP_1 , P_1 = diya hua pehla haal · SP_2 , P_2 = chahiye wala haal.

SOLVED – Asli book questions (verified)

Q (EN 35): ₹207 pe 28% loss $\Rightarrow$ CP

$$207 \times \frac{100}{72} = 287.50 \quad \checkmark \text{ book ans (d)}$$

Q (EN 69): ₹4125 pe 10% gain $\Rightarrow$ 18% ke liye

$$4125 \times \frac{118}{110} = 4425 \quad \checkmark \text{ book ans (b)}$$

Q (EN 71): ₹1729 pe 30% loss $\Rightarrow$ 16% gain ke liye

$$1729 \times \frac{116}{70} = 2865.20 \quad \checkmark \text{ book ans (b)}$$

Q (EN 5): ₹211.20 pe 12% loss; ₹248.40 pe?

$$\frac{248.40 \times 88}{211.20} = 103.5 \Rightarrow 3.5\% \text{ gain} \quad \checkmark \text{ book ans (d)}$$

Q (HI 68): 3 sellers, har ek 25% profit, aakhri ₹1250

$$\frac{1250}{1.25^3} = 640 \quad \checkmark \text{ book ans (a)}$$

Q (EN 40): $x - 15\% \rightarrow \text{Biru} + ₹126 \rightarrow +25\% = ₹1475$

$$(\frac{1475}{1.25} - 126) \div 0.85 = 1240 \quad \checkmark \text{ book ans (a)}$$

Q (EN 105): ₹1,50,000 pe 5% profit se Y ko; Y ne 2% loss pe wapas becha

$$157500 - 154350 = 3150 \text{ gain} \quad \checkmark \text{ book ans (b)}$$

Q (EN 114): 4% aur 10% profit ke SP ka ratio

$$104 : 110 = 52 : 55 \quad \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (80) EN 5, 23, 28–30, 35, 37, 40, 42, 57, 65, 69, 71–75, 79, 81, 83, 84, 86–91, 93–95, 97, 98, 100, 102, 105–107, 109, 112, 114–117, 141, 159, 164, 167 · **HI** 20–22, 30, 37, 51, 52, 61, 62, 67–70, 72, 79, 81, 85–87, 90–97, 99–101, 104, 107, 113

PL-3 — DIFFERENCE MACHINE (₹ farak $\leftrightarrow$ % farak)**PEHCHAN – Question mein ye dikhe**

“difference between 12% gain and 4% loss is ₹28”, “had it been sold for ₹33 more, gain 14%”, “5% more gained by selling for ₹1010 than ₹1000”, “bought 10% less & sold ₹18 more $\Rightarrow$ 40%”, “profit at ₹1875 = loss at ₹1385”, milkman ₹50/L pe loss, ₹60/L pe profit; अंतर, ₹... अधिक प्राप्त होते, ₹... की वृद्धि

$$CP = \frac{\text{₹ farak} \times 100}{\% \text{ farak}} \quad (\text{profit +, loss - : } 12\% \text{ gain vs } 4\% \text{ loss} \Rightarrow 12 - (-4) = 16)$$

- Dono haalat **original CP ke %** mein likho: “10% kam mein khareeda, 40% profit” $\Rightarrow 90 \times 1.4 = 126\%$; phir farak $\Rightarrow$ upar wala formula.
- CP ₹c badhe aur SP ₹s ghate (naya profit p_2): $CP \times \frac{p_1 - p_2}{100} = s + c \times \frac{100 + p_2}{100}$.
- Profit at ₹ S_1 = loss at ₹ S_2 (₹ ya % barabar) $\Rightarrow CP = \frac{S_1 + S_2}{2}$.
- Rate ka farak: quantity = $\frac{\text{total ₹ farak}}{\text{per-unit rate farak}}$.

TERMS – Har letter ka matlab

₹ farak = do SP ka antar · % farak = unke profit % ka antar (dono CP par) · CP = asli kharid.

SOLVED – Asli book questions (verified)**Q (EN 34):** 12% gain aur 4% loss ka farak ₹28

$$\frac{28 \times 100}{16} = 175 \checkmark \text{ book ans (a)}$$

Q (EN 44): ₹33 zyada $\Rightarrow$ 12% se 14%

$$\frac{33 \times 100}{2} = 1650 \checkmark \text{ book ans (c)}$$

Q (EN 47): ₹1010 pe ₹1000 se 5% zyada gain

$$\frac{10 \times 100}{5} = 200 \checkmark \text{ book ans (a)}$$

Q (EN 36): 20% profit; 10% kam khareed, ₹18 zyada bech $\Rightarrow$ 40%

$$90 \times 1.4 = 126\% \text{ vs } 120\% \Rightarrow \frac{18 \times 100}{6} = 300 \checkmark \text{ book ans (c)}$$

Q (EN 39): 28% profit; CP + ₹50, SP – ₹18 $\Rightarrow$ 11.6%

$$CP \times 0.164 = 18 + 50 \times 1.116 \Rightarrow 450 \checkmark \text{ book ans (c)}$$

Q (EN 43): ₹1875 ka profit = ₹1385 ka loss

$$\frac{1875 + 1385}{2} = 1630 \checkmark \text{ book ans (c)}$$

Q (EN 77): ₹1920 ka profit% = ₹1500 ka loss%; 10% profit ke liye

$$CP \ 1710 \Rightarrow 1710 \times 1.1 = 1881 \checkmark \text{ book ans (c)}$$

Q (HI 108): ₹50/L pe ₹2000 loss, ₹60/L pe ₹1500 profit

$$\frac{3500}{10} = 350 \text{ L} \checkmark \text{ book ans (d)}$$

Q (HI 66): 15% loss; 5% kam khareed, ₹195 zyada $\Rightarrow$ 10% profit

$$95 \times 1.1 = 104.5\% \text{ vs } 85\% \Rightarrow \frac{195 \times 100}{19.5} = 1000 \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (30) EN 34, 36, 39, 43–45, 47, 50, 52, 61–64, 68, 77, 80, 92, 132 · HI 55, 58–60, 66, 75–78, 89, 98, 108**PL-4 — ARTICLE-COUNT MACHINE (x ka SP = y ka CP)****PEHCHAN – Question mein ye dikhe**

“SP of 32 items = CP of 38 items”, “profit = SP of 11 m cloth”, “loss = CP of 7 articles”, “CP of 120 pens = SP of x pens at 25% profit”; वस्तुओं के क्रय/विक्रय मूल्य के बराबर

$$SP_{x \text{ items}} = CP_{y \text{ items}} \Rightarrow P\% = \frac{y - x}{x} \times 100 \quad (\text{jitni BECHI, woh neeche})$$

n items bechkar	P/L %	1 item ka price
Profit = SP of k	$\frac{k}{n-k} \times 100$	SP = ₹(kul CP) $\div$ (n – k)
Profit = CP of k	$\frac{k}{n} \times 100$	CP = ₹(kul SP) $\div$ (n + k)
Loss = CP of k	$\frac{k}{n} \times 100$	CP = ₹(kul SP) $\div$ (n – k)
Loss = SP of k	$\frac{k}{n+k} \times 100$	SP = ₹(kul SP) $\div$ n

“CP of x = SP of y, profit P%” $\Rightarrow x : y = (100 + P) : 100$.

TERMS – Har letter ka matlab

x = jitni cheezein BECHI · y = jitni ka CP · n = kul bechi · k = profit/loss jitni cheezon ke barabar.

SOLVED – Asli book questions (verified)**Q (EN 3):** 32 ka SP = 38 ka CP

$$\frac{6}{32} \times 100 = 18.75\% \checkmark \text{ book ans (c)}$$

Q (EN 12): 9 ka SP = 15 ka CP

$$\frac{6}{9} \times 100 = 66.6\% \text{ gain } \checkmark \text{ book ans (b)}$$

Q (EN 16): 33 m bech kar profit = 11 m ka SP

$$\frac{11}{22} \times 100 = 50\% \checkmark \text{ book ans (c)}$$

Q (EN 38): 12 items ₹420 mein, loss = 7 ka CP

$$\frac{420}{12-7} = 84 \checkmark \text{ book ans (a)}$$

Q (EN 54): 18 beds ₹16,800 mein, profit = 3 ka CP

$$\frac{16800}{18+3} = 800 \checkmark \text{ book ans (c)}$$

Q (EN 82): 25 chairs ₹37,500 mein khareedi, profit = 5 ka SP

$$\frac{37500}{25-5} = 1875 \checkmark \text{ book ans (a)}$$

Q (EN 163): 120 pens ka CP = x pens ka SP, 25% profit

$$x = \frac{120 \times 100}{125} = 96 \checkmark \text{ book ans (c)}$$

Q (EN 113): x ka CP = y ka SP, 25% profit $\Rightarrow x : y$

$$125 : 100 = 5 : 4 \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (16) EN 3, 12, 16, 38, 46, 54, 56, 82, 113, 163 · **HI** 5, 28, 32, 33, 45, 84**PL-5 — RATE MACHINE (a items ₹x mein)****PEHCHAN – Question mein ye dikhe**

“buys 8 for ₹5, sells 5 for ₹8”, “sells 10 for ₹1 gaining 30% $\Rightarrow$ ₹1 mein kitne khareede”, per dozen / per kg / per piece, “some rotten / damaged, rest sold at ...”, “how many to sell for net profit ₹45”; की दर से, दर्जन, प्रति kg, टूट गए

$$\text{Khareeda } a \text{ ₹}x \text{ mein, becha } b \text{ ₹}y \text{ mein } \Rightarrow P\% = \left(\frac{a \times y}{b \times x} - 1 \right) \times 100 \quad (\text{cross-multiply})$$

- ₹1 mein n bechkar P% $\Rightarrow$ ₹1 mein khareede = $n \times \frac{100+P}{100}$ (loss: $\times \frac{100-L}{100}$).
- Barabar ginti, do rates $\Rightarrow$ avg CP/item = $\frac{1}{2} \left(\frac{x_1}{a_1} + \frac{x_2}{a_2} \right)$.
- Kharab (rotten/damaged) maal:** kul CP = qty $\times$ rate (+kharcha); kul SP = bachi qty $\times$ rate $\Rightarrow$ PL-1. Target P% ke liye rate = $\frac{CP \times (100+P)/100}{\text{bachi qty}}$.
- Net profit ₹N ke liye kitne bechne = $N \div$ (per item profit).

TERMS – Har letter ka matlab

a, x = khareed (ginti, ₹) · b, y = bikri (ginti, ₹) · dozen = 12.

SOLVED – Asli book questions (verified)**Q (EN 33):** 8 for ₹5 khareed, 5 for ₹8 bech

$$\frac{8 \times 8}{5 \times 5} = 2.56 \Rightarrow 156\% \checkmark \text{ book ans (c)}$$

Q (EN 137): 6 for ₹10 khareed, 4 for ₹6 bech

$$\frac{6 \times 6}{4 \times 10} = 0.9 \Rightarrow 10\% \text{ loss } \checkmark \text{ book ans (b)}$$

Q (EN 156): 5 for ₹4, 4 for ₹5

$$\frac{25}{16} = 1.5625 \Rightarrow 56.25\% \checkmark \text{ book ans (d)}$$

Q (EN 127): 15 lemons ₹3 mein, 60% gain $\Rightarrow$ ₹1 mein kitne khareede

$$5 \times 1.6 = 8 \checkmark \text{ book ans (b)}$$

Q (EN 136): 10 oranges ₹1 mein, 30% gain

$$10 \times 1.3 = 13 \checkmark \text{ book ans (c)}$$

Q (EN 149): 8 for ₹34, 12 for ₹57; net ₹45

$$4.75 - 4.25 = 0.5 \Rightarrow \frac{45}{0.5} = 90 \checkmark \text{ book ans (a)}$$

Q (EN 143): 20 dozen ₹1000, 40 sade; 30% profit ke liye per dozen

$$\frac{1300}{200/12} = 78 \checkmark \text{ book ans (a)}$$

Q (EN 151): 3 for ₹1 aur 2 for ₹1 (barabar); 20% ke liye dozen

$$\text{avg } \frac{5}{12} \Rightarrow \frac{5}{12} \times 1.2 \times 12 = 6 \checkmark \text{ book ans (d)}$$

Q (HI 24): 100 seb ₹700; ₹96 per dozen

$$\frac{100 \times 96}{12 \times 700} = \frac{8}{7} \Rightarrow 14\frac{2}{7}\% \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (37) EN 33, 110, 127, 128, 134–140, 142–144, 147–153, 155–157 · **HI** 24, 31, 35, 39, 41, 42, 44, 48, 53, 54, 63, 102, 106**PL-6 — FALSE WEIGHT MACHINE (nakli baat)****PEHCHAN – Question mein ye dikhe**

dishonest dealer, false weight, “800 g for 1 kg”, “gives 900 gm instead of 1 kg”, “weight 30% kam”, “cheats 7% in buying and selling”, “claims to sell at CP”; बेईमान, बाट, वजन, खोटा

$$P\% = \frac{\text{Asli} - \text{Nakli}}{\text{Nakli}} \times 100 \quad \text{Nakli} = \frac{\text{Asli} \times 100}{100 + P}$$

- Saath mein profit/markup $p\% \Rightarrow \text{Net}\% = \left(\frac{100+p}{100} \times \frac{\text{Asli}}{\text{Nakli}} - 1 \right) \times 100$; ₹a/kg khareed, ₹b/kg bech $\Rightarrow \frac{b}{a} \times \frac{\text{Asli}}{\text{Nakli}}$.
- Weight $x\%$ kam $\Rightarrow \frac{\text{Asli}}{\text{Nakli}} = \frac{100}{100-x}$.
- Khareed aur bikri dono mein $a\%$, $b\%$ fayda $\Rightarrow \left[\left(1 + \frac{a}{100} \right) \left(1 + \frac{b}{100} \right) - 1 \right] \times 100$.

TERMS – Har letter ka matlab

Asli = jitne ka paisa liya (1000 g) · Nakli = asal mein diya (800 g) · p = upar se profit.

SOLVED – Asli book questions (verified)**Q (EN 9):** 1 kg ki jagah 800 g

$$\frac{200}{800} \times 100 = 25\% \checkmark \text{ book ans (a)}$$

Q (EN 130): 25% gain $\Rightarrow$ 1 kg ki jagah kitna

$$\frac{1000 \times 100}{125} = 800 \text{ g } \checkmark \text{ book ans (a)}$$

Q (EN 133): 20% gain $\Rightarrow$ weight

$$\frac{1000 \times 100}{120} = 833.33 \text{ g } \checkmark \text{ book ans (c)}$$

Q (EN 129): 950 g + MP 15% upar

$$1.15 \times \frac{1000}{950} = 1.2105 \Rightarrow 21 \frac{1}{19}\% \checkmark \text{ book ans (c)}$$

Q (EN 10): ₹18/kg khareed, ₹20/kg bech, 900 g deta

$$\frac{20}{18} \times \frac{1000}{900} = 1.2345 \Rightarrow 23.45\% \checkmark \text{ book ans (c)}$$

Q (HI 16): 25% profit + baat 30% kam

$$\frac{1.25}{0.7} = 1.7857 \Rightarrow 78.57\% \checkmark \text{ book ans (b)}$$

Q (HI 13): khareed pe 30%, bikri pe 30%

$$1.3 \times 1.3 = 1.69 \Rightarrow 69\% \checkmark \text{ book ans (b)}$$

Q (HI 49): 14% gain $\Rightarrow$ 1 kg ki jagah

$$\frac{1000}{1.14} = 877 \text{ g } \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (26) EN 9, 10, 126, 129, 130, 133 · **HI** 2, 7–19, 40, 46, 49, 103 | Ch 6 (Pratishatata): **HI** 69 | Ch 12 (Mishran / Alligation): **EN** 47**PL-7 — SAME-SP MACHINE (do cheezein, ek hi SP)****PEHCHAN – Question mein ye dikhe**

two items sold at the same price / “₹990 each”, ek pe $x\%$ profit doosre pe $y\%$ loss, “selling prices of both are same”, CPs ya net profit/loss poocha; प्रत्येक ₹. . ., विक्रय मूल्य समान

$$\text{Same SP, } x\% \text{ profit \& } x\% \text{ loss} \Rightarrow \text{hamesha LOSS} = \frac{x^2}{100}\%$$

$$\text{Same SP, } a\% \text{ profit, } b\% \text{ loss} \Rightarrow CP_{\text{profit wali}} : CP_{\text{loss wali}} = (100 - b) : (100 + a)$$

- Total CP diya $\Rightarrow$ isi ratio se baanto.
- Net ₹: dono CP (PL-2: $SP \times 100 / (100 \pm P)$) jodo aur $2 \times SP$ se ghatao.

TERMS – Har letter ka matlab

$a = \text{profit } \% \cdot b = \text{loss } \% \cdot \text{ratio mein profit wali ke saamne } (100 - b), \text{ loss wali ke saamne } (100 + a).$

SOLVED – Asli book questions (verified)**Q (EN 119):** ₹990 each; 10% gain, 10% loss

$$\frac{10^2}{100} = 1\% \text{ loss} \checkmark \text{ book ans (c)}$$

Q (EN 120): ₹8,25,000 each; $\pm 9\%$

$$\frac{81}{100} = 0.81\% \text{ loss} \checkmark \text{ book ans (b)}$$

Q (EN 53): kul ₹6,50,000; A 20% profit, B 25% loss, same SP

$$75 : 120 = 5 : 8 \Rightarrow A = 2,50,000 \checkmark \text{ book ans (d)}$$

Q (EN 60): kul ₹900; 25% profit, 25% loss, same SP

$$75 : 125 = 3 : 5 \Rightarrow 337.5, 562.5 \checkmark \text{ book ans (d)}$$

Q (EN 66): kul ₹480; 15% loss, 19% profit, same SP

$$\text{loss wali : profit wali} = 119 : 85 = 7 : 5 \Rightarrow 280, 200 \checkmark \text{ book ans (a)}$$

Q (EN 122): kul ₹41,000; 20% profit, 15% loss, same SP $\Rightarrow$ net

$$85 : 120 \Rightarrow 17000 \rightarrow 20400; \text{kul SP } 40800 \Rightarrow ₹200 \text{ loss} \checkmark \text{ book ans (b)}$$

Q (EN 125): ₹12,000 each; 20% profit, 10% loss

$$24000 - (10000 + 13333.33) = 666.67 \text{ profit} \checkmark \text{ book ans (c)}$$

Q (HI 26): ₹20,000 each; $\pm 15\%$

$$17391.3 + 23529.4 - 40000 = 920.7 \approx 921 \text{ loss} \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (16) EN 53, 60, 66, 70, 103, 104, 119, 120, 122, 125 · **HI** 26, 34, 74, 80 | Ch 6 (Pratishatata): EN 177 · **HI** 103**PL-8 — WEIGHTED PARTS / TWO-ITEM MACHINE****PEHCHAN – Question mein ye dikhe**

“40% goods at 2% loss, rest at 4% profit”, “2/3 part . . ., remaining part”, “total 15% ke liye baaki kitne mein”, do items alag % pe, “total profit ₹. . .”, “neither profit nor loss”, second item ka %, rates ka mixture 2 : 4 : 9; शेष, भाग, मिश्रित, कुल लाभ

$$\text{Overall \%} = \frac{w_1 p_1 + w_2 p_2 + \dots}{w_1 + w_2 + \dots} \quad (w = \text{CP ka hissa, loss} = -)$$

- **Baaki hisse ka %:** $p_{\text{rest}} = \frac{T \cdot W - \sum w_i p_i}{w_{\text{rest}}}$ (T = chahiye overall %, W = kul).
- **Do items, kul CP ₹C, kul profit ₹N:** sab ko chhote % par maano $\Rightarrow$ bacha ₹ = doosre ka CP $\times$ (% gap)/100.
- **Break-even:** ek ka profit ₹ = doosre ka loss ₹. Do haalat ke do equation $\Rightarrow$ Cramer (NS-9).
- **Mixture:** avg CP = $\frac{\sum \text{qty} \times \text{rate}}{\sum \text{qty}} \Rightarrow$ phir PL-2.

TERMS – Har letter ka matlab

w = har hisse ka CP (ya fraction / ginti) · p = us hisse ka profit % · T = target overall %.

SOLVED – Asli book questions (verified)

Q (EN 101): ₹15,000; 45% pe 40% gain, 25% pe 10% loss, baaki CP pe
 $0.45 \times 40 - 0.25 \times 10 = 15.5\% \Rightarrow 2325$ ✓ **book ans (b)**

Q (EN 59): 2/3 pe 6% profit, baaki 3% loss, profit ₹540
 $4 - 1 = 3\% \Rightarrow \frac{540}{0.03} = 18000$ ✓ **book ans (b)**

Q (EN 49): 40% pe 2% loss, baaki 4% profit; ₹250
 $-0.8 + 2.4 = 1.6\% \Rightarrow 15625$ ✓ **book ans (d)**

Q (EN 99): 60 bags @₹400; 18 pe 8%; kul 16.4% $\Rightarrow$ baaki ka rate
 $\frac{984-144}{42} = 20\% \Rightarrow 480$ ✓ **book ans (b)**

Q (EN 168): ₹3 lakh; 25% pe -25%, 40% pe +25%; kul 15%
 baaki 35% pe $\frac{15-3.75}{0.35} = 32.14\% \Rightarrow 1,38,750$ ✓ **book ans (b)**

Q (HI 83): 50 items ₹550; 1/5 pe 10% loss; kul 20%
 baaki 40 pe 27.5% $\Rightarrow 11 \times 1.275 = 14.025$ ✓ **book ans (d)**

Q (EN 48): 2 Samsung + 3 Mi = ₹40,200; 10%, 20%; profit ₹5640
 sab 10% $\Rightarrow 4020$; bacha 1620 = 3 Mi $\times 10\% \Rightarrow 5400$ ✓ **book ans (a)**

Q (EN 121): ₹400 pe 25% profit; doosri 10% loss pe, na profit na loss
 profit 80 = 10% of CP₂ $\Rightarrow 800$ ✓ **book ans (c)**

Q (HI 82): ₹50, ₹40, ₹30 wale seb 2 : 4 : 9, 30% profit
 $\frac{530}{15} \times 1.3 = 45.9$ ✓ **book ans (a)**

BOOK KE SAWAAL (30) EN 21, 48, 49, 51, 58, 59, 96, 99, 101, 121, 123, 145, 146, 165, 166, 168 · **HI** 23, 27, 43, 71, 73, 82, 83, 88, 105, 109, 112, 114 | Ch 6
 (Pratishatata): EN 171 · **HI** 98

PL-9 — MP – DISCOUNT – PROFIT CHAIN**PEHCHAN – Question mein ye dikhe**

marked / printed / labelled / list price, MRP, discount, “marks 25% above CP and allows 12%”, “discount and profit both 20%”,
 “buys at 24% discount of printed price”; अंकित मूल्य, छूट

$$\frac{MP}{CP} = \frac{100 + P}{100 - D} \quad P\% = \frac{M(100 - D)}{100} - 100 \quad (M = MP, CP \text{ ke } \% \text{ mein})$$

- Successive discounts: $SP = MP \left(1 - \frac{d_1}{100}\right) \left(1 - \frac{d_2}{100}\right)$.
- MP se d₁% kam mein khareeda, MP se d₂% zyada mein becha $\Rightarrow P\% = \frac{100+d_2}{100-d_1} \times 100 - 100$.
- MP : CP = a : b $\Rightarrow M = \frac{a}{b} \times 100$ · CP = MP ka c% $\Rightarrow M = \frac{100}{c} \times 100$.

TERMS – Har letter ka matlab

MP = marked/list price · D = discount % · P = profit % · M = MP jab CP = 100.

SOLVED – Asli book questions (verified)

Q (EN 158): 20% discount ke baad bhi 8% profit; CP ₹500 $\Rightarrow$ MP

$$500 \times \frac{108}{80} = 675 \quad \checkmark \text{ book ans (d)}$$

Q (EN 131): 4% discount, CP ₹120, 10% profit $\Rightarrow$ MP

$$120 \times \frac{110}{96} = 137.50 \quad \checkmark \text{ book ans (b)}$$

Q (EN 67): MP ₹1280, 10% discount, 20% profit $\Rightarrow$ CP

$$1280 \times \frac{90}{120} = 960 \quad \checkmark \text{ book ans (b)}$$

Q (EN 15): 25% upar mark, 12% chhoot

$$125 \times 0.88 = 110 \Rightarrow 10\% \quad \checkmark \text{ book ans (a)}$$

Q (EN 7): MP pe 18% profit; 10% discount

$$118 \times 0.9 = 106.2 \Rightarrow 6.2\% \quad \checkmark \text{ book ans (a)}$$

Q (EN 161): discount = profit = 20% $\Rightarrow$ MP kitna upar

$$\frac{120}{80} = 1.5 \Rightarrow 50\% \quad \checkmark \text{ book ans (a)}$$

Q (EN 24): CP = MP ka 75%, 15% discount

$$\frac{85}{75} = 1.1333 \Rightarrow 13.33\% \quad \checkmark \text{ book ans (b)}$$

Q (EN 14): printed se 24% kam khareeda, 20% zyada becha

$$\frac{120}{76} = 1.5789 \Rightarrow 57\frac{17}{19}\% \quad \checkmark \text{ book ans (b)}$$

Q (EN 13): MP : CP = 5 : 3; 25% + 10% discounts

$$\frac{5 \times 0.75 \times 0.9}{3} = 1.125 \Rightarrow 12.5\% \quad \checkmark \text{ book ans (a)}$$

Q (HI 110): price +20%, phir 5% + 5% chhoot

$$1.2 \times 0.95^2 = 1.083 \Rightarrow 8.3\% \quad \checkmark \text{ book ans (d)}$$

Q (EN 162): 25% phir 20% discount, final ₹18,000 $\Rightarrow$ label

$$\frac{18000}{0.75 \times 0.8} = 30000 \quad \checkmark \text{ book ans (c)}$$

BOOK KESAWAAL (17) EN 7, 13–15, 19, 22, 24, 67, 76, 111, 124, 131, 158, 161, 162 · **HI** 65, 110

PL-10 — “100 MAAN LO” CONDITION MACHINE**PEHCHAN – Question mein ye dikhe**

“CP went up 50%, SP same $\Rightarrow$ new profit %”, “sold at 60% of this price $\Rightarrow$ 20% loss”, “SP doubled $\Rightarrow$ profit tripled”, “% profit on CP = 20% on SP”, “production cost +15%”, “had he purchased 15% less”, SP –50% & sale +700%; दोगुना, मूल विक्रय मूल्य

$$CP = 100 \text{ maan lo} \Rightarrow SP = 100 \pm P \Rightarrow \text{jo badle, use badlo} \Rightarrow \text{naya } P\% = \frac{SP' - CP'}{CP'} \times 100$$

- SP ke f hisse (fraction/%) par q% loss/profit $\Rightarrow$ poora $SP = \frac{100 \pm q}{f}$ (CP = 100 par).
- $SP \times m$ karne par profit $\times k \Rightarrow SP = \frac{k-1}{k-m} CP$ (m = 2, k = 3 $\Rightarrow$ SP = 2CP $\Rightarrow$ 100%).
- Profit SP ka r% $\Rightarrow$ CP par $\frac{r}{100-r} \times 100\%$.
- Bikri (quantity) badhne se % nahi badalta: sirf per-item CP, SP dekho.

TERMS – Har letter ka matlab

CP', SP' = badlaav ke baad · f = SP ka hissa · m = SP kitne guna · k = profit kitne guna.

SOLVED – Asli book questions (verified)**Q (EN 4):** 74% profit; CP +50%, SP same

$$\frac{174-150}{150} = 16\% \checkmark \text{ book ans (b)}$$

Q (EN 6): 30% profit; CP +25%, SP +10%

$$\frac{143-125}{125} = 14.4\% \checkmark \text{ book ans (c)}$$

Q (EN 17): 140% profit; SP –50%, bikri +700%

$$240 \times 0.5 = 120 \Rightarrow 20\% \checkmark \text{ book ans (c)}$$

Q (EN 8): SP ka 60% pe 20% loss $\Rightarrow$ asli SP pe

$$SP = \frac{80}{0.6} = 133.3 \Rightarrow 33\frac{1}{3}\% \checkmark \text{ book ans (b)}$$

Q (HI 25): SP ka 7/12 pe 16% loss; SP ka 80% pe?

$$SP = \frac{84}{7/12} = 144 \Rightarrow 144 \times 0.8 = 115.2 \Rightarrow 15.2\% \checkmark \text{ book ans (b)}$$

Q (HI 36): SP double $\Rightarrow$ profit teen guna

$$SP = \frac{3-1}{3-2} CP = 2CP \Rightarrow 100\% \checkmark \text{ book ans (c)}$$

Q (EN 160): CP par kitna % = SP par 20%

$$\frac{20}{80} \times 100 = 25\% \checkmark \text{ book ans (c)}$$

Q (EN 85): ₹24,750 pe 10%; cost +15%, 15% profit chahiye

$$\frac{24750}{1.1} \times 1.15 \times 1.15 = 29756.25 \checkmark \text{ book ans (c)}$$

Q (EN 78): ₹70,000 + ₹5000; 15% kam khareedta to 15% profit

$$(59500 + 5000) \times 1.15 = 74175 \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (9) EN 4, 6, 8, 17, 78, 85, 160 · HI 25, 36**COVERAGE CHECK – Har sawaal ka formula**

Formula	Naam	EN Qs	HI Qs
PL-1	BASIC P/L % (CP, SP, kharcha, fraction)	11	9
PL-2	CP $\leftrightarrow$ SP MULTIPLIER + SP TRANSFER + CHAIN	47	33
PL-3	DIFFERENCE MACHINE (₹ farak $\leftrightarrow$ % farak)	18	12
PL-4	ARTICLE-COUNT MACHINE (x ka SP = y ka CP)	10	6
PL-5	RATE MACHINE (a items ₹x mein)	24	13
PL-6	FALSE WEIGHT MACHINE (nakli baat)	6	18
PL-7	SAME-SP MACHINE (do cheezein, ek hi SP)	10	4
PL-8	WEIGHTED PARTS / TWO-ITEM MACHINE	16	12
PL-9	MP – DISCOUNT – PROFIT CHAIN	15	2
PL-10	“100 MAAN LO” CONDITION MACHINE	7	2
$\rightarrow$ NS-9	TWO-NUMBER + EQUATION MACHINE (Ch 1)	4	0
$\rightarrow$ PC-2	MULTIPLIER MACHINE (badha / ghata / ulta) (Ch 6)	0	1
$\rightarrow$ PC-4	POPULATION / GROWTH MACHINE (Ch 6)	0	1
$\rightarrow$ RP-5	PROPORTION MACHINE (a : b :: c : d) (Ch 9)	0	1
Total (book ke sawaal)		168/168 ✓	114/114 ✓
$\rightarrow$ NS-9: EN 41, 108, 118, 154 · $\rightarrow$ PC-2: HI 56 · $\rightarrow$ PC-4: HI 57 · $\rightarrow$ RP-5: HI 64			

Chapter 8

Discount (Batta / Chhoot) बट्टा / छूट

4 formulas · English book 76 sawaal · Hindi book 79 sawaal · 32 solved examples ✓

DS-1 — BASIC DISCOUNT (MP, SP, D%) + TAX / COMMISSION

PEHCHAN – Question mein ye dikhe

marked / list / labelled price aur SP diye $\Rightarrow$ discount %, “discount of 21% $\Rightarrow$ SP”, “paid ₹28,473 after 25% discount $\Rightarrow$ MP”, “saved ₹200 at 25%”, VAT / GST / sales tax / commission, “₹3600 at 10% discount $\Rightarrow$ at 15%?”; अंकित मूल्य, सूची मूल्य, छूट, विक्रय मूल्य

$$D = MP - SP \quad D\% = \frac{MP - SP}{MP} \times 100 \quad SP = MP \times \frac{100 - d}{100} \quad MP = SP \times \frac{100}{100 - d}$$

- ₹s bachaye (d% par) $\Rightarrow MP = \frac{s \times 100}{d}$; “₹s bachaye, ₹k kharch kiye” $\Rightarrow d\% = \frac{s}{s+k} \times 100$.
- Tax / VAT / GST** (discount ke baad lagta hai): Bill = $MP \left(1 - \frac{d}{100}\right) \left(1 + \frac{t}{100}\right)$; ulta: divide. Commission c% $\Rightarrow SP = MP \left(1 - \frac{c}{100}\right)$.
- SP transfer:** $SP_2 = SP_1 \times \frac{100 - d_2}{100 - d_1}$ · kai cheezein $\Rightarrow$ kul MP par discount.

TERMS – Har letter ka matlab

MP = marked/list price (hamesha 100% wala) · d = discount % · t = tax % · discount hamesha MP par.

SOLVED – Asli book questions (verified)

Q (EN 11): MP ₹2850, 21% discount

$$2850 \times 0.79 = 2251.50 \quad \checkmark \text{ book ans (d)}$$

Q (EN 13): 25% discount ke baad ₹28,473 diye

$$\frac{28473}{0.75} = 37964 \quad \checkmark \text{ book ans (d)}$$

Q (EN 22): 25% discount pe ₹200 bache

$$\frac{200 \times 100}{25} = 800 \quad \checkmark \text{ book ans (c)}$$

Q (EN 45): MP ₹1880, SP ₹1598

$$\frac{282}{1880} \times 100 = 15\% \quad \checkmark \text{ book ans (c)}$$

Q (EN 39): ₹6500, 5% discount phir 4% sales tax

$$6500 \times 0.95 \times 1.04 = 6422 \quad \checkmark \text{ book ans (c)}$$

Q (HI 56): MP ₹x, 20% chhoot, 12.5% VAT $\Rightarrow$ ₹432

$$\frac{432}{0.8 \times 1.125} = 480 \quad \checkmark \text{ book ans (b)}$$

Q (EN 62): ₹3600 pe 10% discount; 15% pe?

$$3600 \times \frac{85}{90} = 3400 \quad \checkmark \text{ book ans (d)}$$

Q (EN 50): ₹5 bachaye, ₹45 kharche

$$\frac{5}{50} \times 100 = 10\% \quad \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (39) EN 11, 13, 18, 22, 23, 25, 38, 39, 41–43, 45–47, 50, 52, 53, 55, 56, 62, 64, 71–74 · **HI** 2, 10, 32, 40, 46, 51, 56, 66, 67, 69, 70, 72, 75, 77

DS-2 — SUCCESSIVE DISCOUNT MACHINE

PEHCHAN – Question mein ye dikhe

successive / consecutive / additional / further discounts, “single discount equivalent to 10%, 15%, 20%”, “one discount 10%, find other”, “which shop is cheapest”, “difference between 32% and 20% + 20%”, “10% every 4 months”; क्रमिक, क्रमागत, एकल छूट, समतुल्य

Do discounts: $a + b - \frac{ab}{100}$ Teen: $\left[1 - \left(1 - \frac{a}{100}\right)\left(1 - \frac{b}{100}\right)\left(1 - \frac{c}{100}\right)\right] \times 100$

- $SP = MP \left(1 - \frac{a}{100}\right)\left(1 - \frac{b}{100}\right) \cdots$ (order se farak nahi padta).
- **Doosra discount:** $x\% = \left(1 - \frac{SP}{MP(1 - \frac{a}{100})}\right) \times 100$.
- **Compare:** $(100 - a)(100 - b)$ jiska sabse **chhota** $\Rightarrow$ sabse sasta (bada single discount).
- Single d% vs successive (net e%) ka ₹ farak $\Rightarrow MP = \frac{\text{₹ farak} \times 100}{|e - d|}$.
- Discount phir badhotri $(-20\%, +10\%) \Rightarrow \left(1 - \frac{20}{100}\right)\left(1 + \frac{10}{100}\right)$.

TERMS – Har letter ka matlab

a, b, c = ek ke baad ek discounts · x = dhoondhna wala discount · net = kul single discount.

SOLVED – Asli book questions (verified)

Q (EN 2): 10%, 15%, 20% ka single
 $1 - 0.9 \times 0.85 \times 0.8 = 0.388 \Rightarrow 38.8\%$ ✓ **book ans (d)**

Q (EN 9): 30%, 20%, 10% ka single
 $1 - 0.7 \times 0.8 \times 0.9 = 49.6\%$ ✓ **book ans (c)**

Q (HI 6): 10% aur 20%
 $10 + 20 - 2 = 28\%$ ✓ **book ans (b)**

Q (EN 5): 32% vs 20% + 20% ka farak ₹25
 net 36% $\Rightarrow \frac{25 \times 100}{4} = 625$ ✓ **book ans (b)**

Q (EN 6): ₹200 $\rightarrow$ ₹150, ek 10% $\Rightarrow$ doosra
 $1 - \frac{150}{200} = 16.66\%$ ✓ **book ans (d)**

Q (EN 10): ₹120 $\rightarrow$ ₹90, doosra 9% $\Rightarrow$ pehla
 $1 - \frac{90}{120 \times 0.91} = 17.58\%$ ✓ **book ans (b)**

Q (EN 30): har 4 mahine 10%; Dec mein ₹25,515
 Jan $\rightarrow$ Dec 3 discounts $\Rightarrow \frac{25515}{0.9^3} = 35000$ ✓ **book ans (b)**

Q (EN 51): weekend 60% + Sunday 10% = ₹36; Tuesday?
 $MP = \frac{36}{0.4 \times 0.9} = 100 \Rightarrow 100 - 36 = 64$ ✓ **book ans (d)**

Q (HI 58): ₹58,000 $\rightarrow$ ₹42,000; 5%, 10%, x%
 $1 - \frac{42000}{58000 \times 0.95 \times 0.9} = 15.3\%$ ✓ **book ans (d)**

BOOK KE SAWAAL (50) EN 2–10, 12, 16, 21, 26, 30–34, 36, 40, 51, 57, 68 · **HI** 1, 3, 5, 6, 8, 12, 14, 18, 25–28, 30, 31, 34, 35, 37, 39, 42–44, 53, 58, 61, 63, 68, 76

DS-3 — MP – DISCOUNT – PROFIT CHAIN (= PL-9)

PEHCHAN – Question mein ye dikhe

discount ke saath profit/loss/cost price, “marks 20% above CP, allows discount, profit 8%”, “to gain 25% after 10% discount, mark at?”, “how much % above CP should MP be”, CP badla to naya discount; लाभ, क्रय मूल्य से ... अधिक अंकित

$$\frac{MP}{CP} = \frac{100 + P}{100 - D} \quad P\% = \frac{M(100 - D)}{100} - 100 \quad D\% = \frac{M - (100 + P)}{M} \times 100$$

- $M = MP$ jab $CP = 100$ (MP “x% above CP” $\Rightarrow M = 100 + x$).
- Successive discounts $\Rightarrow (100 - D)$ ki jagah product $100\left(1 - \frac{a}{100}\right)\left(1 - \frac{b}{100}\right)$.
- “Discount % = Profit %”, “SP = 3 × discount” jaise $\Rightarrow CP = 100$ (ya $D = 1$) maan kar seedha likho.

TERMS – Har letter ka matlab

$M = MP$ (CP = 100 par) · P = profit % · D = discount %.

SOLVED – Asli book questions (verified)**Q (EN 15):** CP ₹360, 10% discount, 25% gain $\Rightarrow$ MP

$$360 \times \frac{125}{90} = 500 \quad \checkmark \text{ book ans (c)}$$

Q (EN 17): 5% discount, 10% profit $\Rightarrow$ MP kitna upar

$$\frac{110}{95} = 1.1579 \Rightarrow 15\frac{15}{19}\% \quad \checkmark \text{ book ans (d)}$$

Q (EN 20): 20% upar mark, profit 8% $\Rightarrow$ discount

$$\frac{120-108}{120} = 10\% \quad \checkmark \text{ book ans (c)}$$

Q (EN 28): 22% discount ke baad 17% profit

$$\frac{117}{78} = 1.5 \Rightarrow 50\% \quad \checkmark \text{ book ans (a)}$$

Q (EN 29): ₹1680 pe 16% loss; 8% discount ke baad 15% profit $\Rightarrow$ MP

$$CP \ 2000 \Rightarrow 2000 \times \frac{115}{92} = 2500 \quad \checkmark \text{ book ans (c)}$$

Q (EN 66): 25% upar mark; na profit na loss $\Rightarrow$ max discount

$$\frac{125-100}{125} = 20\% \quad \checkmark \text{ book ans (b)}$$

Q (HI 7): 25% + 10% discount pe 25% profit; CP +30% $\Rightarrow$ naya single discount

$$M = \frac{125}{0.675} = 185.19; \text{ chahiye } 130 \times 1.25 = 162.5 \Rightarrow 12.25\% \quad \checkmark \text{ book ans (c)}$$

Q (HI 15): X, Y ₹600 each; X: 25% + 15% pe ₹216 profit; Y: 46%

$$MP = \frac{816}{0.6375} = 1280 \Rightarrow Y = 691.2 \Rightarrow 15.2\% \quad \checkmark \text{ book ans (a)}$$

Q (EN 14): SP = 3 $\times$ discount, discount % = profit % $\Rightarrow$ D : CP

$$MP = 4D \Rightarrow 25\%; CP = \frac{3D}{1.25} = 2.4D \Rightarrow 5 : 12 \quad \checkmark \text{ book ans (c)}$$

Q (EN 19): 12 ka CP = 9 ka SP; 10 pe discount = 5 pe profit

$$CP \ 1, SP \ \frac{4}{3}, D = \frac{1}{6} \Rightarrow \frac{1/6}{3/2} = 11.11\% \quad \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (53) EN 1, 14, 15, 17, 19, 20, 28, 29, 37, 48, 54, 58–61, 63, 65–67, 69, 70, 75, 76 · HI 4, 7, 9, 15–17, 19, 21–24, 29, 33, 36, 38, 41, 45, 47–50, 57, 59, 60, 62, 64, 65, 73, 78, 79**DS-4 — FREE-ITEM SCHEME (Buy x Get y)****PEHCHAN – Question mein ye dikhe**

“buy 5 get 3 free”, “3 saree lene par 2 free”, schemes compare (free vs successive), free items + discount + profit; मुफ्त, खरीदें ... पाएं

$$\text{Buy } x \text{ Get } y \text{ free} \Rightarrow \text{Discount \%} = \frac{y}{x+y} \times 100$$

- Compare: har scheme ka single discount % nikalo (successive ke liye DS-2) $\Rightarrow$ bada = zyada chhoot.
- Free + discount d% + profit P%: customer x ka paisa $x \cdot MP(1 - \frac{d}{100})$ deta hai, $x + y$ cheezein leta hai $\Rightarrow x M(1 - \frac{d}{100}) = (x + y)(100 + P)$.

TERMS – Har letter ka matlab x = jitni ke paise diye · y = jitni free mili · M = MP (CP = 100 par).**SOLVED – Asli book questions (verified)****Q (HI 52):** 5 khareedo 3 free

$$\frac{3}{8} \times 100 = 37.5\% \quad \checkmark \text{ book ans (a)}$$

Q (HI 54): 3 saree par 2 free

$$\frac{2}{5} \times 100 = 40\% \quad \checkmark \text{ book ans (c)}$$

Q (HI 13): 30% + 20% vs 2 khareedo 1 free $\Rightarrow$ farak

$$44\% - 33\frac{1}{3}\% = 10\frac{2}{3}\% \quad \checkmark \text{ book ans (b)}$$

Q (HI 11): (I) 30% + 35%, (II) 6 + 4 free, (III) 5 + 6 free $\Rightarrow$ max

$$I \ 54.5\%, II \ 40\%, III \ \frac{6}{11} = 54.55\% \Rightarrow \text{sirf III} \quad \checkmark \text{ book ans (d)}$$

Q (HI 55): 5% chhoot + 14 par 2 free, 33% profit $\Rightarrow$ MP kitna upar

$$14 \times 0.95M = 16 \times 133 \Rightarrow M = 160 \Rightarrow 60\% \quad \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (7) HI 11, 13, 20, 52, 54, 55 | Ch 28 (Vividh): EN 79

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
DS-1	BASIC DISCOUNT (MP, SP, D%) + TAX / COMMISSION	25	14
DS-2	SUCCESSIVE DISCOUNT MACHINE	23	27
DS-3	MP – DISCOUNT – PROFIT CHAIN (= PL-9)	23	30
DS-4	FREE-ITEM SCHEME (Buy x Get y)	0	6
→ PC-1	PERCENT BASICS (of · kitna % · poor number) (Ch 6)	1	1
→ PC-4	POPULATION / GROWTH MACHINE (Ch 6)	2	0
→ PC-5	REVERSE + CONSUMPTION MACHINE (Ch 6)	1	0
→ PL-1	BASIC P/L % (CP, SP, kharcha, fraction) (Ch 7)	1	1
Total (book ke sawaal)		76/76 ✓	79/79 ✓
→ PC-1: EN 49 HI 74 · → PC-4: EN 24, 27 · → PC-5: EN 35 · → PL-1: EN 44 HI 71			

Chapter 9

Ratio & Proportion (Anupat aur Samanupat) अनुपात और समानुपात

8 formulas · English book 178 sawaal · Hindi book 162 sawaal · 59 solved examples ✓

RP-1 — RATIO KA BATWARA (share / ek hissa / farak)

PEHCHAN – Question mein ye dikhe

“₹1470 in 3 : 4”, “y’s share ₹2400, total?”, “difference of B and C is ₹5940”, boys/girls ratio se ginti ya %, sum & difference se ratio, units (kg-g, din-ghanta), sum of squares / product diya; अनुपात में बाँटा, भाग, अंतर

$$\text{Total } T \text{ ko } a : b : c \text{ mein } \Rightarrow \text{hissa} = T \times \frac{a}{a+b+c} \quad 1 \text{ unit} = \frac{\text{diya hua hissa}}{\text{uska ratio}} = \frac{\text{farak}}{a-b}$$

- Sum S, difference D $\Rightarrow$ bada : chhota = $(S + D) : (S - D)$; “farak = sum ka x%” $\Rightarrow (100 + x) : (100 - x)$.
- Group % = $\frac{a}{a+b} \times 100$ · ratio simplify: HCF se bhaag (pehle same unit).
- Squares ka sum $\Rightarrow k^2 = \frac{\text{sum}}{a^2+b^2+c^2}$ · product $\Rightarrow k^3 = \frac{\text{product}}{abc}$.

TERMS – Har letter ka matlab

a, b, c = ratio ke terms · k = 1 unit · T = kul rakam/ginti.

SOLVED – Asli book questions (verified)

Q (EN 2): ₹1470 ko 3 : 4, Mohan (4)

$$1470 \times \frac{4}{7} = 840 \text{ ✓ book ans (d)}$$

Q (EN 5): 4 : 3, y ka ₹2400 $\Rightarrow$ total

$$\frac{2400}{3} \times 7 = 5600 \text{ ✓ book ans (c)}$$

Q (EN 57): 2 : 3 : 5, B-C farak ₹5940

$$\frac{5940}{2} \times 10 = 29700 \text{ ✓ book ans (d)}$$

Q (HI 12): 7 : 5, ek hissa doosre se 14 zyada

$$\frac{14}{2} \times 12 = 84 \text{ ✓ book ans (c)}$$

Q (EN 53): avg salary ₹65,000, ratio 15 : 11 $\Rightarrow$ wife

$$130000 \times \frac{11}{26} = 55000 \text{ ✓ book ans (d)}$$

Q (EN 15): sum 80, diff 8 $\Rightarrow$ ratio

$$88 : 72 = 11 : 9 \text{ ✓ book ans (d)}$$

Q (EN 11): farak = sum ka 30%

$$130 : 70 = 13 : 7 \text{ ✓ book ans (b)}$$

Q (EN 73): 3 : 7 : 8, squares ka sum 7808 $\Rightarrow$ chhoti

$$k^2 = \frac{7808}{122} = 64 \Rightarrow 24 \text{ ✓ book ans (a)}$$

Q (EN 75): 3 : 5 : 2, product 10290 $\Rightarrow$ badi

$$k^3 = \frac{10290}{30} = 343 \Rightarrow 35 \text{ ✓ book ans (b)}$$

BOOK KE SAWAAL (73) EN 1, 2, 4, 5, 8–11, 13, 15, 16, 22, 24, 25, 43, 52–54, 56, 57, 59, 72–75, 172, 175, 176 · **HI** 1–3, 5–8, 12–14, 16–20, 30–41, 47, 49, 50, 56–60, 147, 153 | Ch 10 (Saajhedaari): EN 2, 4, 9–12, 23 | Ch 11 (Samay aur Karya): EN 189

RP-2 — RATIO JODNA (A : B, B : C $\Rightarrow$ A : B : C)

PEHCHAN – Question mein ye dikhe

“a : b = 32 : 35 and b : c = 21 : 32”, “red : green ... while pink : red ...”, copper : zinc & nickel : zinc, A : B : C : D chain, “A is 4 times B, B is 16 times C”; और B : C, जबकि

$$A : B = a : b, B : C = c : d \Rightarrow A : B : C = ac : bc : bd \quad A : C = ac : bd$$

- Common term (B) ko LCM par barabar karo, baaki usi guna se badhao.
- **Chain:** $\frac{A}{D} = \frac{A}{B} \cdot \frac{B}{C} \cdot \frac{C}{D}$ (numerators ka product : denominators ka product).
- Ulta diya ho (C : B, D : B) to pehle palat do.

TERMS – Har letter ka matlab

a : b, c : d = diye hue do ratios · B = common term.

SOLVED – Asli book questions (verified)

Q (EN 28): a : b = 32 : 35, b : c = 21 : 32

$$a : c = 32 : 21 : 35 : 32 = 3 : 5 \quad \checkmark \text{ book ans (c)}$$

Q (EN 26): R : G = 15 : 32, P : R = 18 : 25 $\Rightarrow$ G : P

$$R = 75 \Rightarrow G = 160, P = 54 \Rightarrow 80 : 27 \quad \checkmark \text{ book ans (a)}$$

Q (EN 47): a : b = 7 : 9, b : c = 5 : 11 $\Rightarrow$ a : b : c

$$35 : 45 : 99 \quad \checkmark \text{ book ans (d)}$$

Q (EN 48): a : b = 3 : 5, c : b = 3 : 2, c : d = 5 : 6 $\Rightarrow$ a : d

$$\frac{3}{5} \cdot \frac{2}{3} \cdot \frac{5}{6} = \frac{1}{3} \quad \checkmark \text{ book ans (c)}$$

Q (EN 39): A : B = 3 : 4, B : C = 6 : 5 $\Rightarrow$ A : (A + C)

$$18 : 24 : 20 \Rightarrow 18 : 38 = 9 : 19 \quad \checkmark \text{ book ans (c)}$$

Q (EN 36): a : b = 2 : 3, a : c = 10 : 21 $\Rightarrow$ b : c

$$a = 10 \Rightarrow b = 15 \Rightarrow 15 : 21 = 5 : 7 \quad \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (43) EN 26–28, 30–34, 36, 37, 39–42, 44–50, 55, 58, 67–69, 166, 167 · **HI** 21, 22, 24–27, 29, 44, 45, 48, 51–54 | Ch 10 (Saajhedaari): **EN** 5

RP-3 — BARABARI SE RATIO (x% of A = y% of B, $\frac{1}{2}A = \frac{1}{3}B$)
PEHCHAN – Question mein ye dikhe

“10% of x = 15% of y”, “ $\frac{1}{3}$ of A = $\frac{3}{4}$ of B = $\frac{1}{6}$ of C”, “ $2A = 3B = 5C$ ”, “ $A/4 = B/5 = C/6$ ”, “A 15% more than B, B 20% less than C”, parts ko alag % se badhana; का...% = का...%, गुना, भाग बराबर

$$\frac{1}{p}A = \frac{1}{q}B = \frac{1}{r}C \Rightarrow A : B : C = p : q : r \quad pA = qB = rC \Rightarrow A : B : C = \frac{1}{p} : \frac{1}{q} : \frac{1}{r}$$

- $x\%$ of $A = y\%$ of $B \Rightarrow A : B = y : x$ (ulta).
- “x% zyada / kam” wali chain: jiske saath compare ho use 100 maano $\Rightarrow$ baaki likho.
- Har hissa alag % badhe: $a(1 + \frac{p}{100}) : b(1 + \frac{q}{100})$; sab mein same % $\Rightarrow$ ratio same.
- $\frac{1}{a} : \frac{1}{b} : \frac{1}{c} \Rightarrow$ LCM se guna karke poore number.

TERMS – Har letter ka matlab

p, q, r = barabari wale fractions/guna · x, y = percents.

SOLVED – Asli book questions (verified)**Q (EN 7):** boys ka 15% = girls ka $\frac{1}{3}$

$$B : G = \frac{1}{3} : 0.15 = 20 : 9 \quad \checkmark \text{ book ans (a)}$$

Q (EN 20): pehle ka 25% = doosre ke 50% ka 3 guna

$$0.25A = 1.5B \Rightarrow 6 : 1 \quad \checkmark \text{ book ans (b)}$$

Q (EN 63): 15% x = 25% y = 50% z

$$\frac{1}{15} : \frac{1}{25} : \frac{1}{50} = 10 : 6 : 3 \quad \checkmark \text{ book ans (b)}$$

Q (EN 60): A, B se 15% zyada; B, C se 20% kam

$$C \ 100, B \ 80, A \ 92 \Rightarrow 23 : 20 : 25 \quad \checkmark \text{ book ans (a)}$$

Q (EN 66): ₹3900 ko $\frac{1}{2} : \frac{1}{3} : \frac{1}{4} \Rightarrow K$ ($\frac{1}{3}$)

$$6 : 4 : 3 \Rightarrow 3900 \times \frac{4}{13} = 1200 \quad \checkmark \text{ book ans (c)}$$

Q (HI 43): ₹651; $2A = 3B = 5C \Rightarrow C$

$$15 : 10 : 6 \Rightarrow 651 \times \frac{6}{31} = 126 \quad \checkmark \text{ book ans (b)}$$

Q (EN 178): ₹13,680; pehla = teesre ka $\frac{3}{5}$, doosra : teesra = 4 : 7

$$21 : 20 : 35 \Rightarrow 13680 \times \frac{21}{76} = 3780 \quad \checkmark \text{ book ans (a)}$$

Q (EN 86): 4 : 5; male +80%, female +60%

$$4 \times 1.8 : 5 \times 1.6 = 9 : 10 \quad \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (35) EN 7, 14, 20, 29, 35, 38, 60–66, 71, 78, 83, 86, 174, 177, 178 · **HI** 9, 11, 23, 28, 42, 43, 46, 160, 162 | Ch 10 (Saajhedaari): **EN** 3, 7, 24, 25 | Ch 28 (Vividh): **EN** 26, 51

RP-4 — RATIO CHANGE MACHINE (jodna / ghatana / dena)**PEHCHAN – Question mein ye dikhe**

“2 : 3, 4 jodne par 7 : 10”, “what should be added to 7 : 11 to make 4 : 5”, “A gives ₹12 to B”, “8 more boys admitted”, “12 green balls added”, milk / sugar / zinc milaya, “50 male & 100 female shifted”; जोड़ा, घटाया, दिए, प्रवेश

$$\frac{ak + x}{bk + y} = \frac{c}{d} \Rightarrow k = \frac{cy - dx}{ad - bc} \quad (\text{ghatana ho to } x, y \text{ negative})$$

- Dono mein same x: $k = \frac{x(c-d)}{ad-bc} \cdot a : b$ ko $c : d$ banane ke liye jodo $x = \frac{bc-ad}{d-c}$.
- **Lena-dena** (total same): $T = \frac{\text{₹ diya}}{\left| \frac{a}{a+b} - \frac{c}{c+d} \right|}$.
- **Sirf ek badla** (milk/zinc/balls add): na badalne wale ko fix rakho $\Rightarrow$ naya ratio usi se $\Rightarrow$ farak = joda gaya.

TERMS – Har letter ka matlab

$a : b$ = purana ratio · $c : d$ = naya · x, y = jo joda (+) / ghataya (−) · k = 1 unit.

SOLVED – Asli book questions (verified)

Q (EN 79): 2 : 3, dono mein 4 $\Rightarrow$ 7 : 10; farak

$$k = \frac{4(7-10)}{20-21} = 12 \Rightarrow 12 \checkmark \text{ book ans (c)}$$

Q (EN 82): 7 : 11 mein kya jodein ki 4 : 5

$$x = \frac{44-35}{1} = 9 \checkmark \text{ book ans (b)}$$

Q (EN 77): 3 : 4, dono + ₹3000 $\Rightarrow$ 18 : 23; A ka naya

$$k = \frac{3000(18-23)}{69-72} = 5000 \Rightarrow 18000 \checkmark \text{ book ans (d)}$$

Q (EN 88): 4 : 7; -50 M, -100 F $\Rightarrow$ 7 : 12

$$k = \frac{7(-100)-12(-50)}{48-49} = 100 \Rightarrow 400 \checkmark \text{ book ans (b)}$$

Q (HI 63): 2 : 7; pehla +14, doosra -6 $\Rightarrow$ 4 : 3; sum

$$k = \frac{4(-6)-3(14)}{6-28} = 3 \Rightarrow 27 \checkmark \text{ book ans (c)}$$

Q (EN 89): 9 : 5, Arun ₹12 deta $\Rightarrow$ 4 : 3

$$T = \frac{12}{\frac{9}{14}-\frac{4}{7}} = 168 \Rightarrow 108 \checkmark \text{ book ans (c)}$$

Q (EN 101): 4 : 7, Jen ₹1 deti $\Rightarrow$ 7 : 12

$$T = \frac{1}{\frac{7}{19}-\frac{4}{11}} = 209 \checkmark \text{ book ans (a)}$$

Q (EN 100): red : green 15 : 26, +12 green $\Rightarrow$ 1 : 2

$$30k - 26k = 12 \Rightarrow k = 3 \Rightarrow 45 \checkmark \text{ book ans (c)}$$

Q (EN 107): 25 L, milk : water 4 : 1 $\Rightarrow$ 16 : 1

$$\text{water } 5 \Rightarrow \text{milk } 80 \Rightarrow 80 - 20 = 60 \checkmark \text{ book ans (c)}$$

Q (HI 61): ₹4 aur ₹7, dono ko same $\Rightarrow$ 7 : 9

$$x = \frac{49-36}{2} = 6.50 \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (57) EN 6, 17, 70, 76, 77, 79–82, 84, 85, 87–114 · HI 55, 61–73, 159 | Ch 10 (Saajhedaari): EN 13 | Ch 28 (Vividh): EN 70 · HI 55

RP-5 — PROPORTION MACHINE (a : b :: c : d)**PEHCHAN – Question mein ye dikhe**

“a : b :: c : x”, fourth / third / mean proportional, continued proportion, compound ratio, duplicate / sub-duplicate, “x and y in direct proportion”, “what added to each of 4, 8, 12, 22 makes them proportional”; समानुपात, चतुर्थानुपाती, तृतीयानुपाती, मध्यानुपाती

$$a : b :: c : d \iff ad = bc \quad 4\text{th} = \frac{bc}{a} \quad 3\text{rd} = \frac{b^2}{a} \quad \text{Mean} = \sqrt{ab}$$

- Continued a, b, c $\Rightarrow b^2 = ac$ · a, b, c, d continued $\Rightarrow \frac{a}{d} = \frac{a^3}{b^3}$.
- Compound ratio = sab ka product · duplicate $a^2 : b^2$, sub-duplicate $\sqrt{a} : \sqrt{b}$, triplicate $a^3 : b^3$.
- Direct proportion: $\frac{y_1}{x_1} = \frac{y_2}{x_2}$ · a, b, c, d ko proportional banane ke liye jodo $x = \frac{bc-ad}{a+d-b-c}$.

TERMS – Har letter ka matlab

a, b, c, d = pehla...chautha term · extremes (a, d) ka product = means (b, c) ka product.

SOLVED – Asli book questions (verified)**Q (EN 118):** 10 aur 50 ka third proportional

$$\frac{50^2}{10} = 250 \quad \checkmark \text{ book ans (c)}$$

Q (EN 126): 15, 10, 12 ka fourth proportional

$$\frac{10 \times 12}{15} = 8 \quad \checkmark \text{ book ans (d)}$$

Q (EN 146): 2 aur 98 ka mean proportional

$$\sqrt{196} = 14 \quad \checkmark \text{ book ans (d)}$$

Q (EN 145): 0.16 aur 0.64 ka mean

$$\sqrt{0.1024} = 0.32 \quad \checkmark \text{ book ans (d)}$$

Q (EN 116): 20, 50 ka third : 9, 16 ka mean

$$125 : 12 \quad \checkmark \text{ book ans (d)}$$

Q (EN 128): 45 : 75, 3 : 5, 51 : 68, 256 : 81 ka compound

$$\frac{64}{75} \quad \checkmark \text{ book ans (a)}$$

Q (EN 134): 4, 8, 12, 22 mein kya jodein

$$x = \frac{96-88}{26-20} = \frac{4}{3} \quad \checkmark \text{ book ans (a)}$$

Q (EN 3): direct proportion: x = 37 pe y = 92.5; x = 16 pe

$$92.5 \times \frac{16}{37} = 40 \quad \checkmark \text{ book ans (b)}$$

Q (HI 103): 145 ko 2 : 3 (a, b); c = a, b ka mean

$$\sqrt{58 \times 87} = 71 \quad \checkmark \text{ book ans (d)}$$

BOOK KESAWAAL (107) EN 3, 18, 115–146, 171, 173 · **HI** 74–124, 126–140, 161 | Ch 7 (Labb aur Hani): **HI** 64 | Ch 11 (Samay aur Karya): **EN** 205 | Ch 26 (Sankhyiki aur Prayikta): **EN** 212 · **HI** 92

RP-6 — COINS / NOTES MACHINE**PEHCHAN – Question mein ye dikhe**

₹1, 50 paise, 25 paise coins ratio mein, total ₹. . ., “₹50 aur ₹100 ke kul 50 note, ₹4000”, “equal number of ₹50, ₹100, ₹200 notes”, coins ka %; सिक्के, नोट, पैसे

$$k = \frac{\text{Total ₹}}{a v_1 + b v_2 + c v_3} \Rightarrow \text{coins} = ak, bk, ck$$

- v = har coin ki value ₹ mein (50 paise = 0.5, 25 paise = 0.25) · ek type ki ginti di $\Rightarrow k = \text{ginti} \div \text{uska ratio}$.
- Do tarah ke note, kul N note, kul ₹V:** chhote wale = $\frac{v_2 N - V}{v_2 - v_1}$.
- Coins ka % diya $\Rightarrow 100$ coins ki value nikaalo $\Rightarrow$ unitary.

TERMS – Har letter ka matlab

a : b : c = coins ki GINTI ka ratio · v₁, v₂, v₃ = unki value · k = 1 unit.

SOLVED – Asli book questions (verified)**Q (EN 147):** ₹2, ₹5, ₹10 = 5 : 7 : 8, total ₹1250 $\Rightarrow$ ₹5 coins

$$k = \frac{1250}{125} = 10 \Rightarrow 70 \quad \checkmark \text{ book ans (c)}$$

Q (EN 152): 25p, 10p, 5p = 1 : 2 : 3, ₹30 $\Rightarrow$ 5p

$$k = \frac{30}{0.6} = 50 \Rightarrow 150 \quad \checkmark \text{ book ans (c)}$$

Q (EN 154): 50p, 25p, 10p = 5 : 4 : 3, ₹171

$$k = \frac{171}{3.8} = 45 \Rightarrow 225, 180, 135 \quad \checkmark \text{ book ans (b)}$$

Q (EN 158): ₹210; 20% ₹5, 25% ₹10, 15% ₹2, baaki ₹1

$$100 \text{ coins} = ₹420 \Rightarrow 50 \text{ coins} \Rightarrow ₹1 \text{ wale } 20 \quad \checkmark \text{ book ans (d)}$$

Q (HI 142): ₹50 + ₹100 ke 50 note = ₹4000

$$\frac{5000-4000}{50} = 20 \quad \checkmark \text{ book ans (d)}$$

Q (HI 141): ₹50, ₹100, ₹200 barabar ginti, ₹5250

$$\frac{5250}{350} = 15 \Rightarrow 45 \quad \checkmark \text{ book ans (b)}$$

BOOK KESAWAAL (25) EN 147, 148, 150–161 · **HI** 141–146, 148–152

RP-7 — INCOME – EXPENDITURE – SAVINGS RATIO

PEHCHAN – Question mein ye dikhe

incomes ratio + expenses ratio + savings ₹... , “savings 2 : 3, spend ₹35,400 & ₹48,600”, “each saves ₹10,000”, income : expenditure 9 : 5; आय, व्यय, बचत

$$\text{Income } a : b, \text{ Kharcha } c : d, \text{ Bachat } S_1, S_2 \Rightarrow k = \frac{S_1 d - S_2 c}{ad - bc} \quad (\text{Income} = ak, bk)$$

- Dono barabar bachayen (S) $\Rightarrow k = \frac{S(d-c)}{ad-bc}$.
- Income ratio + **savings** ratio, kharcha ₹ diye $\Rightarrow$ wahi formula (S ki jagah kharcha, c : d = savings ratio).
- Income : Kharcha = a : b $\Rightarrow$ Bachat = Income $\times \frac{a-b}{a}$.

TERMS – Har letter ka matlab

I = income · E = kharcha · S = bachat · I = E + S.

SOLVED – Asli book questions (verified)

Q (EN 12): income 7 : 5, exp 9 : 7, bachat 1700, 1100

$$k = \frac{11900-9900}{49-45} = 500 \Rightarrow 3500, 2500 \quad \checkmark \text{ book ans (d)}$$

Q (EN 163): income 5 : 7, savings 2 : 3, kharcha 35,400 & 48,600

$$k = \frac{106200-97200}{15-14} = 9000 \Rightarrow 18000 \quad \checkmark \text{ book ans (a)}$$

Q (EN 164): income 7 : 5, bachat 12,000 & 9000, exp 17 : 12 $\Rightarrow$ kul exp

$$k = 9000 \Rightarrow 63000 + 45000 - 21000 = 87000 \quad \checkmark \text{ book ans (c)}$$

Q (HI 157): income 5 : 7, dono ₹10,000 bachate, exp 2 : 3

$$k = \frac{10000(3-2)}{15-14} = 10000 \Rightarrow 50000 \quad \checkmark \text{ book ans (b)}$$

Q (EN 51): income : exp = 9 : 5, income ₹27,000

$$27000 \times \frac{4}{9} = 12000 \quad \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (9) EN 12, 51, 162–164 · HI 154–157

RP-8 — RATIO ALGEBRA (componendo – dividendo)

PEHCHAN – Question mein ye dikhe

“(m + n) : (m – n) = 7 : 3 $\Rightarrow$ (m³ + n³) : (m³ – n³)”, “(3x + 2y) : (3x – 2y) = 5 : 3”, “a³ + b³ : a³ – b³ = 185 : 158”, “(a + b) : (b + c) : (c + a) = 6 : 7 : 8”, a/2 = b/3 = c/5 = k, powers ka ratio

$$\frac{x+y}{x-y} = \frac{p}{q} \iff \frac{x}{y} = \frac{p+q}{p-q} \quad (\text{componendo-dividendo})$$

- Ratio mil gaya (x : y = a : b) $\Rightarrow$ **x = a, y = b rakh do** kisi bhi expression mein.
- $\frac{a}{2} = \frac{b}{3} = \frac{c}{5} = k \Rightarrow a = 2k, b = 3k, c = 5k$.
- (a+b) : (b+c) : (c+a) = p : q : r, a+b+c = S $\Rightarrow c = S \cdot \frac{q+r-p}{p+q+r}$.
- Linear: $\frac{2x+7y}{4x-3y} = \frac{8}{3} \Rightarrow$ cross-multiply $\Rightarrow x : y$.

TERMS – Har letter ka matlab

p : q = diya ratio · x, y = dhoondhne wale · k = common value.

SOLVED – Asli book questions (verified)

Q (EN 19): $(m + n) : (m - n) = 7 : 3 \Rightarrow (m^3 + n^3) : (m^3 - n^3)$

$m : n = 10 : 4 = 5 : 2 \Rightarrow 133 : 117$ ✓ **book ans (a)**

Q (EN 21): $(3x + 2y) : (3x - 2y) = 5 : 3 \Rightarrow x : y$

$3x : 2y = 8 : 2 \Rightarrow x : y = 8 : 3$ ✓ **book ans (d)**

Q (EN 23): $a^3 + b^3 : a^3 - b^3 = 185 : 158$

$a^3 : b^3 = 343 : 27 \Rightarrow 7 : 3$ ✓ **book ans (d)**

Q (HI 15): $x : y = 7 : 3 \Rightarrow (x^2 + y^2) : (x^2 - y^2)$

$58 : 40 = 29 : 20$ ✓ **book ans (b)**

Q (EN 165): $(a+b) : (b+c) : (c+a) = 6 : 7 : 8, a + b + c = 14$

$c = 14 \times \frac{7+8-6}{21} = 6$ ✓ **book ans (c)**

Q (HI 125): $(2x + 7y) : (4x - 3y) = 8 : 3$

$6x + 21y = 32x - 24y \Rightarrow 45 : 26$ ✓ **book ans (a)**

BOOK KESAWAAL (10) EN 19, 21, 23, 165, 168, 170 · HI 10, 15, 125, 158

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
RP-1	RATIO KA BATWARA (share / ek hissa / farak)	28	37
RP-2	RATIO JODNA ($A : B, B : C \Rightarrow A : B : C$)	28	14
RP-3	BARABARI SE RATIO ($x\%$ of $A = y\%$ of $B, \frac{1}{2}A = \frac{1}{3}B$)	20	9
RP-4	RATIO CHANGE MACHINE (jodna / ghatana / dena)	39	15
RP-5	PROPORTION MACHINE ($a : b :: c : d$)	36	67
RP-6	COINS / NOTES MACHINE	14	11
RP-7	INCOME – EXPENDITURE – SAVINGS RATIO	5	4
RP-8	RATIO ALGEBRA (componendo – dividendo)	6	4
→ NS-9	TWO-NUMBER + EQUATION MACHINE (Ch 1)	1	1
→ DF-4	FRACTION OPERATION MACHINE ($+$ $-$ $\times$ $\div$, “kya jodein (Ch 2)	1	0
Total (book ke sawaal)		178/178 ✓	162/162 ✓

→ **DF-4:** EN 169 · → **NS-9:** EN 149 HI 4

Chapter 10

Partnership (Saajhedaari) साझेदारी

2 formulas · English book 54 sawaal · Hindi book 25 sawaal · 17 solved examples ✓

PT-1 — CAPITAL × TIME MACHINE (profit ka batwara)

PEHCHAN – Question mein ye dikhe

“A ₹4000, B ₹8000 . . . left after 6 months, profit ₹34,000”, joined after x months, withdrew ₹5000 after 3 months, cows × months (rent), days × wages, “half profit equally, rest in ratio”, charity % pehle; निवेश, लाभ में हिस्सा, महीने बाद

$$\text{Profit ratio} = C_1T_1 : C_2T_2 : C_3T_3 \quad \text{Hissa} = \text{Profit} \times \frac{C_iT_i}{\sum CT}$$

- Beech mein capital badla $\Rightarrow$ tukdon ka jod: $30000 \times 3 + 25000 \times 9$. Join after m months $\Rightarrow T = 12 - m$ (2 saal $\Rightarrow 24 - m$).
- Investment “A = 3B, B = $\frac{2}{3}C$ ” $\Rightarrow$ pehle ratio (RP-3), phir batwara. Cows × months, days × wage — same machine.
- Pehle x% alag (charity / chalane wale ka hissa / aadha barabar), baaki ratio mein.

TERMS – Har letter ka matlab

C = capital · T = mahine (kitne samay paisa laga) · Σ = sab ka jod.

SOLVED – Asli book questions (verified)

Q (EN 26): ₹4000, ₹8000, ₹6000; Sharma 6 mahine baad gaya; 8 mahine ka ₹34,000

$$24 : 64 : 48 \Rightarrow 34000 \times \frac{64}{136} = 16000 \quad \checkmark \text{ book ans (d)}$$

Q (EN 30): ₹20,000 & ₹30,000; Hareesh ne 3 mahine baad ₹5000 nikale

$$240000 : (90000 + 225000) = 16 : 21 \quad \checkmark \text{ book ans (a)}$$

Q (EN 38): ₹12,000, ₹6000; C ₹15,000 8 mahine baad; 2 saal ka ₹33,600

$$288 : 144 : 240 \Rightarrow 33600 \times \frac{240}{672} = 12000 \quad \checkmark \text{ book ans (b)}$$

Q (EN 47): P ₹25,000×6, Q ₹44,000×5, R ₹50,000×3; ₹1,04,000

$$150 : 220 : 150 \Rightarrow 104000 \times \frac{150}{520} = 30000 \quad \checkmark \text{ book ans (d)}$$

Q (HI 8): 20 : 15 : 12; C ne 3 mahine baad aadha nikala; ₹3,23,000

$$240 : 180 : (36 + 54) \Rightarrow 323000 \times \frac{90}{510} = 57000 \quad \checkmark \text{ book ans (a)}$$

Q (EN 18): rent ₹415; 25 cows × 6, 40 × 8, 30 × 12

$$150 : 320 : 360 \Rightarrow 415 \times \frac{320}{830} = 160 \quad \checkmark \text{ book ans (d)}$$

Q (EN 27): A = 3B, B = $\frac{2}{3}C$; ₹6600

$$6 : 2 : 3 \Rightarrow 6600 \times \frac{2}{11} = 1200 \quad \checkmark \text{ book ans (b)}$$

Q (EN 40): ₹1400 & ₹1800; aadha barabar, aadha ratio mein; farak ₹47

$$\frac{P}{2} \times \frac{2}{16} = 47 \Rightarrow P = 752 \quad \checkmark \text{ book ans (a)}$$

Q (HI 15): 7 : 8; 7% charity; A ka ₹12,600

$$12600 \times \frac{15}{7} \div 0.93 = 29032.26 \quad \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (44) EN 8, 17–19, 26–47 · HI 6–15, 17, 19–25

PT-2 — UNKNOWN CAPITAL / TIME (ulta PT-1)

PEHCHAN – Question mein ye dikhe

“profit shared 3 : 2, how much did B invest”, “after how many months did Sohan join”, “how long was P’s money used”, ratio of investments from profit & time, “find C’s capital”, value of x; पूंजी में योगदान, कितने महीने बाद

$$\frac{C_1 T_1}{C_2 T_2} = \frac{p_1}{p_2} \Rightarrow C_2 = \frac{C_1 T_1 p_2}{p_1 T_2} \quad T_2 = \frac{C_1 T_1 p_2}{p_1 C_2}$$

- Join kab kiya: T_2 nikaalo $\Rightarrow$ join after $12 - T_2$ mahine.
- Profit ratio aur samay diye $\Rightarrow$ investment ratio $= \frac{p_1}{T_1} : \frac{p_2}{T_2} : \frac{p_3}{T_3}$.
- Samay same $\Rightarrow$ capital ratio = profit ratio (capital $\propto$ profit).

TERMS – Har letter ka matlab

$p_1 : p_2$ = profit ratio · C = capital · T = samay (mahine).

SOLVED – Asli book questions (verified)

Q (EN 16): P ₹3500 $\times$ 12; Q 5 mahine baad; Q ka profit 50% zyada

$$C_Q = \frac{42000 \times 1.5}{7} = 9000 \quad \checkmark \text{ book ans (c)}$$

Q (EN 14): A ₹28,000 $\times$ 12; B 5 mahine baad, C 7 baad; 4 : 2 : 3

$$B = \frac{336000 \times 2}{4 \times 7} = 24000, \quad C = 50400 \quad \checkmark \text{ book ans (a)}$$

Q (HI 16): A ₹7250 $\times$ 12; B 3 mahine baad; 8 : 9

$$B = \frac{87000 \times 9}{8 \times 9} = 10875 \quad \checkmark \text{ book ans (a)}$$

Q (EN 49): Raja ₹4400 $\times$ 12, Ranga ₹2400; ₹1200 mein Raja ₹800

$$T = \frac{52800 \times 400}{800 \times 2400} = 11 \Rightarrow 1 \text{ mahine baad} \quad \checkmark \text{ book ans (c)}$$

Q (EN 51): ₹1,00,000 $\times$ 12; Sohan ₹40,000; 3 : 1

$$T = \frac{1200000}{3 \times 40000} = 10 \Rightarrow 2 \quad \checkmark \text{ book ans (b)}$$

Q (EN 52): 5 : 13; Q 6 mahine; profit 25 : 26 $\Rightarrow$ P ka samay

$$T_P = \frac{13 \times 6 \times 25}{26 \times 5} = 15 \quad \checkmark \text{ book ans (b)}$$

Q (EN 48): samay 6, 12, 14; profit 5 : 4 : 7 $\Rightarrow$ investment

$$\frac{5}{6} : \frac{4}{12} : \frac{7}{14} = 5 : 2 : 3 \quad \checkmark \text{ book ans (a)}$$

Q (EN 6): kul capital ₹33,000; profit ₹15,000 mein A 4500, B 5500 $\Rightarrow$ C

$$33000 \times \frac{5000}{15000} = 11000 \quad \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (15) EN 1, 6, 14, 16, 48–54 · HI 1–3, 16

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
PT-1	CAPITAL $\times$ TIME MACHINE (profit ka batwara)	26	18
PT-2	UNKNOWN CAPITAL / TIME (ulta PT-1)	11	4
$\rightarrow$ NS-9	TWO-NUMBER + EQUATION MACHINE (Ch 1)	3	2
$\rightarrow$ PC-3	NET CHANGE MACHINE (do/teen change ek saath) (Ch 6)	1	1
$\rightarrow$ RP-1	RATIO KA BATWARA (share / ek hissa / farak) (Ch 9)	7	0
$\rightarrow$ RP-2	RATIO JODNA ($A : B, B : C \Rightarrow A : B : C$) (Ch 9)	1	0
$\rightarrow$ RP-3	BARABARI SE RATIO ($x\%$ of $A = y\%$ of $B, \frac{1}{2}A = \frac{1}{3}B$) (Ch 9)	4	0
$\rightarrow$ RP-4	RATIO CHANGE MACHINE (jodna / ghatana / dena) (Ch 9)	1	0
Total (book ke sawaal)		54/54 ✓	25/25 ✓
$\rightarrow$ NS-9: EN 20–22 HI 5, 18 · $\rightarrow$ PC-3: EN 15 HI 4 · $\rightarrow$ RP-1: EN 2, 4, 9–12, 23 · $\rightarrow$ RP-2: EN 5 · $\rightarrow$ RP-3: EN 3, 7, 24, 25 · $\rightarrow$ RP-4: EN 13			

Chapter 11

Work & Time (Samay aur Karya) समय और कार्य

7 formulas · English book 206 sawaal · Hindi book 106 sawaal · 49 solved examples ✓

WT-1 — LCM / RATE MACHINE (saath mein, akele, kitna kaam)

PEHCHAN – Question mein ye dikhe

“A 15 din, B 20 din $\Rightarrow$ saath mein”, “A + B = 6 din, A = 9 $\Rightarrow$ B”, 3 log alag-alag, “40% kaam 8 din mein”, “6 din saath kaam $\Rightarrow$ kitna bacha”, “ $\frac{3}{5}$ kaam kitne din mein”, builder–destroyer (minus), kaun pehle khatam karega; साथ मिलकर, अकेले, कार्य का भाग

$$\text{Kul kaam} = \text{LCM}(\text{din}) \quad \text{Saath} = \frac{ab}{a+b} \quad \text{Akela} = \frac{aT}{a-T} \quad (T = \text{saath ka time})$$

- Teen log: $\frac{1}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}$ · teesra dhoondho: $\frac{1}{c} = \frac{1}{T} - \frac{1}{a} - \frac{1}{b}$.
- Hissa diya: “f kaam d din mein” $\Rightarrow$ poora = $\frac{d}{f}$ din · f kaam saath mein = $f \times T$.
- t din saath $\Rightarrow$ kaam = $t \left(\frac{1}{a} + \frac{1}{b} \right)$, bacha = $1 - \text{usse}$. Tod-phod wala (destroyer) $\Rightarrow$ uska rate **minus**.
- Pages/boxes/eggs per minute $\Rightarrow$ rate seedhe jodo-ghatao.

TERMS – Har letter ka matlab

a, b, c = akele din · T = saath ka time · $1/a$ = ek din ka kaam (rate).

SOLVED – Asli book questions (verified)

Q (EN 5): A 15, B 20 $\Rightarrow$ saath

$$\frac{15 \times 20}{35} = 8\frac{4}{7} \text{ din } \checkmark \text{ book ans (c)}$$

Q (EN 10): saath 6, Ranga 18 $\Rightarrow$ Raju

$$\frac{18 \times 6}{18-6} = 9 \checkmark \text{ book ans (a)}$$

Q (EN 1): Paras 40% in 8 din ($\Rightarrow$ 20); saath 10% roz ($\Rightarrow$ 10) $\Rightarrow$ Deepti

$$\frac{20 \times 10}{20-10} = 20 \checkmark \text{ book ans (d)}$$

Q (EN 30): A 8, B 10, C 20 $\Rightarrow$ saath

$$\frac{1}{\frac{1}{8} + \frac{1}{10} + \frac{1}{20}} = 3\frac{7}{11} \checkmark \text{ book ans (a)}$$

Q (EN 34): Meenu 16, Simi 12, teeno 4 $\Rightarrow$ Raj

$$\frac{1}{\frac{1}{4} - \frac{1}{16} - \frac{1}{12}} = \frac{48}{5} \checkmark \text{ book ans (c)}$$

Q (EN 111): P 12, Q 16; 4 din saath $\Rightarrow$ bacha

$$1 - 4 \times \frac{7}{48} = \frac{5}{12} \checkmark \text{ book ans (c)}$$

Q (EN 115): A+B 10, C 14 $\Rightarrow$ $\frac{3}{5}$ kaam

$$\frac{3}{5} \times \frac{70}{12} = \frac{7}{2} \checkmark \text{ book ans (b)}$$

Q (EN 193): builder 20 h, destroyer 50 h; 30 h baad destroyer bahar

$$30 \left(\frac{1}{20} - \frac{1}{50} \right) = 0.9 \Rightarrow +2 = 32 \text{ h } \checkmark \text{ book ans (d)}$$

BOOK KESAWAAL (67) EN 1, 3, 5, 6, 9–12, 14, 20–22, 26, 28–31, 34, 36, 37, 40–42, 45, 48, 58–61, 64, 102–108, 110, 111, 113, 115, 116, 172, 191–193, 195, 199 · HI 1, 2, 5–8, 11–13, 15, 17–19, 23, 25, 28, 38, 96 | Ch 28 (Vividh): EN 29

WT-2 — PAIR-SUM MACHINE (A+B, B+C, C+A)

PEHCHAN – Question mein ye dikhe

“A and B 18 din, B and C 15 din, A and C 12 din”, teeno saath, ek akela, “A, B, C 81 din; A+B 97.2; B+C 162 $\Rightarrow$ B”; A और B ... B और C ... C और A

$$\text{Teeno saath} = \frac{2}{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}} \quad \text{A akela} = \frac{1}{\frac{1}{T} - \frac{1}{(B+C)}}$$

- Jodon ke rates jodo = $2 \times (A + B + C)$ ka rate.
- Teeno ka T + do jode diye $\Rightarrow$ beech wala: $\frac{1}{B} = \frac{1}{A+B} + \frac{1}{B+C} - \frac{1}{T}$.
- Teesra jod: $\frac{1}{P+R} = \frac{2}{T} - \frac{1}{P+Q} - \frac{1}{Q+R}$.

TERMS – Har letter ka matlab

x, y, z = teeno jodon ke din · T = teeno saath ka time.

SOLVED – Asli book questions (verified)

Q (EN 44): A+B 10, B+C 15, A+C 18 $\Rightarrow$ teeno

$$\frac{2}{\frac{20}{90}} = 9 \quad \checkmark \text{ book ans (c)}$$

Q (EN 46): 10, 20, 30 $\Rightarrow$ teeno

$$\frac{2 \times 60}{11} = \frac{120}{11} \quad \checkmark \text{ book ans (a)}$$

Q (EN 54): A+B 10, B+C 12, C+A 15 $\Rightarrow$ A

$$\text{teeno } \frac{1}{8} \Rightarrow \frac{1}{\frac{1}{8} - \frac{1}{12}} = 24 \quad \checkmark \text{ book ans (b)}$$

Q (EN 57): A+B 40, B+C 56, C+A 70 $\Rightarrow$ C

$$\text{teeno } \frac{1}{35} \Rightarrow \frac{1}{\frac{1}{35} - \frac{1}{40}} = 280 \quad \checkmark \text{ book ans (d)}$$

Q (EN 56): ABC 81, AB 97.2, BC 162 $\Rightarrow$ B

$$\frac{1}{97.2} + \frac{1}{162} - \frac{1}{81} \Rightarrow 243 \quad \checkmark \text{ book ans (a)}$$

Q (EN 53): P+Q 8, Q+R 12, PQR 6 $\Rightarrow$ P+R

$$\frac{2}{6} - \frac{1}{8} - \frac{1}{12} = \frac{1}{8} \Rightarrow 8 \quad \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (36) EN 32, 38, 43, 44, 46, 47, 49, 51–57, 62, 63, 65–69, 184, 200, 204 · HI 3, 4, 9, 10, 14, 20–22, 24, 26, 27, 99

WT-3 — EFFICIENCY RATIO MACHINE (twice as good, x% more efficient)**PEHCHAN – Question mein ye dikhe**

“A is twice as good as B, together 16 din”, “P takes 50% more time than Q”, “B 25% more efficient”, “3 guna achha aur 20 din kam leta”, efficiency ka ratio, kaam ka ratio; कुशल, गुना, अनुपात

$$A = k \times B, \text{ saath } T \Rightarrow B = T(k+1), \quad A = \frac{T(k+1)}{k} \quad \text{Time} \propto \frac{1}{\text{Efficiency}}$$

- x% zyada efficient $\Rightarrow$ time = $t \times \frac{100}{100+x}$ (x% kam $\Rightarrow \frac{100}{100-x}$).
- “50% zyada time” $\Rightarrow$ time 3 : 2 $\Rightarrow$ efficiency 2 : 3.
- k guna achha aur d din kam $\Rightarrow$ dheema = $\frac{dk}{k-1}$, tez = $\frac{d}{k-1}$.
- Kaam ka hissa $\propto$ efficiency; $M_1 D_1 = M_2 D_2$ se efficiency ratio (M : W).

TERMS – Har letter ka matlab

k = kitne guna tez · T = saath ka time · x = % zyada efficiency.

SOLVED – Asli book questions (verified)

Q (EN 16): Himanshu $2 \times$ Ankit, saath $16 \Rightarrow$ Ankit

$$16 \times 3 = 48 \checkmark \text{ book ans (b)}$$

Q (EN 19): Ishan $2 \times$ Kamal, saath $29 \Rightarrow$ Kamal

$$29 \times 3 = 87 \checkmark \text{ book ans (c)}$$

Q (EN 24): P, Q se 50% zyada time; saath $18 \Rightarrow$ Q

$$\text{eff } 2 : 3 \Rightarrow 18 \times \frac{5}{3} = 30 \checkmark \text{ book ans (a)}$$

Q (EN 163): Sudhir $4.5 \times$ Aarav, saath $8 \Rightarrow$ Aarav

$$8 \times 5.5 = 44 \checkmark \text{ book ans (c)}$$

Q (EN 165): A 10 din, B 25% zyada efficient

$$10 \times \frac{100}{125} = 8 \checkmark \text{ book ans (d)}$$

Q (EN 171): A 20, B 75% zyada efficient

$$20 \times \frac{100}{175} = 11\frac{3}{7} \checkmark \text{ book ans (b)}$$

Q (EN 178): Raju 3 guna achha, 20 din kam $\Rightarrow$ Ravi

$$\frac{20 \times 3}{2} = 30 \checkmark \text{ book ans (c)}$$

Q (EN 100): $20 W \times 16 = 16 M \times 15 \Rightarrow M : W$

$$320 : 240 = 4 : 3 \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (33) EN 15, 16, 18, 19, 23–25, 27, 99–101, 109, 161, 163, 165, 171, 173–175, 177–179, 185–188, 203 · **HI** 74, 75, 86, 87, 90, 97

WT-4 — LEAVE / JOIN MACHINE (beech mein chhoda / juda)**PEHCHAN – Question mein ye dikhe**

“A left 3 days before completion”, “A leaves after 6 days, B finishes”, “worked 21 days then B finished in 5”, “after 10 days Sundar joined”, “B finished remaining in 23 days, after how many days A left”, “A does $1/5$ alone, B rest, total 22 days”; छोड़ दिया, शामिल, शेष कार्य

$$\text{Har aadmi ka kaam} = \frac{\text{uske din}}{\text{uska akela time}}, \text{ sab ka jod} = 1$$

- **A, k din pehle chhod de** (B aakhir tak): Kul $= \frac{(a+k)b}{a+b}$ din.
- Saath shuru, A gaya, B ne baaki y din mein kiya $\Rightarrow$ saath ke din $= \frac{1-y}{\frac{1}{a} + \frac{1}{b}}$.
- A ne x din kiya, B ne y din mein khatam $\Rightarrow \frac{x}{a} + \frac{y}{b} = 1$ (B nikaalo).
- A ne f hissa akele, B ne baaki; kul D din $\Rightarrow fA + (1-f)B = D$ aur $\frac{1}{A} + \frac{1}{B} = \frac{1}{T} \Rightarrow$ **options daal kar check** ($f = \frac{1}{2} \Rightarrow A + B = 2D, AB = T(A + B)$).

TERMS – Har letter ka matlab

a, b = akele time · k = kitne din pehle chhoda · x, y = kaam kiye din.

SOLVED – Asli book questions (verified)**Q (EN 124):** A 8, B 12; A 3 din pehle chhodta

$$\frac{(8+3) \times 12}{20} = 6\frac{3}{5} \checkmark \text{ book ans (a)}$$

Q (EN 132): Janaki 45, Mansi 72; Janaki 33 din pehle

$$\frac{78 \times 72}{117} = 48 \checkmark \text{ book ans (b)}$$

Q (EN 133): A 5, B 10; A 4 din pehle chhodta $\Rightarrow$ A ke din

$$\frac{9 \times 10}{15} = 6 \Rightarrow 6 - 4 = 2 \checkmark \text{ book ans (c)}$$

Q (EN 119): A 45, B 40; B ne baaki 23 din mein

$$\frac{1 - \frac{23}{40}}{\frac{1}{45} + \frac{1}{40}} = 9 \checkmark \text{ book ans (d)}$$

Q (EN 128): A 24; A ne 21 din, B ne baaki 5 din $\Rightarrow$ saath

$$B = 40 \Rightarrow \frac{24 \times 40}{64} = 15 \checkmark \text{ book ans (c)}$$

Q (EN 123): Shyam 16, Sundar 24; 10 din baad Sundar juda

$$10 + \frac{6/16}{5/48} = 13\frac{3}{5} \checkmark \text{ book ans (a)}$$

Q (EN 160): saath 12; Rathin 5 din pehle gaya, kul 15 $\Rightarrow$ Bratin

$$\frac{10}{12} + \frac{5}{B} = 1 \Rightarrow 30 \checkmark \text{ book ans (a)}$$

Q (EN 156): saath 12; A aadha, B aadha, kul 25; B tez

$$A + B = 50, AB = 600 \Rightarrow 20, 30 \Rightarrow 20 \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (72) EN 2, 13, 17, 50, 112, 114, 118–150, 152, 154–156, 158–160, 164, 167, 170, 196–198 · **HI** 16, 29–37, 45–48, 50–52, 70, 71, 88**WT-5 — ALTERNATE DAY / CYCLE MACHINE****PEHCHAN – Question mein ye dikhe**

“work on alternate days starting with A”, “P day 1, Q day 2, R day 3 cycle”, “B and C assist A every third day”, pairs (A, B), (B, C), (C, A) ki cycle; एक-एक दिन छोड़कर, वैकल्पिक, बारी-बारी, प्रत्येक तीसरे दिन

$$\text{Kul kaam} = \text{LCM}, \quad 1 \text{ cycle ka kaam} = \sum \text{efficiency} \Rightarrow \text{poori cycles} = \left\lfloor \frac{\text{LCM}}{\text{cycle kaam}} \right\rfloor$$

- Poori cycles ke baad bacha kaam $\Rightarrow$ agle din wale ki efficiency se (fraction din).
- Dhyan:** aakhri cycle ke beech mein hi kaam khatam ho sakta hai — cycle se pehle check karo.
- Har teesre din madad $\Rightarrow$ cycle = 3 din: A, A, (A + B + C).

TERMS – Har letter ka matlab

LCM = kul kaam units · efficiency = LCM $\div$ din · cycle = jitne din baad pattern dohraaye.

SOLVED – Asli book questions (verified)**Q (EN 7):** A 10, B 12, alternate, A se shuru

$$\text{LCM 60: } 6 + 5 = 11/2 \text{ din} \Rightarrow 5 \text{ cycle} = 55 \Rightarrow 10\frac{5}{6} \checkmark \text{ book ans (d)}$$

Q (EN 4): A 30, B 40, alternate, B se shuru

$$\text{LCM 120: } 3 + 4 \Rightarrow 17 \text{ cycle} = 119 \Rightarrow 34\frac{1}{3} \checkmark \text{ book ans (c)}$$

Q (EN 35): P 9, Q 18, R 12 cycle

$$\text{LCM 36: } 4 + 2 + 3 = 9 \Rightarrow 4 \text{ cycle} = 12 \text{ din} \checkmark \text{ book ans (d)}$$

Q (EN 33): A 20; B 30, C 60 har teesre din

$$\text{LCM 60: } 3 + 3 + 6 = 12/3 \text{ din} \Rightarrow 15 \text{ din} \checkmark \text{ book ans (a)}$$

Q (HI 54): X 8, Y 12, alternate, X se

$$\text{LCM 24: } 3 + 2 \Rightarrow 4 \text{ cycle} = 20, +3, +\frac{1}{2} \Rightarrow 9\frac{1}{2} \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (29) EN 4, 7, 8, 33, 35, 39, 157, 169, 190 · **HI** 39, 40, 53–69, 72

WT-6 — M·D·H = WORK MACHINE (men, women, days, hours)**PEHCHAN – Question mein ye dikhe**

“12 men 24.5 din $\Rightarrow$ 5 walls 49 din”, “15 men or 25 women”, “6M + 8W 14 din; 8M + 12W 10 din”, men beech mein chhodein/judein, hours per day, food for soldiers / students (provisions); पुरुष, महिलाएं, व्यक्ति, घंटे प्रतिदिन

$$\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$$

- “x men or y women” $\Rightarrow x M = y W$ (ek ko doosre mein badlo).
- Do groups barabar kaam: $D_1(aM + bW) = D_2(cM + dW) \Rightarrow M : W$ efficiency $\Rightarrow$ kaam units.
- Beech mein log gaye/aaye: bacha kaam = $M \times$ (bache din) $\Rightarrow$ naye logon se divide.
- Khana (provisions): $\log \times \text{din} = \text{constant}$ (jitne din beet gaye, pehle ghatao).

TERMS – Har letter ka matlab

M = log · D = din · H = ghante/din · W = kaam (walls, bulbs, kuan . . .).

SOLVED – Asli book questions (verified)

Q (EN 70): 12 men 24.5 din 1 wall $\Rightarrow$ 5 walls 49 din

$$\frac{12 \times 24.5 \times 5}{49} = 30 \checkmark \text{ book ans (b)}$$

Q (EN 71): 6M + 8W 14 din; 8M + 12W 10 din $\Rightarrow$ 1 man

$$84m + 112w = 80m + 120w \Rightarrow m = 2w \Rightarrow 140 \checkmark \text{ book ans (a)}$$

Q (EN 78): 15 men or 25 women, 22 din $\Rightarrow$ 9M + 18W

$$9M = 15W \Rightarrow 33W \Rightarrow \frac{25 \times 22}{33} = 16\frac{2}{3} \checkmark \text{ book ans (a)}$$

Q (EN 86): 8 men $\times$ 9 h $\times$ 20 din $\Rightarrow$ 7 men $\times$ 10 h

$$\frac{8 \times 9 \times 20}{70} = \frac{144}{7} \checkmark \text{ book ans (d)}$$

Q (EN 90): 36 men 24 din; 8 din baad 4 gaye

$$\frac{36 \times 16}{32} = 18 \checkmark \text{ book ans (a)}$$

Q (EN 80): M men 50 din; 5 zyada $\Rightarrow$ 10 din kam

$$50M = 40(M + 5) \Rightarrow 20 \checkmark \text{ book ans (d)}$$

Q (EN 151): 1200 soldiers 28 din khana; 4 din baad kuch gaye $\Rightarrow$ 32 din

$$1200 \times 24 = x \times 32 \Rightarrow 900 \Rightarrow 300 \text{ gaye } \checkmark \text{ book ans (a)}$$

Q (HI 82): 8M + 13W 12 din; 10M + 14W 10 din $\Rightarrow$ 1 man = ? women

$$96m + 156w = 100m + 140w \Rightarrow m = 4w \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (60) EN 70–98, 151, 153, 194, 201, 202, 206 · HI 41–44, 49, 73, 76–85, 89, 91–95, 98, 100, 101

WT-7 — WAGES MACHINE (paisa $\propto$ kaam)**PEHCHAN – Question mein ye dikhe**

“A and B undertake work for ₹600 . . . with help of C in 2 days $\Rightarrow$ C’s share”, “A 5 din, B 8 din, ₹6760 $\Rightarrow$ A ka hissa”, skilled/unskilled wages ratio; मजदूरी, हिस्सा, धनराशि

$$\text{Hissa} = \text{Kul ₹} \times \frac{\text{uska kiya kaam}}{\text{poora kaam}} \quad C \text{ ka hissa} = ₹ \times \left[1 - T \left(\frac{1}{a} + \frac{1}{b} \right) \right]$$

- Saath poora kaam kiya $\Rightarrow$ hissa $\propto$ efficiency $\propto \frac{1}{\text{din}}$ (A : B = b : a).
- Log $\times$ wage-rate $\Rightarrow$ ratio (12 $\times$ 9 : 14 $\times$ 5 : 10 $\times$ 4).

TERMS – Har letter ka matlab

T = sab ne saath kitne din kaam kiya · a, b = akele din.

SOLVED – Asli book questions (verified)

Q (EN 166): ₹600; A 4, B 6; C ke saath 2 din $\Rightarrow$ C
 $600 \left[1 - 2 \times \frac{5}{12} \right] = 100$ ✓ **book ans (a)**

Q (EN 168): ₹15,000; 15, 18; C ke saath 6 din $\Rightarrow$ C
 $15000 \times \frac{24}{90} = 4000$ ✓ **book ans (a)**

Q (EN 180): ₹350; Rajesh 4, Mahesh 3 $\Rightarrow$ Rajesh
 $\frac{1}{4} : \frac{1}{3} = 3 : 4 \Rightarrow 150$ ✓ **book ans (d)**

Q (EN 183): ₹6760; A 5, B 8 $\Rightarrow$ A
 $6760 \times \frac{8}{13} = 4160$ ✓ **book ans (a)**

Q (HI 102): ₹800; C 24, B 8; A ke saath 3 din $\Rightarrow$ A
 $800 \left[1 - 3 \times \frac{4}{24} \right] = 400$ ✓ **book ans (c)**

Q (EN 162): 12, 14, 10 workers, wage 9 : 5 : 4, ₹13,189 $\Rightarrow$ unskilled
 $108 : 70 : 40 \Rightarrow 13189 \times \frac{40}{218} = 2420$ ✓ **book ans (b)**

BOOK KESAWAAL (15) EN 162, 166, 168, 176, 180–183 · **HI** 102–106 | Ch 28 (Vividh): **EN** 71 · **HI** 56

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
WT-1	LCM / RATE MACHINE (saath mein, akele, kitna kaam)	48	18
WT-2	PAIR-SUM MACHINE (A+B, B+C, C+A)	24	12
WT-3	EFFICIENCY RATIO MACHINE (twice as good, x% more efficient)	27	6
WT-4	LEAVE / JOIN MACHINE (beech mein chhoda / juda)	52	20
WT-5	ALTERNATE DAY / CYCLE MACHINE	9	20
WT-6	M·D·H = WORK MACHINE (men, women, days, hours)	35	25
WT-7	WAGES MACHINE (paisa $\propto$ kaam)	8	5
$\rightarrow$ DF-8	FRACTION EQUATION MACHINE (ansh-har wale sawaal) (Ch 2)	1	0
$\rightarrow$ RP-1	RATIO KA BATWARA (share / ek hissa / farak) (Ch 9)	1	0
$\rightarrow$ RP-5	PROPORTION MACHINE ($a : b :: c : d$) (Ch 9)	1	0
Total (book ke sawaal)		206/206 ✓	106/106 ✓
$\rightarrow$ DF-8: EN 117 · $\rightarrow$ RP-1: EN 189 · $\rightarrow$ RP-5: EN 205			

Chapter 12

Alligation / Mixture (Mishran / Alligation) मिश्रण / एलीगेशन

4 formulas · English book 58 sawaal · Hindi book 16 sawaal · 23 solved examples ✓

MX-1 — ALLIGATION CROSS RULE (kis ratio mein milaayein)

PEHCHAN – Question mein ye dikhe

“in what ratio should ₹45 and ₹52 sugar be mixed”, “30% and 60% solution $\Rightarrow$ 40%”, “how much 80% juice must be mixed with 36 L of 25%”, profit ke saath mixture price, “what part replaced”, “ratio of first blend and second blend”; किस अनुपात में मिलाएँ, अन्य किसम का मूल्य

$$\frac{\text{Sasta}}{\text{Mehenga}} = \frac{d - m}{m - c} \quad \begin{matrix} c & d \\ d - m & m - c \end{matrix}$$

- m = mean (mixture) value; selling price + profit diya $\Rightarrow m = SP \times \frac{100}{100+P}$ (pehle CP banao).
- Ratio wale mixtures (4 : 3, 6 : 7) $\Rightarrow$ ek hi component ka fraction lo ($\frac{4}{7}, \frac{6}{13}$) aur usi par cross rule.
- **Kitna hissa replace hua** (40% $\rightarrow$ 26% using 19%) $\Rightarrow$ replaced = $\frac{40-26}{40-19}$.
- Ek quantity di ho $\Rightarrow$ doosri = di hui $\times$ ratio.

TERMS – Har letter ka matlab

c = sasta (kam %) · d = mehenga (zyada %) · m = milane ke baad wala (beech ka) · m hamesha c aur d ke beech.

SOLVED – Asli book questions (verified)

Q (EN 10): 80% juice kitna milayein 36 L ke 25% mein $\Rightarrow$ 60%

$25 : 80 \Rightarrow (80 - 60) : (60 - 25) = 20 : 35 \Rightarrow 36 \times \frac{35}{20} = 63$ ✓ **book ans (a)**

Q (EN 28): ₹45 & ₹52 sugar; ₹55.20 pe 15% profit

$m = \frac{55.2}{1.15} = 48 \Rightarrow (52 - 48) : (48 - 45) = 4 : 3$ ✓ **book ans (a)**

Q (EN 38): ₹50 & ₹60 rice; ₹70 pe 20% profit

$m = 58\frac{1}{3} \Rightarrow 1\frac{2}{3} : 8\frac{1}{3} = 1 : 5$ ✓ **book ans (c)**

Q (EN 14): 40% water wala doodh, 19% wale se replace $\Rightarrow$ 26%

replaced = $\frac{40-26}{40-19} = \frac{2}{3}$ ✓ **book ans (a)**

Q (EN 25): blend 1: apple 20%; blend 2: apple 70%; chahiye 4 : 3

$m = \frac{4}{7} = 57\frac{1}{7}\% \Rightarrow 12\frac{6}{7} : 37\frac{1}{7} = 9 : 26$ ✓ **book ans (d)**

Q (HI 3): 6 kg ₹48 + 3 kg ? $\Rightarrow$ ₹60

$6 \times 12 = 3(x - 60) \Rightarrow x = 84$ ✓ **book ans (c)**

Q (EN 21): honey : water 4 : 3 (3.5 L) + 6 : 7 (x) $\Rightarrow$ 1 : 1

$\frac{4}{7}, \frac{6}{13}$ vs $\frac{1}{2} \Rightarrow 7 : 13 \Rightarrow 3.5 \times \frac{13}{7} = 6.5$ ✓ **book ans (b)**

BOOK KE SAWAAL (18) EN 10, 14, 21, 25, 28, 30, 31, 37–43, 45 · HI 3, 5, 9

MX-2 — WEIGHTED MIX (mixtures jodna, price, profit)

PEHCHAN – Question mein ye dikhe

“2 L of 3 : 2 + 3 L of 2 : 3 $\Rightarrow$ new ratio”, equal quantities of two alloys, 4 vessels 80%, 75% . . . , “30 kg @ ₹30 + 40 kg @ ₹28 sold @ ₹28”, rate for 30% profit, parts of alloy (13 : 6 : 1, zinc 90 kg), average price; मिश्र धातु, मिश्रण का मूल्य

$$\text{Component} = \sum (\text{quantity} \times \text{us mein hissa}) \quad \text{Avg price} = \frac{\sum q_i p_i}{\sum q_i}$$

- Barabar quantity $\Rightarrow$ fractions seedhe jodo ($1/3 + 1/4 \dots$). Ratio mein milaye $\Rightarrow$ har ratio ko us quantity se guna.
- Mixture ka profit: kul SP vs kul CP (PL-1); bechne ka rate = avg CP $\times \frac{100+P}{100}$.
- Alloy ka ek part diya (zinc 90 kg in 13 : 6 : 1) $\Rightarrow$ RP-1 unit method.

TERMS – Har letter ka matlab

q = quantity $\cdot$ p = price / % $\cdot$ hissa = component ka fraction us mixture mein.

SOLVED – Asli book questions (verified)

Q (EN 24): 2 L (3 : 2) + 3 L (2 : 3)

milk 1.2 + 1.2 = 2.4, water 2.6 $\Rightarrow$ 12 : 13 ✓ **book ans (b)**

Q (EN 27): gold : platinum 5 : 2 aur 5 : 7, barabar

$\frac{5}{7} + \frac{5}{12} : \frac{2}{7} + \frac{7}{12} = 95 : 73$ ✓ **book ans (c)**

Q (EN 34): 4 vessels 80, 75, 60, 50% milk $\Rightarrow$ water : milk

milk 2.65 $\Rightarrow$ 1.35 : 2.65 = 27 : 53 ✓ **book ans (b)**

Q (EN 32): water : squash 4 : 1 aur 3 : 1, 1 : 2 mein

$0.8 + 1.5 : 0.2 + 0.5 = 23 : 7$ ✓ **book ans (b)**

Q (EN 3): 30 kg @30 + 40 kg @28, becha @28

CP 2020, SP 1960 $\Rightarrow$ 3% loss ✓ **book ans (c)**

Q (EN 48): 30 kg @13.56 + 20 kg @18.15; 30% profit ka rate

avg 15.40 $\Rightarrow \times 1.3 \approx ₹20$ ✓ **book ans (a)**

Q (EN 58): 274 kg @252, 197 @280, 54 @316; becha @283.50

SP 148837.5 – CP 141272 = 7565.50 profit ✓ **book ans (a)**

BOOK KE SAWAAL (31) EN 2, 3, 5, 8, 24, 26, 27, 29, 32–34, 44, 46, 48–50, 52, 54–58 · **HI** 1, 4, 8, 10–12, 15, 16 | Ch 28 (Vividh): **EN** 4

MX-3 — ADD / EVAPORATE MACHINE (ek cheez badli, doosri fix)

PEHCHAN – Question mein ye dikhe

“how much water to add to 90 ml of 38% $\Rightarrow$ 17.1%”, “2 L water evaporated”, “add 10 L alcohol $\Rightarrow$ 3 : 5”, “pure milk added to reduce water to 2%”, “milk : water 3 : 1 $\Rightarrow$ 2 : 3, water add”, “9 L mixture replaced by 10 L of B”; मिलाया, वाष्पित

Jo nahi badla, woh fix: $C_1 V_1 = C_2 V_2$ (sirf paani jodo/udao $\Rightarrow$ solute same)

- Paani jodna: $V_2 = \frac{C_1 V_1}{C_2} \Rightarrow \text{add} = V_2 - V_1$ · udaya (evaporate) $\Rightarrow C_2 = \frac{C_1 V_1}{V_2}$.
- Ratio a : b $\Rightarrow$ c : d sirf B jodkar: A fix $\Rightarrow$ naya B = $A \times \frac{d}{c} \Rightarrow \text{add} = \text{naya} - \text{purana}$ (RP-4).
- Kuch nikaal kar doosra daala: nikaalne mein dono ratio se kam honge $\Rightarrow$ phir jo daala usme jodo.

TERMS – Har letter ka matlab

C = concentration / % $\cdot$ V = volume $\cdot$ A = jo component nahi badla.

SOLVED – Asli book questions (verified)**Q (EN 1):** 90 ml 38% $\Rightarrow$ 17.1% ke liye paani

$$V_2 = \frac{90 \times 38}{17.1} = 200 \Rightarrow 110 \text{ ml} \checkmark \text{ book ans (c)}$$

Q (EN 4): 80 L, 3 : 1 $\Rightarrow$ 2 : 3, paani

$$\text{milk } 60 \text{ fix} \Rightarrow \text{water } 60 \times \frac{3}{2} = 90 \Rightarrow 70 \checkmark \text{ book ans (c)}$$

Q (EN 6): 10 L se 2 L paani uda, ab 6% namak

$$\frac{8 \times 6}{10} = 4.8\% \checkmark \text{ book ans (a)}$$

Q (EN 11): 8 L mein 4% paani $\Rightarrow$ 2% ke liye pure milk

$$\text{paani } 0.32 \text{ fix} \Rightarrow \frac{0.32}{0.02} = 16 \Rightarrow 8 \text{ L} \checkmark \text{ book ans (c)}$$

Q (EN 12): alcohol : water 5 : 4, +9 L water $\Rightarrow$ 4 : 5

$$k = \frac{4 \times 9}{25 - 16} = 4 \Rightarrow 20 \checkmark \text{ book ans (d)}$$

Q (EN 19): 30 L 5% namak $\Rightarrow$ 3%

$$\frac{1.5}{0.03} = 50 \Rightarrow 20 \text{ L} \checkmark \text{ book ans (a)}$$

Q (EN 18): A : B 7 : 5; 9 L nikaal kar 10 L B $\Rightarrow$ 7 : 9

$$9(7x - 5.25) = 7(5x + 6.25) \Rightarrow x = 3.25 \Rightarrow 22.75 \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (21) EN 1, 4, 6, 7, 9, 11–13, 15–20, 22, 36, 51 · **HI** 2, 6, 13, 14**MX-4 — REPLACEMENT MACHINE (baar-baar nikaalo, paani daalo)****PEHCHAN – Question mein ye dikhe**

“from 50 L milk, 10 L taken out and replaced by water, repeated three times”, “process repeated once more”; बार-बार निकालकर पानी

$$\text{Bacha shuddh} = \text{Shuru} \times \left(1 - \frac{x}{V}\right)^n$$

- x = har baar nikaala, V = bartan ki poori capacity, n = kitni baar.
- Paani = V – bacha doodh $\Rightarrow$ ratio.

TERMS – Har letter ka matlab

$$V = \text{total volume} \cdot x = \text{har baar nikala} \cdot n = \text{baar}.$$

SOLVED – Asli book questions (verified)**Q (EN 35):** 50 L milk, 10 L $\times$ 3 baar $\Rightarrow$ water : milk

$$\text{milk} = 50 \left(\frac{4}{5}\right)^3 = 25.6 \Rightarrow 24.4 : 25.6 = 61 : 64 \checkmark \text{ book ans (c)}$$

Q (EN 53): 80 L, 8 L $\times$ 3 baar

$$80 \times 0.9^3 = 58.32 \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (2) EN 35, 53**COVERAGE CHECK – Har sawaal ka formula**

Formula	Naam	EN Qs	HI Qs
MX-1	ALLIGATION CROSS RULE (kis ratio mein milaayein)	15	3
MX-2	WEIGHTED MIX (mixtures jodna, price, profit)	22	8
MX-3	ADD / EVAPORATE MACHINE (ek cheez badli, doosri fix)	17	4
MX-4	REPLACEMENT MACHINE (baar-baar nikaalo, paani daalo)	2	0
$\rightarrow$ PC-1	PERCENT BASICS (of $\cdot$ kitna % $\cdot$ poor number) (Ch 6)	1	1
$\rightarrow$ PL-6	FALSE WEIGHT MACHINE (nakli baat) (Ch 7)	1	0
Total (book ke sawaal)		58/58 ✓	16/16 ✓
$\rightarrow$ PC-1: EN 23 HI 7 $\cdot$ $\rightarrow$ PL-6: EN 47			

Chapter 13

Pipe & Cistern (Nal aur Tanki) नल और टंकी

6 formulas · English book 155 sawaal · Hindi book 95 sawaal · 36 solved examples ✓

PP-1 — NET RATE MACHINE (bharna +, khaali karna −)

PEHCHAN – Question mein ye dikhe

“A 21 h, B 18 h $\Rightarrow$ saath”, fill + empty pipe saath, “A, B fill; C empties $\Rightarrow$ kitne mein bharegi”, adhi/ $\frac{2}{3}$ bhari tanki se shuru, third pipe dhoondho, pipes ki speed ka ratio (twice as fast), “15 pipes 9 min $\Rightarrow$ 20 pipes”, holes (dono khaali); पाइप, नल, टंकी भरना/खाली करना

$$\frac{1}{T} = \sum \frac{1}{\text{fill}} - \sum \frac{1}{\text{empty}} \quad \text{do: } \frac{ab}{a+b} \quad \text{fill } a, \text{ empty } b: \frac{ab}{b-a} \quad \text{teen: } \frac{XYZ}{YZ + XZ - XY}$$

- Tanki pehle se f bhari $\Rightarrow$ bacha $(1 - f) \times T$; khaali karni ho to f $\times$ (emptying time).
- LCM units: fill pipe +, empty pipe $\Rightarrow$ net units/hour $\Rightarrow$ LCM $\div$ net.
- Speed ratio (A = 2B, C = 2A) $\Rightarrow$ units maano $\Rightarrow$ kul kaam = units $\times$ T (WT-3). k pipes $\Rightarrow$ pipes $\times$ time = constant.

TERMS – Har letter ka matlab

a, b = pipes ke apne time · T = saath ka time · fill = bharne wala, empty = khaali karne wala (leak/hole).

SOLVED – Asli book questions (verified)

Q (EN 1): A 21 h, B 18 h $\Rightarrow$ saath

$$\frac{21 \times 18}{39} = 9 \frac{9}{13} \text{ h } \checkmark \text{ book ans (a)}$$

Q (EN 23): fill 20 min, empty 60 min

$$\frac{20 \times 60}{40} = 30 \checkmark \text{ book ans (d)}$$

Q (EN 48): A 5, B 8 fill; C 10 empty

$$\frac{1}{\frac{1}{5} + \frac{1}{8} - \frac{1}{10}} = 4 \frac{4}{9} \checkmark \text{ book ans (a)}$$

Q (EN 52): teeno 12 h; 10 h, 15 h fill $\Rightarrow$ drain

$$\frac{1}{\frac{1}{10} + \frac{1}{15} - \frac{1}{12}} = 12 \checkmark \text{ book ans (d)}$$

Q (EN 62): A 7, C 10.5 fill; B 5.25 empty $\Rightarrow$ $\frac{2}{3}$ tanki

$$\text{net } \frac{1}{21} \Rightarrow \frac{2}{3} \times 21 = 14 \checkmark \text{ book ans (a)}$$

Q (EN 32): fill 7.8, empty 19.5; aadhi khaali tanki

$$0.5 \times \frac{7.8 \times 19.5}{11.7} = 6.5 \checkmark \text{ book ans (d)}$$

Q (EN 145): Z = 3Y, Y = 2X; teeno 5 h $\Rightarrow$ X

$$1 + 2 + 6 = 9 \text{ units } \Rightarrow 45 \Rightarrow \text{X } 45 \text{ h } \checkmark \text{ book ans (d)}$$

Q (EN 135): 5 pipes 80 min $\Rightarrow$ 8 pipes

$$\frac{5 \times 80}{8} = 50 \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (138) EN 1, 3–6, 8, 10–21, 23–28, 30–37, 39, 42, 43, 45–50, 52–56, 58, 60, 62–73, 75–78, 80, 81, 83, 126, 135–140, 142, 144–150, 152–154 · HI 1–6, 9–15, 17, 18, 20–24, 28–38, 40, 43, 45–56, 88–95

PP-2 — LEAK MACHINE (rasaav)

PEHCHAN – Question mein ye dikhe

“fills in 9 h but due to leak takes 1 h more”, “1 h 30 min but with leak 2 h 15 min”, “both pipes . . . leak, 30 minutes more $\Rightarrow$ leak empties in?”; रिसाव, छिद्र

$$\text{Leak khaali karega} = \frac{a \times b}{b - a} \quad (a = \text{bina leak, } b = \text{leak ke saath})$$

- “x ghante zyada laga” $\Rightarrow b = a + x$.
- Do pipes + leak: pehle $a = \text{saath ka time (PP-1)}$, phir yahi formula.

TERMS – Har letter ka matlab

$a = \text{normal time} \cdot b = \text{leak ke saath time} \cdot \text{leak ka apna time} = \text{khaali karne ka.}$

SOLVED – Asli book questions (verified)

Q (EN 29): 9 h, leak se 10 h

$$\frac{9 \times 10}{1} = 90 \checkmark \text{ book ans (d)}$$

Q (EN 38): 1.5 h, leak se 2.25 h

$$\frac{1.5 \times 2.25}{0.75} = 4.5 \text{ h } \checkmark \text{ book ans (b)}$$

Q (EN 44): 4 h, leak se 5 h

$$\frac{20}{1} = 20 \checkmark \text{ book ans (a)}$$

Q (EN 51): A 6, B 30 (saath 5); leak se 30 min zyada

$$\frac{5 \times 5.5}{0.5} = 55 \checkmark \text{ book ans (d)}$$

Q (EN 86): X 21, Y 24 (saath 11.2); leak se 48 min zyada

$$\frac{11.2 \times 12}{0.8} = 168 \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (16) EN 29, 38, 40, 41, 44, 51, 86, 91, 109 · **HI** 16, 19, 26, 27, 39, 41, 42

PP-3 — $\sqrt{(ab)}$ TRICK (saath T, ek a zyada, doosra b zyada)

PEHCHAN – Question mein ye dikhe

“together 20 h; A fills 9 h faster than B”, “big pipe 12 min less than small, together $45/4$ min”, “Q takes 6 hours more than P, together 4”, teen inlets (first two together = third); जल्दी भरता है, घंटे अधिक

$$A = T + a, B = T + b \Rightarrow T^2 = a \times b \quad (\text{sirf farak diya: } b - a = d \Rightarrow a = \frac{-d + \sqrt{d^2 + 4T^2}}{2})$$

- Farak d aur saath T diya $\Rightarrow$ options se check: $\frac{1}{A} + \frac{1}{A+d} = \frac{1}{T}$ (sabse fast).
- Teen inlets: “pehle do saath = teesra akela” $\Rightarrow T = \text{teesre ka time} \Rightarrow$ wahi trick.

TERMS – Har letter ka matlab

$T = \text{saath ka time} \cdot a, b = \text{har pipe T se kitna zyada leta hai} \cdot d = \text{do pipes ka farak.}$

SOLVED – Asli book questions (verified)

Q (EN 2): saath 20 h; A, B se 9 h tez $\Rightarrow B$

$$ab = 400, b - a = 9 \Rightarrow 16, 25 \Rightarrow B = 45 \checkmark \text{ book ans (a)}$$

Q (EN 7): saath $45/4$ min; bada 12 min kam $\Rightarrow$ bada

$$a = \frac{-12 + \sqrt{144 + 506.25}}{2} = 6.75 \Rightarrow 18 \checkmark \text{ book ans (c)}$$

Q (EN 9): saath 6 h; A 5 h tez $\Rightarrow A$

$$ab = 36, b - a = 5 \Rightarrow 4, 9 \Rightarrow A = 10 \checkmark \text{ book ans (b)}$$

Q (EN 22): saath 4 h; Q 6 h zyada $\Rightarrow P$

$$ab = 16, b - a = 6 \Rightarrow 2, 8 \Rightarrow P = 6 \checkmark \text{ book ans (b)}$$

Q (EN 59): teesra = pehle do saath; farak 9 aur 4

$$T = \sqrt{9 \times 4} = 6 \Rightarrow \text{pehla } 6 + 9 = 15 \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (5) EN 2, 7, 9, 22, 59

PP-4 — OPEN / CLOSE MACHINE (beech mein band / chalu)

PEHCHAN – Question mein ye dikhe

“after how much time should A be turned off so that tank fills in 16 min”, “B closed after some time, filled in half an hour”, “both for 4 min then A turned off”, “A closed 4 minutes before the tank is filled”, “all three for 2 h then C closed”, “A for some time then B, total 19 h”; बंद कर दिया, चालू किया

$$\sum \frac{\text{pipe kitni der khula}}{\text{uska akela time}} = 1 \quad (\text{khaali karne wala } -)$$

- **A, k min pehle band:** $Kul = \frac{(a+k)b}{a+b}$ (WT-4 wala).
- Pehle t tak saath $\Rightarrow$ bhara $t \left(\frac{1}{a} + \frac{1}{b} \right) \Rightarrow$ baaki $\div$ jo pipe chal raha.
- Pehle A akela t, phir saath $\Rightarrow \frac{t}{a} +$ saath wala hissa.

TERMS – Har letter ka matlab

a, b = pipes ka akela time · k = kitne min pehle band · t = kitni der khula.

SOLVED – Asli book questions (verified)

Q (EN 85): A 16, B 32; kul 16 min $\Rightarrow$ A kab band

$$\frac{t}{16} + \frac{16}{32} = 1 \Rightarrow t = 8 \quad \checkmark \text{ book ans (a)}$$

Q (EN 87): A 36, B 48; 30 min mein bhari $\Rightarrow$ B kab band

$$\frac{30}{36} + \frac{t}{48} = 1 \Rightarrow 8 \quad \checkmark \text{ book ans (c)}$$

Q (EN 89): A 12, B 16; 4 min saath, phir B

$$1 - \frac{7}{12} = \frac{5}{12} \Rightarrow \frac{5}{12} \times 16 = \frac{20}{3} \quad \checkmark \text{ book ans (b)}$$

Q (EN 96): A 12, B 16; A 4 min pehle band

$$\frac{16 \times 16}{28} = 9 \text{ min } 8 \text{ s} \quad \checkmark \text{ book ans (a)}$$

Q (EN 97): A + B 24; 8 h saath, phir A akela, kul 28

$$\text{A ne } 20 \text{ h mein } \frac{2}{3} \Rightarrow 30 \quad \checkmark \text{ book ans (a)}$$

Q (EN 104): ABC 4 h, AB 9 h; 3 h baad sirf C

$$C = 7.2 \text{ h; bacha } \frac{1}{4} \Rightarrow 1.8 \text{ h} \quad \checkmark \text{ book ans (b)}$$

Q (EN 112): A 15, B 25; pehle A phir B, kul 19 $\Rightarrow$ A

$$\frac{t}{15} + \frac{19-t}{25} = 1 \Rightarrow 9 \quad \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (58) EN 57, 61, 74, 79, 82, 84, 85, 87–89, 92–108, 110–116, 151 · HI 7, 25, 44, 57–76

PP-5 — ALTERNATE HOUR MACHINE (ek ghanta bharo, ek ghanta khaali)

PEHCHAN – Question mein ye dikhe

“inlet and outlet opened alternately one hour each”, “starting with pipe B ... full tank ... time to empty”, “P and Q opened one by one for an hour”; बारी-बारी, एक-एक घंटे, एकान्तर

$$2 \text{ ghante ka net} = \frac{1}{a} - \frac{1}{b} \Rightarrow \text{jab tak bacha} \leq \text{agle ghante ka kaam, tab tak cycles}$$

- **Trap:** aakhri ghante mein bharne wala pipe akela hi khatam kar deta hai — use cycle mein mat giniye.
- LCM units: fill +x, empty -y $\Rightarrow$ n cycles tak level $n(x - y)$ jab tak $n(x - y) + x \geq \text{LCM}$.
- Do fill pipes baari-baari $\Rightarrow$ WT-5 jaisa.

TERMS – Har letter ka matlab

a = bharne ka time · b = khaali karne ka · x, y = LCM units per ghanta.

SOLVED – Asli book questions (verified)**Q (EN 118):** inlet 15 h, outlet 21 h, baari-baariLCM 105: $+7, -5 \Rightarrow 49$ cycles = 98 $\Rightarrow +1 = 99$ h ✓ **book ans (d)****Q (EN 125):** A fill 18, B empty 30LCM 90: $+5, -3 \Rightarrow 43$ cycles = 86 $\Rightarrow 4/5$ h $\Rightarrow 86$ h 48 min ✓ **book ans (d)****Q (EN 124):** bhari tanki; A empty 32, B fill 40 (A se)LCM 160: $-5, +4 \Rightarrow 155$ cycles = 310 $\Rightarrow +1 = 311$ ✓ **book ans (d)****Q (EN 127):** bhari tanki; A empty 28, B fill 35LCM 140: 135 cycles = 270 $\Rightarrow 271$ ✓ **book ans (b)****Q (EN 90):** P 6, Q 4 baari-baari, P seLCM 12: $2 + 3 \Rightarrow 2$ cycles = 10 $\Rightarrow P$ 1 h $\Rightarrow 5$ h ✓ **book ans (d)****BOOK KE SAWAAL (19)** EN 90, 117–125, 127 · **HI** 77–84**PP-6 — CAPACITY / LITRE MACHINE****PEHCHAN – Question mein ye dikhe**

“another pipe takes out 20 litres per hour . . . capacity”, “pipe C empties 3 gallons per minute”, “80% full has 2 L more than 66 $\frac{2}{3}$ %”, “3/5 full, 20 L used $\Rightarrow 7/12$ ”, buckets, tank volume $\div$ per-day use, solution ka hissa pipes se; लीटर, बाल्टी, धारिता

$$\text{Capacity } C : \frac{C}{a} \pm r = \frac{C}{T} \Rightarrow C = \frac{r}{\left| \frac{1}{T} - \frac{1}{a} \right|} \quad C = \frac{\text{₹/litre farak}}{\text{fraction farak}}$$

- Litres/hour wala pipe $r \Rightarrow$ baaki rates ko C/time likho $\Rightarrow C$ nikaalo.
- Do fractions ka farak = litres ka farak $\Rightarrow C = \frac{L}{f_1 - f_2}$.
- Bucket: bucket $\times$ ginti = constant; volume ($\text{m}^3 \times 1000 = \text{litre}$) $\div$ roz ka kharcha = din.
- Kisi pipe ka hissa (solution) = uska rate $\div$ kul rate.

TERMS – Har letter ka matlab

C = tanki ki capacity · r = litres per hour · f = bhara hua fraction.

SOLVED – Asli book questions (verified)**Q (EN 128):** P 20 h mein khaali; Q 20 L/h; dono 12 h $C \left(\frac{1}{12} - \frac{1}{20} \right) = 20 \Rightarrow 600$ ✓ **book ans (d)****Q (EN 129):** hole 8 h; pipe 6 L/h bharta; ab 12 h mein khaali $C \left(\frac{1}{8} - \frac{1}{12} \right) = 6 \Rightarrow 144$ ✓ **book ans (c)****Q (EN 134):** A 25, B 30 min; C 3 gal/min nikaale; saath 15 min $C \left(\frac{1}{25} + \frac{1}{30} - \frac{1}{15} \right) = 3 \Rightarrow 450$ ✓ **book ans (a)****Q (EN 132):** 3/5 bhara, 20 L use $\Rightarrow 7/12$ $\frac{20}{\frac{3}{5} - \frac{7}{12}} = 1200$ ✓ **book ans (a)****Q (EN 155):** Sita ki bucket $3 \times$ Ramu; Sita 60 baar $\frac{60 \times 3}{3+1} = 45$ ✓ **book ans (a)****Q (EN 141):** P 30, Q 20, R 10 min; 3 min baad U (R) ka hissa $\frac{1/10}{1/30+1/20+1/10} = \frac{6}{11}$ ✓ **book ans (c)****BOOK KE SAWAAL (14)** EN 128–134, 141, 143, 155 · **HI** 8, 85–87

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
PP-1	NET RATE MACHINE (bharna +, khaali karna –)	85	53
PP-2	LEAK MACHINE (rasaav)	9	7
PP-3	$\sqrt{(ab)}$ TRICK (saath T, ek a zyada, doosra b zyada)	5	0
PP-4	OPEN / CLOSE MACHINE (beech mein band / chalu)	35	23
PP-5	ALTERNATE HOUR MACHINE (ek ghanta bharo, ek ghanta khaali)	11	8
PP-6	CAPACITY / LITRE MACHINE	10	4
Total (book ke sawaal)		155/155 ✓	95/95 ✓

Chapter 14

Simple Interest (Sadharan Byaj) साधारण ब्याज

6 formulas · English book 200 sawaal · Hindi book 76 sawaal · 43 solved examples ✓

SI-1 — PRT MACHINE (SI, Amount, P, R, T)

PEHCHAN – Question mein ye dikhe

SI / amount / principal / rate / time — koi ek poocha, baaki diye; mahine / din (73 days, 7 Feb se 20 April), “5 paise per rupee per month”, SI ka ratio; साधारण ब्याज, मूलधन, मिश्रधन, दर, समय

$$SI = \frac{P \times R \times T}{100} \quad A = P + SI = P \left(1 + \frac{RT}{100}\right) \quad P = \frac{A \times 100}{100 + RT}$$

- $R = \frac{SI \times 100}{P \times T}$, $T = \frac{SI \times 100}{P \times R}$ · T mahine mein $\Rightarrow \div 12$, din mein $\Rightarrow \div 365$ (pehla din chhodo, aakhri gino).
- “x paise per rupee per month” $\Rightarrow R = 12x\%$ saalana.
- Same P par SI ka ratio = $R_1 T_1 : R_2 T_2$.

TERMS – Har letter ka matlab

P = mooldhan · R = saalana dar % · T = saal · A = mishradhan (amount).

SOLVED – Asli book questions (verified)

Q (EN 2): ₹48,750, 16%, 73 din

$$\frac{48750 \times 16 \times 73}{36500} = 1560 \text{ ✓ book ans (a)}$$

Q (EN 9): ₹2000, 8.25%, 7 Feb $\rightarrow$ 20 Apr 2022

$$21 + 31 + 20 = 72 \text{ din} \Rightarrow \frac{2000 \times 8.25 \times 72}{36500} \approx 33 \text{ ✓ book ans (d)}$$

Q (EN 14): ₹10,000, 6 mahine, 5 paise/rupee/mahina

$$10000 \times 0.05 \times 6 = 3000 \text{ ✓ book ans (c)}$$

Q (EN 7): $12\frac{1}{2}\%$ pe 2 saal mein ₹8250 $\Rightarrow$ SI

$$P = \frac{8250 \times 100}{125} = 6600 \Rightarrow 1650 \text{ ✓ book ans (a)}$$

Q (EN 38): 14.5%, 6 saal, A = ₹13,464 $\Rightarrow$ P

$$\frac{13464 \times 100}{187} = 7200 \text{ ✓ book ans (a)}$$

Q (EN 60): ₹1500 mahina chahiye, 12% $\Rightarrow$ capital

$$\frac{18000 \times 100}{12} = 150000 = 1.5 \text{ lakh } \text{ ✓ book ans (b)}$$

Q (EN 159): ₹7800, 2 saal 8 mahine, SI ₹1976 $\Rightarrow$ R

$$\frac{1976 \times 100}{7800 \times \frac{8}{3}} = 9.5\% \text{ ✓ book ans (c)}$$

Q (EN 165): ₹850, 3.6%, SI ₹25.5 $\Rightarrow$ mahine

$$\frac{25.5 \times 100 \times 12}{850 \times 3.6} = 10 \text{ ✓ book ans (c)}$$

BOOK KE SAWAAL (132) EN 2–9, 11–18, 20–25, 27–35, 37, 38, 43, 44, 47–49, 53, 58, 60, 62, 72, 75, 76, 79, 83, 89, 90, 93–98, 100–103, 105, 106, 112–114, 122, 123, 129, 130, 137, 142, 146–150, 152, 153, 158, 159, 162, 164–166, 168, 169, 171–173, 185, 186, 188 · **HI** 1–13, 16, 17, 19, 20, 24, 32–34, 36–38, 41, 44, 47, 48, 50, 54–58, 60, 61, 64, 65, 67, 68 | Ch 15 (Chakraviddhi Byaj): EN 202

SI-2 — DO AMOUNT MACHINE (T₁ pe A₁, T₂ pe A₂)

PEHCHAN – Question mein ye dikhe

“amounts to ₹24,500 in 8 years and to ₹32,000 in 13 years”, “₹720 after 2 years and ₹1020 after a further 5 years”, maturity value after 3 and 5 years; ... वर्ष में ₹ ... और ... वर्ष में ₹ ...

$$1 \text{ saal ka SI} = \frac{A_2 - A_1}{T_2 - T_1} \quad P = A_1 - T_1 \times (1 \text{ saal ka SI}) \quad R = \frac{1 \text{ saal SI}}{P} \times 100$$

TERMS – Har letter ka matlab

$A_1, A_2 =$ do amounts · $T_1, T_2 =$ unke saal.

SOLVED – Asli book questions (verified)

Q (EN 39): 8 saal ₹24,500, 13 saal ₹32,000 $\Rightarrow P$
 $\frac{7500}{5} = 1500 \Rightarrow 24500 - 12000 = 12500$ ✓ **book ans (a)**

Q (EN 111): 2 saal ₹1250, 5 saal ₹2000 $\Rightarrow R$
 $250/\text{saal}, P = 750 \Rightarrow 33.33\%$ ✓ **book ans (b)**

Q (EN 124): 6 saal ₹12,000, 9 saal ₹15,000 $\Rightarrow R$
 $1000/\text{saal}, P = 6000 \Rightarrow 16\frac{2}{3}\%$ ✓ **book ans (a)**

Q (EN 61): 5 saal ₹2340, 2 saal ₹2016 $\Rightarrow P$
 $108/\text{saal} \Rightarrow 2016 - 216 = 1800$ ✓ **book ans (b)**

Q (EN 199): 4 saal ₹20,720, 6 saal ₹24,080 $\Rightarrow P, R$
 $1680/\text{saal} \Rightarrow P \text{ 14000}, R \text{ 12\%}$ ✓ **book ans (c)**

BOOK KE SAWAAL (13) EN 39, 61, 65, 77, 107, 111, 118, 124, 141, 155, 199 · **HI** 43, 45

SI-3 — RATE / TIME CHANGE MACHINE (farak se P)**PEHCHAN – Question mein ye dikhe**

“had the rate been 2% higher, ₹84 more”, “15 months at 7.5% exceeds 8 months at 12.5% by ₹32.50”, “difference between 12% for 4 years and 9% for 5 years is ₹412.50”, “if invested for 5 years instead of 4, ₹375 more”, difference of rates; दर . . . % अधिक होती, अंतर, कम है

$$P = \frac{\Delta SI \times 100}{R_1 T_1 - R_2 T_2} \quad (\text{sirf rate badla: } P = \frac{\Delta SI \times 100}{\Delta R \times T})$$

- Amount mein farak = SI mein farak (P same).
- Rate ka farak: $\Delta R = \frac{\Delta SI \times 100}{P T}$.

TERMS – Har letter ka matlab

$\Delta SI =$ do haalat ke interest ka farak · $R_1 T_1 - R_2 T_2 =$ unke (rate $\times$ time) ka farak.

SOLVED – Asli book questions (verified)

Q (EN 1): 1 saal, A = ₹1484; rate +2% $\Rightarrow$ ₹26.50 zyada $\Rightarrow SI$
 $P = \frac{26.5 \times 100}{2} = 1325 \Rightarrow 159$ ✓ **book ans (d)**

Q (EN 42): 3 saal, rate +2% $\Rightarrow$ ₹84 zyada
 $\frac{84 \times 100}{6} = 1400$ ✓ **book ans (c)**

Q (EN 46): 18 mahine 5.5% – 14 mahine 6% = ₹62.50
 $\frac{6250}{8.25 - 7} = 5000$ ✓ **book ans (c)**

Q (EN 80): $12\% \times 4 - 9\% \times 5 = ₹412.50$
 $\frac{41250}{48 - 45} = 13750$ ✓ **book ans (d)**

Q (EN 85): 5 saal A ₹5250; rate +2% $\Rightarrow$ ₹5600
 $\frac{350 \times 100}{10} = 3500$ ✓ **book ans (d)**

Q (EN 91): 7.5%, 4 ki jagah 5 saal $\Rightarrow$ ₹375 zyada
 $\frac{375 \times 100}{7.5} = 5000$ ✓ **book ans (b)**

Q (EN 120): ₹1200, 3 saal, SI farak ₹10.80 $\Rightarrow$ rate farak
 $\frac{1080}{3600} = 0.3\%$ ✓ **book ans (d)**

BOOK KE SAWAAL (34) EN 1, 41, 42, 45, 46, 52, 63, 66–68, 71, 74, 78, 80, 85, 86, 91, 99, 104, 120, 135, 196, 200 · **HI** 14, 18, 21, 23, 25, 29, 30, 72 | Ch 15 (Chakraviddhi Byaj): EN 54, 207 · **HI** 69

SI-4 — n-GUNA / R = T MACHINE

PEHCHAN – Question mein ye dikhe

“doubles in 8 years”, “becomes 3 times in 15 years $\Rightarrow$ 5 times?”, “SI = $\frac{3}{4}$ of sum in 15 years”, “rate and time numerically equal”, “SI = $\frac{9}{16}$ of P”, “increase its value by 50%”, “how many times in 12 years”; दोगुनी, तिगुनी, गुना, मूलधन का ... भाग

$$n \text{ guna } T \text{ saal mein } \Rightarrow R = \frac{(n-1) \times 100}{T} \quad T_2 = T_1 \times \frac{n_2 - 1}{n_1 - 1}$$

- SI = P ka f hissa $\Rightarrow R \times T = 100f$ · R = T $\Rightarrow R = \sqrt{100f}$ (ya SI = P $\Rightarrow R^2 = \frac{SI \times 100}{P}$).
- T saal baad kitne guna: $1 + \frac{RT}{100}$.

TERMS – Har letter ka matlab

n = kitne guna (double = 2) · f = SI, P ka kitna hissa · R = T wale mein dono barabar.

SOLVED – Asli book questions (verified)

Q (EN 109): 8 saal mein double

$$\frac{100}{8} = 12.5\% \checkmark \text{ book ans (d)}$$

Q (EN 115): 8 saal mein triple

$$\frac{200}{8} = 25\% \checkmark \text{ book ans (c)}$$

Q (EN 170): 15 saal mein 3 guna $\Rightarrow$ 5 guna

$$15 \times \frac{4}{2} = 30 \checkmark \text{ book ans (b)}$$

Q (EN 183): 7 saal mein 4 guna $\Rightarrow$ 16 guna

$$7 \times \frac{15}{3} = 35 \checkmark \text{ book ans (d)}$$

Q (EN 116): ₹24,000, R = T, SI ₹19,440

$$R^2 = \frac{19440 \times 100}{24000} = 81 \Rightarrow 9\% \checkmark \text{ book ans (d)}$$

Q (EN 128): 15 saal mein SI = $\frac{3}{4}$ P

$$\frac{75}{15} = 5\% \checkmark \text{ book ans (c)}$$

Q (EN 151): SI = P/36, R = T

$$R = \sqrt{\frac{100}{36}} = \frac{5}{3}\% \checkmark \text{ book ans (b)}$$

Q (EN 194): 5 saal mein double $\Rightarrow$ 8 saal mein kitne guna

$$R \ 20\% \Rightarrow 1 + 1.6 = 2\frac{3}{5} \checkmark \text{ book ans (b)}$$

Q (HI 40): 8 saal mein SI = $\frac{11}{25}$ P

$$\frac{44}{8} = 5\frac{1}{2}\% \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (54) EN 87, 108–110, 115–117, 119, 121, 125, 127, 128, 131–134, 139, 140, 144, 151, 154, 160, 161, 167, 170, 174–184, 187, 189–191, 194, 195, 198 · HI 22, 39, 40, 46, 49, 51, 59, 62, 63, 66, 69

SI-5 — SPLIT / MIX INVESTMENT MACHINE

PEHCHAN – Question mein ye dikhe

“divide ₹6600 into two parts so that SI equal”, “some at 5%, rest at 8%, total interest ₹1812”, “ $\frac{1}{2}$ capital at 5%, $\frac{1}{3}$ at 8%, rest at 10%”, rates badalte saal (6% 3 saal, 9% agle 5 ...), equal SI at 4%, 6%, 8%, interchanged amounts, alag amounts same rate; दो भागों में, शेष, समान ब्याज

$$\text{SI barabar: } P_1 : P_2 : P_3 = \frac{1}{R_1 T_1} : \frac{1}{R_2 T_2} : \frac{1}{R_3 T_3} \quad \text{Mix: Income} = C \times \frac{\sum f_i r_i}{100}$$

- Do rates par kul P, kul SI diya $\Rightarrow$ sab ko chhote rate par maano $\Rightarrow$ bacha SI $\div$ (rate farak $\times$ T) = bade rate wala hissa (alligation).
- Rate saal-dar-saal badle: $SI = \frac{P(r_1 t_1 + r_2 t_2 + \dots)}{100}$.
- Alag P, same R: $R = \frac{SI \times 100}{\sum P_i T_i}$ · interchanged: jod aur ghata kar dono amounts.

TERMS – Har letter ka matlab

f = capital ka hissa · r = uski dar · P₁, P₂ = bhaag.

SOLVED – Asli book questions (verified)

Q (EN 26): ₹6600; $10\% \times 3 = 9\% \times 4$ (SI barabar)

$P_1 : P_2 = 36 : 30 = 6 : 5 \Rightarrow 3600$ ✓ **book ans (a)**

Q (EN 40): ₹9200; 5% aur 8%, 3 saal, kul ₹1812 $\Rightarrow$ 5% wala

sab 5% $\Rightarrow$ 1380; bacha $432 \div 9\% = 4800 \Rightarrow 4400$ ✓ **book ans (c)**

Q (EN 50): $6\% \times 3, 9\% \times 5, 13\% \times 3$; kul ₹8160

$\frac{8160 \times 100}{18 + 45 + 39} = 8000$ ✓ **book ans (c)**

Q (EN 57): $\frac{2}{3}$ @6%, $\frac{1}{5}$ @10%, baaki @15%; ₹600

$\frac{60000}{4 + 2 + 2} = 7500$ ✓ **book ans (d)**

Q (EN 73): 4%, 6%, 8% pe SI barabar; kul ₹2600

$\frac{1}{4} : \frac{1}{6} : \frac{1}{8} = 6 : 4 : 3 \Rightarrow 1200$ ✓ **book ans (d)**

Q (EN 36): 12% & 10% $\Rightarrow$ ₹125; ulta $\Rightarrow$ ₹128 $\Rightarrow$ 10% wala

jod: 1150, farak 150 $\Rightarrow$ 650 ✓ **book ans (d)**

Q (EN 70): ₹16,000 @8% + x @18% = kul 10%

$16000 : x = 8 : 2 \Rightarrow x = 4000 \Rightarrow 20000$ ✓ **book ans (b)**

Q (EN 126): ₹7200 $\times 3 +$ ₹8400 $\times 4 =$ ₹4968

$\frac{496800}{55200} = 9\%$ ✓ **book ans (d)**

BOOK KESAWAAL (33) EN 26, 36, 40, 50, 51, 54–57, 64, 69, 70, 73, 81, 82, 84, 88, 92, 126, 136, 143, 145, 157 · **HI** 26–28, 31, 52, 53, 70, 71, 73 | Ch 6
(Pratishatata): EN 180

SI-6 — INSTALMENT / PRESENT WORTH / MISC**PEHCHAN – Question mein ye dikhe**

“annual instalment to discharge ₹9600 in 5 years”, “owes ₹8000 due in one year, settles after 2 months”, “SI added every six months $\Rightarrow$ effective rate”, “in how many years debts equal”, borrow at 6% lend at 9%, principal tripled after 5 years; किस्त, निपटान

$$\text{Instalment } x = \frac{100 D}{100 n + \frac{R n(n-1)}{2}} \quad \text{Present worth} = \frac{A \times 100}{100 + RT}$$

- Har chhe mahine SI jude $\Rightarrow$ effective $= \left[\left(1 + \frac{r}{200} \right)^2 - 1 \right] \times 100$.
- Do karz barabar kab: $P_1 \left(1 + \frac{R_1 t}{100} \right) = P_2 \left(1 + \frac{R_2 t}{100} \right) \Rightarrow t = \frac{P_2 - P_1}{\frac{P_1 R_1 - P_2 R_2}{100}}$.
- Udhaar le kar aage diya $\Rightarrow$ saalana fayda $= P \times (\text{rate farak})/100$. Beech mein P badla $\Rightarrow$ tukdon ka SI jodo.

TERMS – Har letter ka matlab

D = karz · n = kitni kishtein (saal) · A = aage dene wali rakam · PW = aaj ki keemat.

SOLVED – Asli book questions (verified)**Q (EN 193):** ₹9600, 5 saal, 10% $\Rightarrow$ kishit

$$\frac{960000}{500+100} = 1600 \quad \checkmark \text{ book ans (b)}$$

Q (EN 192): ₹8000 1 saal baad dena, 2 mahine mein chukana, 9%

$$\frac{8000}{1+0.09 \times \frac{10}{12}} \approx 7442 \quad \checkmark \text{ book ans (a)}$$

Q (EN 156): 12% SI har 6 mahine jude

$$1.06^2 - 1 = 12.36\% \quad \checkmark \text{ book ans (d)}$$

Q (EN 163): ₹80,000 @24% vs ₹91,000 @20% $\Rightarrow$ barabar kab

$$\frac{11000}{19200-18200} = 11 \text{ saal} \quad \checkmark \text{ book ans (a)}$$

Q (EN 197): ₹3000 6% pe liya, 9% pe diya $\Rightarrow$ saal ka fayda

$$3000 \times 3\% = 90 \quad \checkmark \text{ book ans (a)}$$

Q (EN 19): 10 saal SI ₹600; 5 saal baad P teen guna

$$300 + 900 = 1200 \quad \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (14) EN 19, 59, 156, 163, 192, 193, 197 · **HI** 15, 35, 42, 74–76 | *Ch 15 (Chakraviddhi Byaj):* **HI** 65**COVERAGE CHECK – Har sawaal ka formula**

Formula	Naam	EN Qs	HI Qs
SI-1	PRT MACHINE (SI, Amount, P, R, T)	91	40
SI-2	DO AMOUNT MACHINE (T ₁ pe A ₁ , T ₂ pe A ₂)	11	2
SI-3	RATE / TIME CHANGE MACHINE (farak se P)	23	8
SI-4	n-GUNA / R = T MACHINE	43	11
SI-5	SPLIT / MIX INVESTMENT MACHINE	23	9
SI-6	INSTALMENT / PRESENT WORTH / MISC	7	6
$\rightarrow$ CI-2	CI – SI DIFFERENCE / RELATION MACHINE (Ch 15)	2	0
Total (book ke sawaal)		200/200 ✓	76/76 ✓

 $\rightarrow$ **CI-2:** EN 10, 138

Chapter 15

Compound Interest (Chakravridhi Byaj) चक्रवृद्धि ब्याज

6 formulas · English book 206 sawaal · Hindi book 70 sawaal · 43 solved examples ✓

CI-1 — AMOUNT MACHINE (A, P, CI; half-yearly, quarterly, alag rates)

PEHCHAN – Question mein ye dikhe

amount / principal / CI poocha; half-yearly, quarterly, 8-monthly; “3 years 3 months”, alag saal alag rate (6%, 12%, 18%), “interest for the third year”, second year ka CI; चक्रवृद्धि ब्याज, मिश्रधन, अर्ध-वार्षिक, तिमाही

$$A = P \left(1 + \frac{r}{100}\right)^n \quad CI = A - P \quad P = \frac{A}{\left(1 + \frac{r}{100}\right)^n}$$

Compounding	rate	time
Half-yearly	r/2	2n
Quarterly	r/4	4n
8-monthly	2r/3	3n/2

- **Adhoore saal** (3 saal 3 mahine): $P(1+r)^3 \left(1 + \frac{r}{100} \cdot \frac{3}{12}\right)$ · alag rates: $P(1+r_1)(1+r_2)(1+r_3)$.
- k-th saal ka CI = $P \cdot \frac{r}{100} \left(1 + \frac{r}{100}\right)^{k-1}$.
- **Percent ladder**: 2 saal ka CI% = $2r + \frac{r^2}{100}$ · 3 saal = $3r + \frac{3r^2}{100} + \frac{r^3}{10^3}$ (10% $\Rightarrow$ 21%, 33.1%).

TERMS – Har letter ka matlab

P = mooldhan · r = saalana dar % · n = saal (ya periods) · A = mishradhan.

SOLVED – Asli book questions (verified)

Q (EN 1): 10% pe 2 saal mein ₹10,164 $\Rightarrow$ P

$$\frac{10164}{1.21} = 8400 \text{ ✓ book ans (b)}$$

Q (EN 15): ₹2000, 40%; teesre saal ka CI

$$2000 \times 0.4 \times 1.4^2 = 1568 \text{ ✓ book ans (c)}$$

Q (EN 10): doosre saal ka CI ₹154, 10% $\Rightarrow$ P

$$\frac{154}{0.1 \times 1.1} = 1400 \text{ ✓ book ans (a)}$$

Q (EN 8): 6%, 12%, 18%; 3 saal mein ₹20,000 $\Rightarrow$ P

$$\frac{20000}{1.06 \times 1.12 \times 1.18} = 14276.58 \text{ ✓ book ans (a)}$$

Q (EN 24): 8% half-yearly, 1½ saal, ₹1,96,851 $\Rightarrow$ P

$$\frac{196851}{1.04^3} = 175000 \text{ ✓ book ans (b)}$$

Q (EN 26): ₹13,500, 20% quarterly, 6 mahine

$$13500 \times 1.05^2 = 14883.75 \text{ ✓ book ans (b)}$$

Q (EN 35): 15% 8-monthly, 2 saal $\Rightarrow$ ₹98,494; yearly pe?

$$P = \frac{98494}{1.1^3} = 74000 \Rightarrow 74000 \times 1.15^2 = 97865 \text{ ✓ book ans (a)}$$

Q (EN 61): ₹1,00,000, 20%, 3 saal 3 mahine $\Rightarrow$ CI

$$172800 \times 1.05 - 100000 = 81440 \text{ ✓ book ans (a)}$$

Q (EN 70): ₹25,000; 4%, 5%, 6% $\Rightarrow$ CI

$$25000(1.04 \times 1.05 \times 1.06 - 1) = 3938 \text{ ✓ book ans (c)}$$

BOOK KE SAWAAL (110) EN 1, 2, 5, 6, 8–10, 12, 15, 18, 21–29, 31–33, 35–37, 39–47, 49–53, 55–58, 60–66, 68–77, 79, 81–98, 100, 198 · **HI** 1, 2, 4, 5, 8, 10–25, 27–34

CI-2 — CI – SI DIFFERENCE / RELATION MACHINE

PEHCHAN – Question mein ye dikhe

“difference between CI and SI for 2 years at 9% is ₹97.2”, 3 saal ka farak, “SI ₹500 for 2 years ⇒ CI”, “CI ₹696.30, SI ₹660 ⇒ rate”, “CI for 3 years ₹3783 ⇒ SI”; चक्रवृद्धि और साधारण ब्याज का अंतर

$$2 \text{ saal: } D = P \left(\frac{r}{100} \right)^2 \quad 3 \text{ saal: } D = \frac{P r^2 (300 + r)}{10^6}$$

- 2 saal: $CI = SI \left(1 + \frac{r}{200} \right) \cdot \text{rate } r = \frac{2(CI-SI)}{SI} \times 100$.
- Kisi bhi n: CI se P nikaalo (CI-1 ladder), phir $SI = \frac{PrT}{100}$ (ya ulta).
- Half-yearly vs SI: $P \left[\left(1 + \frac{r}{200} \right)^{2n} - 1 \right] - \frac{Prn}{100}$.

TERMS – Har letter ka matlab

$D = CI - SI$ · $r = \text{dar}$ · $P = \text{mooldhan}$.

SOLVED – Asli book questions (verified)

Q (EN 132): 2 saal 9%, farak ₹97.2 ⇒ P

$$\frac{97.2 \times 10^4}{81} = 12000 \quad \checkmark \text{ book ans (d)}$$

Q (EN 134): ₹5000, 10%, 3 saal ⇒ farak

$$\frac{5000 \times 100 \times 310}{10^6} = 155 \quad \checkmark \text{ book ans (d)}$$

Q (EN 144): 10%, 3 saal, farak ₹620 ⇒ P

$$\frac{620 \times 10^6}{100 \times 310} = 20000 \quad \checkmark \text{ book ans (d)}$$

Q (EN 137): ₹24,000, 2 saal, farak ₹60 ⇒ r

$$r^2 = \frac{60 \times 10^4}{24000} = 25 \Rightarrow 5\% \quad \checkmark \text{ book ans (c)}$$

Q (EN 119): SI ₹660, CI ₹696.30 (2 saal) ⇒ r

$$\frac{2 \times 36.3}{660} \times 100 = 11\% \quad \checkmark \text{ book ans (b)}$$

Q (EN 112): SI ₹500, 2 saal, 10% ⇒ CI

$$500 \times 1.05 = 525 \quad \checkmark \text{ book ans (b)}$$

Q (EN 107): CI 3 saal 5% = ₹3783 ⇒ SI

$$P = \frac{3783}{0.157625} = 24000 \Rightarrow 3600 \quad \checkmark \text{ book ans (c)}$$

Q (EN 145): ₹1000, 10%, 4 saal ⇒ farak

$$464.10 - 400 = 64.10 \quad \checkmark \text{ book ans (a)}$$

Q (EN 147): 10% half-yearly, 2 saal; farak ₹124.05 ⇒ P

$$0.21550625P - 0.2P = 124.05 \Rightarrow 8000 \quad \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (75) EN 67, 99, 101–104, 106–123, 127–161 · **HI** 6, 7, 35, 36, 38, 43–51 | Ch 14 (Sadharan Byaj): EN 10, 138

CI-3 — RATE MACHINE (dar nikaalo)

PEHCHAN – Question mein ye dikhe

“₹10,000 → ₹11,025 in 2 years ⇒ rate”, “interest in two successive years ₹225 and ₹236.25”, “₹9680 after 2 years, ₹10648 after 3”, “₹6400 in 2 years, ₹8100 in 4”, 8 guna in 3 saal ⇒ rate, sum aur rate dono; ब्याज की दर ज्ञात

$$1 + \frac{r}{100} = \left(\frac{A}{P} \right)^{1/n} \quad \text{Lagataar saal: } r = \frac{A_2 - A_1}{A_1} \times 100 = \frac{I_2 - I_1}{I_1} \times 100$$

- A_m aur A_n ($m > n$): $\left(\frac{A_m}{A_n} \right)^{\frac{1}{m-n}} = 1 + \frac{r}{100} \Rightarrow P = \frac{A_n}{(1+r)^n}$.
- Pehchano: $1.21 = 1.1^2$, $1.331 = 1.1^3$, $1.1025 = 1.05^2$, $1.44 = 1.2^2$, $1.2544 = 1.12^2$, $1.265625 = 1.125^2$.

TERMS – Har letter ka matlab

A/P = kitne guna hua · n = saal · I_1, I_2 = lagataar saalon ka interest.

SOLVED – Asli book questions (verified)**Q (EN 163):** lagataar saal ka interest ₹225, ₹236.25

$$\frac{11.25}{225} = 5\% \checkmark \text{ book ans (d)}$$

Q (EN 168): 2 saal ₹9680, 3 saal ₹10,648

$$\frac{968}{9680} = 10\% \checkmark \text{ book ans (a)}$$

Q (EN 175): 2 saal ₹6400, 4 saal ₹8100

$$\sqrt{\frac{8100}{6400}} = 1.125 \Rightarrow 12.5\% \checkmark \text{ book ans (d)}$$

Q (EN 166): ₹57,600 → ₹72,900, 1 saal half-yearly

$$1.265625 = 1.125^2 \Rightarrow 12.5\% \text{ per half} \Rightarrow 25\% \checkmark \text{ book ans (d)}$$

Q (EN 181): ₹305 → ₹670, 3 saal

$$\left(\frac{670}{305}\right)^{1/3} = 1.3 \Rightarrow 30\% \checkmark \text{ book ans (a)}$$

Q (EN 197): 2 saal ₹3025, 3 saal ₹3327.5 ⇒ P, r

$$r \ 10\% \Rightarrow P = \frac{3025}{1.21} = 2500 \checkmark \text{ book ans (d)}$$

BOOK KESAWAAL (29) EN 162–181, 197, 200 · HI 3, 52–57**CI-4 — TIME / n-GUNA MACHINE****PEHCHAN – Question mein ye dikhe**

“doubles in 6 years ⇒ 8 times?”, “3 times in 10 years ⇒ 9 times”, “in how many years ₹10,000 → ₹13,310 at 10%”, quarterly mein kitna samay; कितने वर्ष, गुनी

$$k \text{ guna } T \text{ saal mein} \Rightarrow k^m \text{ guna } mT \text{ saal mein}$$

- Time: $\frac{A}{P}$ ko $(1 + \frac{r}{100})$ ki power pehchano ($1.331 = 1.1^3$, $2.8561 = 1.3^4$, $1.050625 = 1.025^2$).
- Half-yearly / quarterly periods ko 2 / 4 se bhaag karke saal.

TERMS – Har letter ka matlab

k = kitne guna · T = uska samay · m = power ($8 = 2^3 \Rightarrow m = 3$).

SOLVED – Asli book questions (verified)**Q (EN 183):** 6 saal mein 2 guna ⇒ 8 guna

$$8 = 2^3 \Rightarrow 18 \checkmark \text{ book ans (a)}$$

Q (EN 184): 10 saal mein double ⇒ 16 guna

$$16 = 2^4 \Rightarrow 40 \checkmark \text{ book ans (c)}$$

Q (HI 60): 10 saal mein 3 guna ⇒ 9 guna

$$9 = 3^2 \Rightarrow 20 \checkmark \text{ book ans (c)}$$

Q (EN 185): ₹20,000 → ₹26,620 at 10%

$$1.331 = 1.1^3 \Rightarrow 3 \checkmark \text{ book ans (c)}$$

Q (EN 195): ₹10,000 → ₹28,561 at 30%

$$2.8561 = 1.3^4 \Rightarrow 4 \checkmark \text{ book ans (c)}$$

Q (EN 187): ₹3200 → ₹3362 at 10% quarterly

$$1.050625 = 1.025^2 \Rightarrow 2 \text{ quarter} = \frac{1}{2} \text{ saal} \checkmark \text{ book ans (d)}$$

BOOK KESAWAAL (24) EN 34, 38, 182–196 · HI 9, 58–63**CI-5 — INSTALMENT / SPLIT / PRESENT WORTH****PEHCHAN – Question mein ye dikhe**

“annual equal payment to discharge ₹28,700 due in 2 years at 5%”, “repaid in 2 equal instalments”, “divide ₹22,100 so that amount after 18 years = after 20 years”, present worth, ₹200 at the beginning of each year, part payment every year, two schemes NSC/PPF; कित, बराबर कित

$$\text{Kisht } x = \frac{P(1+i)^n i}{(1+i)^n - 1} \quad \left(n = 2 : x = \frac{P(1+i)^2}{2+i} \right) \quad \left(i = \frac{r}{100} \right)$$

- Present worth = $\frac{A}{(1+i)^n}$ · kishton se P: $P = \frac{x}{1+i} + \frac{x}{(1+i)^2} + \dots$.
- **Split** (A ka n_1 saal baad = B ka n_2 saal baad): $A : B = (1+i)^{n_2-n_1} : 1$.
- Har saal shuru mein x jama: $x(1+i) + x(1+i)^2 + \dots$ · har saal ke ant mein p chukaya: bacha = $P(1+i)^n - p[(1+i)^{n-1} + \dots + 1]$.
- Do schemes (14%, 11%): sab ko chhoti scheme par maano $\Rightarrow$ bacha CI $\div$ (factor farak).

TERMS – Har letter ka matlab

$x = \text{har kisht} \cdot i = r/100 \cdot n = \text{kishton ki ginti} \cdot P = \text{karz}$.

SOLVED – Asli book questions (verified)

Q (EN 7): ₹28,700, 2 saal, 5% $\Rightarrow$ kisht

$$\frac{28700 \times 1.1025}{2.05} = 15435 \quad \checkmark \text{ book ans (d)}$$

Q (EN 16): ₹16,400, 2 kisht, 5%

$$\frac{16400 \times 1.1025}{2.05} = 8820 \quad \checkmark \text{ book ans (d)}$$

Q (EN 11): ₹338 2 saal baad, 4% $\Rightarrow$ present worth

$$\frac{338}{1.04^2} = 312.50 \quad \checkmark \text{ book ans (d)}$$

Q (EN 4): ₹22,100; Monali 18 saal = Timir 20 saal, 10%

$$M : T = 1.21 : 1 \Rightarrow 22100 \times \frac{1.21}{2.21} = 12100 \quad \checkmark \text{ book ans (b)}$$

Q (EN 201): ₹3364; A 5 saal = B 7 saal, 5%

$$A : B = 1.1025 : 1 \Rightarrow 1764 \quad \checkmark \text{ book ans (c)}$$

Q (EN 17): ₹200 har saal shuru mein, 5%, 2 saal

$$210 + 220.5 = 430.5 \quad \checkmark \text{ book ans (b)}$$

Q (HI 64): ₹11,550, 10%, 2 barabar kisht

$$\frac{11550 \times 1.21}{2.1} = 6655 \quad \checkmark \text{ book ans (d)}$$

Q (EN 14): ₹10,000: NSC 14%, PPF 11%, 2 saal CI ₹2726 $\Rightarrow$ PPF

$$\text{sab } 11\% \Rightarrow 2321; \text{bacha } 405 \div 0.0675 = 6000 \text{ NSC} \Rightarrow \text{PPF } 4000 \quad \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (16) EN 4, 7, 11, 13, 14, 16, 17, 30, 48, 59, 201 · HI 26, 64, 66–68

CI-6 — SI vs CI / YEARLY vs HALF-YEARLY MACHINE

PEHCHAN – Question mein ye dikhe

“borrowed at 9% SI, invested at 10% CI for 3 years, profit ₹1952”, “lent at same rate CI $\Rightarrow$ profit”, “half-yearly gives ₹482 more than annual”, “second scheme interest how much more”, who paid more; साधारण पर उधार, चक्रवृद्धि पर दिया

$$\text{Fayda} = P \left[\left(1 + \frac{r_2}{100} \right)^n - 1 - \frac{r_1 n}{100} \right] \quad \text{Half-yearly} - \text{yearly} = P \left[\left(1 + \frac{r}{200} \right)^{2n} - \left(1 + \frac{r}{100} \right)^n \right]$$

- Same rate, 2 saal: fayda = CI – SI = $P \left(\frac{r}{100} \right)^2$ (CI-2).
- Farak diya $\Rightarrow P = \text{farak} \div \text{bracket wala factor}$.

TERMS – Har letter ka matlab

$r_1 = \text{SI wali dar} \cdot r_2 = \text{CI wali dar} \cdot n = \text{saal}$.

SOLVED – Asli book questions (verified)

Q (EN 3): 40%, 2 saal; half-yearly se ₹4544 zyada $\Rightarrow P$

$$P(1.2^4 - 1.4^2) = 0.1136P = 4544 \Rightarrow 40000 \checkmark \text{ book ans (b)}$$

Q (EN 19): 20%, 2 saal; half-yearly ₹482 zyada

$$P(1.1^4 - 1.2^2) = 0.0241P \Rightarrow 20000 \checkmark \text{ book ans (b)}$$

Q (EN 20): 9% SI pe liya, 10% CI pe lagaya, 3 saal fayda ₹1952

$$P(1.331 - 1 - 0.27) = 0.061P \Rightarrow 32000 \checkmark \text{ book ans (b)}$$

Q (EN 124): ₹75,000 15% SI liya, 15% CI pe diya, 2 saal

$$75000 \times 0.0225 = 1687.5 \checkmark \text{ book ans (d)}$$

Q (EN 126): ₹100: 10% SI vs 10% half-yearly, 1 saal (paise mein)

$$10.25 - 10 = 0.25 \text{ ₹} = 25 \text{ paise} \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (18) EN 3, 19, 20, 80, 105, 124–126, 199, 203, 205, 206 · **HI** 37, 39–42, 70

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
CI-1	AMOUNT MACHINE (A, P, CI; half-yearly, quarterly, alag rates)	81	29
CI-2	CI – SI DIFFERENCE / RELATION MACHINE	59	14
CI-3	RATE MACHINE (dar nikaalo)	22	7
CI-4	TIME / n-GUNA MACHINE	17	7
CI-5	INSTALMENT / SPLIT / PRESENT WORTH	11	5
CI-6	SI vs CI / YEARLY vs HALF-YEARLY MACHINE	12	6
$\rightarrow PC-4$	POPULATION / GROWTH MACHINE (Ch 6)	1	0
$\rightarrow SI-1$	PRT MACHINE (SI, Amount, P, R, T) (Ch 14)	1	0
$\rightarrow SI-3$	RATE / TIME CHANGE MACHINE (farak se P) (Ch 14)	2	1
$\rightarrow SI-6$	INSTALMENT / PRESENT WORTH / MISC (Ch 14)	0	1
Total (book ke sawaal)		206/206 ✓	70/70 ✓
$\rightarrow PC-4$: EN 204 · $\rightarrow SI-1$: EN 202 · $\rightarrow SI-3$: EN 54, 207 HI 69 · $\rightarrow SI-6$: HI 65			

Chapter 16

Problems on Age (Aayu Sambandhi Prashn) आयु संबंधी प्रश्न

4 formulas · English book 241 sawaal · Hindi book 122 sawaal · 26 solved examples ✓

AG-1 — DO-PAL RATIO MACHINE (t₁ pe a : b, t₂ pe c : d)

PEHCHAN – Question mein ye dikhe

“2 years ago 4 : 5, 3 years hence 5 : 6”, “father 3 times now, after 10 years twice”, “6 years ago 6 times, 3 years hence 3 times”, present ages ka ratio poocha, “7 : 9 five years ago, 5 : 6 after 10 years”; अनुपात, गुनी, पहले ... बाद

$$\text{Unit } k = \frac{(t_2 - t_1)(c - d)}{a d - b c} \Rightarrow t_1 \text{ par umar } ak, bk$$

- “m guna” = ratio m : 1. Pehle wala (ago) samay = −, baad wala (hence) = +, abhi = 0.
- Jawab ka samay alag ho to ak mein $(t - t_1)$ jodo.
- Shortcut: umar ka **farak kabhi nahi badalta** $\Rightarrow$ dono ratios ke $(a - b)$ ko barabar karke bhi units milte hain.

TERMS – Har letter ka matlab

$a : b = t_1$ samay ka ratio · $c : d = t_2$ ka · $k = 1$ unit (t_1 par).

SOLVED – Asli book questions (verified)

Q (EN 5): abhi 3 guna, 10 saal baad 2 guna $\Rightarrow$ beti

$$k = \frac{10(2-1)}{3-2} = 10 \checkmark \text{ book ans (d)}$$

Q (EN 2): 6 saal pehle 6 guna, 3 saal baad 3 guna $\Rightarrow$ beti

$$k = \frac{9 \times 2}{3} = 6 \Rightarrow 6 + 6 = 12 \checkmark \text{ book ans (b)}$$

Q (EN 15): 2 saal pehle 4 : 5, 3 saal baad 5 : 6 $\Rightarrow$ Pranav

$$k = \frac{5(-1)}{24-25} = 5 \Rightarrow 25 + 2 = 27 \checkmark \text{ book ans (d)}$$

Q (EN 27): 9 : 10 abhi, 12 saal baad 13 : 14

$$k = \frac{12(-1)}{126-130} = 3 \Rightarrow 27 \checkmark \text{ book ans (a)}$$

Q (EN 32): abhi 6 : 7, 15 saal pehle 9 : 11 $\Rightarrow$ Mahendra

$$t_1 = -15: k = \frac{15(-1)}{63-66} = 5 \Rightarrow 45 + 15 = 60 \checkmark \text{ book ans (b)}$$

Q (EN 189): 5 saal pehle 7 : 9, 10 saal baad 5 : 6 $\Rightarrow$ abhi ka ratio

$$k = 5 \Rightarrow 35, 45 \Rightarrow 40 : 50 = 4 : 5 \checkmark \text{ book ans (d)}$$

Q (EN 119): 5 saal baad 3 guna, 5 saal pehle 7 guna $\Rightarrow$ dono

$$k = \frac{10 \times 2}{4} = 5 \Rightarrow 35, 5 \Rightarrow \text{abhi } 40, 10 \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (123) EN 2, 5, 12, 13, 15, 16, 23–25, 27, 30–33, 44–48, 51–53, 55–57, 64, 65, 67, 68, 70, 72, 82, 89, 91, 93, 95, 103, 104, 107, 114, 117, 119–123, 126–128, 136, 139, 143, 147, 149, 153, 155–160, 179, 181, 184–186, 189–191, 193–195, 198–200, 205–207, 212, 232 · **HI** 8, 10, 15, 16, 18, 21, 23, 26, 33–36, 41, 57, 60, 61, 63, 66–68, 71, 73, 75, 76, 78, 84–86, 88, 92, 98, 99, 103, 105, 107, 108, 110–112, 115, 117, 118, 120

AG-2 — FARAK MACHINE (umar ka antar kabhi nahi badalta)

PEHCHAN – Question mein ye dikhe

“Ritesh 12 years older; 17 years ago 4 times”, “difference 20; 5 years ago 5 times”, “Kohli 3 years younger, ratio 7 : 8”, “mother’s age at daughter’s birth 28, ratio 9 : 2”, “I was as old as you are now when you were born”; वर्ष बढ़ा/छोटा, अंतर, जन्म के समय

$$\text{Farak } d, \text{ kisi samay } A = m \times B \Rightarrow B_{\text{us samay}} = \frac{d}{m-1} \quad \text{ratio } a : b \Rightarrow 1 \text{ unit} = \frac{d}{a-b}$$

- Janm ke samay maa/pita ki umar = farak (d).
- “Main tumhari abhi ki umar ka tha jab tum paida hue” $\Rightarrow$ pita = 2 × beta (abhi).
- Us samay ki umar mein bache hue saal jodo/ghatao.

TERMS – Har letter ka matlab

$d = \text{umar ka farak} \cdot m = \text{kitne guna} \cdot a : b = \text{ratio}.$

SOLVED – Asli book questions (verified)

Q (EN 14): farak 20; 5 saal pehle 5 guna $\Rightarrow$ Arun

$$\frac{20}{4} = 5 \Rightarrow \text{Kiran 10, Arun 30} \checkmark \text{ book ans (c)}$$

Q (EN 1): 12 saal bada; 17 saal pehle 4 guna $\Rightarrow$ Mohit

$$\frac{12}{3} = 4 \Rightarrow 4 + 17 = 21 \checkmark \text{ book ans (d)}$$

Q (EN 4): 3 saal chhota, ratio 7 : 8

$$1 \text{ unit} = 3 \Rightarrow 21 \checkmark \text{ book ans (d)}$$

Q (EN 63): farak 16; 6 saal pehle 3 guna $\Rightarrow$ chhota

$$\frac{16}{2} = 8 \Rightarrow 14 \checkmark \text{ book ans (c)}$$

Q (EN 35): ratio 3 : 8, janm ke samay maa 25

$$\frac{25}{5} = 5 \Rightarrow 40 \checkmark \text{ book ans (b)}$$

Q (EN 201): “jab tum paida hue main tumhari umar ka tha”, pita 40 $\Rightarrow$ beta 5 saal pehle

$$\text{beta } 20 \Rightarrow 15 \checkmark \text{ book ans (d)}$$

Q (EN 36): farak 24; 2 saal pehle pita = beta (abhi) ka 2 guna

$$S + 22 = 2S \Rightarrow S = 22 \Rightarrow F = 46 \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (86) EN 1, 4, 7, 14, 22, 26, 28, 35–38, 42, 49, 50, 58, 59, 63, 66, 73, 77, 88, 94, 96, 102, 106, 108, 112, 140–142, 151, 154, 164–169, 171–174, 177, 188, 197, 201, 210, 214, 215, 221, 222, 233 · **HI** 3, 4, 6, 7, 11–13, 17, 19, 24, 29–31, 43, 50, 51, 56, 59, 64, 69, 70, 72, 87, 93–95, 104, 106, 113, 114, 116, 121–123

AG-3 — SUM / AVERAGE MACHINE (jod har saal badhta hai)**PEHCHAN – Question mein ye dikhe**

“sum 30; after 5 years ratio 3 : 2”, “8 years ago total 56; ratio 4 : 7 : 9”, average age of family, “children born at intervals of 3 years, sum 60”, successive years, “after 5 years average 33”; योग, औसत, अंतराल पर जन्मे

n logon ka jod: t saal baad $= S + nt$, t saal pehle $= S - nt \Rightarrow$ ratio se baanto (RP-1)

- Average $\times$ logon ki ginti = jod $\cdot$ naya member = naya jod – purana jod.
- Barabar antar (AP) mein n bachche: beech wala $= \frac{S}{n} \Rightarrow$ sabse chhota = beech $- \frac{(n-1)d}{2}$.
- Sum S aur difference $D \Rightarrow$ bada $= \frac{S+D}{2} \cdot$ do groups ke average (boys 15.4, girls 18.2, sab 16.8) $\Rightarrow$ MX-1 alligation.

TERMS – Har letter ka matlab

$S = \text{abhi ka jod} \cdot n = \text{kitne log} \cdot t = \text{saal} \cdot d = \text{do bachchon ka antar}.$

SOLVED – Asli book questions (verified)

Q (EN 10): jod 30; 5 saal baad 3 : 2 $\Rightarrow$ A

$$40 \times \frac{3}{5} = 24 \Rightarrow 19 \checkmark \text{ book ans (d)}$$

Q (EN 34): jod 45; 5 saal pehle 6 : 1 $\Rightarrow$ pita

$$35 \times \frac{6}{7} = 30 \Rightarrow 35 \checkmark \text{ book ans (c)}$$

Q (EN 41): 4 : 7 : 9; 8 saal pehle jod 56 $\Rightarrow$ sabse bada

$$80 \times \frac{9}{20} = 36 \checkmark \text{ book ans (c)}$$

Q (EN 29): 10 saal pehle pati-patni avg 42; ab parivaar (3) avg 39 $\Rightarrow$ beta

$$117 - 104 = 13 \checkmark \text{ book ans (b)}$$

Q (EN 18): patni 4 saal chhoti; 5 saal baad avg 33

$$\text{jod abhi } 56 \Rightarrow \frac{56-4}{2} = 26 \checkmark \text{ book ans (c)}$$

Q (EN 218): 5 bachche, 3 saal antar, jod 60 $\Rightarrow$ chhota

$$\text{beech } 12 \Rightarrow 12 - 6 = 6 \checkmark \text{ book ans (a)}$$

Q (EN 162): 7 bachche lagataar saal; bade 3 ka jod 93 $\Rightarrow$ chhote 3

$$93 - 3 \times 4 = 81 \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (91) EN 8–11, 18, 19, 21, 29, 34, 39–41, 43, 54, 60, 61, 81, 84–86, 97, 98, 111, 125, 130, 131, 135, 145, 146, 150, 152, 162, 163, 170, 175, 176, 178, 180, 183, 196, 202–204, 208, 218, 219, 223, 224, 226, 234, 235, 237, 239–241 · **HI** 1, 2, 5, 9, 14, 20, 22, 25, 27, 28, 37, 38, 40, 42, 44–49, 53, 55, 58, 62, 65, 74, 77, 79, 81–83, 97, 100–102, 119

AG-4 — EQUATION / PRODUCT / FRACTION MACHINE

PEHCHAN – Question mein ye dikhe

“after 6 years product 231”, “sum 30, product 221”, “ $\frac{2}{3}$ of my age = $\frac{3}{4}$ of cousin’s”, “sum = $7 \times$ difference”, LCM / HCF of ages, “after 11 years = $5 \times$ (5 years before)”, “ $3 \times P - Q = 25$ ”; गुणनफल, भाग के बराबर

Ek variable: sabse chhote ko x maano $\Rightarrow$ ek line ki equation **Product:** factor pairs se

- Sum S , product $P \Rightarrow$ dono umar = roots ($\frac{S \pm \sqrt{S^2 - 4P}}{2}$) ya factor pair jiska jod S ho.
- $\frac{p}{q}A = \frac{r}{s}B \Rightarrow A : B = \frac{r}{s} : \frac{p}{q} \Rightarrow$ phir AG-2/AG-1.
- Do equations ($2x + 3y = 106$, $4y - 3x = 11$) $\Rightarrow$ Cramer (NS-9) · options daal kar check bhi fast hai.

TERMS – Har letter ka matlab

x = anjaan umar · S, P = jod, gunanfal.

SOLVED – Asli book questions (verified)

Q (EN 20): Varun 3 guna; 6 saal baad gunanfal 231

$231 = 21 \times 11 \Rightarrow x + 6 = 11 \Rightarrow$ Varun 15 ✓ **book ans (a)**

Q (EN 101): $\frac{2}{3}$ meri = $\frac{3}{4}$ cousin; mera (-3) = cousin $(+1)$

$9 : 8 \Rightarrow 9k - 3 = 8k + 1 \Rightarrow k = 4 \Rightarrow 36$ ✓ **book ans (a)**

Q (EN 99): jod = $7 \times$ farak; 5 saal baad jod = $9 \times$ farak $\Rightarrow$ bada

$2D = 10 \Rightarrow D = 5, S = 35 \Rightarrow 20$ ✓ **book ans (c)**

Q (EN 133): jod 30, gunanfal 221

17, 13 ✓ **book ans (a)**

Q (EN 92): 11 saal baad = $5 \times$ (5 saal pehle)

$x + 11 = 5(x - 5) \Rightarrow 9$ ✓ **book ans (d)**

BOOK KE SAWAAL (60) EN 3, 6, 17, 20, 62, 69, 71, 74–76, 78–80, 83, 87, 90, 92, 99–101, 105, 109, 110, 113, 115, 116, 118, 124, 129, 132–134, 137, 138, 144, 148, 161, 182, 187, 192, 209, 211, 213, 216, 220, 227–231, 238 · **HI** 32, 39, 52, 80, 89–91, 96, 109

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
AG-1	DO-PAL RATIO MACHINE (t ₁ pe a : b, t ₂ pe c : d)	80	43
AG-2	FARAK MACHINE (umar ka antar kabhi nahi badalta)	52	34
AG-3	SUM / AVERAGE MACHINE (jod har saal badhta hai)	55	36
AG-4	EQUATION / PRODUCT / FRACTION MACHINE	51	9
$\rightarrow$ DF-8	FRACTION EQUATION MACHINE (ansh-har wale sawaal) (Ch 2)	3	0
Total (book ke sawaal)		241/241 ✓	122/122 ✓
$\rightarrow$ DF-8: EN 217, 225, 236			

Chapter 17

Average (Ausat) औसत

6 formulas · English book 116 sawaal · Hindi book 82 sawaal · 38 solved examples ✓

AV-1 — SERIES AVERAGE (lagataar / AP numbers)

PEHCHAN – Question mein ye dikhe

consecutive numbers / even / odd, first n natural / odd / even numbers, first 20 multiples of 7, squares of first 10, primes / composites ka average, numbers between 12 and 39 divisible by 5; क्रमागत, गुणज, प्राकृतिक, अभाज्य

$$\text{AP ka average} = \frac{\text{pehla} + \text{aakhri}}{2} = \text{beech wala}$$

Series (pehle n)	Average
Natural 1, 2, ..., n	$\frac{n+1}{2}$
Odd	n
Even	$n + 1$
k ke multiples	$\frac{k(n+1)}{2}$
Squares $1^2 \dots n^2$	$\frac{(n+1)(2n+1)}{6}$
Cubes	$\frac{n(n+1)^2}{4}$

Consecutive (gap d) aur average $A \Rightarrow$ beech = A, sabse chhota = $A - (n-1)d/2$ · primes 2, 3, 5, 7, 11, 13, 17, 19; composites 4, 6, 8, 9, 10, 12, 14, 15 (yaad).

TERMS – Har letter ka matlab

n = kitne terms · k = kiska multiple · d = common gap.

SOLVED – Asli book questions (verified)

Q (EN 2): 11 ke pehle 12 multiples

$$\frac{11 \times 13}{2} = 71.5 \text{ ✓ book ans (c)}$$

Q (EN 6): pehle 10 natural ke squares ka mean

$$\frac{11 \times 21}{6} = \frac{77}{2} \text{ ✓ book ans (d)}$$

Q (EN 7): pehle 10 odd

$$= n = 10 \text{ ✓ book ans (b)}$$

Q (EN 14): 7 ke pehle 20 multiples

$$\frac{7 \times 21}{2} = 73.5 \text{ ✓ book ans (c)}$$

Q (EN 13): 5 lagataar even, avg 40 $\Rightarrow$ sabse chhota

$$40 - 4 = 36 \text{ ✓ book ans (b)}$$

Q (HI 7): pehle 35 natural

$$\frac{36}{2} = 18 \text{ ✓ book ans (c)}$$

Q (HI 16): 12 aur 39 ke beech 5 se divisible

$$15 \dots 35 \Rightarrow \frac{15+35}{2} = 25 \text{ ✓ book ans (c)}$$

BOOK KE SAWAAL (24) EN 1–3, 5–8, 11–20 · HI 2–4, 7, 10, 16, 17

AV-2 — TOTAL MACHINE (Total = Avg $\times$ n)

PEHCHAN – Question mein ye dikhe

“misread 16 instead of 58”, “find the missing / 6th value”, “5 numbers avg 15, four are 10, 12, 14, 18”, “each number divided by 2”, table ka average, family monthly spending; त्रुटिवश, शेष संख्या

$$\text{Total} = \text{Average} \times n \quad \text{Missing} = nA - \sum (\text{baaki}) \quad \text{Sahi avg} = A + \frac{\text{sahi} - \text{galat}}{n}$$

- Har number mein $+k \Rightarrow \text{avg} +k$; $\times k \Rightarrow \text{avg} \times k$; $\div k \Rightarrow \text{avg} \div k$.
- Har value k kam/zyada napi gayi $\Rightarrow \text{avg}$ bhi k kam/zyada.
- Umar wale: t saal baad har member $+t \Rightarrow \text{total} + n \cdot t$ (naya bachcha 0 se shuru).

TERMS – Har letter ka matlab

$A = \text{average}$ · $n = \text{ginti}$ · $\text{galat/sahi} = \text{misread value}$.

SOLVED – Asli book questions (verified)

Q (EN 4): 7 ka avg 53; 58 ki jagah 16 padha

$$53 + \frac{42}{7} = 59 \quad \checkmark \text{ book ans (c)}$$

Q (EN 25): 40 ka avg 68; 25 ko 45 padha

$$68 - \frac{20}{40} = 67.5 \quad \checkmark \text{ book ans (d)}$$

Q (EN 99): 20 ka avg 50; 13, 24 ko 31, 42 likha

$$50 - \frac{36}{20} = 48.2 \quad \checkmark \text{ book ans (b)}$$

Q (EN 70): 5 mahine ka kharcha; 6 ka avg ₹4500 $\Rightarrow$ 6th

$$27000 - 24800 = 2200 \quad \checkmark \text{ book ans (d)}$$

Q (EN 63): 6 ka avg 17.5; paanch diye

$$105 - 90 = 15 \quad \checkmark \text{ book ans (b)}$$

Q (EN 54): har seb 20 g kam tola, avg 104 g

$$104 + 20 = 124 \quad \checkmark \text{ book ans (b)}$$

BOOK KESAWAAL (69) EN 4, 9, 22, 25, 45, 49, 50, 52, 54, 56, 58, 60, 61, 63, 67, 70, 76, 85–97, 99–101, 103, 104, 114–116 · **HI** 1, 6, 8, 11–15, 19, 48, 51, 52, 55–57, 60, 64, 65, 68, 71–74, 76–82 | Ch 28 (Vividh): EN 28

AV-3 — COMBINED / WEIGHTED AVERAGE (do group)

PEHCHAN – Question mein ye dikhe

class A 40 students avg 23, class B 45 avg 22; boys : girls 7 : 3 with averages; “9 boys, girls avg 15, overall 14.28 $\Rightarrow$ total”, sections, batches, ek group ka avg nikaalo; लड़के-लड़कियाँ, समूह, बैच, पूरी कक्षा

$$\bar{A} = \frac{n_1 a_1 + n_2 a_2 + \dots}{n_1 + n_2 + \dots} \quad \frac{n_1}{n_2} = \frac{a_2 - \bar{A}}{\bar{A} - a_1} \quad (\text{alligation})$$

- Doosre group ka avg: $a_2 = \frac{(n_1 + n_2)\bar{A} - n_1 a_1}{n_2}$.
- Ratio diya (boys : girls) $\Rightarrow$ ratio hi n ki jagah daalo.

TERMS – Har letter ka matlab

$n = \text{group ki ginti}$ · $a = \text{group ka avg}$ · $\bar{A} = \text{sab ka avg}$.

SOLVED – Asli book questions (verified)**Q (EN 23):** 100 @25 + 50 @70

$$\frac{2500+3500}{150} = 40 \checkmark \text{ book ans (d)}$$

Q (EN 27): 20 @50%, 40 @60%, 60 @70%

$$\frac{1000+2400+4200}{120} = 63\% \checkmark \text{ book ans (d)}$$

Q (EN 29): 9 boys @13, girls @15, sab 14.28 $\Rightarrow$ kul

$$\text{girls} = 9 \times \frac{1.28}{0.72} = 16 \Rightarrow 25 \checkmark \text{ book ans (b)}$$

Q (EN 31): 18 boys @16; 30 sab @18.1 $\Rightarrow$ girls

$$\frac{543-288}{12} = 21.25 \checkmark \text{ book ans (a)}$$

Q (EN 77): 12 @8, n @10, sab 9.2 $\Rightarrow$ n

$$12 \times \frac{1.2}{0.8} = 18 \checkmark \text{ book ans (a)}$$

Q (HI 26): sab 70; boys girls se 20% zyada; ratio 3 : 2 $\Rightarrow$ girls

$$\frac{3(1.2g)+2g}{5} = 70 \Rightarrow g = 62.5 \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (32) EN 21, 23, 27, 29, 31, 35–38, 40, 48, 68, 74, 77, 98 · **HI** 25–30, 32, 34, 38–41, 43, 61, 69, 70, 75**AV-4 — JOIN / LEAVE / REPLACE MACHINE****PEHCHAN – Question mein ye dikhe**

“teacher’s age included, average increases by 1”, “a student leaves, average drops by 2”, “one man of 72 kg replaced, average +2”, “new student scored how much more than average”, “two students dropped”; शामिल, स्थान पर, छोड़ दिया

$$\text{Aaya} = A + (n + 1) d \quad \text{Gaya} = A - (n - 1) d \quad \text{Badla} = \text{purana} + n d$$

- d = avg mein badlav (badha +, ghata –), A = purana avg, n = purani ginti.
- Kai log aaye/gaye: naya total – purana total.

TERMS – Har letter ka matlab

A = pehle ka avg · n = pehle kitne · d = avg kitna badla.

SOLVED – Asli book questions (verified)**Q (EN 47):** 35 students avg 15; teacher se +1

$$15 + 36 \times 1 = 51 \checkmark \text{ book ans (b)}$$

Q (EN 46): 14 ka avg 20; ek gaya, avg –2

$$20 + 13 \times 2 = 46 \checkmark \text{ book ans (d)}$$

Q (EN 72): 6 ka avg +2; 72 kg wala badla

$$72 + 6 \times 2 = 84 \checkmark \text{ book ans (a)}$$

Q (EN 62): 25 ka avg 50; ek aaya, +0.5

$$50 + 26 \times 0.5 = 63 \checkmark \text{ book ans (d)}$$

Q (EN 43): 7 students; 8th aaya, avg +2 $\Rightarrow$ kitna zyada

$$8 \times 2 = 16 \checkmark \text{ book ans (a)}$$

Q (EN 65): 10 ka avg 16; 2 gaye, 16.25

$$160 - 130 = 30 \checkmark \text{ book ans (b)}$$

Q (HI 50): 4 logon mein 53 kg wala badla, +3

$$53 + 12 = 65 \checkmark \text{ book ans (c)}$$

Q (EN 69): 6 ka avg 16; ek nikaala $\Rightarrow$ 17 (avg +1)

$$16 - 5 \times 1 = 11 \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (32) EN 24, 26, 30, 32, 34, 41, 43, 44, 46, 47, 51, 57, 59, 62, 64, 65, 69, 71, 72, 102 · **HI** 33, 35, 37, 44, 46, 47, 49, 50, 53, 54, 58, 59

AV-5 — OVERLAP / PART-SUM MACHINE

PEHCHAN – Question mein ye dikhe

“mean of 21 is 42; first 11 mean 50, last 11 mean 35 $\Rightarrow$ 11th”, “first 6 and last 6”, “first 22 and last 22 $\Rightarrow$ 23rd”, “average of P, Q, R 58; P & Q 54; Q & R 48 $\Rightarrow$ Q”, 5 subjects first two / last two $\Rightarrow$ third, range; पहली ... अंतिम ...

$$\text{Common (overlap)} = k_1 a_1 + k_2 a_2 - N A \quad \text{Beech ka (gap)} = N A - (k_1 a_1 + k_2 a_2)$$

- Pair averages: $Q = 2(P, Q \text{ avg}) + 2(Q, R \text{ avg}) - 3(P, Q, R \text{ avg})$.
- w, x, y, z: $(x + y + z) - (w + x + y) = z - w = \text{range} = 3(\text{bada avg} - \text{chhota avg})$.

TERMS – Har letter ka matlab

N, A = sab ki ginti, avg $\cdot$ k₁, a₁ = pehla hissa $\cdot$ k₂, a₂ = aakhri hissa.

SOLVED – Asli book questions (verified)

Q (EN 73): 21 ka 42; pehle 11 ka 50, aakhri 11 ka 35 $\Rightarrow$ 11th
 $550 + 385 - 882 = 53$ ✓ **book ans (b)**

Q (EN 75): 11 ka 44; pehle 6 ka 39, aakhri 6 ka 48 $\Rightarrow$ 6th
 $234 + 288 - 484 = 38$ ✓ **book ans (c)**

Q (EN 84): 45 ka 23; pehle 22 ka 18, aakhri 22 ka 21 $\Rightarrow$ 23rd
 $1035 - 858 = 177$ ✓ **book ans (c)**

Q (EN 53): P, Q, R 58; P, Q 54; Q, R 48 $\Rightarrow$ Q
 $108 + 96 - 174 = 30$ ✓ **book ans (c)**

Q (EN 39): 5 subjects 75; pehle 2 ka 65, aakhri 2 ka 85 $\Rightarrow$ teesra
 $375 - 130 - 170 = 75$ ✓ **book ans (c)**

Q (EN 106): chhote 3 ka 22, bade 3 ka 28 $\Rightarrow$ range
 $3 \times 6 = 18$ ✓ **book ans (b)**

BOOK KE SAWAAL (29) EN 10, 39, 53, 55, 66, 73, 75, 78, 82–84, 105–113 $\cdot$ HI 5, 9, 18, 20–24, 42

AV-6 — REQUIRED SCORE / INNINGS MACHINE

PEHCHAN – Question mein ye dikhe

batsman innings, “Shreya 12 tests avg 24, to equal Reena’s 26 in 16 $\Rightarrow$ remaining 4”, “what should he score in 11th innings”, “after 10th innings average increased by 8”; पारी, बनाना चाहिए

$$\text{Chahiye} = \text{Target avg} \times \text{kul ginti} - \text{abhi tak ka total}$$

- Nayi pari x ke baad avg d badha, naya avg B (n pariyan) $\Rightarrow x = B + (n - 1)d$.
- Do nayi pariyan (ek doosri se k zyada): dono ka jod = naya total – purana $\Rightarrow$ baant do.

TERMS – Har letter ka matlab

Target avg = jo chahiye $\cdot$ kul ginti = pariyan/tests total.

SOLVED – Asli book questions (verified)

Q (EN 28): Reena 16×26 ; Shreya $12 \times 24 \Rightarrow$ baaki 4 ka avg

$$\frac{416-288}{4} = 32 \checkmark \text{ book ans (b)}$$

Q (EN 33): 12×25 vs $8 \times 23 \Rightarrow$ baaki 4

$$\frac{300-184}{4} = 29 \checkmark \text{ book ans (b)}$$

Q (HI 62): 10vi pari mein x; avg +8, naya 28

$$28 + 9 \times 8 = 100 \checkmark \text{ book ans (d)}$$

Q (HI 63): $7 \times 28 + 3 \times 32$; 11vi mein kitna $\Rightarrow$ avg 33

$$363 - 292 = 71 \checkmark \text{ book ans (a)}$$

Q (EN 79): 12 pari avg 55, 14 pari avg 60; 14th = 13th + 20 $\Rightarrow$ 13th

$$840 - 660 = 180 \Rightarrow 80 \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (13) EN 28, 33, 42, 79–81 · HI 31, 36, 45, 62, 63, 66, 67

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
AV-1	SERIES AVERAGE (lagataar / AP numbers)	17	7
AV-2	TOTAL MACHINE (Total = Avg $\times$ n)	38	30
AV-3	COMBINED / WEIGHTED AVERAGE (do group)	15	17
AV-4	JOIN / LEAVE / REPLACE MACHINE	20	12
AV-5	OVERLAP / PART-SUM MACHINE	20	9
AV-6	REQUIRED SCORE / INNINGS MACHINE	6	7
Total (book ke sawaal)		116/116 ✓	82/82 ✓

Chapter 18

Speed, Time & Distance (Chaal, Samay aur Doori) चाल, समय और दूरी

7 formulas · English book 149 sawaal · Hindi book 78 sawaal · 64 solved examples ✓

ST-1 — BASIC $D = S \times T$ + UNITS + TUKDE

PEHCHAN – Question mein ye dikhe

find speed / time / distance, km/h $\leftrightarrow$ m/s, “how much in 1 hour / 24 minutes”, round of a square field, diagonal of a lawn, km per litre, speed reduced by 12%, “same distance in 6 hours $\Rightarrow$ speed”, remaining 36 km after 5/12 and 1/3, “remaining distance in remaining time”, speed badhti hai har 2 ghante; शेष दूरी शेष समय, प्रति लीटर

$$D = S \times T \quad \text{km/h} \xrightarrow{\times \frac{5}{18}} \text{m/s} \quad \text{m/s} \xrightarrow{\times \frac{18}{5}} \text{km/h}$$

- **Same doori, naya samay** $\Rightarrow$ naya speed $= \frac{S_1 T_1}{T_2}$.
- **Baaki doori $\div$ baaki samay** = zaroori speed (1/4 doori 1/3 samay mein $\Rightarrow$ 3/4 doori 2/3 samay mein).
- **Hisse** (1/4 + 1/2 + ...) jodo; bacha fraction = diya km $\Rightarrow$ Total = km $\div$ bacha fraction.
- **Kai tukde**: kul samay = Σ (doori $\div$ speed); speeds ek-doesre ke guna hon to sab ek x mein likho.
- **Har 2 ghante speed +k** $\Rightarrow$ tukdon ki doori AP: $\frac{n}{2}$ (pehla + aakhri).
- Field ka chakkar = perimeter $\div$ speed (square 4a, 1 hectare = 10000 m²); diagonal $\sqrt{l^2 + b^2}$.

TERMS – Har letter ka matlab

D = doori · S = speed · T = samay (units same rakho: km-h ya m-s).

SOLVED – Asli book questions (verified)

Q (EN 3): 57.6 km in 48 min $\Rightarrow$ m/s

$$\frac{57.6}{0.8} = 72 \text{ km/h} \times \frac{5}{18} = 20 \text{ ✓ book ans (d)}$$

Q (EN 4): 8 h @75; same doori 6 h mein

$$\frac{75 \times 8}{6} = 100 \text{ ✓ book ans (b)}$$

Q (EN 35): 275 km in 5 h; speed $-5 \Rightarrow$ 250 km kitne mein

$$\frac{250}{55-5} = 5 \text{ h ✓ book ans (a)}$$

Q (EN 47): 25 hectare square field ka chakkar @10 km/h

$$\text{side } 500 \text{ m} \Rightarrow 2 \text{ km} \div 10 = 12 \text{ min ✓ book ans (d)}$$

Q (EN 51): 60 $\times$ 40 lawn ka diagonal @3 km/h

$$\frac{72.1}{5/6} = 86.5 \text{ s ✓ book ans (d)}$$

Q (EN 67): 5/12 train + 1/3 bus + 36 km cycle

$$\text{bacha } \frac{1}{4} = 36 \Rightarrow 144 \text{ ✓ book ans (b)}$$

Q (HI 1): 150 km 12 h mein; 2/3 doori 5/8 samay mein

$$\frac{50}{4.5} = 11.1 \text{ ✓ book ans (d)}$$

Q (HI 38): pehle 2 h 80 km, har 2 h +4 km/h $\Rightarrow$ 10 h

$$80, 88, \dots, 112 \Rightarrow \frac{5}{2}(80 + 112) = 480 \text{ ✓ book ans (a)}$$

Q (EN 26): 120 steamer + 450 train + 60 scooter = 13.5 h; train = 3 $\times$ scooter = 1.5 $\times$ steamer

$$\frac{120 \cdot 1.5 + 450 + 60 \cdot 3}{x} = 13.5 \Rightarrow x = 60 \text{ ✓ book ans (a)}$$

BOOK KE SAWAAL (62) EN 3–6, 11, 26, 27, 33, 35, 38, 47, 51, 55, 59, 60, 64–67, 69, 71, 102 · HI 1–7, 13, 16, 17, 19, 20, 22, 26, 30, 36–39 | Ch 19 (Railgaadi): EN 3–6, 8, 9, 12, 14, 15, 17–20, 138, 149, 153 · HI 1–3, 5, 6

ST-2 — LATE-EARLY / TIME-GAP MACHINE

PEHCHAN – Question mein ye dikhe

“at 30 km/h 5 min late, at 40 km/h 3 min early $\Rightarrow$ distance”, “two men at 6 and 8 km/h, one takes 40 min more”, “4 km/h slower $\Rightarrow$ 2 hours more $\Rightarrow$ speed”, “+10 $\Rightarrow$ 90 min early, -10 $\Rightarrow$ 150 min late”, race: A beats B by 3 s; देरी से, पहले पहुँच

$$D = \frac{S_1 S_2}{S_2 - S_1} \times \Delta t \quad \Delta t = \begin{cases} t_1 + t_2 & \text{ek late, ek early} \\ |t_1 - t_2| & \text{dono late / dono early / seedha farak} \end{cases}$$

Ulta (D diya, speed poochi): $S(S \pm \Delta S) = \frac{D \cdot \Delta S}{\Delta t} \Rightarrow$ aisi do sankhya jinka farak ΔS

- Δt ghante mein (min $\div$ 60).
- Sahi (on-time) samay = $\frac{D}{S_1}$ - late.
- Do alag badlav (+a se t_1 jaldi, -b se t_2 der) $\Rightarrow \frac{S+a}{S-b} = \frac{a t_2}{b t_1}$ se S, phir $D = \frac{S(S+a)t_1}{a}$.

TERMS – Har letter ka matlab

$S_1 < S_2$ do speeds $\cdot \Delta t$ = samay ka farak (ghante) $\cdot \Delta S$ = speed ka farak.

SOLVED – Asli book questions (verified)

Q (EN 77): 30 km/h 5 min late, 40 km/h 3 min early

$$\frac{30-40}{10} \times \frac{8}{60} = 16 \checkmark \text{ book ans (d)}$$

Q (EN 63): 2.5 km/h 6 min late, 3.5 km/h 6 min early

$$\frac{8.75}{1} \times \frac{12}{60} = 1.75 \checkmark \text{ book ans (c)}$$

Q (EN 34): 50 $\Rightarrow$ 10 min late, 60 $\Rightarrow$ 5 min late; sahi samay?

$$D = 300 \times \frac{5}{60} = 25 \Rightarrow 30 - 10 = 20 \text{ min } \checkmark \text{ book ans (c)}$$

Q (EN 72): 6 aur 8 km/h, 40 min farak

$$\frac{48}{2} \times \frac{2}{3} = 16 \checkmark \text{ book ans (b)}$$

Q (EN 85): 45 ki jagah 54 se chale to 12 min ki deri poori

$$\frac{45-54}{9} \times \frac{12}{60} = 54 \checkmark \text{ book ans (b)}$$

Q (EN 14): 96 km; 4 km/h dheere $\Rightarrow$ 2 h zyada

$$S(S-4) = \frac{96 \cdot 4}{2} = 192 = 16 \times 12 \Rightarrow 16 \checkmark \text{ book ans (c)}$$

Q (EN 13): 36 km; +4 km/h se 6 min ki deri poori

$$S(S+4) = \frac{36 \cdot 4}{0.1} = 1440 = 36 \times 40 \Rightarrow 36 \checkmark \text{ book ans (c)}$$

Q (EN 90): +10 $\Rightarrow$ 90 min pehle, -10 $\Rightarrow$ 150 min late

$$\frac{S+10}{S-10} = \frac{2.5}{1.5} \Rightarrow S = 40, D = \frac{40 \cdot 50 \cdot 1.5}{10} = 300 \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (39) EN 13, 14, 16, 22, 24, 34, 58, 63, 72, 74, 77, 78, 80, 82, 85, 87-90, 97-99, 101, 108 **HI** 8, 21, 34, 42, 44-46, 52-54 | Ch 19
(Railgaadi): EN 10, 11, 144, 155 **HI** 79

ST-3 — RATIO MACHINE (speed $\propto$ 1/samay)

PEHCHAN – Question mein ye dikhe

“at 3/4 of usual speed 15 min late $\Rightarrow$ usual time”, “speed +30% $\Rightarrow$ 24 min less”, “speeds 4 : 5, A takes 20 min more”, ratio of speeds 18 : 12 $\Rightarrow$ ratio of time, “half distance in double time”, “12 h; +5 km/h $\Rightarrow$ 9 h”, “1/5 distance at 5/6 of usual speed”, “doubles his speed”, “7.5 km more at 9 km/h instead of 4”, cross at 220 km of 300 km; सामान्य चाल की 2/5, अनुपात

$$\text{Same } D : \frac{S_1}{S_2} = \frac{T_2}{T_1} \quad \text{Same } T : \frac{D_1}{D_2} = \frac{S_1}{S_2}$$

$$\text{Speed} \times \frac{a}{b} \Rightarrow \text{Time} \times \frac{b}{a} \Rightarrow \boxed{\text{Farak} = T \left| \frac{b}{a} - 1 \right|}$$

- Speed ratio $a : b \Rightarrow$ time ratio $b : a$; **1 part = diya farak** (min ya km/h).
- Tukdon mein alag speed-factor k_i (doori hissa p_i): naya samay = $T \sum \frac{p_i}{k_i}$.
- Same samay mein zyada doori: $T = \frac{\text{extra doori}}{\text{speed ka farak}}$.
- Saath chale, beech mein mile: doori ratio = speed ratio.

TERMS – Har letter ka matlab

T = usual samay · a/b = speed kitne guna hui · k = speed-factor.

SOLVED – Asli book questions (verified)

Q (EN 42): $3/4$ speed pe 15 min late $\Rightarrow$ usual

$$T \left(\frac{4}{3} - 1 \right) = 15 \Rightarrow 45 \checkmark \text{ book ans (c)}$$

Q (HI 14): $2/5$ speed pe 30 min late

$$T \left(\frac{5}{2} - 1 \right) = 30 \Rightarrow 20 \checkmark \text{ book ans (a)}$$

Q (EN 28): speed +30% $\Rightarrow$ 24 min kam

$$T \left(1 - \frac{10}{13} \right) = 24 \Rightarrow 104 \text{ min} = 1 \text{ h } 44 \text{ min} \checkmark \text{ book ans (d)}$$

Q (EN 30): speeds $4 : 5$, A 20 min zyada; A double speed pe?

$$\text{times } 5 : 4, 1 \text{ part} = 20 \Rightarrow A = 100 \Rightarrow 50 \text{ min} \checkmark \text{ book ans (c)}$$

Q (EN 104): 12 h; +5 km/h $\Rightarrow$ 9 h

$$\text{speeds } 3 : 4, 1 \text{ part} = 5 \Rightarrow 15 \times 12 = 180 \checkmark \text{ book ans (d)}$$

Q (EN 84): speed $1/10$ kam $\Rightarrow$ 18 min zyada; usual 50 km/h

$$\text{time} + 1/9 \Rightarrow T = 162 \text{ min} \Rightarrow 50 \times 2.7 = 135 \checkmark \text{ book ans (a)}$$

Q (EN 95): 4 ki jagah 9 km/h $\Rightarrow$ 7.5 km zyada

$$T = \frac{7.5}{5} = 1.5 \Rightarrow 4 \times 1.5 = 6 \checkmark \text{ book ans (d)}$$

Q (EN 39): 7 am–8:30; $1/5$ doori $5/6$ speed, baaki $6/5$ speed

$$90 \left(\frac{1}{5} \cdot \frac{6}{5} + \frac{4}{5} \cdot \frac{5}{6} \right) = 81.6 \text{ min} \Rightarrow 8:21 \checkmark \text{ book ans (c)}$$

Q (EN 145): speeds $18 : 12 \Rightarrow$ time ratio

$$12 : 18 = 2 : 3 \checkmark \text{ book ans (c)}$$

Q (EN 149): 300 km, 220 km par cross

$$220 : 80 = 11 : 4 \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (29) EN 1, 2, 8, 28, 30, 39, 42, 57, 84, 95, 104, 144–146, 148, 149 · **HI** 14, 18, 50 | Ch 19 (Railgaadi): EN 2, 7, 13, 152, 154, 156, 157 · **HI** 80 | Ch 28 (Vividh): EN 37, 46

ST-4 — AVERAGE SPEED (Kul doori $\div$ Kul samay)

PEHCHAN – Question mein ye dikhe

average speed for the whole journey, goes at x returns at y , equilateral triangle ke 3 sides, $2/5$ distance at 50 rest at 60, “excluding stoppages 45, including 27”, “average 60 chahiye $\Rightarrow$ baaki ki speed”, aana-jaana 5 h $\Rightarrow$ doori, same distance at 3, 4, 8 km/h; औसत चाल, वापस लौटा, आधी दूरी

$$\text{Avg} = \frac{\text{Kul doori}}{\text{Kul samay}}$$

Case	Average speed
Barabar doori, 2 speeds (aana-jaana)	$\frac{2xy}{x+y}$
Barabar doori, 3 speeds	$\frac{3xyz}{xy+yz+zx}$
Doori ke hisse p, q, r	$\frac{1}{\frac{p}{x} + \frac{q}{y} + \frac{r}{z}}$
Barabar samay	$\frac{x+y}{2}$ (warna $\frac{t_1x+t_2y}{t_1+t_2}$)

- **Ulta:** avg A, jaate $x \Rightarrow$ lautte $y = \frac{Ax}{2x-A}$; baaki hisse ki speed = $(A \times \text{kul samay} - \text{ho chuki doori}) \div \text{bacha samay}$.
- **Doori poochi:** aana-jaana total T $\Rightarrow$ ek taraf $D = \frac{Txy}{x+y}$; hisson wali journey $\Rightarrow D = T \times \text{avg}$.
- **Stoppage:** har ghante ruke = $\frac{\text{bina ruke-ruk ke}}{\text{bina ruke}} \times 60 \text{ min}$.

TERMS – Har letter ka matlab

x, y, z = alag hisson ki speed · p, q, r = doori ke fraction · A = average.

SOLVED – Asli book questions (verified)

Q (EN 112): 45 jaana, 36 aana

$$\frac{2 \cdot 45 \cdot 36}{81} = 40 \checkmark \text{ book ans (c)}$$

Q (EN 124): triangle ke sides 300, 200, 240

$$\frac{3 \cdot 300 \cdot 200 \cdot 240}{180000} = 240 \checkmark \text{ book ans (a)}$$

Q (EN 116): 2/5 @50, baaki @60

$$\frac{1}{0.008+0.01} = 55\frac{5}{9} \checkmark \text{ book ans (d)}$$

Q (EN 140): 1 h @50 + 2 h @70

$$\frac{50+140}{3} = 63.33 \checkmark \text{ book ans (b)}$$

Q (EN 32): bina stoppage 45, stoppage sahit 27

$$\frac{18}{45} \times 60 = 24 \text{ min} \checkmark \text{ book ans (a)}$$

Q (HI 57): avg 15, jaate 20 $\Rightarrow$ lautte

$$\frac{15 \cdot 20}{40-15} = 12 \checkmark \text{ book ans (d)}$$

Q (EN 111): 800 km, pehle 300 @50, avg 60 chahiye

$$\frac{500}{\frac{40}{3}-6} = \frac{750}{11} \checkmark \text{ book ans (d)}$$

Q (EN 94): 60 jaana, 40 aana, kul 5 h $\Rightarrow$ ek taraf

$$\frac{5 \cdot 60 \cdot 40}{100} = 120 \checkmark \text{ book ans (a)}$$

Q (EN 70): 14 h; 1/3 @40, baaki @60 $\Rightarrow$ doori

$$\frac{14}{\frac{1}{120} + \frac{1}{90}} = 720 \checkmark \text{ book ans (a)}$$

Q (EN 105): barabar doori @3, 4, 8; kul 42.5 min

$$d \left(\frac{1}{3} + \frac{1}{4} + \frac{1}{8} \right) = \frac{17}{24} \Rightarrow d = 1, \text{ kul } 3 \text{ km} \checkmark \text{ book ans (d)}$$

Q (EN 117): 160 km, 3¼ h mein, 10 + 5 min ruki

$$\frac{160}{3} = 53\frac{1}{3} \checkmark \text{ book ans (b)}$$

BOOK KESAWAAL (93) EN 31, 32, 68, 70, 75, 83, 94, 96, 100, 103, 105, 106, 110–143 · **HI** 29, 31–33, 35, 41, 49, 51, 55–78 | Ch 19 (Railgaadi): **EN** 16, 135–137, 139, 140, 142, 143, 145, 148 · **HI** 4, 7, 74, 78 | Ch 28 (Vividh): **EN** 40

ST-5 — RELATIVE SPEED (milna / peecha)

PEHCHAN – Question mein ye dikhe

start towards each other $\Rightarrow$ meet after, thief–police chase, “30 min later police starts”, circular track opposite direction, “reaches Kalka and returns, meets 20 km from Kalka”, buses at interval of 15 min – woman from opposite side meets every 10 min, fog mein 100 m tak dikhi 4 min, Train A 5 am–9 am & B 6:30–10 am; पीछा, चोर, विपरीत दिशा, एक दूसरे की ओर

$$\text{Milne ka samay} = \frac{\text{Gap}}{S_1 + S_2} \text{ (aamne-saamne)} \quad \text{Pakadne ka samay} = \frac{\text{Gap}}{S_1 - S_2} \text{ (peeche)}$$

- **Head start:** gap = pehle wale ki speed $\times$ deri (+ pehle se doori).
- **Jaa kar laut ke mile** (doori D, milan x km pehle): tez ne D + x, dheema D - x $\Rightarrow$ speed ratio (D+x) : (D-x); dono ne milkar 2D.
- **Buses har T min** speed v; saamne se aata aadmi har t min mile $\Rightarrow$ uski speed = $\frac{v(T-t)}{t}$.
- **Kohra:** relative speed = dikhne ki doori $\div$ dikhne ka samay.
- **Clock-time trains:** doori = 1 maano, speed = 1 $\div$ (safar ke ghante).
- Lambi cheez (train/bus) kisi ko cross kare $\Rightarrow$ Train chapter (TR).

TERMS – Har letter ka matlab

Gap = beech ki doori $\cdot$ S₁, S₂ = dono ki speed.

SOLVED – Asli book questions (verified)

Q (EN 29): 240 km, speeds 5 : 7, B = 84

$$\frac{240}{60+84} \text{ h} = 100 \text{ min} \quad \checkmark \text{ book ans (b)}$$

Q (EN 36): A 1:00 @40, B 1:40 @50

$$\text{gap } \frac{80}{3} \Rightarrow \frac{80/3}{10} \text{ h} = 160 \text{ min} \quad \checkmark \text{ book ans (d)}$$

Q (EN 81): chor @60, police 30 min baad @80 $\Rightarrow$ police ki doori

$$\frac{30}{20} = 1.5 \text{ h} \times 80 = 120 \quad \checkmark \text{ book ans (d)}$$

Q (EN 107): 400 m aage; chor 10, police 15 $\Rightarrow$ chor kitna bhaga

$$\frac{0.4}{5} \text{ h} \times 10 = 0.8 \text{ km} \quad \checkmark \text{ book ans (b)}$$

Q (EN 37): 2 km circular, 9 aur 11 ulta disha

$$\frac{2}{20} \text{ h} = 6 \text{ min} \quad \checkmark \text{ book ans (d)}$$

Q (EN 10): 95 km; Rishi 5 dheema; Kanpur se 25 km par mile

$$120 : 70 = 12 : 7, \text{ farak } 5 \Rightarrow 7 \quad \checkmark \text{ book ans (c)}$$

Q (EN 92): 21 km; 3 aur 4 km/h; laut ke mile

$$2D = 42 \div 7 = 6 \text{ h} \Rightarrow 3 \times 6 = 18 \quad \checkmark \text{ book ans (d)}$$

Q (EN 12): buses 25 km/h har 15 min; aurat har 10 min

$$\frac{25(15-10)}{10} = 12.5 \quad \checkmark \text{ book ans (c)}$$

Q (EN 17): aadmi 3 km/h; gaadi 100 m tak 4 min dikhi

$$3 + \frac{0.1}{4/60} = \frac{9}{2} \quad \checkmark \text{ book ans (a)}$$

Q (EN 43): chor 100; police 75 (1 h), 30 min repair, phir 120

$$\text{gap } 25 + 50 = 75 \Rightarrow 75/20 = 3:45 \Rightarrow \text{kul } 5:15 \quad \checkmark \text{ book ans (c)}$$

Q (EN 52): A 5–9 am, B 6:30–10 am

$$6:30 \text{ tak A ne } 3/8; \frac{5/8}{1/4+2/7} = \frac{7}{6} \text{ h} \Rightarrow 7:40 \quad \checkmark \text{ book ans (a)}$$

BOOK KESAWAAL (21) EN 7, 9, 10, 12, 17, 20, 21, 23, 29, 36, 37, 43, 52, 61, 79, 81, 92, 107 **HI** 10, 40, 48

ST-6 — MEETING KE BAAD ($\checkmark$ rule)

PEHCHAN – Question mein ye dikhe

“they meet and after meeting take 9 h and 16 h to reach”, “Parvej takes 12 h after meeting, Gautam 3 h”, ratio of speeds / speed of other / time to meet / total journey time; मिलने के बाद

$$\frac{S_1}{S_2} = \sqrt{\frac{t_2}{t_1}}$$

$$\text{Milne tak ka samay} = \sqrt{t_1 t_2}$$

- t_1 = milne ke BAAD pehle wale ka samay, t_2 = doosre ka.
- Kisi ka poora safar = $\sqrt{t_1 t_2}$ + uska baad wala samay.

TERMS – Har letter ka matlab

t_1, t_2 = milne ke baad bacha samay · S_1, S_2 = speeds.

SOLVED – Asli book questions (verified)

Q (EN 147): baad mein 9 h aur 16 h

$$\sqrt{16} : \sqrt{9} = 4 : 3 \quad \checkmark \text{ book ans (b)}$$

Q (EN 19): Parvej 60, baad mein 12 h; Gautam 3 h

$$60\sqrt{\frac{12}{3}} = 120 \quad \checkmark \text{ book ans (a)}$$

Q (EN 25): Peter 30, 13.5 h; Paul 6 h

$$30\sqrt{\frac{13.5}{6}} = 45 \quad \checkmark \text{ book ans (c)}$$

Q (EN 44): baad mein 54 aur 24 min; Q wale ka poora safar

$$\sqrt{54 \cdot 24} = 36 \Rightarrow 36 + 24 = 60 \quad \checkmark \text{ book ans (d)}$$

Q (HI 15): $B = 1.5$ A; A baad mein 5.4 h $\Rightarrow$ P

$$5.4 \times \left(\frac{2}{3}\right)^2 = 2.4 \quad \checkmark \text{ book ans (d)}$$

Q (EN 18): A 55; baad mein 5 h aur 6 h $\Rightarrow$ B

$$55\sqrt{\frac{5}{6}} = \frac{55\sqrt{30}}{6} \quad \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (8) EN 18, 19, 25, 44, 147 · HI 11, 12, 15

ST-7 — DO SAWARI / DO SPEED KA BATWARA**PEHCHAN – Question mein ye dikhe**

“64 km in 7 h, partly on foot @7, partly by bicycle @12”, “some time at 75 and rest at 55”, walk one way & ride back 3 h 45, walk both ways 4 h 20, “10 km by car rest by train; exact opposite 0.5 h more”, partly by air partly by train, 16 km tanga swap; पैदल और सवारी, कुछ समय के लिए, संयोजन

$$\text{Tez sawari par samay} = \frac{D - S_{dheema} \times T}{S_{tez} - S_{dheema}} \quad (\text{alligation})$$

$$\text{Walk+Ride} = T_1, \text{ Walk+Walk} = T_2 \Rightarrow \text{Ride+Ride} = 2T_1 - T_2$$

$$\text{Swap (a km} \leftrightarrow \text{b km): } \Delta t = |a - b| \cdot \left| \frac{1}{x} - \frac{1}{y} \right|$$

- Doori poochi $\Rightarrow$ us sawari ka samay $\times$ uski speed.
- Air/train: “sab air se $\Rightarrow$ train-samay ka k hissa bachta, h ghante pehle” $\Rightarrow$ train samay = h/k.

TERMS – Har letter ka matlab

D = kul doori · T = kul samay · x, y = do speeds · a, b = swap hone wale tukde.

SOLVED – Asli book questions (verified)**Q (EN 40):** 64 km, 7 h; paidal 7, cycle 12

$$\frac{64-49}{5} = 3 \text{ h} \checkmark \text{ book ans (b)}$$

Q (EN 48): 300 km, 5 h; 75 aur 55

$$\frac{300-275}{20} = 1.25 \text{ h} = 1 \text{ h } 15 \text{ min} \checkmark \text{ book ans (c)}$$

Q (EN 76): 61 km, 9 h; paidal 4, cycle 9 $\Rightarrow$ paidal doori

$$\text{cycle } \frac{61-36}{5} = 5 \text{ h} \Rightarrow \text{paidal } 4 \text{ h} \times 4 = 16 \checkmark \text{ book ans (b)}$$

Q (EN 62): 220 km, 4.5 h; train 60, bus 40 $\Rightarrow$ bus samay

$$\text{train } \frac{220-180}{20} = 2 \Rightarrow \text{bus } 2.5 \text{ h} \checkmark \text{ book ans (d)}$$

Q (EN 45): walk + cycle 3:45, walk + walk 4:20

$$7:30 - 4:20 = 3:10 \checkmark \text{ book ans (a)}$$

Q (EN 50): walk + ride 10 h; ride + ride 5 h $\Rightarrow$ walk + walk

$$2 \times 10 - 5 = 15 \checkmark \text{ book ans (a)}$$

Q (EN 91): 10 km car baaki train; ulta 0.5 h zyada; train 50, car 40

$$(D - 20) \left(\frac{1}{40} - \frac{1}{50} \right) = 0.5 \Rightarrow 120 \checkmark \text{ book ans (c)}$$

Q (HI 9): 40 km: 16 paidal @4 + tanga; ulta 1 h zyada

$$8 \left(\frac{1}{4} - \frac{1}{v} \right) = 1 \Rightarrow v = 8 \checkmark \text{ book ans (b)}$$

Q (EN 86): 480 km, 4 h; sab air $\Rightarrow$ train ka 4/5 bache, 2 h pehle

$$\text{train } 2.5 \text{ h, air } 1.5 \text{ h; air speed } 240 \Rightarrow \text{train doori } 120 \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (23) EN 15, 40, 45, 46, 48–50, 53, 54, 56, 62, 73, 76, 86, 91 · **HI** 9, 23–25, 27, 28, 43, 47**COVERAGE CHECK – Har sawaal ka formula**

Formula	Naam	EN Qs	HI Qs
ST-1	BASIC D = S $\times$ T + UNITS + TUKDE	22	19
ST-2	LATE–EARLY / TIME-GAP MACHINE	24	10
ST-3	RATIO MACHINE (speed $\propto$ 1/samay)	16	3
ST-4	AVERAGE SPEED (Kul doori $\div$ Kul samay)	46	32
ST-5	RELATIVE SPEED (milna / peecha)	18	3
ST-6	MEETING KE BAAD ($\sqrt{}$ rule)	5	3
ST-7	DO SAWARI / DO SPEED KA BATWARA	15	8
$\rightarrow$ TR-2	PLATFORM / BRIDGE / TUNNEL (Ch 19)	2	0
$\rightarrow$ TR-5	TRAIN AUR CHALTA AADMI / GAADI (Ch 19)	1	0
Total (book ke sawaal)		149/149 ✓	78/78 ✓

 $\rightarrow$ **TR-2:** EN 41, 109 · $\rightarrow$ **TR-5:** EN 93

Chapter 19

Train (Railgaadi) रेलगाड़ी

6 formulas · English book 159 sawaal · Hindi book 80 sawaal · 48 solved examples ✓

TR-1 — POLE / KHAMBHA / KHADA AADMI

PEHCHAN – Question mein ye dikhe

crosses a pole / post / signal / telegraph post / tree / man standing, “length of the train”, “time to cross a pole”, poles 50 m apart
ginna, 21 posts in 1 min; खंभे, सिग्नल पोस्ट

$$\text{Samay} = \frac{L}{S} \quad L = S \times t \quad (S \text{ m/s mein: km/h} \times \frac{5}{18})$$

- Khada aadmi / pole / signal = **bina lambai** ki cheez $\Rightarrow$ sirf train ki lambai L paar karni.
- Khambhe ginna**: n khambhe $\Rightarrow (n - 1)$ gap; shuru ka khambha bhi gina jaye to $+1$.
- Do trains ke L aur t diye $\Rightarrow$ speed ratio = $\frac{L_1}{t_1} : \frac{L_2}{t_2}$.

TERMS – Har letter ka matlab

L = train ki lambai (m) · S = speed (m/s) · t = samay (s).

SOLVED – Asli book questions (verified)

Q (EN 21): 72 km/h, pole 12 s $\Rightarrow$ lambai

$$20 \times 12 = 240 \text{ m} \quad \checkmark \text{ book ans (d)}$$

Q (EN 25): 300 m, pole 5 s $\Rightarrow$ speed

$$60 \text{ m/s} \times \frac{18}{5} = 216 \quad \checkmark \text{ book ans (c)}$$

Q (EN 26): 180 m @90 km/h $\Rightarrow$ post

$$\frac{180}{25} = 7.2 \text{ s} \quad \checkmark \text{ book ans (d)}$$

Q (EN 146): 1 min mein 21 posts, 50 m door

$$20 \text{ gap} \times 50 = 1 \text{ km/min} = 60 \text{ km/h} \quad \checkmark \text{ book ans (a)}$$

Q (EN 147): 45 km/h, 4 h, poles 50 m door, pehla bhi gino

$$\frac{180000}{50} + 1 = 3601 \quad \checkmark \text{ book ans (d)}$$

Q (HI 76): M pole 30 s, N 105 s; M ki lambai N ki aadhi

$$\frac{1/2}{30} : \frac{1}{105} = 7 : 4 \quad \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (21) EN 21–30, 33–36, 93, 146, 147 · HI 8, 9, 11, 76

TR-2 — PLATFORM / BRIDGE / TUNNEL

PEHCHAN – Question mein ye dikhe

crosses a platform / bridge / tunnel / canal of length P , “length of the platform / train”, “pole to the end of the train, distance of pole from front”, man walks across a bridge; प्लेटफॉर्म, पुल, सुरंग

$$\text{Samay} = \frac{L + P}{S} \quad P = S t - L \quad L = S t - P$$

- Lambi cheez (platform, pul, surang) ki lambai **jodo**; aadmi/car jaisi chhoti cheez ki lambai 0.
- “Pole se pehle d doori, aakhri sire ke paar hone tak” $\Rightarrow d + L = S t$.

TERMS – Har letter ka matlab

P = platform / pul / surang ki lambai · L = train ki lambai.

SOLVED – Asli book questions (verified)**Q (EN 1):** 725 m train, 235 m surang, 48 s

$$\frac{960}{48} = 20 \text{ m/s} = 72 \text{ km/h} \checkmark \text{ book ans (b)}$$

Q (EN 52): 110 m @90 km/h, 180 m pul

$$\frac{290}{25} = 11.6 \text{ s} \checkmark \text{ book ans (b)}$$

Q (EN 60): 72 km/h, 250 m platform, 26 s $\Rightarrow$ train

$$20 \times 26 - 250 = 270 \checkmark \text{ book ans (b)}$$

Q (EN 63): 162 m @54 km/h, 44 s $\Rightarrow$ platform

$$15 \times 44 - 162 = 498 \checkmark \text{ book ans (d)}$$

Q (EN 32): 110 m @36 km/h, 53 s $\Rightarrow$ pole ki doori

$$10 \times 53 - 110 = 420 \checkmark \text{ book ans (b)}$$

Q (EN 48): aadmi 10 km/h, pul $7\frac{1}{2}$ min mein

$$\frac{10000}{60} \times 7.5 = 1250 \text{ m} \checkmark \text{ book ans (b)}$$

Q (HI 13): 150 m @25 km/h, 100 m surang

$$\frac{250}{125/18} = 36 \text{ s} \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (43) EN 1, 32, 48, 50, 52–55, 59–66, 68, 70–74, 76, 78–80, 82 · **HI** 10, 13–19, 21–26 | Ch 18 (Chaal, Samay aur Doori): **EN** 41, 109**TR-3 — EK TRAIN, DO CROSSING (pole + platform / 2 platform)****PEHCHAN – Question mein ye dikhe**

“crosses a pole in 8 s and a platform in 32 s”, “platform 200 m in 40 s and a man in 20 s”, “130 m platform in 14.5 s and 245 m in 20.25 s”, “crosses 100 m platform in 60 s $\Rightarrow$ pole time”, bridge 30 s $\Rightarrow$ platform ke beech khada aadmi; खंभे और प्लेटफॉर्म

$$S = \frac{P_2 - P_1}{t_2 - t_1} \quad (\text{pole / khada aadmi ho to } P = 0) \quad L = S t_1 - P_1$$

$$\text{Pole } t_0 + \text{platform } t : \quad L = \frac{P t_0}{t - t_0} \quad P = S(t - t_0)$$

- Pehle speed nikalo (lambai ka farak $\div$ samay ka farak), phir jo poocha.
- Aadmi chal raha ho to relative speed lo (TR-5).

TERMS – Har letter ka matlab

P_1, P_2 = do platform / pul (pole = 0) · t_1, t_2 = unke samay.

SOLVED – Asli book questions (verified)**Q (EN 57):** 69 km/h; pole 8 s, platform 32 s $\Rightarrow$ platform

$$\frac{69 \cdot 5}{18} \times 24 = 460 \checkmark \text{ book ans (a)}$$

Q (EN 49): 200 m platform 40 s, aadmi 20 s $\Rightarrow$ train

$$\frac{200 \cdot 20}{20} = 200 \checkmark \text{ book ans (c)}$$

Q (EN 67): 130 m 14.5 s, 245 m 20.25 s $\Rightarrow$ speed

$$\frac{115}{5.75} = 20 \text{ m/s} = 72 \checkmark \text{ book ans (d)}$$

Q (EN 51): 50 m 15 s, 20 m 10 s $\Rightarrow$ train

$$S = \frac{30}{5} = 6 \Rightarrow 90 - 50 = 40 \checkmark \text{ book ans (a)}$$

Q (EN 31): 45 km/h, 100 m platform 60 s $\Rightarrow$ pole

$$L = 750 - 100 = 650 \Rightarrow \frac{650}{12.5} = 52 \text{ s} \checkmark \text{ book ans (d)}$$

Q (EN 90): 200 m train, 300 m pul 30 s $\Rightarrow$ beech khada aadmi

$$S = \frac{50}{3} \Rightarrow \frac{200}{50/3} = 12 \text{ s} \checkmark \text{ book ans (c)}$$

Q (HI 31): 1950 m train pole 150 s $\Rightarrow$ 338 m platform

$$\frac{2288}{13} = 176 \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (32) EN 31, 49, 51, 56–58, 67, 69, 75, 77, 81, 83–88, 90–92, 141 · **HI** 20, 27–31, 33–36, 75

TR-4 — DO TRAINS (aamne-saamne / ek disha)

PEHCHAN – Question mein ye dikhe

two trains of lengths L_1, L_2 at S_1, S_2 cross each other (opposite / same direction), “length of the second train”, equal speed / equal length, “man sitting in the slow train”, combined speed, same direction 30 s & opposite 6 s $\Rightarrow$ speeds; दो रेलगाड़ियाँ, समानांतर पटरियों, सापेक्ष चाल

$$t = \frac{L_1 + L_2}{S_1 + S_2} \text{ (ulti disha)}$$

$$t = \frac{L_1 + L_2}{S_1 - S_2} \text{ (ek disha)}$$

- **Doosri train mein baitha aadmi** $\Rightarrow$ sirf crossing wali train ki lambai: $L = S_{rel} \times t$.
- Barabar speed, ulti disha $\Rightarrow 2S = \frac{L_1+L_2}{t}$; barabar lambai $\Rightarrow 2L = S_{rel} \times t$.
- Ek disha t_1 , ulti disha $t_2 \Rightarrow \frac{S_1}{S_2} = \frac{t_1+t_2}{t_1-t_2}$.
- Combined speed $= \frac{L_1+L_2}{t}$ (km/h ke liye $\times 18/5$).

TERMS – Har letter ka matlab

L_1, L_2 = dono trains · S_1, S_2 = speeds · S_{rel} = jodo (ulti) / ghatao (ek disha).

SOLVED – Asli book questions (verified)

Q (EN 104): 80 m + 120 m, 30 & 20 m/s ulti

$$\frac{200}{50} = 4 \text{ s } \checkmark \text{ book ans (b)}$$

Q (EN 94): 200 m + 300 m, 120 & 100 km/h ek disha

$$\frac{500}{50/9} = 90 \text{ s} = 1.5 \text{ min } \checkmark \text{ book ans (d)}$$

Q (EN 107): 250 m @100, doosri @62 ulti, 10 s

$$45 \times 10 - 250 = 200 \checkmark \text{ book ans (c)}$$

Q (EN 108): barabar speed, 120 + 140 m, 10 s

$$2S = 26 \Rightarrow 13 \text{ m/s } \checkmark \text{ book ans (a)}$$

Q (HI 42): barabar lambai, 56 & 46 m/s ek disha, 26 s

$$2L = 10 \times 26 \Rightarrow 130 \checkmark \text{ book ans (c)}$$

Q (EN 102): 110 & 90 km/h ek disha; dheemi train ke aadmi ko 18 s

$$\frac{20 \cdot 5}{18} \times 18 = 100 \text{ m } \checkmark \text{ book ans (c)}$$

Q (EN 127): 153 + 127 m, 7.2 s $\Rightarrow$ combined speed

$$\frac{280}{7.2} \times \frac{18}{5} = 140 \checkmark \text{ book ans (b)}$$

Q (EN 118): dono 120 m; pole 10 s aur 15 s $\Rightarrow$ ulti crossing

$$\frac{240}{12+8} = 12 \text{ s } \checkmark \text{ book ans (a)}$$

Q (EN 150): 200 + 400 m; ek disha 30 s, ulti 6 s

$$S_1 - S_2 = 20, S_1 + S_2 = 100 \text{ m/s } \Rightarrow 216 \text{ \& } 144 \checkmark \text{ book ans (a)}$$

Q (EN 158): P, Q ko 60 s; P ka yatri 40 s

$$L_Q = 40x, L_P = 20x \Rightarrow 1 : 2 \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (51) EN 47, 94, 95, 98, 101, 102, 104, 105, 107–109, 114–116, 118–121, 123–127, 129, 130, 134, 150, 158 · **HI** 37, 39, 41–47, 50, 51, 53, 56–58, 60, 62–64, 67–70

TR-5 — TRAIN AUR CHALTA AADMI / GAADI

PEHCHAN – Question mein ye dikhe

train crosses a man walking / running / cycling at s km/h (same or opposite direction), overtakes two persons at 2 and 4 km/h in 9 and 10 s, rider: opposite 2 s, same 5 s, bus overtakes a truck 150 m ahead, fog mein gaadi dikhi; चल रहे व्यक्ति, दौड़ते व्यक्ति

$$t = \frac{L}{S \mp s} \quad (\text{ek disha } -, \text{ ulthi disha } +)$$

$$\text{Do log, ek disha: } L = \frac{(s_2 - s_1)t_1 t_2}{t_2 - t_1} \quad S = \frac{s_2 t_2 - s_1 t_1}{t_2 - t_1}$$

- Do log ulthi disha: $S = \frac{s_2 t_2 - s_1 t_1}{t_1 - t_2}$.
- Rider: ulthi disha t_1 , ek disha $t_2 \Rightarrow s = \frac{S(t_2 - t_1)}{t_1 + t_2}$.
- Pole t_0 + ulthi disha ka aadmi $t \Rightarrow S t_0 = (S + s) t$.

TERMS – Har letter ka matlab

s = aadmi ki speed · S = train ki speed · L = train ki lambai.

SOLVED – Asli book questions (verified)

Q (EN 43): 110 m @60, aadmi 6 ulthi

$$\frac{110}{66.5/18} = 6 \text{ s} \quad \checkmark \text{ book ans (b)}$$

Q (EN 38): 210 m, 12 s, aadmi 4.5 ulthi $\Rightarrow$ train

$$63 - 4.5 = 58.5 \quad \checkmark \text{ book ans (a)}$$

Q (EN 44): 125 m, 10 s, aadmi 5 ek disha

$$45 + 5 = 50 \quad \checkmark \text{ book ans (a)}$$

Q (EN 37): 2 & 4 km/h log, 9 & 10 s $\Rightarrow$ lambai

$$\frac{2 \cdot \frac{5}{18} \cdot 90}{1} = 50 \text{ m} \quad \checkmark \text{ book ans (d)}$$

Q (HI 38): 15 & 35 m/s log, 20 & 40 s

$$\frac{20 \cdot 800}{20} = 800 \text{ m} \quad \checkmark \text{ book ans (a)}$$

Q (EN 46): 4.5 & 6.3 km/h, 6 & 6.4 s $\Rightarrow$ speed

$$\frac{40.32 - 27}{0.4} = 33.3 \quad \checkmark \text{ book ans (b)}$$

Q (EN 40): ulthi disha 4 & 10 m/s, 10 & 8 s

$$\frac{80 - 40}{2} = 20 \text{ m/s} \quad \checkmark \text{ book ans (b)}$$

Q (EN 151): 70 km/h; rider ulthi 2 s, ek disha 5 s

$$\frac{70 \cdot 3}{7} = 30 \quad \checkmark \text{ book ans (a)}$$

Q (EN 159): pole 20 s; cyclist 5 ulthi, 18 s

$$20S = 18(S + 5) \Rightarrow 45 \quad \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (21) EN 37–46, 89, 103, 131, 151, 159 · **HI** 12, 32, 38, 71, 77 | Ch 18 (Chaal, Samay aur Doori): **EN** 93

TR-6 — TRAINS KA MILNA / PEECHA (bina lambai)

PEHCHAN – Question mein ye dikhe

two trains start towards each other, “one has travelled 30 km more than the other $\Rightarrow$ distance”, meet at 110 km from one station $\Rightarrow$ ratio, second train leaves 2 h later & catches in 4 h, clock-time departures (5 am–9 am), “126 km apart after 2 hours”; अधिक दूरी तय, प्रस्थान, बजे

$$\text{Milna} = \frac{\text{Gap}}{S_1 + S_2} \quad \text{Peecha} = \frac{\text{Gap}}{S_1 - S_2}$$

$$x \text{ km zyada chali} \Rightarrow D = \frac{x(S_1 + S_2)}{S_1 - S_2} \quad h \text{ ghante baad chali, } t \text{ mein pakda} \Rightarrow S_2 = \frac{S_1(h + t)}{t}$$

- Milan point ek station se x km $\Rightarrow$ speed ratio $x : (D - x)$.
- Clock-time trains: doori = 1, speed = $1 \div (\text{safar ke ghante})$; der se chali to pehle gap ghatao.
- t ghante baad beech ki doori $d \Rightarrow S_1 + S_2 = \frac{D-d}{t}$.
- Milne tak doori ka farak = $(S_1 - S_2) \times \text{samay}$.

TERMS – Har letter ka matlab

$D = \text{stations ki doori} \cdot x = \text{kitna zyada chali} \cdot h = \text{head start (ghante)}.$

SOLVED – Asli book questions (verified)

Q (EN 106): 30 & 35 km/h, ek ne 30 km zyada

$$\frac{30-65}{5} = 390 \checkmark \text{ book ans (c)}$$

Q (EN 111): 85 & 105, 200 km zyada

$$\frac{200-190}{20} = 1900 \checkmark \text{ book ans (d)}$$

Q (EN 112): 200 km, 110 km par mile

$$110 : 90 = 11 : 9 \checkmark \text{ book ans (a)}$$

Q (EN 99): 40 km/h; 2 h baad doosri, 4 h mein pakda

$$\frac{40-6}{4} = 60 \checkmark \text{ book ans (a)}$$

Q (EN 96): 14:30 @75, 16:30 @90 $\Rightarrow$ kahan milengi

$$\text{gap } 150 \div 15 = 10 \text{ h} \times 90 = 900 \checkmark \text{ book ans (d)}$$

Q (EN 122): 80 & 95 km/h, 12 h baad mile $\Rightarrow$ farak

$$15 \times 12 = 180 \checkmark \text{ book ans (d)}$$

Q (EN 133): 8 am @40; 1 h baad B se @50; 220 km

$$9 \text{ am tak gap } 180 \div 90 = 2 \text{ h} \Rightarrow 11 \text{ am} \checkmark \text{ book ans (b)}$$

Q (EN 128): 380 km; 2 h baad 126 km door; ek 7 tez

$$S_1 + S_2 = 127 \Rightarrow 67, 60 \checkmark \text{ book ans (d)}$$

Q (HI 54): A 6 am–2 pm, B 8 am–3 pm

$$8 \text{ am tak A ne } \frac{1}{4}; \frac{3/4}{1/8+1/7} = 2.8 \text{ h} \Rightarrow 10:48 \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (26) EN 96, 97, 99, 100, 106, 110–113, 117, 122, 128, 132, 133 · **HI** 40, 48, 49, 52, 54, 55, 59, 61, 65, 66, 72, 73

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
TR-1	POLE / KHAMBHA / KHADA AADMI		17 4
TR-2	PLATFORM / BRIDGE / TUNNEL		27 14
TR-3	EK TRAIN, DO CROSSING (pole + platform / 2 platform)		21 11
TR-4	DO TRAINS (aamne-saamne / ek disha)		28 23
TR-5	TRAIN AUR CHALTA AADMI / GAADI		15 5
TR-6	TRAINS KA MILNA / PEECHA (bina lambai)		14 12
$\rightarrow ST-1$	$BASIC D = S \times T + UNITS + TUKDE$ (Ch 18)		16 5
$\rightarrow ST-2$	$LATE-EARLY / TIME-GAP MACHINE$ (Ch 18)		4 1
$\rightarrow ST-3$	$RATIO MACHINE$ (speed $\propto 1/\text{samay}$) (Ch 18)		7 1
$\rightarrow ST-4$	$AVERAGE SPEED$ (Kul doori $\div$ Kul samay) (Ch 18)		10 4
Total (book ke sawaal)		159/159 ✓	80/80 ✓

$\rightarrow$ **ST-1:** EN 3–6, 8, 9, 12, 14, 15, 17–20, 138, 149, 153 HI 1–3, 5, 6 · $\rightarrow$ **ST-2:** EN 10, 11, 144, 155 HI 79 · $\rightarrow$ **ST-3:** EN 2, 7, 13, 152, 154, 156, 157 HI 80 · $\rightarrow$ **ST-4:** EN 16, 135–137, 139, 140, 142, 143, 145, 148 HI 4, 7, 74, 78

Chapter 20

Boat & Stream (Naav aur Dhaara) नाव और धारा

4 formulas · English book 47 sawaal · Hindi book 13 sawaal · 26 solved examples ✓

BS-1 — D / U MACHINE (dhaara ke saath / khilaaf)

PEHCHAN – Question mein ye dikhe

speed of boat in still water / speed of stream, “28 km downstream in 1 h 45 min”, “16 km downstream in 2 h and upstream in 4 h”, upstream 40 & still 55 $\Rightarrow$ downstream, time to swim 26 km downstream, return in same time; धारा की दिशा, धारा के विपरीत, शांत जल

$$\boxed{D = B + S} \quad \boxed{U = B - S} \quad \Rightarrow \quad B = \frac{D + U}{2} \quad S = \frac{D - U}{2}$$

- Pehle D aur U nikalo: **doori $\div$ samay** (downstream wali trip se D, upstream se U).
- Koi do cheezein (B, S, D, U mein se) diye $\Rightarrow$ baaki dono turant.
- Samay poocha: downstream $\frac{d}{B+S}$, upstream $\frac{d}{B-S}$.

TERMS – Har letter ka matlab

B = boat (still water) · S = stream · D = downstream speed · U = upstream speed.

SOLVED – Asli book questions (verified)

Q (EN 2): B 14; 28 km downstream $1\frac{3}{4}$ h

$$D = 16 \Rightarrow S = 2 \quad \checkmark \text{ book ans (a)}$$

Q (EN 4): B 11; 19 km upstream 2 h

$$U = 9.5 \Rightarrow S = 1.5 \quad \checkmark \text{ book ans (c)}$$

Q (EN 8): 16 km: down 2 h, up 4 h

$$\frac{8-4}{2} = 2 \quad \checkmark \text{ book ans (b)}$$

Q (EN 13): 100 km: up 16 h, down 10 h

$$\frac{10-6.25}{2} = 1.875 \quad \checkmark \text{ book ans (b)}$$

Q (EN 15): 5 h mein 4 km up, 16 km down

$$\frac{3.2-0.8}{2} = 1.2 \quad \checkmark \text{ book ans (c)}$$

Q (EN 24): U 40, B 55 $\Rightarrow$ D

$$2 \times 55 - 40 = 70 \quad \checkmark \text{ book ans (b)}$$

Q (EN 21): 150 km down 7.5 h, S 5; wapas usi samay mein

$$U \text{ chahiye } 20 \Rightarrow B = 20 + 5 = 25 \quad \checkmark \text{ book ans (c)}$$

Q (EN 41): S 3, B 5; 26 km downstream

$$\frac{26}{8} = 3\frac{1}{4} \text{ h} \quad \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (27) EN 2–5, 7, 8, 10, 11, 13, 15, 17, 20, 21, 24–26, 28–30, 41, 42 · **HI** 3, 4, 6, 8, 10, 12

BS-2 — TIME-RATIO MACHINE (up = n $\times$ down)

PEHCHAN – Question mein ye dikhe

“upstream takes twice / three times the time of downstream”, time ratio 4 : 1, “8 h 48 min up, 4 h down $\Rightarrow$ ratio of boat to stream”, stream 3 km/h, down 3 h & up 4 h $\Rightarrow$ boat, “12 h down, 24 h back $\Rightarrow$ still water time”, t_1 & t_2 general; दोगुना समय, अनुपात

$$t_{up} = n \times t_{down} \Rightarrow \boxed{\frac{B}{S} = \frac{n+1}{n-1}} \quad \frac{D}{U} = n$$

$$B = S \cdot \frac{t_u + t_d}{t_u - t_d} \quad \text{Still water ka samay} = \frac{2t_u t_d}{t_u + t_d}$$

- B diya $\Rightarrow S = B \cdot \frac{n-1}{n+1}$; S diya $\Rightarrow B = S \cdot \frac{n+1}{n-1}$.
- Same doori ke samay ka ratio = speeds ka ulta ratio.

TERMS – Har letter ka matlab

n = upstream samay $\div$ downstream samay $\cdot t_u, t_d$ = up / down samay.

SOLVED – Asli book questions (verified)

Q (EN 1): B 3; up = 2 $\times$ down

$$S = 3 \cdot \frac{1}{3} = 1 \quad \checkmark \text{ book ans (d)}$$

Q (EN 6): B 28/3; up = 3 $\times$ down

$$\frac{28}{3} \cdot \frac{2}{4} = \frac{14}{3} \quad \checkmark \text{ book ans (d)}$$

Q (EN 35): up 8 h 48 min, down 4 h $\Rightarrow B : S$

$$n = 2.2 \Rightarrow 3.2 : 1.2 = 8 : 3 \quad \checkmark \text{ book ans (b)}$$

Q (EN 38): down 4 h, wapas 6 h $\Rightarrow S : B$

$$n = 1.5 \Rightarrow B : S = 5 : 1 \quad \checkmark \text{ book ans (a)}$$

Q (EN 37): up = $n \times$ down $\Rightarrow x : y$

$$\frac{n+1}{n-1} \quad \checkmark \text{ book ans (b)}$$

Q (EN 23): S 3; down 3 h, up 4 h

$$3 \cdot \frac{4+3}{4-3} = 21 \quad \checkmark \text{ book ans (d)}$$

Q (EN 44): down 12 h, wapas 24 h $\Rightarrow$ still water

$$\frac{2 \cdot 12 \cdot 24}{36} = 16 \text{ h} \quad \checkmark \text{ book ans (d)}$$

BOOK KESAWAAL (15) EN 1, 6, 16, 18, 19, 23, 27, 35–39, 44 $\cdot$ HI 2, 9

BS-3 — AANA-JAANA (round trip) MACHINE

PEHCHAN – Question mein ye dikhe

goes downstream and returns / reaches its destination and returns, total time given $\Rightarrow$ stream / distance, “15 h in still water, 16 h in flowing river”, average speed of the round trip, who returns first (cycle vs boat), t hours more upstream $\Rightarrow$ distance; वापस आने में कुल समय, औसत चाल

$$\text{Kul samay } T = \frac{d}{B+S} + \frac{d}{B-S} = \frac{2dB}{B^2-S^2} \quad \Rightarrow \quad \boxed{d = \frac{T(B^2-S^2)}{2B}}$$

$$\text{Round-trip avg speed} = \frac{B^2-S^2}{B} \quad \text{Up mein } t \text{ ghante zyada} \Rightarrow d = \frac{t(B^2-S^2)}{2S}$$

- Still water mein $T_0 = \frac{2d}{B} \Rightarrow \frac{T}{T_0} = \frac{B^2}{B^2-S^2}$.
- Tukde alag hon (4 km down + 8 km up) $\Rightarrow$ avg = kul doori $\div$ kul samay.

TERMS – Har letter ka matlab

d = ek taraf ki doori $\cdot T$ = aana + jaana ka kul samay.

SOLVED – Asli book questions (verified)**Q (EN 14):** B 20; 364 km jaana-aana 40 h

$$400 - S^2 = \frac{364 \cdot 40}{40} \Rightarrow S = 6 \checkmark \text{ book ans (d)}$$

Q (EN 12): still 15 h, bahti nadi 16 h; B – S = 15

$$\frac{B^2}{B^2 - S^2} = \frac{16}{15} \Rightarrow B = 4S \Rightarrow S = 5 \checkmark \text{ book ans (d)}$$

Q (EN 33): B 35, S 5 $\Rightarrow$ avg

$$\frac{1225 - 25}{35} = 34.28 \checkmark \text{ book ans (b)}$$

Q (EN 34): B 40, S 10 $\Rightarrow$ avg

$$\frac{1500}{40} = 37.5 \checkmark \text{ book ans (c)}$$

Q (EN 43): Ram cycle 12; Gopi boat B 10, S 4

$$\text{Gopi avg } \frac{84}{10} = 8.4 < 12 \Rightarrow \text{Ram } \checkmark \text{ book ans (b)}$$

Q (EN 46): B 12, S 3; 135 km up aur wapas

$$\frac{135}{9} + \frac{135}{15} = 24 \checkmark \text{ book ans (a)}$$

Q (EN 32): 4 km down 15 min mein, 8 km up; D = 2U

$$D = 16, U = 8 \Rightarrow \frac{12}{0.25+1} = 9.6 \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (13) EN 9, 12, 14, 31–34, 43, 45–47 · **HI** 5, 7**BS-4 — DO TRIP, DO EQUATION****PEHCHAN – Question mein ye dikhe**

“63 km down + 30 km up in 7 h; 28 km down + 48 km up in 6 h”, “40 km down and 12 km up in 4 h, stream 4”, up 12.6 km in 21 min with B : S = 7 : 2 $\Rightarrow$ total time; कुल समय

$$\frac{a_1}{D} + \frac{b_1}{U} = T_1, \quad \frac{a_2}{D} + \frac{b_2}{U} = T_2 \quad \left(p = \frac{1}{D}, q = \frac{1}{U} \right)$$

- p, q ko cross-multiply (Cramer) se nikalo: $p = \frac{T_1 b_2 - T_2 b_1}{a_1 b_2 - a_2 b_1}$, $q = \frac{a_1 T_2 - a_2 T_1}{a_1 b_2 - a_2 b_1}$.
- Options wale sawal: option ka B daal ke check karo (sabse tez).
- B : S ratio + ek speed diya $\Rightarrow D : U = (B+S) : (B-S)$ se dono.

TERMS – Har letter ka matlab

a, b = down / up doori · T = kul samay · D, U = down / up speed.

SOLVED – Asli book questions (verified)**Q (EN 40):** 63 down + 30 up = 7 h; 28 down + 48 up = 6 h $\Rightarrow$ 35 down + 27 up

$$D = 14, U = 12 \Rightarrow 2.5 + 2.25 = 4 \text{ h } 45 \text{ min } \checkmark \text{ book ans (d)}$$

Q (HI 1): 30 up + 44 down = 10 h; 40 up + 55 down = 13 h

$$U = 5, D = 11 \Rightarrow S = 3 \checkmark \text{ book ans (b)}$$

Q (EN 22): S 4; 40 down + 12 up = 4 h

$$B = 12: \frac{40}{16} + \frac{12}{8} = 4 \checkmark \checkmark \text{ book ans (b)}$$

Q (HI 13): 12.6 km up 21 min mein; B : S = 7 : 2

$$U = 36 = 5k \Rightarrow D = 9k = 64.8 \Rightarrow 1.2 + 0.6 = 1.8 \text{ h } \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (5) EN 22, 40 · **HI** 1, 11, 13**COVERAGE CHECK – Har sawaal ka formula**

Formula	Naam	EN Qs	HI Qs
BS-1	D / U MACHINE (dhaara ke saath / khilaaf)	21	6
BS-2	TIME-RATIO MACHINE (up = n $\times$ down)	13	2
BS-3	AANA-JAANA (round trip) MACHINE	11	2
BS-4	DO TRIP, DO EQUATION	2	3
Total (book ke sawaal)		47/47 ✓	13/13 ✓

Chapter 21

Mensuration (Kshetramiti) क्षेत्रमिति

10 formulas · English book 354 sawaal · Hindi book 279 sawaal · 87 solved examples ✓

MN-1 — TRIANGLE (tribhuj) PACK

PEHCHAN – Question mein ye dikhe

area of triangle with sides / base-height, equilateral / isosceles / right triangle, hypotenuse, altitude, inradius / circumradius, similar triangles (area ratio), midpoints triangle, triangle inequality, regular hexagon; त्रिभुज, समबाहु, समकोण, कर्ण, षट्भुज

Cheez	Formula
Area	$\frac{1}{2}bh$ · Heron $\sqrt{s(s-a)(s-b)(s-c)}$, $s = \frac{a+b+c}{2}$
Equilateral (a)	Area $\frac{\sqrt{3}}{4}a^2$, height $\frac{\sqrt{3}}{2}a$, $r = \frac{a}{2\sqrt{3}}$, $R = \frac{a}{\sqrt{3}}$
Isosceles (a, a, b)	$\frac{b}{4}\sqrt{4a^2 - b^2}$ · Right-isosceles: area = $\frac{\text{hyp}^2}{4}$
Right (p, b, h)	$\frac{1}{2}pb$, $r = \frac{p+b-h}{2}$, $R = \frac{h}{2}$ · triplets 3-4-5, 5-12-13, 8-15-17, 7-24-25, 20-21-29, 9-40-41
Koi bhi	$r = \frac{\text{Area}}{s}$, $R = \frac{abc}{4\text{Area}}$, altitude = $\frac{2\text{Area}}{\text{base}}$

- **Similar triangles:** side ratio $k \Rightarrow$ height / median / perimeter ratio k , area ratio k^2 .
- Midpoints jodne se bana triangle = $\frac{1}{4}$ area; $BD = DE = EC \Rightarrow$ har hissa $\frac{1}{3}$ area.
- Teesri side c : $|a - b| < c < a + b$. Regular hexagon = 6 equilateral: $\frac{3\sqrt{3}}{2}a^2$.
- Andar ghoomta circle (radius ρ) ka centre $\Rightarrow$ side $a - 2\sqrt{3}\rho$ ka equilateral; $\sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$.

TERMS – Har letter ka matlab

a, b, c = sides · s = semi-perimeter · r = inradius · R = circumradius.

SOLVED – Asli book questions (verified)

Q (EN 3): hyp 13, ek side 5

doosri $12 \Rightarrow \frac{1}{2} \cdot 5 \cdot 12 = 30$ ✓ **book ans (c)**

Q (HI 23): sides 32, 60, 68

$8-15-17 \times 4 \Rightarrow \frac{1}{2} \cdot 32 \cdot 60 = 960$ ✓ **book ans (a)**

Q (EN 37): 72, 30, 78 m; 20 paise / m²

$5-12-13 \times 6 \Rightarrow 1080 \times 0.20 = ₹216$ ✓ **book ans (c)**

Q (EN 6): equilateral area $25\sqrt{3}$

$a^2 = 100 \Rightarrow 10$ ✓ **book ans (d)**

Q (EN 22): inradius 8, perimeter 64

$r \cdot s = 8 \times 32 = 256$ ✓ **book ans (c)**

Q (EN 13): right triangle 24 & 7 $\Rightarrow$ circumradius

$\frac{25}{2} = 12.5$ ✓ **book ans (a)**

Q (EN 30): similar 1 : 3, bada 72

$\frac{72}{9} = 8$ ✓ **book ans (b)**

Q (EN 36): 5-12-13 ke midpoints ka triangle

$\frac{30}{4} = 7.5$ ✓ **book ans (c)**

Q (EN 38): 16, 12, 20 $\Rightarrow$ altitude

$\frac{2 \times 96}{20} = 9.6$ ✓ **book ans (b)**

Q (EN 18): sides 4 aur 10 $\Rightarrow$ teesri a

$6 < a < 14$ ✓ **book ans (a)**

BOOK KE SAWAAL (90) EN 1-51 · HI 1-34, 258, 259, 265, 274, 278

MN-2 — 4-SIDED PACK (rectangle, square, rhombus, parallelogram, trapezium)**PEHCHAN – Question mein ye dikhe**

rectangle length / breadth / diagonal / perimeter, square diagonal, rhombus with diagonals / side, parallelogram base $\times$ height, trapezium parallel sides, quadrilateral with diagonal & offsets, nearest-integer limits, max area with a rope, crossing diagonally at a speed; आयत, वर्ग, समचतुर्भुज, समांतर चतुर्भुज, समलंब

Shape	Area	Aur
Rectangle	$l b$	$P = 2(l+b)$, $d = \sqrt{l^2 + b^2}$, $(l+b)^2 = d^2 + 2A$
Square	$a^2 = \frac{d^2}{2}$	$P = 4a$, $d = a\sqrt{2}$
Rhombus	$\frac{1}{2}d_1d_2$	$a^2 = \left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2$, $P = 4a$; angle $120^\circ/60^\circ \Rightarrow \frac{\sqrt{3}}{2}a^2$
Parallelogram	base $\times$ height	triangle same base & parallels = $\frac{1}{2}$ parallelogram
Trapezium	$\frac{1}{2}(a+b)h$	quadrilateral: $\frac{1}{2}d(h_1 + h_2)$

- Fixed perimeter $P \Rightarrow$ max area square: $\left(\frac{P}{4}\right)^2$.
- “Nearest integer” $\Rightarrow$ har side ± 0.5 (upper: $+0.5$, lower: -0.5).
- Diagonal paar karna: samay = diagonal $\div$ speed. Wire se shape badli $\Rightarrow$ perimeter same.
- Rotational symmetry: square 4, rectangle 2, equilateral triangle 3.

TERMS – Har letter ka matlab

l , b = length, breadth $\cdot d$ = diagonal $\cdot A$ = area $\cdot P$ = perimeter.

SOLVED – Asli book questions (verified)

Q (EN 53): rhombus diagonals 48, 64

$$\frac{48 \cdot 64}{2} = 1536 \checkmark \text{ book ans (b)}$$

Q (EN 60): rhombus P 164, d_1 80

$$a \text{ 41, half 40} \Rightarrow 9 \Rightarrow d_2 = 18 \Rightarrow 720 \checkmark \text{ book ans (b)}$$

Q (EN 54): trapezium 1792, h 28, ek side 72

$$\frac{2 \cdot 1792}{28} - 72 = 56 \checkmark \text{ book ans (b)}$$

Q (EN 156): rectangle d 65, ek side 63

$$\text{doosri 16} \Rightarrow P = 158 \checkmark \text{ book ans (d)}$$

Q (EN 175): area 60, P 34 $\Rightarrow$ diagonal

$$\sqrt{17^2 - 120} = 13 \checkmark \text{ book ans (c)}$$

Q (EN 116): square field $2 \text{ km}^2 \Rightarrow$ diagonal

$$\sqrt{2A} = 2 \checkmark \text{ book ans (a)}$$

Q (EN 226): 400 m rope $\Rightarrow$ max area

$$100^2 = 10000 \checkmark \text{ book ans (d)}$$

Q (EN 185): 37 m, 23 m (nearest integer) $\Rightarrow$ upper limit P

$$2(37.5 + 23.5) = 122 \checkmark \text{ book ans (a)}$$

Q (HI 42): diagonal 32, offsets 6 aur 10

$$\frac{1}{2} \cdot 32 \cdot 16 = 256 \checkmark \text{ book ans (c)}$$

Q (EN 136): do squares: areas ka farak 32, sides ka farak 4

$$a + b = 8 \Rightarrow 6, 2 \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (151) EN 52–64, 111, 113, 115, 116, 119, 121–123, 125, 127, 129, 135–137, 142, 143, 147–149, 152, 154–168, 170–172, 174–177, 179–192, 195, 198–202, 204, 206, 207, 209–215, 217–221, 223–227, 349 **HI** 35–57, 74–78, 80, 81, 85, 86, 92, 93, 95–101, 103, 104, 106–109, 111–117, 275

MN-3 — PATH / BORDER / RING + TILES, CARPET, POLES

PEHCHAN – Question mein ye dikhe

path of width w outside / inside a field, two roads crossing in the middle, circular path / ring / racing track, jogging track, border of carpet, mat leaving space, tiles / slabs count, carpet of given width, poles around a field, wire tied in rounds; सड़क, मार्ग, पथ, गलीचा, टाइल, चक्कर

Case	Area
Bahar path (w)	$(l + 2w)(b + 2w) - lb$
Andar path (w)	$lb - (l - 2w)(b - 2w)$
Beech mein do cross roads	$w(l + b) - w^2$
Circular path / ring	$\pi(R^2 - r^2) = \pi(R + r)(R - r)$; width = $\frac{C_1 - C_2}{2\pi}$

- **Tiles / slabs ki ginti** = floor ka area $\div$ ek tile ka area (units same!).
- **Carpet ki lambai** = floor area $\div$ carpet ki chaudi; cost = lambai $\times$ rate.
- Poles: n poles $\Rightarrow$ n gap (band ghera) $\Rightarrow$ perimeter = $n \times$ gap. k chakkar $\Rightarrow$ wire = $k \times P$.
- Cost = rate $\times$ (area ya perimeter).

TERMS – Har letter ka matlab

w = rasta ki chaudi $\cdot$ R , r = bahar, andar radius $\cdot$ C = circumference.

SOLVED – Asli book questions (verified)

Q (EN 124): 18 m square park, 3 m road bahar

$$24^2 - 18^2 = 252 \checkmark \text{ book ans (c)}$$

Q (EN 72): 1386 m² park (r 21), 7 m path andar

$$\frac{22}{7}(441 - 196) = 770 \checkmark \text{ book ans (c)}$$

Q (EN 144): 110 m square, do 5 m roads beech mein

$$5 \cdot 220 - 25 = 1075 \checkmark \text{ book ans (c)}$$

Q (EN 169): 15×11 room, 75 cm carpet @ ₹13/m

$$\frac{165}{0.75} = 220 \text{ m} \times 13 = 2860 \checkmark \text{ book ans (c)}$$

Q (EN 128): 48 poles 5 m door, square

$$P \ 240 \Rightarrow \text{side } 60 \Rightarrow 3600 \checkmark \text{ book ans (b)}$$

Q (EN 91): ring: andar 22, bahar 44 circumference

$$\frac{22}{2\pi} = 3.5 \checkmark \text{ book ans (c)}$$

Q (EN 194): 20×18 ke bahar 2 m track

$$24 \cdot 22 - 360 = 168 \checkmark \text{ book ans (d)}$$

Q (HI 94): 3 chakkar mein 2250 m tar; $l = b + 5$

$$P \ 750 \Rightarrow l + b = 375 \Rightarrow l = 190 \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (46) EN 65, 68, 72, 74, 80, 84, 91, 92, 94, 97, 120, 124, 126, 128, 130, 134, 139, 141, 144, 145, 153, 169, 194, 196, 197, 203, 205, 208, 216, 222 · HI 68, 69, 71, 82, 84, 87–91, 94, 102, 105, 110, 118, 276

MN-4 — CIRCLE PACK (circle, semicircle, sector, wheel, andar-bahar shapes)

PEHCHAN – Question mein ye dikhe

circumference / area / diameter, semicircle / quarter circle perimeter, sector & arc, minute hand sweep, wheel revolutions, wire bent from square into circle, square inside circle, circle inside square / triangle, $C - D = 15$, cows tied at corners; वृत्त, अर्धवृत्त, परिधि, त्रिज्यखंड

$$C = 2\pi r \quad A = \pi r^2 \quad \text{Sector} = \frac{\theta}{360}\pi r^2 \quad \text{Arc} = \frac{\theta}{360}2\pi r$$

Cheez	Formula
Semicircle perimeter	$\pi r + 2r = \frac{36}{7}r \cdot \text{quarter: } \frac{\pi r}{2} + 2r$
Wheel	chakkar = doori $\div 2\pi r$
Square circle mein	diagonal = $2r \Rightarrow \text{area } 2r^2$
Circle square mein	$r = \frac{a}{2} \cdot \text{circle triangle mein: } r = \frac{\text{Area}}{s}$
Triangle circle mein	equilateral: $R = \frac{a}{\sqrt{3}} \cdot \text{right: } R = \frac{\text{hyp}}{2}$

- Wire / same perimeter se shape badli $\Rightarrow$ perimeter barabar rakho.
- $C - D = k \Rightarrow r = \frac{k}{2\pi - 2} = \frac{7k}{30}$. $C = A$ (numerically) $\Rightarrow r = 2$.
- Kono par bandhe janwar (4 kone) $\Rightarrow$ charne wala = ek poora circle; bacha = square $- \pi r^2$.

TERMS – Har letter ka matlab

$r = \text{radius} \cdot d = 2r \cdot \theta = \text{kon (degree)} \cdot \pi = 22/7$.

SOLVED – Asli book questions (verified)

Q (EN 76): area 154 $\Rightarrow$ circumference

$r = 7 \Rightarrow 44$ ✓ **book ans (b)**

Q (EN 78): wheel d 88 cm, 8712 m

$\frac{8712}{2.766} = 3150$ ✓ **book ans (d)**

Q (EN 81): sector r 21, 40°

$\frac{40}{360} \cdot \frac{22}{7} \cdot 441 = 154$ ✓ **book ans (b)**

Q (EN 85): semicircle d 28

$44 + 28 = 72$ ✓ **book ans (c)**

Q (EN 90): C, D se 15 zyada

$\frac{7-15}{30} = 3.5$ ✓ **book ans (b)**

Q (EN 101): square (484) ka wire $\Rightarrow$ circle

wire 88 $\Rightarrow r = 14 \Rightarrow 616$ ✓ **book ans (b)**

Q (EN 132): circle (r 8) mein square

$2r^2 = 128$ ✓ **book ans (b)**

Q (EN 79): minute hand 6 cm, 5 min

$30^\circ \Rightarrow 3\pi = 9.43$ ✓ **book ans (a)**

Q (HI 279): 35 m square ke 4 kono par gaayein

$1225 - \pi(17.5)^2 = 262.5$ ✓ **book ans (c)**

BOOK KE SAWAAL (64) EN 66, 67, 69–71, 73, 75–79, 81–83, 85, 87–90, 93, 95, 96, 98–110, 112, 132, 133, 140, 178, 193, 244 • **HI** 58–67, 70, 72, 73, 79, 83, 136, 260, 263, 267, 279 | *Ch 28 (Vividh):* EN 54 • **HI** 23

MN-5 — SIZE-CHANGE / RATIO MACHINE

PEHCHAN – Question mein ye dikhe

radius increased by 5% $\Rightarrow$ area?, edge +50% $\Rightarrow$ surface?, side doubled / tripled / halved $\Rightarrow$ volume?, ratio of volumes 64 : 1331 $\Rightarrow$ surface ratio, two cylinders same volume with radius 20% less, two cones volume & height ratio $\Rightarrow$ radius ratio, room dimensions changed $\Rightarrow$ painting cost %; गुना, % बढ़ा, अनुपात

$$\text{Lambai} \times k \Rightarrow \text{Area} \times k^2, \quad \text{Volume} \times k^3 \quad \% \text{ change (area)} = 2x + \frac{x^2}{100}$$

- Volume ratio $a^3 : b^3 \Rightarrow$ side ratio $a : b \Rightarrow$ area ratio $a^2 : b^2$.
- Formula mein power dekho: $V_{cyl} \propto r^2 h$, $V_{cone} \propto r^2 h$, $CSA_{cone} \propto r l \Rightarrow$ ratios guna karo.
- Same volume, radius $\times k \Rightarrow$ height $\times \frac{1}{k^2}$.
- Alag-alag guna (1×2 , $b \times \frac{2}{3}$, $h \times \frac{1}{2}$) $\Rightarrow$ naya formula mein daal ke purane se bhaag.

TERMS – Har letter ka matlab

$k = \text{kitne guna} \cdot x = \% \text{ badlaav.}$

SOLVED – Asli book questions (verified)

Q (EN 86): radius $+5\%$

$$10 + 0.25 = 10.25\% \checkmark \text{ book ans (a)}$$

Q (EN 240): edge $+50\% \Rightarrow$ surface

$$1.5^2 = 2.25 \Rightarrow 125\% \checkmark \text{ book ans (c)}$$

Q (EN 278): same volume, radius 20% kam $\Rightarrow$ height

$$\frac{1}{0.64} = 1.5625 \Rightarrow 56.25\% \checkmark \text{ book ans (b)}$$

Q (EN 238): cubes ka volume $64 : 1331 \Rightarrow$ surface

$$4 : 11 \Rightarrow 16 : 121 \checkmark \text{ book ans (a)}$$

Q (EN 291): cylinders: $r : 2 : 3, h : 5 : 3$

$$4 \cdot 5 : 9 \cdot 3 = 20 : 27 \checkmark \text{ book ans (b)}$$

Q (EN 327): cones: $V : 10, h : 2 : 5 \Rightarrow r$

$$r^2 = \frac{V}{h} = \frac{1}{2} : 2 \Rightarrow 1 : 2 \checkmark \text{ book ans (d)}$$

Q (HI 148): har dimension $+100\%$

$$2^3 = 8 \text{ guna} \checkmark \text{ book ans (c)}$$

Q (EN 267): room $3 : 3 : 4; l \times 2, b \times \frac{2}{3}, h \times \frac{1}{2} \Rightarrow$ deewar ka kharcha

$$2h(l + b) : 48 \Rightarrow 32 \Rightarrow 33\frac{1}{3}\% \text{ kam} \checkmark \text{ book ans (b)}$$

BOOK KESAWAAL (30) EN 86, 114, 117, 118, 173, 238–240, 243, 246, 267, 278, 291, 298, 327, 334, 345 · **HI** 127, 138, 148, 159, 183, 209, 215, 218, 235, 257, 261 | Ch 28 (Vividh): EN 67 · **HI** 52

MN-6 — CUBE & CUBOID (ghan, ghanabh)**PEHCHAN – Question mein ye dikhe**

volume / total surface / lateral surface / diagonal of cube or cuboid, painting cost, cubes joined end to end, areas of three adjacent faces, boxes inside a store, water level rise (persons dipping), earth dug & spread, wooden box thickness, walls & ceiling painting, cube edges vs square perimeter; घन, घनाभ, दीवारें

	Volume	TSA	LSA (4 deewar)	Diagonal
Cube a	a^3	$6a^2$	$4a^2$	$a\sqrt{3}$
Cuboid	lbh	$2(lb + bh + hl)$	$2h(l + b)$	$\sqrt{l^2 + b^2 + h^2}$

- Teen faces $x, y, z \Rightarrow V = \sqrt{xyz}$. Edges ka jod: cube $12a$, cuboid $4(l+b+h)$.
- n cube ek line mein jude $\Rightarrow TSA = (4n + 2)a^2$.
- Pani ka level = daala gaya volume $\div$ base area; mitti faili $\Rightarrow$ height = volume $\div$ bachi zameen.
- Kitne box / cube $\Rightarrow$ bada volume $\div$ chhota volume. Moti deewar (t) $\Rightarrow$ bahar ka V – andar ka V.
- Deewar + chhat = $2h(l+b) + lb$; khula tank = LSA + base.

TERMS – Har letter ka matlab

$a = \text{cube ki side} \cdot l, b, h = \text{cuboid} \cdot TSA = \text{kul surface} \cdot LSA = \text{side walls.}$

SOLVED – Asli book questions (verified)**Q (EN 229):** painting ₹588 @ ₹2/cm²TSA 294 ⇒ a 7 ⇒ 343 ✓ **book ans (a)****Q (EN 252):** ratio 4 : 3 : 5, edges 144 ⇒ TSA12, 9, 15 ⇒ 2(108+135+180)=846 ✓ **book ans (b)****Q (EN 274):** faces 20, 32, 40 $\sqrt{25600} = 160$ ✓ **book ans (c)****Q (EN 257):** 500 log × 0.04 m³, pond 80 × 50 $\frac{20}{4000} \text{ m} = 0.5 \text{ cm}$ ✓ **book ans (d)****Q (EN 259):** 4 × 2 × 1.5 tank, 22 × 14 field mein faili $\frac{12}{300} \text{ m} = 4 \text{ cm}$ ✓ **book ans (d)****Q (EN 272):** do cube (216) jode $(4 \cdot 2 + 2) \cdot 36 = 360$ ✓ **book ans (a)****Q (EN 247):** side +1 ⇒ volume +169 $3a^2 + 3a + 1 = 169 \Rightarrow 7$ ✓ **book ans (b)****Q (HI 141):** 20 × 25 × 10 store, 2 × 1.25 × 1 box $\frac{5000}{2.5} = 2000$ ✓ **book ans (d)****Q (EN 248):** cube ke edges = square ka P; V = area12a = 4s ⇒ s = 3a; $a^3 = 9a^2 \Rightarrow a 9, s 27$ ✓ **book ans (c)****BOOK KE SAWAAL (84)** EN 131, 138, 146, 150, 151, 228–236, 241, 242, 247–257, 259–266, 268–274 · **HI** 119–126, 128–135, 139–142, 144–147, 149, 150, 152–158, 160–164, 264 | Ch 28 (Vividh): EN 2, 24, 52**MN-7 — CYLINDER (belan) + PIPE****PEHCHAN – Question mein ye dikhe**

volume / curved / total surface of cylinder, tank capacity ⇒ depth, r + h given with TSA, ratio CSA : TSA ⇒ h : r, hollow pipe painting / weight, sheet rolled into cylinder, wire volume; बेलन, पाइप, टंकी

$$V = \pi r^2 h \quad CSA = 2\pi rh \quad TSA = 2\pi r(r + h) \quad \frac{CSA}{TSA} = \frac{h}{h + r}$$

- **Hollow** (R bahar, r andar): $V = \pi(R^2 - r^2)h$; saari surface = $2\pi Rh + 2\pi rh + 2\pi(R^2 - r^2)$.
- Sheet mod ke cylinder: jo side gol hui = $2\pi r$, doosri = h.
- r + h diya aur TSA ⇒ $2\pi r(r + h)$ se seedha r.

TERMS – Har letter ka matlab

r = radius · h = height · R = bahar ka radius.

SOLVED – Asli book questions (verified)**Q (EN 275):** base circumference 176, h 12 $\Rightarrow$ TSA

$$r \ 28 \Rightarrow 176 \times 40 = 7040 \checkmark \text{ book ans (c)}$$

Q (EN 277): $r + h = 39$, TSA 1716 $\Rightarrow$ V

$$r \ 7, h \ 32 \Rightarrow 4928 \checkmark \text{ book ans (d)}$$

Q (EN 285): d 10, TSA 330 $\Rightarrow$ h

$$5 + h = 10.5 \Rightarrow 5.5 \checkmark \text{ book ans (a)}$$

Q (EN 292): CSA : TSA = 3 : 7 $\Rightarrow$ h : r

$$h : (h+r) = 3 : 7 \Rightarrow 3 : 4 \checkmark \text{ book ans (d)}$$

Q (EN 293): h 14, CSA 660 $\Rightarrow$ V

$$r \ 7.5 \Rightarrow 2475 \checkmark \text{ book ans (c)}$$

Q (EN 313): 22×10 sheet, chhoti side height

$$C \ 22 \Rightarrow r \ 3.5 \Rightarrow 385 \checkmark \text{ book ans (a)}$$

Q (EN 276): 10 m pipe, d 15 & 17 cm, ₹0.15 / cm^2

$$\pi(17000 + 15000 + 32) \times 0.15 = 15100.80 \checkmark \text{ book ans (b)}$$

Q (HI 192): wire d 2 cm, 7 m

$$\pi \cdot 1 \cdot 700 = 2200 \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (69) EN 275–277, 279–290, 292–297, 299, 302–306, 309, 311, 313–315 · **HI** 165–175, 177–182, 184, 185, 187–201, 203, 205, 277**MN-8 — CONE (shanku) + FRUSTUM****PEHCHAN – Question mein ye dikhe**

volume / curved / total surface of cone, slant height, conical tent cloth of given width, sector (120°) turned into cone, frustum slant height, volume = surface numerically; शंकु, तिर्यक ऊँचाई

$$V = \frac{1}{3}\pi r^2 h \quad l = \sqrt{r^2 + h^2} \quad CSA = \pi r l \quad TSA = \pi r(r + l)$$

- **Sector (R, θ) se cone:** $l = R$, $r = \frac{\theta}{360}R$.
- **Frustum:** slant = $\sqrt{h^2 + (R - r)^2}$, $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$.
- Tent ka kapda: lambai = CSA $\div$ kapde ki chaudi.

TERMS – Har letter ka matlab

r = base radius · h = height · l = slant height.

SOLVED – Asli book questions (verified)**Q (EN 317):** l 13, base $144\pi \Rightarrow$ V

$$r \ 12, h \ 5 \Rightarrow 240\pi \checkmark \text{ book ans (c)}$$

Q (EN 319): r 5, h 12 $\Rightarrow$ TSA

$$l \ 13 \Rightarrow \pi \cdot 5 \cdot 18 = 90\pi \checkmark \text{ book ans (b)}$$

Q (EN 320): d 20, l 10.5 $\Rightarrow$ CSA

$$\frac{22}{7} \cdot 10 \cdot 10.5 = 330 \checkmark \text{ book ans (d)}$$

Q (EN 332): h 24, l 25 $\Rightarrow$ CSA

$$r \ 7 \Rightarrow 550 \checkmark \text{ book ans (d)}$$

Q (EN 322): frustum h 8, radii 3, 9

$$\sqrt{64 + 36} = 10 \checkmark \text{ book ans (a)}$$

Q (EN 325): sector r 21, 120°

$$21 \cdot \frac{120}{360} = 7 \checkmark \text{ book ans (c)}$$

Q (EN 333): tent d 16, l 5.6, kapda 4 m chaudi

$$\frac{140.8}{4} = 35.2 \text{ m} \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (14) EN 316–320, 322–325, 329, 332, 333 · **HI** 207, 208

MN-9 — SPHERE & HEMISPHERE (gola, ardh-gola)

PEHCHAN – Question mein ye dikhe

surface area / volume of sphere, hemisphere total surface / volume, hollow spherical shell weight, $V \div SA$, radius $+2 \Rightarrow$ surface $+704$, capsule; गोला, अर्धगोला

	Volume	Surface
Sphere	$\frac{4}{3}\pi r^3$	$4\pi r^2$
Hemisphere	$\frac{2}{3}\pi r^3$	CSA $2\pi r^2$, TSA $3\pi r^2$
Hollow (R, r)	$\frac{4}{3}\pi(R^3 - r^3)$	hollow hemisphere TSA $2\pi R^2 + 2\pi r^2 + \pi(R^2 - r^2)$

- $\frac{V}{SA} = \frac{r}{3}$. Weight = volume $\times$ density.

TERMS – Har letter ka matlab

r = radius · R = bahar ka radius.

SOLVED – Asli book questions (verified)

Q (EN 338): SA 38.5 $\Rightarrow$ r

$$r^2 = 3.0625 \Rightarrow 1.75 \checkmark \text{ book ans (d)}$$

Q (EN 340): hemisphere TSA 1848 $\Rightarrow$ diameter

$$3\pi r^2 \Rightarrow r = 14 \Rightarrow 28 \checkmark \text{ book ans (d)}$$

Q (EN 343): hemisphere TSA $108\pi \Rightarrow V$

$$r = 6 \Rightarrow 144\pi \checkmark \text{ book ans (c)}$$

Q (EN 347): d 42 $\Rightarrow V$

$$\frac{4}{3} \cdot \frac{22}{7} \cdot 9261 = 38808 \checkmark \text{ book ans (c)}$$

Q (EN 346): shell 8 & 9 cm, 6 g/cm³

$$\frac{4}{3}\pi \cdot 217 \cdot 6 = 5456 \text{ g} \checkmark \text{ book ans (a)}$$

Q (HI 232): $V \div SA = 9$

$$r = 27 \checkmark \text{ book ans (c)}$$

Q (HI 249): $r + 2 \Rightarrow SA + 704$

$$16\pi(r + 1) = 704 \Rightarrow 13 \checkmark \text{ book ans (a)}$$

Q (HI 227): hollow hemisphere 10 & 5 $\Rightarrow$ TSA

$$200\pi + 50\pi + 75\pi = 325\pi \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (37) EN 312, 338, 340, 343, 344, 346, 347 · HI 204, 206, 214, 216, 217, 219–221, 223–225, 227, 230–233, 236–239, 241–249, 252

MN-10 — MELT & RECAST + COMBINATION + PRISM / PYRAMID

PEHCHAN – Question mein ye dikhe

melted and recast into balls / cones / wire / coins / cubes, liquid of a bowl filled in bottles, cavity hollowed out, tent = cylinder + cone, hemisphere on cylinder / cone, rocket, prism, pyramid, pipe flow per minute, cone & sphere of equal volume; पिघलाकर, ढाला, प्रिज्म, पिरामिड, तंबू

Volume same rehta hai

$$n = \frac{V_{bada}}{V_{chhota}}$$

$$\text{Wire ki lambai} = \frac{V}{\pi r^2}$$

- Gole se gole: $n = \left(\frac{R}{r}\right)^3$; kai gole mila ke ek: $R^3 = r_1^3 + r_2^3 + \dots$
- Combination:** volume jodo/ghatao; surface = sirf dikhne wale hisse (jodne wali jagah hatao).
- Prism:** $V = \text{base area} \times h$, side surface = base perimeter $\times h$. **Pyramid:** $V = \frac{1}{3} \text{base} \times h$.
- Pipe flow:** har second = cross-section area $\times$ speed.
- Barabar volume $\Rightarrow$ dono formule barabar rakh ke ratio.

TERMS – Har letter ka matlab

$V = \text{volume} \cdot R, r = \text{bada, chhota radius} \cdot n = \text{kitne bane.}$

SOLVED – Asli book questions (verified)

Q (EN 342): 6, 8, 10 cm gole mila ke ek

$$R^3 = 1728 \Rightarrow 12 \checkmark \text{ book ans (b)}$$

Q (EN 350): r 6 se r 0.3 ki goliyan

$$20^3 = 8000 \checkmark \text{ book ans (c)}$$

Q (EN 308): 42 cm gola $\Rightarrow$ 7 mm wire

$$\frac{\frac{4}{3} \cdot 21^3}{0.35^2} \text{ cm} = 1008 \text{ m} \checkmark \text{ book ans (a)}$$

Q (EN 339): r 24 hemisphere bowl $\Rightarrow$ bottles r 6, h 8

$$\frac{9216}{288} = 32 \checkmark \text{ book ans (a)}$$

Q (EN 321): cylinder r 3, h 5 $\Rightarrow$ cones h 1, r 1 mm

$$\frac{45}{0.01/3} = 13500 \checkmark \text{ book ans (a)}$$

Q (EN 351): 7 cm cube mein cone cavity (h 7, r 3)

$$343 - 66 = 277 \checkmark \text{ book ans (b)}$$

Q (EN 330): tent: cylinder (10, 25) + cone (10, l 15), +20%

$$650\pi \times 1.2 = 2451.4 \checkmark \text{ book ans (d)}$$

Q (HI 254): prism base 8 : 15 : 17, h 21, side surface 840

$$P \ 40 \Rightarrow 8, 15, 17 \Rightarrow 60 \times 21 = 1260 \checkmark \text{ book ans (b)}$$

Q (HI 255): pyramid 16×12 , lateral edge 26

$$h \ 24 \Rightarrow \frac{1}{3} \cdot 192 \cdot 24 = 1536 \checkmark \text{ book ans (a)}$$

Q (HI 273): pipe 10.4 cm^2 , 54 km/h, 60% bhara, 5 min

$$10.4 \times 1500 \times 0.6 \times 300 \text{ cm}^3 = 2808 \text{ L} \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (55) EN 237, 245, 258, 300, 301, 307, 308, 310, 321, 326, 328, 330, 331, 335–337, 339, 341, 342, 348, 350–354 · HI 137, 143, 151, 176, 186, 202, 210–213, 222, 226, 228, 229, 234, 240, 250, 251, 253–256, 262, 266, 268–273

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
MN-1	TRIANGLE (tribhuj) PACK	51	39
MN-2	4-SIDED PACK (rectangle, square, rhombus, parallelogram, tra	96	55
MN-3	PATH / BORDER / RING + TILES, CARPET, POLES	30	16
MN-4	CIRCLE PACK (circle, semicircle, sector, wheel, andar-bahar	42	20
MN-5	SIZE-CHANGE / RATIO MACHINE	17	11
MN-6	CUBE & CUBOID (ghan, ghanabh)	42	39
MN-7	CYLINDER (belan) + PIPE	32	37
MN-8	CONE (shanku) + FRUSTUM	12	2
MN-9	SPHERE & HEMISPHERE (gola, ardh-gola)	7	30
MN-10	MELT & RECAST + COMBINATION + PRISM / PYRAMID	25	30
Total (book ke sawaal)		354/354 ✓	279/279 ✓

Chapter 22

Algebra (Beejganit) बीजगणित

9 formulas · English book 294 sawaal · Hindi book 268 sawaal · 72 solved examples ✓

AL-1 — SERIES MACHINE (AP / GP)

PEHCHAN – Question mein ye dikhe

nth term / sum of AP, sum of first n odd / even numbers, how many numbers between 1000 and 3000 divisible by 7, GP nth term / ratio, $1/(5 \times 6) + 1/(6 \times 7) + \dots$, AM & GM of two numbers, sum of cubes; समांतर श्रेणी, गुणोत्तर श्रेणी

	AP (a, d)	GP (a, r)
n-th term	$a + (n - 1)d$	ar^{n-1}
Sum	$\frac{n}{2}(\text{pehla} + \text{aakhri}) = \frac{n}{2}[2a + (n - 1)d]$	$a \frac{r^n - 1}{r - 1}$
Terms ki ginti	$n = \frac{\text{aakhri} - \text{pehla}}{d} + 1$	$r^{n-1} = \frac{\text{aakhri}}{a}$

- Pehle n odd ka jod $= n^2$; even $= n(n + 1)$; $1^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$.
- A se B tak d se divisible: $\left\lfloor \frac{B}{d} \right\rfloor - \left\lfloor \frac{A-1}{d} \right\rfloor$.
- $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$ (telescoping). AM $= \frac{a+b}{2}$, GM $= \sqrt{ab} \Rightarrow a + b = 2 \cdot \text{AM}$, $ab = \text{GM}^2$.
- AP ke 3 terms: a - d, a, a + d (sum se a turant).

TERMS – Har letter ka matlab

a = pehla term · d = common difference · r = common ratio · n = terms.

SOLVED – Asli book questions (verified)

Q (EN 1): pehle 25 odd ka jod

$$25^2 = 625 \text{ ✓ book ans (b)}$$

Q (EN 8): 3, 8, 13, ... ka 50th term

$$3 + 49 \times 5 = 248 \text{ ✓ book ans (b)}$$

Q (EN 2): 1000–3000 ke beech 7 se divisible

$$428 - 142 = 286 \text{ ✓ book ans (c)}$$

Q (EN 10): GP a 2, r 3 $\Rightarrow$ 5th

$$2 \cdot 3^4 = 162 \text{ ✓ book ans (d)}$$

Q (EN 34): $6 + 11 + \dots + 71$

$$n = 14 \Rightarrow 7 \times 77 = 539 \text{ ✓ book ans (a)}$$

Q (EN 29): $1 + 4 + 7 + \dots + X = 782$

$$\frac{n}{2}(3n - 1) = 782 \Rightarrow n = 23, X = 67 \text{ ✓ book ans (d)}$$

Q (EN 7): $1/(5 \times 6) + \dots$ (20 terms)

$$\frac{1}{5} - \frac{1}{25} = 0.16 \text{ ✓ book ans (a)}$$

Q (EN 13): $(8 + 27 + \dots + 1000) + (2 + 4 + \dots + 20)$

$$55^2 - 1 + 110 = 3134 \text{ ✓ book ans (b)}$$

Q (EN 23): GP: 3 terms ka jod : 6 ka = 125 : 152

$$1 + r^3 = \frac{152}{125} \Rightarrow r = \frac{3}{5} \text{ ✓ book ans (d)}$$

Q (EN 33): $7 + 14 + 21 + \dots = 952$

$$n(n + 1) = 272 \Rightarrow 16 \text{ ✓ book ans (a)}$$

BOOK KE SAWAAL (47) EN 1–35 · HI 1–12

AL-2 — HCF / LCM OF EXPRESSIONS

PEHCHAN – Question mein ye dikhe

HCF / LCM of monomials ($15x^3y^4$, $12x^2y^5$), of polynomials ($x^4 - y^4$, $x^8 - y^8$), HCF is $(x + a) \Rightarrow a$; लघुत्तम समापवर्त्य, महत्तम समापवर्तक

HCF = common factors, sabse CHHOTI power **LCM** = saare factors, sabse BADI power

- Numbers ka HCF / LCM alag, variables ka alag.
- Polynomial: pehle factorise karo — $a^2 - b^2 = (a - b)(a + b)$, $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$, middle-term split.

TERMS – Har letter ka matlab

power = ghaat (x^3 mein 3).

SOLVED – Asli book questions (verified)

Q (EN 42): LCM of $15x^3y^4$, $12x^2y^5$

$60x^3y^5$ ✓ book ans (a)

Q (EN 45): HCF of $6xy^2z$, $8x^2y^3z^2$, $12x^3y^3z^3$

$2xy^2z$ ✓ book ans (d)

Q (EN 43): HCF of $a^3 + b^3$, $(a + b)^2$, $a^2 - b^2$

$(a + b)$ ✓ book ans (a)

Q (EN 38): HCF of $2x^2 + 5x - 12$, $x^2 + x - 12$ is $(x + a)$

$(2x - 3)(x + 4)$, $(x + 4)(x - 3) \Rightarrow a = 4$ ✓ book ans (c)

Q (EN 37): $X = 3^2 \cdot 2$, $Y = 2^3 \cdot 3$, $Z = 3^2 \cdot 2^2 \Rightarrow \text{LCM}(XY, YZ, ZX)$

$2^5 \cdot 3^4 = 2592$ ✓ book ans (d)

BOOK KE SAWAAL (16) EN 36–47 · HI 14–17

AL-3 — LINEAR PAIR (do seedhi equations)

PEHCHAN – Question mein ye dikhe

value of k for no solution / infinite / unique solution / coincident lines, “ $5x + 7y = 19$, $7x + 5y = 17 \Rightarrow x$ ”, $x + 2y = 27$ & $x - 2y = -1$, decimal equations, $2^x(x+y) = 16$ & $64^x(x-y) = 2$; रेखिक समीकरण युग्म, हल

Condition	Hal
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	ek (unique)
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	koi nahi (parallel)
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	anant (coincident)

- Solve: coefficients ulte hon ($5, 7 / 7, 5$) $\Rightarrow$ **jodo aur ghatao**: $x + y$ aur $x - y$ turant.
- Warna Cramer: $x = \frac{c_1b_2 - c_2b_1}{a_1b_2 - a_2b_1}$, $y = \frac{a_1c_2 - a_2c_1}{a_1b_2 - a_2b_1}$.

TERMS – Har letter ka matlab

$a_1x + b_1y + c_1 = 0$ aur $a_2x + b_2y + c_2 = 0$.

SOLVED – Asli book questions (verified)**Q (EN 51):** $2x + 3y + 11 = 0$, $6x + ky + 33 = 0$ coincident

$$\frac{2}{6} = \frac{3}{k} \Rightarrow 9 \checkmark \text{ book ans (b)}$$

Q (EN 56): $16x - 12y + 9$, $12x + ky - 11$: no solution

$$\frac{16}{12} = \frac{-12}{k} \Rightarrow -9 \checkmark \text{ book ans (c)}$$

Q (EN 57): $20x + 5y + 11$, $50x - ky - 9$: no solution

$$\frac{20}{50} = \frac{5}{-k} \Rightarrow -12.5 \checkmark \text{ book ans (b)}$$

Q (EN 60): $3x + y = 1$, $px + 2y = 5$: no solution

$$\frac{3}{p} = \frac{1}{2} \Rightarrow 6 \checkmark \text{ book ans (b)}$$

Q (EN 259): $5x + 7y = 19$, $7x + 5y = 17$

$$x + y = 3, y - x = 1 \Rightarrow x = 1 \checkmark \text{ book ans (a)}$$

Q (EN 260): $x + 2y = 27$, $x - 2y = -1$

$$4y = 28 \Rightarrow 7 \checkmark \text{ book ans (c)}$$

Q (EN 265): $0.08x + 0.04y = 10$, $0.2(x - 1) + 0.4y = 24.8$

$$2x + y = 250, x + 2y = 125 \Rightarrow x = 125 \checkmark \text{ book ans (a)}$$

Q (EN 187): infinite solutions: $2x - (a - 4)y = 2b + 1$, $4x - (a - 1)y = 5b - 1$

$$\frac{2}{4} = \frac{a-4}{a-1} = \frac{2b+1}{5b-1} \Rightarrow a = 7, b = 3 \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (23) EN 51–61, 187, 203, 259, 260, 262, 265 • HI 31, 33–36 | Ch 28 (Vividh): EN 31**AL-4 — DO-VARIABLE + x + 1/x MACHINE****PEHCHAN – Question mein ye dikhe**

if $x + 1/x = 3 \Rightarrow x^2 + 1/x^2, x^3 + 1/x^3, x^4 + 1/x^4; x - 1/x; a + b \& ab \Rightarrow a^2 + b^2, a^3 + b^3; a - b \& a^2 + b^2 \Rightarrow ab; x = \sqrt{3} + \sqrt{2} \Rightarrow x^2 + 1/x^2; x^4 + 1/x^4 = 194 \Rightarrow x^3 + 1/x^3$

Diya k	Square	Cube
$x + \frac{1}{x} = k$	$x^2 + \frac{1}{x^2} = k^2 - 2$	$x^3 + \frac{1}{x^3} = k^3 - 3k$
$x - \frac{1}{x} = k$	$x^2 + \frac{1}{x^2} = k^2 + 2$	$x^3 - \frac{1}{x^3} = k^3 + 3k$
$a + b, ab$	$a^2 + b^2 = (a + b)^2 - 2ab$	$a^3 + b^3 = (a + b)^3 - 3ab(a + b)$
$a - b, ab$	$a^2 + b^2 = (a - b)^2 + 2ab$	$a^3 - b^3 = (a - b)^3 + 3ab(a - b)$

- Ulta chalna: $x^4 + \frac{1}{x^4} = m \Rightarrow x^2 + \frac{1}{x^2} = \sqrt{m+2} \Rightarrow x + \frac{1}{x} = \sqrt{\sqrt{m+2}+2}$.
- $x^2 - kx + 1 = 0 \Rightarrow x + \frac{1}{x} = k$. $x = p + \sqrt{q} \Rightarrow \frac{1}{x} = \text{conjugate}$ (jab $p^2 - q = 1$).
- $(a + b)^2 - (a - b)^2 = 4ab \cdot x^6 + \frac{1}{x^6} = \left(x^3 + \frac{1}{x^3}\right)^2 - 2$.

TERMS – Har letter ka matlab

k = diya hua jod / farak.

SOLVED – Asli book questions (verified)

Q (EN 64): $x + 1/x = 3 \Rightarrow x^2 + 1/x^2$

$9 - 2 = 7$ ✓ **book ans (a)**

Q (EN 66): $x + 1/x = 42 \Rightarrow x^3 + 1/x^3$

$42^3 - 126 = 73962$ ✓ **book ans (b)**

Q (EN 67): $a + b = 56, (a - b)^2 = 496 \Rightarrow ab$

$\frac{3136-496}{4} = 660$ ✓ **book ans (d)**

Q (EN 69): $a + b = 48, ab = 56 \Rightarrow a^3 + b^3$

$48^3 - 3 \cdot 56 \cdot 48 = 102528$ ✓ **book ans (c)**

Q (EN 78): $x^4 + 1/x^4 = 1154 \Rightarrow x + 1/x$

$\sqrt{1156} = 34 \Rightarrow \sqrt{36} = 6$ ✓ **book ans (b)**

Q (EN 95): $x^4 + 1/x^4 = 194 \Rightarrow x^3 + 1/x^3$

$14 \Rightarrow 4 \Rightarrow 64 - 12 = 52$ ✓ **book ans (c)**

Q (EN 116): $x - 1/x = 3 \Rightarrow x^4 + 1/x^4$

$11^2 - 2 = 119$ ✓ **book ans (b)**

Q (EN 124): $a + 1/a = -3 \Rightarrow a^6 + 1/a^6$

$\text{cube} - 18 \Rightarrow 324 - 2 = 322$ ✓ **book ans (b)**

Q (EN 49): $x + y = 8, xy = 15 \Rightarrow x^4 + y^4$

$34^2 - 2 \cdot 225 = 706$ ✓ **book ans (d)**

Q (EN 293): $a^2 + b^2 = 80, a - b = 4 \Rightarrow ab$

$\frac{80-16}{2} = 32$ ✓ **book ans (d)**

BOOK KE SAWAAL (88) EN 49, 50, 64, 66–71, 75, 76, 78, 79, 93–95, 97, 98, 100–102, 104, 106, 109, 110, 112–132, 244, 246, 263, 290, 292, 293 · **HI** 29, 48–52, 62–65, 75, 76, 78, 79, 81, 83, 87–91, 93, 94, 97, 103, 104, 107–116

AL-5 — TEEN-VARIABLE IDENTITY PACK (a + b + c)**PEHCHAN – Question mein ye dikhe**

$a + b + c$ and $ab + bc + ca \Rightarrow a^3 + b^3 + c^3 - 3abc, a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc, a(a + b + c) = 45 \dots \Rightarrow a^2 + b^2 + c^2, a^2 + b^2 + c^2 = ab + bc + ca, \text{sum of squares} = 0, (a - b)^3 + (b - c)^3 + (c - a)^3$

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$a^3 + b^3 + c^3 - 3abc = (a + b + c) [a^2 + b^2 + c^2 - (ab + bc + ca)]$$

- $a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc$ (jaise $(a-b) + (b-c) + (c-a) = 0$).
- **Squares ka jod = 0** $\Rightarrow$ har bracket 0 ($x^2 + y^2 - 4x + 8y + 20 = 0 \Rightarrow (x-2)^2 + (y+4)^2 = 0$).
- $a^2 + b^2 + c^2 = ab + bc + ca \Rightarrow a = b = c$.
- $x(x+y+z) = p, y(\dots) = q, z(\dots) = r \Rightarrow$ **sab jodo:** $(x + y + z)^2 = p + q + r$.
- Value daalne wale sawal: $\frac{a^3+b^3+c^3-3abc}{ab+bc+ca-a^2-b^2-c^2} = -(a + b + c)$.

TERMS – Har letter ka matlab

$S = a + b + c \cdot Q = ab + bc + ca.$

SOLVED – Asli book questions (verified)**Q (EN 65):** $p + q + r = 13$, $pq + qr + rp = 30$ $13(169 - 90) = 1027$ ✓ **book ans (c)****Q (EN 87):** $a + b + c = 9$, $a^2 + b^2 + c^2 = 29$ $Q = 26 \Rightarrow 9(29 - 26) = 27$ ✓ **book ans (b)****Q (EN 80):** $a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = ? \times abc$ 3 ✓ **book ans (c)****Q (EN 81):** $a(a + b + c) = 45$, $b(\dots) = 75$, $c(\dots) = 105$ $S = 15 \Rightarrow 3, 5, 7 \Rightarrow 83$ ✓ **book ans (a)****Q (EN 92):** $a^2 + b^2 + c^2 + 3 = 2(a + b + c)$ har $(a - 1)^2 = 0 \Rightarrow 3$ ✓ **book ans (d)****Q (HI 20):** $x^2 + y^2 - 4x + 8y + 20 = 0 \Rightarrow x^2 + y^2$ $x^2 + y^2 - 4x + 8y + 20 = 0 \Rightarrow 20$ ✓ **book ans (c)****Q (HI 46):** $x + y + z = 11$, $xy + yz + zx = 42$ $121 - 84 = 37$ ✓ **book ans (b)****Q (EN 89):** $a^5, b^4, c^8 \Rightarrow$ bada fraction $-(5 + 4 + 8) = -17$ ✓ **book ans (c)****Q (EN 245):** $x(x+y+z) = 30$, $y(\dots) = 64$, $z(\dots) = 50 \Rightarrow 2(x+y+z)$ $\sqrt{144} = 12 \Rightarrow 24$ ✓ **book ans (c)****BOOK KE SAWAAL (37)** EN 62, 65, 74, 77, 80–92, 96, 99, 105, 107, 108, 111, 245, 250, 256 · **HI** 20, 37, 46, 56, 74, 84, 95, 96, 105, 106, 244**AL-6 — REMAINDER / FACTOR THEOREM****PEHCHAN – Question mein ye dikhe**

remainder when $P(x)$ is divided by $(x - 2) / (3x + 2) / (x^3 - 2)$, same remainder $\Rightarrow a$, $(x - 3)$ is a factor $\Rightarrow k$, divisible by $x + 2$, two remainder conditions $\Rightarrow a + b$; शेषफल, विभाजित, गुणनखंड

$$P(x) \div (x - a) \Rightarrow \text{Remainder} = P(a) \quad (x - a) \text{ factor} \iff P(a) = 0$$

- $(px + q)$ se bhaag $\Rightarrow x = -\frac{q}{p}$ daalo.
- $(x^3 - 2)$ se bhaag $\Rightarrow$ har jagah $x^3 = 2$ daalo.
- Do shartein $(a, b) \Rightarrow$ dono equations likh ke **jodo** ($a + b$ poocha ho to seedha).
- $(x^2 - 1)$ factor $\Rightarrow P(1) = 0$ aur $P(-1) = 0$.

TERMS – Har letter ka matlab $P(x) = \text{polynomial} \cdot a = \text{divisor ka zero.}$

SOLVED – Asli book questions (verified)**Q (EN 136):** $27x^3 - 9x^2 + 3x - 8 \div (3x + 2)$ $x = -2/3 \Rightarrow -8 - 4 - 2 - 8 = -22$ ✓ **book ans (a)****Q (EN 145):** $4x^3 - 2x^2 + 5x - 8 \div (x - 2)$ $32 - 8 + 10 - 8 = 26$ ✓ **book ans (b)****Q (EN 137):** $2x^m + x^3 - 3x^2 - 26 \div (x - 2) = 994$ $2^{m+1} = 1024 \Rightarrow 9$ ✓ **book ans (b)****Q (EN 139):** $4x^6 - 5x^3 - 3 \div (x^3 - 2)$ $16 - 10 - 3 = 3$ ✓ **book ans (a)****Q (EN 144):** $3x^2 + ax + 4$ divisible by $x - 8$ $192 + 8a + 4 = 0 \Rightarrow -24.5$ ✓ **book ans (a)****Q (EN 134):** same remainder by $(x + 1)$ $-4 + a + 3 + 1 = 1 - 1 - 1 + 6 \Rightarrow 5$ ✓ **book ans (c)****Q (EN 161):** $x^4 - 2x^3 + 3x^2 - x + k$, multiple of $(x - 3)$ $81 - 54 + 27 - 3 + k = 0 \Rightarrow -51$ ✓ **book ans (b)****Q (EN 153):** $x^2 + ax + b \div (x - 4) = 32$, $x^2 + bx + a \div (x - 4) = 35$ jodo: $32 + 5(a + b) = 67 \Rightarrow 7$ ✓ **book ans (d)****BOOK KE SAWAAL (41)** EN 134–154, 156–161, 163, 164, 168 · **HI** 117–127**AL-7 — EXPAND / SIMPLIFY / FACTORISE****PEHCHAN – Question mein ye dikhe**simplify $x(2x - 5) + 6(x^2 - 4) + 18$, expand $(c - 3)^3$, factorise $x^2 + 11xy + 24y^2$, square root of $(x^2 + 4x + 4)(x^2 + 6x + 9)$, $9x = 55^2 - 35^2$, coefficient of a term, which option is equivalent; सरल कीजिए, विस्तारित, गुणनखंड

Identity	Khula roop
$(a \pm b)^2$	$a^2 \pm 2ab + b^2$
$a^2 - b^2$	$(a - b)(a + b)$
$(a \pm b)^3$	$a^3 \pm 3a^2b + 3ab^2 \pm b^3$
$a^3 \pm b^3$	$(a \pm b)(a^2 \mp ab + b^2)$

- **Factorise** $x^2 + px + q$: aise do number jinka jod p, guna q (middle-term split).
- Option wale sawal: $x = 1$ (ya 2) daal ke dono taraf ki value milao — 10 second.
- Like terms (same power) hi jodte / ghatate hain.

TERMS – Har letter ka matlab

like terms = same variable, same power.

SOLVED – Asli book questions (verified)**Q (EN 133):** $x(2x - 5) + 6(x^2 - 4) + 18$ $8x^2 - 5x - 6$ ✓ **book ans (c)****Q (EN 155):** $x^2 + 11xy + 24y^2$ $8 \times 3 = 24, 8 + 3 = 11 \Rightarrow (x + 8y)(x + 3y)$ ✓ **book ans (c)****Q (EN 166):** $x^2 + x - 42$ $(x - 6)(x + 7)$ ✓ **book ans (c)****Q (HI 42):** $9x = 55^2 - 35^2$ $\frac{90 \cdot 20}{9} = 200$ ✓ **book ans (b)****Q (EN 294):** $40x^2 = 334^2 - 134^2$ $\frac{468 \cdot 200}{40} = 2340$ ✓ **book ans (a)****Q (HI 21):** $x = 2 \Rightarrow (-x)^2 \cdot (-x)^3$ $-x^5 = -32$ ✓ **book ans (b)****Q (HI 268):** $(3x - 2y)^2(x^2 + y^3)^2$ mein x^4y^3 ka coefficient $9x^2 \cdot 2x^2y^3 = 18$ ✓ **book ans (b)****BOOK KE SAWAAL (71)** EN 48, 63, 72, 73, 103, 133, 155, 162, 165–167, 169–171, 247–249, 251, 253, 257, 261, 266, 294 · **HI** 13, 18, 21–27, 38–40, 42, 44, 47, 53–55, 57–61, 66–70, 72, 73, 77, 80, 82, 85, 86, 92, 98–101, 129, 138, 141, 143, 144, 234, 268 | Ch 28 (Vividh): EN 10**AL-8 — QUADRATIC MACHINE (Sum – Product – D)****PEHCHAN – Question mein ye dikhe**

sum / product of roots, nature of roots, equal roots $\Rightarrow$ k, equation whose roots are -2 and 3 , other root when one is given, roots in ratio $1 : 2$, $\alpha - \beta = 1 \Rightarrow m$, $\alpha^2 + \beta^2$, $\alpha/\beta + \beta/\alpha$, roots α^2 , $1/\alpha$, no real solution $\Rightarrow$ range of k, quadratic inequality, common root; द्विघात समीकरण, मूल, विविक्तकर

$$ax^2 + bx + c = 0 : \quad \alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a} \quad D = b^2 - 4ac$$

D	Roots
> 0	real, alag
$= 0$	barabar ($b^2 = 4ac$)
< 0	real nahi

- **Equation from roots:** $x^2 - (\text{jod})x + \text{guna} = 0$.
- $\alpha^2 + \beta^2 = S^2 - 2P$, $(\alpha - \beta)^2 = S^2 - 4P$, $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{S^2 - 2P}{P}$, $\alpha^3 + \beta^3 = S^3 - 3PS$.
- Ek root diya $\Rightarrow$ doosra = $S - \text{diya}$ (ya $P \div \text{diya}$). Roots ratio $m : n \Rightarrow \frac{S^2}{P} = \frac{(m+n)^2}{mn}$.
- $a + b + c = 0 \Rightarrow$ roots 1 aur c/a . $ax^2 + bx + c < 0$ ($a > 0$) $\Rightarrow$ x dono roots ke beech.
- Highest power 2 ho (simplify ke baad) tabhi quadratic.

TERMS – Har letter ka matlab $S = \alpha + \beta$, $P = \alpha\beta$, $D = \text{discriminant}$.

SOLVED – Asli book questions (verified)**Q (EN 176):** $2y^2 - 6y - 7 = 0$ ka jod

$$\frac{6}{2} = 3 \checkmark \text{ book ans (c)}$$

Q (EN 180): $x^2 - kx + 169$ equal roots

$$k^2 = 676 \Rightarrow \pm 26 \checkmark \text{ book ans (b)}$$

Q (EN 178): $2x^2 - 3x + a = 0$, roots 1 : 2

$$3\alpha = \frac{3}{2}, 2\alpha^2 = \frac{a}{2} \Rightarrow a = 1 \checkmark \text{ book ans (b)}$$

Q (EN 202): $x^2 - 19x + 88$, ek root 8

$$19 - 8 = 11 \checkmark \text{ book ans (b)}$$

Q (EN 198): $x^2 - 8x + k$: squares ka jod 40

$$64 - 2k = 40 \Rightarrow 12 \checkmark \text{ book ans (a)}$$

Q (EN 199): $x^2 - 5x + m$, $\alpha - \beta = 1$

$$25 - 4m = 1 \Rightarrow 6 \checkmark \text{ book ans (b)}$$

Q (EN 214): $3x^2 - 5x + 2 \Rightarrow a/b + b/a$

$$\frac{25/9 - 4/3}{2/3} = \frac{13}{6} \checkmark \text{ book ans (b)}$$

Q (EN 191): roots -2 aur 3

$$x^2 - x - 6 = 0 \checkmark \text{ book ans (b)}$$

Q (EN 209): $x^2 + kx + k = 0$ ka real hal nahi

$$k^2 - 4k < 0 \Rightarrow 0 < k < 4 \checkmark \text{ book ans (b)}$$

Q (HI 209): $x^2 - 11x + 18 \Rightarrow \alpha^2 + \beta^2$

$$121 - 36 = 85 \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (208) EN 172–186, 188–202, 204–243, 252 · **HI** 28, 30, 32, 102, 128, 130–137, 139, 140, 142, 145–233, 235–243, 245–267**AL-9 — VENN MACHINE (sets)****PEHCHAN – Question mein ye dikhe**

students speaking English / Hindi / Bengali, failed in Maths / Physics / Chemistry, like tea & coffee, neither A nor B, only English, cricket & kabaddi, survey of newspapers, total population

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A \cup B \cup C) = A + B + C - (AB + BC + CA) + ABC$$

- **Neither** = Total – $(A \cup B)$. **Only A** = $A - (A \cap B)$.
- % wale sawal: total 100 maano; failed in both se passed = $100 - (A + B - \text{both})$.
- Sirf / only numbers diye hon $\Rightarrow$ Venn ke chhote hisse seedhe jodo.

TERMS – Har letter ka matlab $A \cap B$ = dono · $A \cup B$ = kam se kam ek.**SOLVED – Asli book questions (verified)****Q (EN 283):** 14 ne A, 13 ne B, 2 ne dono

$$14 + 13 - 2 = 25 \checkmark \text{ book ans (d)}$$

Q (EN 282): 15 English, 5 dono $\Rightarrow$ only English

$$15 - 5 = 10 \checkmark \text{ book ans (b)}$$

Q (EN 289): 5000 mein 3200 tea, 2500 coffee, 1500 dono $\Rightarrow$ neither

$$5000 - 4200 = 800 \checkmark \text{ book ans (a)}$$

Q (EN 288): 5000; coffee 3500, tea 2500, dono 2000 $\Rightarrow$ neither

$$5000 - 4000 = 1000 \checkmark \text{ book ans (b)}$$

Q (EN 277): 500: cricket 350, kabaddi 125, neither 75 $\Rightarrow$ dono %

$$475 - 425 = 50 \Rightarrow 10\% \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (24) EN 267–289 | Ch 28 (Vividh): EN 75

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
AL-1	SERIES MACHINE (AP / GP)	35	12
AL-2	HCF / LCM OF EXPRESSIONS	12	4
AL-3	LINEAR PAIR (do seedhi equations)	17	5
AL-4	DO-VARIABLE + x + 1/x MACHINE	52	36
AL-5	TEEN-VARIABLE IDENTITY PACK (a + b + c)	26	11
AL-6	REMAINDER / FACTOR THEOREM	30	11
AL-7	EXPAND / SIMPLIFY / FACTORISE	23	47
AL-8	QUADRATIC MACHINE (Sum – Product – D)	71	137
AL-9	VENN MACHINE (sets)	23	0
→ IS-3	SURD SIMPLIFY + LIKE-SURD MACHINE (Ch 3)	0	1
→ IS-6	POWER LAWS (ghatank ke niyam) (Ch 3)	4	2
→ SIM-7	RATIO SUBSTITUTION MACHINE (Ch 4)	1	2
Total (book ke sawaal)		294/294 ✓	268/268 ✓
→ IS-3: HI 71 · → IS-6: EN 254, 255, 258, 291 HI 19, 41 · → SIM-7: EN 264 HI 43, 45			

Chapter 23

Trigonometry (Trikonmiti) त्रिकोणमिति

8 formulas · English book 221 sawaal · Hindi book 124 sawaal · 62 solved examples ✓

TG-1 — IDENTITY TRIANGLE + “ $\theta = 45^\circ$ DAALO” TRICK

PEHCHAN – Question mein ye dikhe

simplify / find the value of $(1 + \sin A)/(1 - \sin A)$, $\sec A(1 - \cos A)(\operatorname{cosec} A + \cot A)$, $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2$, $(1 - \cos^2 \theta)(\cot^2 \theta + 1)$, $\tan \theta + 1/\tan \theta$, which identity is not true; सरल कीजिए, का मान ज्ञात कीजिए

$$\sin^2 \theta + \cos^2 \theta = 1 \quad \sec^2 \theta - \tan^2 \theta = 1 \quad \csc^2 \theta - \cot^2 \theta = 1$$

- **10-second trick:** jahan answer θ ka expression ho, $\theta = 45^\circ$ ($\sin = \cos = 1/\sqrt{2}$, $\tan = 1$) ya $\theta = 30^\circ$ daalo aur option se milao. Do option same aayein to doosra angle daalo.
- $\frac{1+\sin A}{1-\sin A} = (\sec A + \tan A)^2$, $\frac{1-\cos A}{1+\cos A} = (\csc A - \cot A)^2$.
- $\tan \theta + \cot \theta = \sec \theta \csc \theta = \frac{1}{\sin \theta \cos \theta}$; $\sin^6 + \cos^6 = 1 - 3 \sin^2 \cos^2$.
- Sab ko $\sin$, $\cos$ mein badlo $\Rightarrow$ cancel.

TERMS – Har letter ka matlab

$\theta = \text{kon} \cdot \operatorname{cosec} = \csc$.

SOLVED – Asli book questions (verified)

Q (EN 14): $1/(1 + \sin \theta) + 1/(1 - \sin \theta)$

$\frac{2}{\cos^2 \theta} = 2 \sec^2 \theta$ ✓ **book ans (c)**

Q (EN 105): $(1 - \cos^2 \theta)(\cot^2 \theta + 1) - 1$

$\sin^2 \theta \csc^2 \theta - 1 = 0$ ✓ **book ans (a)**

Q (EN 81): $\sin^2 + \cos^2 - (\sec^2 - \tan^2) + \tan \theta \cos \theta - \sin \theta$

$1 - 1 + \sin - \sin = 0$ ✓ **book ans (d)**

Q (EN 80): $2\sec^2 A + 4\operatorname{cosec}^2 A - 2\tan^2 A - 4\cot^2 A$

$2 + 4 = 6$ ✓ **book ans (d)**

Q (HI 22): $(1 + \cot^2)(1 + \cos)(1 - \cos) + (1 - \sin)(1 + \sin)(1 + \tan^2)$

$1 + 1 = 2$ ✓ **book ans (d)**

Q (HI 37): $(1 + \tan^2 \theta)(1 - \sin \theta)(1 + \sin \theta)$

$\sec^2 \theta \cos^2 \theta = 1$ ✓ **book ans (b)**

Q (EN 3): $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2$

$\theta = 45^\circ$: $4.5 + 4.5 = 9 = 7 + 1 + 1 \Rightarrow 7 + \cot^2 \theta + \tan^2 \theta$ ✓ **book ans (a)**

Q (HI 36): $\tan^6 A + 3\tan^2 A \sec^2 A + 1$

$(\sec^2 - \tan^2 = 1)$ ka cube $\Rightarrow \sec^6 A$ ✓ **book ans (c)**

BOOK KE SAWAAL (73) EN 1, 3–7, 10, 14, 15, 22, 23, 26, 29, 33, 80, 81, 85–87, 89, 99, 102, 105, 106, 108–111 · HI 2–6, 8–15, 18, 19, 21, 22, 24, 26–32, 35–39, 42, 45, 49–51, 53, 55–58, 63, 89, 90, 94, 99

TG-2 — PAIR TRICKS ($\sec + \tan = k$, $\tan + \cot = k \dots$)

PEHCHAN – Question mein ye dikhe

$\sec \theta + \tan \theta = k \Rightarrow \sec \theta - \tan \theta / \sec \theta / \sin \theta$, $\operatorname{cosec} A - \cot A = 3$, $\tan \theta + \cot \theta = 5 \Rightarrow \tan^2 + \cot^2$, $\tan^4 \theta + \tan^2 \theta = 11 \Rightarrow \sec^4 \theta - \sec^2 \theta$, $\sin \theta + \operatorname{cosec} \theta = 2$, $a \cos \theta - b \sin \theta = c$

$$\sec \theta + \tan \theta = k \Rightarrow \sec \theta - \tan \theta = \frac{1}{k}, \quad \sec \theta = \frac{k + \frac{1}{k}}{2}, \quad \tan \theta = \frac{k - \frac{1}{k}}{2}, \quad \sin \theta = \frac{k^2 - 1}{k^2 + 1}$$

- Same cosec $\pm$ cot ke saath (cosec = sec ki jagah, cot = tan ki jagah, cos = sin ki jagah).
- $\tan \theta + \cot \theta = k \Rightarrow \tan^2 + \cot^2 = k^2 - 2$ (x + 1/x wala machine).
- $\tan^4 + \cot^4 = \sec^4 - \sec^2 \cdot \cot^4 + \cot^2 = \csc^4 - \csc^2$ (answer wahi number).
- $\sin \theta + \csc \theta = 2 \Rightarrow \theta = 90^\circ$ (sin = 1). $a \cos \theta - b \sin \theta = c \Rightarrow a \sin \theta + b \cos \theta = \pm \sqrt{a^2 + b^2 - c^2}$.
- $\frac{\sec + \tan}{\sec - \tan} = \frac{p}{q} \Rightarrow \sin \theta = \frac{p-q}{p+q}$.

TERMS – Har letter ka matlab

k = diya hua jod.

SOLVED – Asli book questions (verified)

Q (EN 41): $\sec + \tan = 1.25 \Rightarrow \sec - \tan$

$$\frac{1}{1.25} = 0.8 \quad \checkmark \text{ book ans (b)}$$

Q (EN 37): $\sec + \tan = 3.2 \Rightarrow \sec$

$$\frac{3.2+0.3125}{2} = 1.75625 \quad \checkmark \text{ book ans (c)}$$

Q (EN 36): $\tan + \cot = 5 \Rightarrow \tan^2 + \cot^2$

$$25 - 2 = 23 \quad \checkmark \text{ book ans (a)}$$

Q (EN 34): $\sec^4 - \sec^2 = 3 \Rightarrow \tan^4 + \tan^2$

wahi 3 $\checkmark$ book ans (c)

Q (EN 57): $\operatorname{cosec} A - \cot A = 3 \Rightarrow \operatorname{cosec} A + \cot A$

$$\frac{1}{3} \quad \checkmark \text{ book ans (a)}$$

Q (EN 13): $(\sec + \tan)/(\sec - \tan) = 5/3 \Rightarrow \sin$

$$\frac{5-3}{5+3} = \frac{1}{4} \quad \checkmark \text{ book ans (c)}$$

Q (EN 104): $\sin \theta + \operatorname{cosec} \theta = 2 \Rightarrow \sin^8 + \operatorname{cosec}^8$

$$\sin = 1 \Rightarrow 2 \quad \checkmark \text{ book ans (b)}$$

Q (HI 23): $\tan + \cot = 14 \Rightarrow 7\tan^2 + 7\cot^2$

$$7 \times 194 = 1358 \quad \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (41) EN 13, 30, 31, 34–37, 39, 41–44, 46–48, 54, 57, 60, 78, 90, 93–95, 103, 104 · **HI** 1, 7, 17, 23, 34, 54, 68–71, 73, 75–79

TG-3 — VALUE TABLE + EK RATIO SE SAB

PEHCHAN – Question mein ye dikhe

$\tan \theta = 4/3 \Rightarrow \sin \theta + \cos \theta$, $5 \tan \theta = 4 \Rightarrow (3 \sin - 2 \cos)/(2 \sin + 3 \cos)$, $\sin x = 4/5 \Rightarrow \operatorname{cosec} + \cot$, value of $\sin^2 60^\circ + \cos^2 30^\circ + \dots$, find θ if $\sin^2 \beta = \sin 30^\circ$, $\sin(A + B) = \sqrt{3}/2$ & $\cos(A - B) = \sqrt{3}/2 \Rightarrow A, B$; मानक कोण

θ	0°	30°	45°	60°	90°
sin	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$	1
cos	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0
tan	0	$1/\sqrt{3}$	1	$\sqrt{3}$	∞

- **Ek ratio diya** ($\tan = p/b$): triangle banao, $h = \sqrt{p^2 + b^2} \Rightarrow \sin = p/h$, $\cos = b/h$. Triplets: 3-4-5, 5-12-13, 7-24-25, 8-15-17.
- $\frac{a \sin + b \cos}{c \sin + d \cos} \Rightarrow$ upar-neeche cos se bhaag: $\frac{a \tan + b}{c \tan + d}$.
- $\operatorname{cosec} = 1/\sin$, $\sec = 1/\cos$, $\cot = 1/\tan$. Angle poocha $\Rightarrow$ table se ulta padho.

TERMS – Har letter ka matlab

p = perpendicular · b = base · h = hypotenuse.

SOLVED – Asli book questions (verified)**Q (EN 66):** $\tan \theta = 4/3 \Rightarrow \sin + \cos$

$$\frac{4}{5} + \frac{3}{5} = \frac{7}{5} \quad \checkmark \text{ book ans (b)}$$

Q (EN 72): $5 \tan \theta = 4 \Rightarrow (3 \sin - 2 \cos)/(2 \sin + 3 \cos)$

$$\frac{12/5-2}{8/5+3} = \frac{2}{23} \quad \checkmark \text{ book ans (b)}$$

Q (EN 67): $4 \cot \theta = 5 \Rightarrow (5 \sin + 3 \cos)/(5 \sin - 3 \cos)$

$$\tan = 4/5 \Rightarrow \frac{4+3}{4-3} = 7 \quad \checkmark \text{ book ans (c)}$$

Q (EN 71): $\sin \theta = 5/13 \Rightarrow \cos \theta$

$$\frac{12}{13} \quad \checkmark \text{ book ans (b)}$$

Q (EN 70): $\sin x = 4/5 \Rightarrow \operatorname{cosec} x + \cot x$

$$\frac{5}{4} + \frac{3}{4} = 2 \quad \checkmark \text{ book ans (c)}$$

Q (EN 169): $\sin^2 60 + \cos^2 30 + \cot^2 45 + \sec^2 60$

$$3/4 + 3/4 + 1 + 4 = 13/2 \quad \checkmark \text{ book ans (c)}$$

Q (EN 166): $\tan^2 60 - 2 \tan^2 45 - \cot^2 30 + 2 \sin^2 30 + 3/4 \operatorname{cosec}^2 45$

$$3 - 2 - 3 + 1/2 + 3/2 = 0 \quad \checkmark \text{ book ans (b)}$$

Q (EN 117): $\sin^2 \beta - \sin 30^\circ = 0$

$$\sin^2 \beta = 1/2 \Rightarrow 45^\circ \quad \checkmark \text{ book ans (a)}$$

Q (EN 212): $\cos(x - y) = \sqrt{3}/2, \sin(x + y) = 1$

$$x - y = 30, x + y = 90 \Rightarrow 60^\circ, 30^\circ \quad \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (79) EN 2, 8, 9, 18, 27, 28, 38, 40, 45, 49, 51–53, 56, 58, 59, 61–63, 65–72, 74, 76, 91, 92, 96, 101, 113–115, 117, 120, 130, 131, 133, 140, 141, 144–146, 149–151, 166, 168, 169, 212 · **HI** 20, 33, 44, 46–48, 60, 61, 65, 72, 74, 80–83, 91–93, 101, 106–109, 112, 114, 115**TG-4 — COMPLEMENTARY + ALLIED ANGLES (90°, 180°, 360°)****PEHCHAN – Question mein ye dikhe** $\tan 1^\circ \tan 2^\circ \dots \tan 89^\circ, \cos 1^\circ \dots \cos 179^\circ, \sin^2 5^\circ + \sin^2 10^\circ + \sin^2 80^\circ + \sin^2 85^\circ, \sin(48^\circ + \theta) - \cos(42^\circ - \theta), \sec 4A = \operatorname{cosec}(3A - 50^\circ), \tan 2\theta = \cot(\theta + 6^\circ), \sec 33^\circ \operatorname{cosec} 57^\circ \dots, \cos(-780^\circ), \sin 780^\circ, \cos 168^\circ$; पूरक कोण

$$A + B = 90^\circ \Rightarrow \sin A = \cos B, \tan A \tan B = 1, \sec A = \csc B, \sin^2 A + \sin^2 B = 1$$

- **Jodi banao:** $\tan 1^\circ \cdot \tan 89^\circ = 1 \Rightarrow$ poora guna 1; product mein $\cos 90^\circ / \tan 0^\circ$ aaye $\Rightarrow 0$.
- $\sec x = \operatorname{cosec} y, \tan x = \cot y, \sin x = \cos y \Rightarrow x + y = 90^\circ$.
- **Allied:** $180^\circ - \theta$: sin same, cos/tan ulta sign; $180^\circ + \theta$: tan same; $360^\circ + \theta$: sab same; $\cos(-\theta) = \cos \theta$. $780^\circ = 2 \cdot 360^\circ + 60^\circ$.
- $\cos(180^\circ - \theta) = -\cos \theta \Rightarrow \cos 12^\circ + \cos 168^\circ = 0$.

TERMS – Har letter ka matlabcomplementary = jod 90° · supplementary = jod 180° .

SOLVED – Asli book questions (verified)**Q (EN 155):** $\tan 1^\circ \cdot \tan 2^\circ \cdots \tan 89^\circ$ jodiyen = 1 ✓ **book ans (c)****Q (EN 158):** $\cos 1^\circ \cdot \cos 2^\circ \cdots \cos 179^\circ$ $\cos 90^\circ = 0 \Rightarrow 0$ ✓ **book ans (a)****Q (EN 153):** $\sin^2 5^\circ + \sin^2 10^\circ + \sin^2 80^\circ + \sin^2 85^\circ$ $1 + 1 = 2$ ✓ **book ans (c)****Q (EN 154):** $3 \tan 25^\circ \tan 35^\circ \tan 45^\circ \tan 55^\circ \tan 65^\circ$ $3 \times 1 = 3$ ✓ **book ans (b)****Q (EN 161):** $5 \sin 14^\circ \sec 76^\circ + 3 \cot 15^\circ \cot 75^\circ + 2 \tan 45^\circ$ $5 + 3 + 2 = 10$ ✓ **book ans (b)****Q (EN 88):** $\sec 4A = \operatorname{cosec}(3A - 50^\circ) \Rightarrow A + 75^\circ$ $7A - 50 = 90 \Rightarrow A = 20 \Rightarrow 95^\circ$ ✓ **book ans (a)****Q (EN 119):** $\sec 33^\circ \operatorname{cosec} 57^\circ - \cot 57^\circ \tan 33^\circ + \cot 47^\circ \cot 43^\circ$ $(\sec^2 - \tan^2) + 1 = 2$ ✓ **book ans (c)****Q (EN 132):** $\tan 2\theta = \cot(\theta + 6^\circ)$ $3\theta + 6 = 90 \Rightarrow 28^\circ$ ✓ **book ans (d)****Q (EN 163):** $\sin 780^\circ \sin 480^\circ + \cos 120^\circ \sin 30^\circ$ $\frac{3}{4} - \frac{1}{4} = \frac{1}{2}$ ✓ **book ans (d)****Q (HI 95):** $\operatorname{cosec}(47^\circ + \theta) - \sec(43^\circ - \theta) + \tan(61^\circ + \theta) - \cot(29^\circ - \theta) + \dots$ har jodi 0 $\Rightarrow 0$ ✓ **book ans (d)****BOOK KE SAWAAL (40)** EN 24, 75, 88, 116, 118, 119, 121, 124, 127–129, 132, 138, 143, 153–163, 165, 167, 170 · **HI** 16, 43, 52, 95–98, 100, 103–105, 113**TG-5 — COMPOUND + DOUBLE ANGLE****PEHCHAN – Question mein ye dikhe**

$\sin 25^\circ \cos 35^\circ + \cos 25^\circ \sin 35^\circ$, $\tan 15^\circ$, $\tan 75^\circ$, $\sin 15^\circ$, $\operatorname{cosec} 15^\circ$, $(1 + \tan A)(1 + \tan B) = 2$, $\tan \alpha = \frac{1}{2}$ & $\tan \beta = \frac{1}{3} \Rightarrow \alpha + \beta$, $\tan 100^\circ + \tan 125^\circ + \tan 100^\circ \tan 125^\circ$, $\sqrt{\frac{1 + \cos 2A}{1 - \cos 2A}}$, $\cos 2\theta = \sin \theta$, triangle mein $\tan A + \tan B + \tan C$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B \quad \cos(A \pm B) = \cos A \cos B \mp \sin A \sin B \quad \tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

- Yaad: $\tan 15^\circ = 2 - \sqrt{3}$, $\tan 75^\circ = 2 + \sqrt{3}$, $\sin 15^\circ = \cos 75^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}$.
- $\cos 2\theta = 1 - 2\sin^2 \theta = 2\cos^2 \theta - 1$, $\sin 2\theta = 2\sin \theta \cos \theta$, $\frac{1 - \cos 2A}{1 + \cos 2A} = \tan^2 A$.
- **A + B = 45°** $\Leftrightarrow (1 + \tan A)(1 + \tan B) = 2 \Leftrightarrow \tan A + \tan B + \tan A \tan B = 1$ (225° par bhi).
- Triangle: $\tan A + \tan B + \tan C = \tan A \tan B \tan C$.

TERMS – Har letter ka matlab

A, B = do kon.

SOLVED – Asli book questions (verified)**Q (EN 77):** $\tan 15^\circ \cot 75^\circ + \tan 75^\circ \cot 15^\circ$

$$(2 - \sqrt{3})^2 + (2 + \sqrt{3})^2 = 14 \quad \checkmark \text{ book ans (d)}$$

Q (EN 50): $(1 + \tan A)(1 + \tan B) = 2 \Rightarrow \tan(A + B)$

$$A + B = 45^\circ \Rightarrow 1 \quad \checkmark \text{ book ans (a)}$$

Q (EN 64): $\tan \alpha = \frac{1}{2}, \tan \beta = \frac{1}{3} \Rightarrow \alpha + \beta$

$$\frac{\frac{5/6}{1-1/6}}{1-1/6} = 1 \Rightarrow 45^\circ \quad \checkmark \text{ book ans (d)}$$

Q (EN 122): $\tan 100^\circ + \tan 125^\circ + \tan 100^\circ \tan 125^\circ$

$$225^\circ = 180^\circ + 45^\circ \Rightarrow 1 \quad \checkmark \text{ book ans (d)}$$

Q (EN 139): $\cos 12^\circ + \cos 84^\circ + \cos 168^\circ + \cos 96^\circ$

$$\text{joda-joda cancel} \Rightarrow 0 \quad \checkmark \text{ book ans (b)}$$

Q (EN 164): $\cos 5^\circ + \cos 24^\circ + \cos 175^\circ + \cos 204^\circ + \cos 300^\circ$

$$\text{sirf } \cos 300^\circ = \frac{1}{2} \text{ bacha} \quad \checkmark \text{ book ans (a)}$$

Q (EN 25): $\sqrt{\frac{1 + \cos 2A}{1 - \cos 2A}}$

$$\sqrt{\cot^2 A} = \cot A \quad \checkmark \text{ book ans (d)}$$

Q (EN 126): $\sin 25^\circ \cos 35^\circ + \cos 25^\circ \sin 35^\circ$

$$\sin 60^\circ = \sqrt{3}/2 \quad \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (35) EN 11, 12, 16, 17, 19–21, 25, 50, 55, 64, 73, 77, 107, 112, 122, 123, 125, 126, 134–137, 139, 142, 147, 148, 152, 164 · **HI** 66, 67, 84, 102, 110, 111**TG-6 — a sin² + b cos² MACHINE (min / max / angle)****PEHCHAN – Question mein ye dikhe**least / minimum value of $2 \sin^2 \theta + 3 \cos^2 \theta$, $3 \sin^2 \theta + 7 \cos^2 \theta = 6 \Rightarrow \theta$, $p \sin^2 \beta + q \cos^2 \beta = r \Rightarrow \cot^2 \beta$, $3 \sec^2 x - 2 \tan^2 x = 6 \Rightarrow x$

$$a \sin^2 \theta + b \cos^2 \theta : \quad \min = \text{chhota}(a, b), \quad \max = \text{bada}(a, b)$$

$$a \sin^2 \theta + b \cos^2 \theta = r \Rightarrow \cos^2 \theta = \frac{r - a}{b - a}, \quad \tan^2 \theta = \frac{r - b}{a - r}$$

- $a \sec^2 - b \tan^2 = r \Rightarrow \sec^2 = 1 + \tan^2$ daalo: $\tan^2 = \frac{r-a}{a-b}$.
- $a \sin \theta + b \cos \theta$: $\max \sqrt{a^2 + b^2}$, $\min -\sqrt{a^2 + b^2}$.

TERMS – Har letter ka matlab

a, b = coefficients · r = diya hua value.

SOLVED – Asli book questions (verified)**Q (EN 97):** least of $2 \sin^2 + 3 \cos^2$

$$2 \quad \checkmark \text{ book ans (d)}$$

Q (EN 98): minimum of $9 \sin^2 + 10 \cos^2$

$$9 \quad \checkmark \text{ book ans (c)}$$

Q (EN 83): $3 \sin^2 \theta + 7 \cos^2 \theta = 6 \Rightarrow \theta$

$$\cos^2 = \frac{3}{4} \Rightarrow 30^\circ \quad \checkmark \text{ book ans (b)}$$

Q (EN 219): $3 \sec^2 x - 2 \tan^2 x = 6$

$$\tan^2 = 3 \Rightarrow 60^\circ \quad \checkmark \text{ book ans (a)}$$

Q (EN 221): $8 \sec^2 x - 7 \tan^2 x = 11$

$$\tan^2 = 3 \Rightarrow 60^\circ \quad \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (14) EN 82–84, 97, 98, 100, 219, 221 · **HI** 25, 41, 59, 64, 86, 88

TG-7 — HEIGHTS & DISTANCES

PEHCHAN – Question mein ye dikhe

angle of elevation / depression of top of a tower / building / tree, shadow, ladder against a wall, two points with complementary angles, observer's height, tree broken by wind, aeroplane / balloon, two towers; उन्नयन कोण, अवनमन कोण, मीनार, छाया

$$\tan \theta = \frac{\text{Height}}{\text{Base}} \quad 30^\circ : h = \frac{d}{\sqrt{3}} \quad 45^\circ : h = d \quad 60^\circ : h = d\sqrt{3}$$

- **Complementary angles** (a, b doori se) $\Rightarrow h = \sqrt{ab}$.
- $30^\circ \rightarrow 60^\circ$ (ya ulta) chala d $\Rightarrow h = \frac{\sqrt{3}}{2}d$; shadow ka farak d $\Rightarrow h = \frac{d}{\cot \alpha - \cot \beta}$.
- Observer ki height jodo / platform ki height ghatao. Ladder: Pythagoras ($l^2 = h^2 + b^2$).
- Toota ped: tukda + khada hissa = poori height; 60° par $\frac{\text{khada}}{\text{tukda}} = \sin 60^\circ$.

TERMS – Har letter ka matlab

h = unchai · d = zameen par doori · θ = kon.

SOLVED – Asli book questions (verified)

Q (EN 174): 18 m aur 32 m se complementary angles

$$\sqrt{18 \cdot 32} = 24 \quad \checkmark \text{ book ans (d)}$$

Q (EN 179): 1.5 m aadmi, 24.5 m door, 26 m tower

$$\tan = 24.5/24.5 = 1 \Rightarrow 45^\circ \quad \checkmark \text{ book ans (d)}$$

Q (EN 178): $200\sqrt{3}$ m tower, 200 m door se

$$\tan = \sqrt{3} \Rightarrow 60^\circ \quad \checkmark \text{ book ans (b)}$$

Q (EN 198): pillars 5 m, 10 m, 12 m door $\Rightarrow$ tops ki doori

$$\sqrt{144 + 25} = 13 \quad \checkmark \text{ book ans (c)}$$

Q (EN 181): ladder 45° , foot $4\sqrt{2}$ m door

$$4\sqrt{2} \cdot \sqrt{2} = 8 \quad \checkmark \text{ book ans (b)}$$

Q (EN 191): tower $32\sqrt{3}$, depression 60°

$$\frac{32\sqrt{3}}{\sqrt{3}} = 32 \quad \checkmark \text{ book ans (a)}$$

Q (EN 175): 5 m ladder 3 m tak; foot 2.6 m paas

$$\text{base } 1.4 \Rightarrow 4.8 \Rightarrow 1.8 \text{ m upar} \quad \checkmark \text{ book ans (c)}$$

Q (EN 200): shadow 24 m kam, $30^\circ \rightarrow 60^\circ$

$$h = \frac{24}{\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{3}}} = 12\sqrt{3} \quad \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (44) EN 171–205 · HI 116–124

TG-8 — ELIMINATION + EQUATION SOLVE

PEHCHAN – Question mein ye dikhe

$x = r \sin A \cos C$, $y = r \sin A \sin C$, $z = r \cos A \Rightarrow x^2 + y^2 + z^2$, $x = a \sec \theta + b \tan \theta$ & $y = a \tan \theta + b \sec \theta \Rightarrow x^2 - y^2$, $x = a \sin \theta$, $y = b \tan \theta \Rightarrow a^2/x^2 - b^2/y^2$, $x + 1/x = 2 \cos \theta$, $\sqrt{2} \sin(5x - 5)^\circ = \tan 45^\circ$, $2\cos^2\theta - 3\cos\theta + 2 = 2\sin^2\theta$

- **Square karke jodo / ghatao** taaki $\sin^2 + \cos^2 = 1$ ya $\sec^2 - \tan^2 = 1$ bane: $x = r \sin \cos$, $y = r \sin \sin$, $z = r \cos \Rightarrow x^2 + y^2 + z^2 = r^2$.
- $x = a \sec + b \tan$, $y = a \tan + b \sec \Rightarrow x^2 - y^2 = a^2 - b^2$.
- $x = a \sin$, $y = b \tan \Rightarrow \frac{a^2}{x^2} - \frac{b^2}{y^2} = \csc^2 - \cot^2 = 1$.
- **Equation:** sab ko ek ratio mein badlo ($\sin^2 = 1 - \cos^2$), quadratic solve karo, 0° – 90° wala root lo.

TERMS – Har letter ka matlab

r, a, b = constants.

SOLVED – Asli book questions (verified)

Q (EN 207): $x = 3\cos A \cos B$, $y = 3\cos A \sin B$, $z = 3 \sin A$

$3^2 = 9$ ✓ **book ans (a)**

Q (EN 213): $x = a \sin \theta$, $y = b \tan \theta \Rightarrow a^2/x^2 - b^2/y^2$

$\operatorname{cosec}^2 - \cot^2 = 1$ ✓ **book ans (d)**

Q (EN 210): $x + 1/x = 2\cos \theta \Rightarrow x^2 + 1/x^2$

$4\cos^2 \theta - 2 = 2\cos 2\theta$ ✓ **book ans (c)**

Q (EN 206): $\sqrt{2} \sin(5x - 5)^\circ = \tan 45^\circ$

$\sin = 1/\sqrt{2} \Rightarrow 5x - 5 = 45 \Rightarrow x = 10$ ✓ **book ans (d)**

Q (EN 217): $2\cos^2 \theta - 3\cos \theta + 2 = 2\sin^2 \theta$

$4\cos^2 - 3\cos = 0 \Rightarrow \cos = 0 \Rightarrow 90^\circ$ ✓ **book ans (c)**

Q (EN 214): $\sin(A - B) = 1/2$, $\cos(A + B) = 1/2$

$A - B = 30^\circ$, $A + B = 60^\circ \Rightarrow 45^\circ, 15^\circ$ ✓ **book ans (d)**

BOOK KESAWAAL (20) EN 32, 79, 206–211, 213–218, 220 · HI 40, 62, 85, 87 | Ch 28 (Vividh): EN 23

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
TG-1	IDENTITY TRIANGLE + “ $\theta = 45^\circ$ DAALO” TRICK	28	45
TG-2	PAIR TRICKS ($\sec + \tan = k$, $\tan + \cot = k \dots$)	25	16
TG-3	VALUE TABLE + EK RATIO SE SAB	53	26
TG-4	COMPLEMENTARY + ALLIED ANGLES ($90^\circ, 180^\circ, 360^\circ$)	28	12
TG-5	COMPOUND + DOUBLE ANGLE	29	6
TG-6	$a \sin^2 + b \cos^2$ MACHINE (min / max / angle)	8	6
TG-7	HEIGHTS & DISTANCES	35	9
TG-8	ELIMINATION + EQUATION SOLVE	15	4
Total (book ke sawaal)		221/221 ✓	124/124 ✓

Chapter 24

Co-ordinate Geometry (Nirdeshank Jyamiti) निर्देशांक ज्यामिति

5 formulas · English book 49 sawaal · Hindi book 17 sawaal · 31 solved examples ✓

CG-1 — SECTION FORMULA + MIDPOINT + REFLECTION

PEHCHAN – Question mein ye dikhe

point dividing the join of (2, 4) and (7, 9) internally in ratio 1 : 2, ratio in which line $4x + y = 13$ divides a segment, fourth vertex of parallelogram, image of a point in x-axis; विभाजित करने वाला बिंदु

$$P = \left(\frac{m x_2 + n x_1}{m + n}, \frac{m y_2 + n y_1}{m + n} \right) \quad (m : n) \quad \text{Midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- **Criss-cross:** ratio ka pehla number doosre point se guna ($m \cdot x_2$), doosra number pehle point se ($n \cdot x_1$).
- Line $ax + by + c = 0$ kis ratio mein kaatti hai: $k = -\frac{ax_1 + by_1 + c}{ax_2 + by_2 + c}$.
- Parallelogram PQRS: $S = P + R - Q$ (diagonals ek-doesre ko bisect). Centroid = teeno ka average.
- Reflection: x-axis $\Rightarrow (x, -y)$; y-axis $\Rightarrow (-x, y)$; origin $\Rightarrow (-x, -y)$.

TERMS – Har letter ka matlab

$(x_1, y_1), (x_2, y_2)$ = do points · $m : n$ = ratio.

SOLVED – Asli book questions (verified)

Q (EN 7): $(-4, 4), (4, 0)$ ko 3 : 1

$$\left(\frac{12-4}{4}, \frac{0+4}{4} \right) = (2, 1) \quad \checkmark \text{ book ans (b)}$$

Q (EN 8): $(-1, 9), (11, 1)$ ko 3 : 1

$$\left(\frac{33-1}{4}, \frac{3+9}{4} \right) = (8, 3) \quad \checkmark \text{ book ans (d)}$$

Q (EN 12): $(-7, 6), (5, 0)$ ko 1 : 3

$$\left(\frac{5-21}{4}, \frac{0+18}{4} \right) = (-4, 4.5) \quad \checkmark \text{ book ans (b)}$$

Q (EN 5): parallelogram $P(-1, -1), Q(8, 0), R(7, 5) \Rightarrow S$

$$P + R - Q = (-2, 4) \quad \checkmark \text{ book ans (a)}$$

Q (EN 9): $4x + y = 13$ ne $(1, 6), (6, 1)$ ko kis ratio mein

$$-\frac{4+6-13}{24+1-13} = \frac{1}{4} \quad \checkmark \text{ book ans (d)}$$

Q (EN 4): $(7, 8)$ ka x-axis mein image

$$(7, -8) \quad \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (17) EN 3-5, 7-15 · HI 1-5

CG-2 — DISTANCE + TRIANGLE TYPE

PEHCHAN – Question mein ye dikhe

distance between two points, side of square from two consecutive vertices, $(x, 7)$ and $(1, 15)$ are 10 apart, isosceles / equilateral / scalene / right triangle from vertices, equidistant point, B lies on AC, points on x-axis at distance c; दूरी, विषमबाहु

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- Teeno sides nikaalo: do barabar $\Rightarrow$ isosceles, teeno $\Rightarrow$ equilateral, $a^2 + b^2 = c^2 \Rightarrow$ right, sab alag $\Rightarrow$ scalene.
- Equidistant $\Rightarrow$ dono distances ke square barabar karo.
- B, AC par $\Rightarrow AB + BC = AC$. Point (h, k) se x-axis ki doori = $|k|$.

TERMS – Har letter ka matlab

d = doori.

SOLVED – Asli book questions (verified)**Q (EN 39):** square ke consecutive vertices (2, -2), (5, 2)

$$\sqrt{9+16} = 5 \checkmark \text{ book ans (d)}$$

Q (EN 40): (x, 7), (1, 15) ki doori 10

$$(x-1)^2 = 36 \Rightarrow 7, -5 \checkmark \text{ book ans (d)}$$

Q (EN 1): A(1, 2), B(3, 4), C(4, 1)

$$\sqrt{8}, \sqrt{10}, \sqrt{10} \Rightarrow \text{isosceles} \checkmark \text{ book ans (a)}$$

Q (EN 35): (6, 2) aur (4, 6) se barabar door

$$x-2y = -3 \checkmark \text{ book ans (d)}$$

Q (HI 14): AB = x + 3, BC = 2x, AC = 4x - 5, B AC par

$$3x+3 = 4x-5 \Rightarrow 8 \checkmark \text{ book ans (b)}$$

Q (EN 38): (2, 3) se c (< 3) doori par x-axis ke points

$$\text{x-axis 3 door} \Rightarrow 0 \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (11) EN 1, 6, 26, 27, 35, 38-40 · HI 14-16**CG-3 — AREA OF TRIANGLE / QUADRILATERAL****PEHCHAN – Question mein ye dikhe**area of triangle with vertices, area $8 \Rightarrow p$, triangle by axes and $5x + 7y = 35$, quadrilateral ABCD, collinear points, triangle formed by two lines and x-axis; क्षेत्रफल

$$\text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

- Area = 0 $\Rightarrow$ teeno points ek line par (collinear).
- Line $\frac{x}{a} + \frac{y}{b} = 1$ aur axes ka triangle = $\frac{1}{2}ab$.
- Quadrilateral = do triangles ka jod; rhombus jaisa ho to $\frac{1}{2}d_1d_2$.
- Do lines + x-axis: base = x-intercepts ka farak, height = milan-bindu ka y.

TERMS – Har letter ka matlab(x₁, y₁), (x₂, y₂), (x₃, y₃) = teen vertices.**SOLVED – Asli book questions (verified)****Q (EN 16):** (0, 2), (2, 3), (3, 1)

$$\frac{1}{2}|0-2-3| = 2.5 \checkmark \text{ book ans (c)}$$

Q (EN 23): (1, 2), (-4, -3), (4, 1)

$$\frac{1}{2}|-4+4+20| = 10 \checkmark \text{ book ans (b)}$$

Q (HI 7): (3, 4), (5, 7), (-3, 6)

$$\frac{1}{2}|3+10+9| = 11 \checkmark \text{ book ans (c)}$$

Q (EN 18): (3, -2), (2, -3), (p, -4) area 8

$$\frac{1}{2}|p-1| = 8 \Rightarrow 17 \checkmark \text{ book ans (a)}$$

Q (EN 21): axes aur $5x + 7y = 35$

$$\frac{1}{2} \cdot 7 \cdot 5 = \frac{35}{2} \checkmark \text{ book ans (b)}$$

Q (EN 17): A(0, -2), B(2, 1), C(0, 4), D(-2, 1)

$$\text{diagonals } 6, 4 \Rightarrow 12 \checkmark \text{ book ans (b)}$$

Q (EN 28): $x - 5y = 2$, $x + 2y = 9$ aur x-axis

$$A(7, 1), B(2, 0), C(9, 0) \Rightarrow 3.5 \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (18) EN 16-18, 20-25, 28, 29 · HI 6-12

CG-4 — STRAIGHT LINE (slope, equation, angle)

PEHCHAN – Question mein ye dikhe

equation of line through two points / with slope and intercept, perpendicular / parallel lines, intersection of $3x - 2y = 11$ and $x + y = 7$, line passes through $(3, 2) \Rightarrow k$, distance of origin from $4x + 3y = 6$, angle with x-axis, intercepts on axes, graph of $y = 9x^2 + 16$; सरल रेखा, ढलान

$$m = \frac{y_2 - y_1}{x_2 - x_1} = -\frac{a}{b} \quad y - y_1 = m(x - x_1) \quad y = mx + c$$

- Parallel $\Rightarrow$ same slope; perpendicular $\Rightarrow m_1 m_2 = -1$. Angle with x-axis: $\tan \theta = m$.
- Point line par $\Rightarrow (x, y)$ daal ke equation satisfy. Intersection $\Rightarrow$ dono ko solve.
- Origin se doori = $\frac{|c|}{\sqrt{a^2 + b^2}}$; point se $\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$. Intercepts: $x = -c/a$, $y = -c/b$.
- Right angle at A $\Rightarrow AB \perp AC$ (slopes ka guna -1).

TERMS – Har letter ka matlab

m = slope · c = y-intercept.

SOLVED – Asli book questions (verified)

Q (EN 30): $3x - 2y - 11 = 0$ aur $x + y = 7 \Rightarrow 3\alpha + 5\beta$
 $(5, 2) \Rightarrow 25$ ✓ **book ans (c)**

Q (EN 31): $kx - 3y = 6$ through $(3, 2)$
 $3k - 6 = 6 \Rightarrow 4$ ✓ **book ans (a)**

Q (EN 34): $(-2, 5)$ aur $(1, 3)$ se line
 $m = -2/3 \Rightarrow 2x + 3y - 11 = 0$ ✓ **book ans (d)**

Q (EN 41): origin se $4x + 3y = 6$
 $\frac{6}{5}$ ✓ **book ans (d)**

Q (EN 42): $35x - 35y + 15 = 0$ ka x-axis se kon
 $m = 1 \Rightarrow 45^\circ$ ✓ **book ans (c)**

Q (EN 49): $6x - 3y + 10 = 0$, $2x - y + 9 = 0$
dono ka slope $2 \Rightarrow$ parallel ✓ **book ans (b)**

Q (EN 36): $(2, 3)$ se, $(-5, 6) - (-6, 5)$ ke perpendicular
 $m = -1 \Rightarrow x + y - 5 = 0$ ✓ **book ans (a)**

BOOK KESA WAAL (12) EN 2, 19, 30, 31, 34, 36, 37, 41, 42, 49 · HI 13, 17

CG-5 — CIRCLE (centre, radius, tangent)

PEHCHAN – Question mein ye dikhe

equation of circle with centre and radius, radius of $x^2 + y^2 + 7x + 4y + 9 = 0$, length of tangent from a point, position of a point (inside / outside), tangent at a point, angle between circles; वृत्त, स्पर्श रेखा

$$(x - h)^2 + (y - k)^2 = r^2 \quad x^2 + y^2 + 2gx + 2fy + c = 0 : \text{centre}(-g, -f), r = \sqrt{g^2 + f^2 - c}$$

- Tangent ki lambai** (x_1, y_1) se $= \sqrt{S_1}$, jahan $S_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$.
- $S_1 < 0$ andar, $= 0$ circle par, > 0 bahar.
- $x^2 + y^2 = r^2$ par (x_1, y_1) se tangent: $xx_1 + yy_1 = r^2$.
- Do circles ka kon: $\cos \theta = \frac{r_1^2 + r_2^2 - d^2}{2r_1 r_2}$ (d = centres ki doori).

TERMS – Har letter ka matlab

(h, k) = centre · r = radius · S_1 = point daal ke value.

SOLVED – Asli book questions (verified)

Q (EN 32): centre $(-2, 3)$, radius 4

$$x^2 + y^2 + 4x - 6y - 3 = 0 \quad \checkmark \text{ book ans (b)}$$

Q (EN 43): $(2, 3)$ se $x^2 + y^2 = 4$ par tangent

$$\sqrt{4 + 9 - 4} = 3 \quad \checkmark \text{ book ans (b)}$$

Q (EN 44): $(2, 5)$ se tangent $\sqrt{37}$, circle $x^2 + y^2 - 5x + 4y + k$

$$39 + k = 37 \Rightarrow -2 \quad \checkmark \text{ book ans (a)}$$

Q (EN 45): $(1, 2)$ aur $x^2 + y^2 - 3x - 4y + 1 = 0$

$$S_1 = 1 + 4 - 3 - 8 + 1 = -5 < 0 \Rightarrow \text{andar} \quad \checkmark \text{ book ans (d)}$$

Q (EN 47): $x^2 + y^2 + 7x + 4y + 9 = 0$ ka radius

$$\sqrt{\frac{49}{4} + 4 - 9} = \frac{\sqrt{29}}{2} \quad \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (8) EN 32, 33, 43–48

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
CG-1	SECTION FORMULA + MIDPOINT + REFLECTION	12	5
CG-2	DISTANCE + TRIANGLE TYPE	8	3
CG-3	AREA OF TRIANGLE / QUADRILATERAL	11	7
CG-4	STRAIGHT LINE (slope, equation, angle)	10	2
CG-5	CIRCLE (centre, radius, tangent)	8	0
Total (book ke sawaal)		49/49 ✓	17/17 ✓

Chapter 25

Geometry (Jyamiti) ज्यामिति

8 formulas · English book 276 sawaal · Hindi book 207 sawaal · 68 solved examples ✓

GE-1 — LINES & ANGLES (complementary, supplementary, parallel lines)

PEHCHAN – Question mein ye dikhe

complementary / supplementary angles, larger angle is 36° more, angles in ratio, $(6y + 70)^\circ$ & $(3y + 47)^\circ$ supplementary, parallel lines cut by a transversal (alternate, corresponding, co-interior), find x in the figure; पूरक, संपूरक, तिर्यक रेखा, समांतर रेखाएँ

Jodi	Rule
Complementary	jod 90°
Supplementary / linear pair / seedhi rekha	jod 180°
Vertically opposite, alternate, corresponding	barabar
Co-interior (same side, parallel lines)	jod 180°
Ek point ke chaaron taraf	360°

- Ratio $a : b$ wale do angles, jod $J \Rightarrow \frac{a}{a+b}J$ aur $\frac{b}{a+b}J$.
- Jod aur farak diye $\Rightarrow$ bada = (jod + farak)/2.
- Figure mein Z (alternate) aur F (corresponding) shapes dhoondho; zig-zag $\Rightarrow$ andar ke kon = bahar ke konon ka jod.

TERMS – Har letter ka matlab

x = poocha gaya kon · J = jod (90 ya 180).

SOLVED – Asli book questions (verified)

Q (EN 236): supplementary, bada 36° zyada

$$\frac{180-36}{2} = 72^\circ \text{ ✓ book ans (a)}$$

Q (EN 237): complementary: bada = $3 \times$ chhota – 6

$$4x - 6 = 90 \Rightarrow x = 24 \Rightarrow 66^\circ \text{ ✓ book ans (c)}$$

Q (EN 240): complementary $11 : 7 \Rightarrow$ chhota

$$90 \cdot \frac{7}{18} = 35^\circ \text{ ✓ book ans (a)}$$

Q (EN 239): 15° ka complement + 125° ka supplement

$$75 + 55 = 130^\circ \text{ ✓ book ans (c)}$$

Q (EN 244): $(6y + 70)^\circ$ aur $(3y + 47)^\circ$ supplementary

$$9y + 117 = 180 \Rightarrow y = 7 \text{ ✓ book ans (c)}$$

Q (EN 245): $(7x + 5)^\circ$ aur $(x + 5)^\circ$ complementary

$$8x + 10 = 90 \Rightarrow x = 10 \text{ ✓ book ans (a)}$$

Q (EN 241): supplement = $3 \times$ complement + 15

$$180 - x = 285 - 3x \Rightarrow x = 52.5^\circ \text{ ✓ book ans (c)}$$

Q (HI 12): do complementary kon, farak 24°

$$\frac{90+24}{2} = 57^\circ \text{ ✓ book ans (c)}$$

BOOK KE SAWAAL (68) EN 236–246, 249–255, 258–260, 262, 264–267, 271, 272, 274–276 · HI 1–37

GE-2 — TRIANGLE ANGLES (180° , exterior, bisector-meet)

PEHCHAN – Question mein ye dikhe

third angle, angles in ratio $4 : 3 : 5$, isosceles triangle base angles, exterior angle, bisectors of $\angle B$ and $\angle C$ meet at P $\Rightarrow \angle BPC$, incentre / circumcentre / orthocentre angle, $\angle A = 3\angle B$ & $\angle C = 2\angle B$, altitude in right triangle; त्रिभुज के कोण, बहिष्कोण

$$A + B + C = 180^\circ \quad \text{Exterior angle} = \text{do door wale andar ke konon ka jod}$$

Point	$\angle BOC$
Incentre (andar ke bisectors)	$90^\circ + \frac{A}{2}$
Excentre (bahar ke bisectors)	$90^\circ - \frac{A}{2}$
Circumcentre	$2A$
Orthocentre	$180^\circ - A$

- Isosceles: barabar sides ke saamne ke kon barabar $= \frac{180 - \text{vertex}}{2}$.
- Right triangle mein altitude $\Rightarrow$ chhote triangles ke kon original jaise ($\angle ACD = \angle B$).

TERMS – Har letter ka matlab

A = vertex ka kon · O = special point.

SOLVED – Asli book questions (verified)

Q (EN 1): $\angle B, \angle C$ ke bisectors P par; $\angle BPC = 128^\circ$

$$90 + \frac{A}{2} = 128 \Rightarrow A = 76^\circ \quad \checkmark \text{ book ans (b)}$$

Q (EN 5): $\angle A = 75^\circ$, bisectors D par $\Rightarrow \angle BDC$

$$90 + 37.5 = 127.5^\circ \quad \checkmark \text{ book ans (d)}$$

Q (EN 23): incentre O, $\angle BOC = 110^\circ$

$$A = 2(110 - 90) = 40^\circ \quad \checkmark \text{ book ans (a)}$$

Q (EN 3): $AB = AC$, $\angle BAC = 50^\circ$, BC ko D tak $\Rightarrow \angle ACD$

$$\text{base } 65^\circ \Rightarrow 180 - 65 = 115^\circ \quad \checkmark \text{ book ans (a)}$$

Q (EN 6): angles 4 : 3 : 5

$$60^\circ, 45^\circ, 75^\circ \quad \checkmark \text{ book ans (b)}$$

Q (EN 38): $2x, 3x - 8, 5x - 12$

$$10x - 20 = 180 \Rightarrow x = 20 \Rightarrow 88^\circ \quad \checkmark \text{ book ans (b)}$$

Q (EN 22): equilateral, circumcentre O $\Rightarrow \angle BOC$

$$2 \times 60 = 120^\circ \quad \checkmark \text{ book ans (c)}$$

Q (EN 24): C par 90° , $CD \perp AB$, $\angle A = 55^\circ \Rightarrow \angle ACD$

$$90 - 55 = 35^\circ \quad \checkmark \text{ book ans (c)}$$

Q (HI 49): $\angle A = 40^\circ$, $\angle B = 70^\circ \Rightarrow A$ ka bahirkon

$$180 - 40 = 140^\circ \quad \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (54) EN 1, 3–6, 17–19, 21–24, 26, 27, 29, 30, 33, 34, 37–39, 44, 47, 54, 59, 65 · HI 38, 42–44, 46, 49, 51, 53, 54, 61, 62, 65, 66, 69–71, 73, 77, 78, 80, 81, 83, 86–89, 91, 97

GE-3 — SIMILARITY + BPT + ANGLE BISECTOR + MIDPOINT

PEHCHAN – Question mein ye dikhe

DE $\parallel$ BC with AD, DB, AE, EC, similar triangles perimeter / side / area ratio, bisector of $\angle P$ meets QR at S, midpoints $\Rightarrow DE = BC/2$, congruence (SSS, SAS, ASA), two pillars and crossing ropes; समरूप, सर्वांगसम, कोण समद्विभाजक

$$\text{BPT: } DE \parallel BC \Rightarrow \frac{AD}{DB} = \frac{AE}{EC} \quad \text{Bisector: } \frac{BD}{DC} = \frac{AB}{AC}$$

- **Similar:** side ratio $k \Rightarrow$ perimeter / height / median ratio k , area ratio k^2 .
- Midpoints jode $\Rightarrow DE = \frac{1}{2} BC$, $DE \parallel BC$, andar ka triangle $= \frac{1}{4}$ area.
- Same height ke triangles: area $\propto$ base.
- Do khambhe a, b ki ropes cross $\Rightarrow$ milan-bindu ki unchai $= \frac{ab}{a+b}$.
- Congruence: SSS, SAS, ASA, RHS — letters same kram mein milao.

TERMS – Har letter ka matlab

AD, DB, AE, EC = BPT ke tukde $\cdot k$ = side ratio.

SOLVED – Asli book questions (verified)

Q (EN 2): PQ 12, PR 15, QR 18; bisector S $\Rightarrow$ SR

$$18 \cdot \frac{15}{27} = 10 \quad \checkmark \text{ book ans (c)}$$

Q (EN 8): AB 4, AC 5, BC 6; bisector AD $\Rightarrow$ BD

$$6 \cdot \frac{4}{9} = 2.67 \quad \checkmark \text{ book ans (d)}$$

Q (EN 11): D, E midpoints, BC = 6

$$DE = 3 \quad \checkmark \text{ book ans (b)}$$

Q (EN 16): DE $\parallel$ BC: $x/(x-2) = (x+2)/(x-1)$

$$x^2 - x = x^2 - 4 \Rightarrow 4 \quad \checkmark \text{ book ans (c)}$$

Q (EN 25): similar, perimeters 48 & 24, XY 12 $\Rightarrow$ PQ

$$12 \times 2 = 24 \quad \checkmark \text{ book ans (c)}$$

Q (EN 57): BC 4, EF 7, ar(ABC) 144 $\Rightarrow$ ar(DEF)

$$144 \cdot \frac{49}{16} = 441 \quad \checkmark \text{ book ans (c)}$$

Q (EN 61): AB 10, BD 6, DC 7.5 $\Rightarrow$ CA

$$10 \cdot \frac{7.5}{6} = 12.5 \quad \checkmark \text{ book ans (d)}$$

Q (EN 28): AD/BD = 3/5, AC 4 $\Rightarrow$ AE

$$4 \cdot \frac{3}{8} = 1.5 \quad \checkmark \text{ book ans (a)}$$

Q (EN 273): khambhe 8 m aur 12 m, ropes cross

$$\frac{8 \cdot 12}{20} = \frac{24}{5} \text{ m} \quad \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (49) EN 2, 7–9, 11, 15, 16, 25, 28, 35, 36, 41–43, 50, 51, 53, 56–58, 60–63, 273 · **HI** 39–41, 48, 50, 52, 55–59, 63, 64, 67, 68, 72, 74, 75, 82, 84, 85, 94, 96, 99

GE-4 — TRIANGLE LENGTHS: Pythagoras, inradius, circumradius, centroid**PEHCHAN – Question mein ye dikhe**

radius of incircle / circumcircle of 3-4-5 or 9-40-41 triangle, circle of radius 7 circumscribes an equilateral triangle $\Rightarrow$ side, height of equilateral triangle, Pythagorean triplet 8, p, 17, area of circumcircle of right triangle, centroid / median, ratio of altitudes; अंतःवृत्त, परिवृत्त, केन्द्रक, माधिका

Triangle	Inradius r	Circumradius R
Right (a, b, c)	$\frac{a+b-c}{2}$	$\frac{c}{2}$
Equilateral (a)	$\frac{a}{2\sqrt{3}}$	$\frac{a}{\sqrt{3}}$ (R = 2r)
Koi bhi	$\frac{\text{Area}}{s}$	$\frac{abc}{4 \cdot \text{Area}}$

- Pythagoras $c^2 = a^2 + b^2$; triplets 3-4-5, 5-12-13, 8-15-17, 7-24-25, 9-40-41, 13-84-85.
- Equilateral: height $\frac{\sqrt{3}}{2}a$. Centroid median ko 2 : 1 mein baantta hai.
- Altitudes $\propto \frac{1}{\text{side}}$ (same area).

TERMS – Har letter ka matlab

a, b, c = sides (c = hypotenuse) $\cdot s$ = semi-perimeter.

SOLVED – Asli book questions (verified)**Q (EN 12):** 3-4-5 ka inradius : circumradius

$$1 : 2.5 = 2 : 5 \quad \checkmark \text{ book ans (b)}$$

Q (EN 32): right triangle 4 & 3 $\Rightarrow$ inradius

$$\frac{3+4-5}{2} = 1 \quad \checkmark \text{ book ans (a)}$$

Q (EN 64): 9, 40, 41 ka inradius

$$\frac{9+40-41}{2} = 4 \quad \checkmark \text{ book ans (d)}$$

Q (EN 66): 6-8-10 ke circumcircle ka area

$$R = 5 \Rightarrow 25\pi \quad \checkmark \text{ book ans (b)}$$

Q (EN 40): equilateral side $6\sqrt{3} \Rightarrow$ height

$$\frac{\sqrt{3}}{2} \cdot 6\sqrt{3} = 9 \quad \checkmark \text{ book ans (a)}$$

Q (EN 55): 8, p, 17 triplet

$$15 \quad \checkmark \text{ book ans (a)}$$

Q (EN 48): 1.5 & 2 ke right triangle ka circumcircle

$$R = 1.25 \Rightarrow 1.5625\pi \quad \checkmark \text{ book ans (c)}$$

Q (EN 31): $R = 7$, equilateral ka side

$$a = R\sqrt{3} = 7\sqrt{3} \quad \checkmark \text{ book ans (c)}$$

BOOK KE SAWAAL (27) EN 10, 12, 14, 20, 31, 32, 40, 45, 46, 48, 49, 52, 55, 64, 66, 155, 263 · **HI** 45, 60, 76, 79, 90, 92, 93, 95, 98, 100**GE-5 — QUADRILATERALS (parallelogram, rhombus, trapezium, cyclic)****PEHCHAN – Question mein ye dikhe**

angles of quadrilateral in ratio, fourth angle, cyclic quadrilateral opposite angles $3 : 7$, rhombus diagonals $\Rightarrow$ side / perimeter, parallelogram angle 28° less than thrice the smallest, $\angle A = (3x - 25)^\circ$ & $\angle C = (2x + 15)^\circ$, trapezium midline, bisectors of consecutive angles, rotational symmetry; चतुर्भुज, समचतुर्भुज, चक्रीय

$$A + B + C + D = 360^\circ \quad \text{Cyclic: } A + C = B + D = 180^\circ$$

Shape	Khaas baat
Parallelogram	saamne ke kon barabar, paas ke jod 180° , diagonals ek-doesre ko bisect
Rhombus	diagonals $\perp$ bisect: $a^2 = \left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2$, $P = 4a$
Trapezium	midline $= \frac{1}{2}(\text{parallel sides ka jod})$
Rectangle / square	diagonals barabar; square diagonal $= a\sqrt{2}$

- Paas ke do konon ke bisectors ka kon $= \frac{A+B}{2}$ (parallelogram mein 90°).
- Rotational symmetry: square 4, rectangle / rhombus / parallelogram 2, trapezium 1.

TERMS – Har letter ka matlab

d_1, d_2 = diagonals · a = side.

SOLVED – Asli book questions (verified)**Q (EN 67):** cyclic, saamne ke kon $3 : 7$

$$180 \cdot \frac{7}{10} = 126^\circ \checkmark \text{ book ans (d)}$$

Q (EN 70): teen kon 110, 55, 125

$$360 - 290 = 70^\circ \checkmark \text{ book ans (c)}$$

Q (EN 71): $4 : 9 : 11 : 12 \Rightarrow$ sabse bada

$$\frac{360}{36} \cdot 12 = 120^\circ \checkmark \text{ book ans (c)}$$

Q (EN 77): $125^\circ, 35^\circ$, baaki do barabar

$$\frac{200}{2} = 100^\circ \checkmark \text{ book ans (b)}$$

Q (EN 84): rhombus diagonals 96, 110 $\Rightarrow$ perimeter

$$\text{side } 73 \Rightarrow 292 \checkmark \text{ book ans (b)}$$

Q (EN 94): rhombus side 26, ek diagonal 48

$$2\sqrt{676 - 576} = 20 \checkmark \text{ book ans (b)}$$

Q (EN 103): parallelogram: ek kon $= 3 \times$ chhota $- 28$

$$4x - 28 = 180 \Rightarrow 52 \Rightarrow 128^\circ \checkmark \text{ book ans (c)}$$

Q (EN 111): $\angle A = 3x - 25$, $\angle C = 2x + 15$ (saamne)

$$x = 40 \Rightarrow 95^\circ \checkmark \text{ book ans (c)}$$

Q (EN 100): parallelogram ke paas ke konon ke bisectors

$$90^\circ \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (123) EN 67–121, 138, 247, 248, 256, 257, 268, 269 · **HI** 47, 101–160**GE-6 — CIRCLE: CHORD, ANGLES, ARC****PEHCHAN – Question mein ye dikhe**

chord 24 cm at 5 cm from centre $\Rightarrow$ radius, angle at centre $= 2 \times$ angle at circumference, angle in semicircle, PQ diameter & $\angle RPO \Rightarrow \angle RQP$, chords intersecting inside / outside ($AF \cdot BF = CF \cdot DF$), two parallel chords distance, arc length / central angle, equal circles through each other's centres; जीवा, चाप, केंद्र

$$r^2 = d^2 + \left(\frac{c}{2}\right)^2 \quad \angle_{\text{centre}} = 2 \times \angle_{\text{paridhi}} \quad \text{Arc} = \frac{\theta}{360} \cdot 2\pi r$$

- Semicircle mein kon 90° ; same segment ke kon barabar.
- Chords** P par katen (andar ya bahar): $PA \cdot PB = PC \cdot PD$.
- Do parallel chords: ulti taraf $\Rightarrow$ doori $d_1 + d_2$; same taraf $\Rightarrow |d_1 - d_2|$.
- Chord 60° banaye $\Rightarrow$ chord = r. Do barabar circles ek-doosre ke centre se $\Rightarrow$ common chord $r\sqrt{3}$.

TERMS – Har letter ka matlab

c = chord · d = centre se doori · θ = centre par kon.

SOLVED – Asli book questions (verified)**Q (EN 129):** chord 24, centre se 5

$$\sqrt{144 + 25} = 13 \checkmark \text{ book ans (a)}$$

Q (EN 130): $\angle ACB = 46.5^\circ \Rightarrow \angle AOB$

$$93^\circ \checkmark \text{ book ans (b)}$$

Q (EN 131): PQ diameter, $\angle RPO = 42^\circ \Rightarrow \angle RQP$

$$90 - 42 = 48^\circ \checkmark \text{ book ans (a)}$$

Q (EN 136): sector 30° , r 42 $\Rightarrow$ arc

$$\frac{30}{360} \cdot 2 \cdot \frac{22}{7} \cdot 42 = 22 \checkmark \text{ book ans (d)}$$

Q (EN 134): r 10, chords 12 & 16 ulta taraf

$$8 + 6 = 14 \checkmark \text{ book ans (d)}$$

Q (EN 123): bahar F: AF 12, BF 4, CF 16 $\Rightarrow$ CD

$$DF = 48/16 = 3 \Rightarrow 13 \checkmark \text{ book ans (a)}$$

Q (EN 133): r 28, arc 22 $\Rightarrow$ kon

$$45^\circ \checkmark \text{ book ans (c)}$$

Q (EN 142): chord 8 cm, centre par 60°

$$r = 8 \checkmark \text{ book ans (a)}$$

BOOK KE SAWAAL (39) EN 13, 122–137, 139–148, 150–154, 261, 270 · **HI** 161–163, 165, 166**GE-7 — TANGENTS****PEHCHAN – Question mein ye dikhe**

length of tangent from a point at distance d, radius from tangent & distance, two tangents from external point ($\angle PAB \Rightarrow \angle AOB$), tangents inclined at $110^\circ \Rightarrow \angle POA$, chord AB extended meets tangent DE ($DE^2 = DB \cdot DA$), direct / transverse common tangent, alternate segment; स्पर्श रेखा

$$PT^2 = d^2 - r^2 \text{ (tangent } \perp \text{ radius)} \quad PT^2 = PA \cdot PB \text{ (tangent-secant)}$$

- Ek bahar ke point se do tangents barabar; $\angle APB + \angle AOB = 180^\circ$.
- Direct common tangent = $\sqrt{d^2 - (r_1 - r_2)^2}$; transverse = $\sqrt{d^2 - (r_1 + r_2)^2}$; bahar se chhoote circles $\Rightarrow 2\sqrt{r_1 r_2}$.
- Alternate segment: tangent-chord ka kon = doosre segment mein bana kon.

TERMS – Har letter ka matlab

d = centre se doori · r = radius · PA · PB = secant ke tukde.

SOLVED – Asli book questions (verified)**Q (EN 156):** tangent 21, centre se 29 $\Rightarrow$ r

$$\sqrt{841 - 441} = 20 \checkmark \text{ book ans (a)}$$

Q (EN 159): AP 15, r 8 $\Rightarrow$ OP

$$17 \checkmark \text{ book ans (c)}$$

Q (EN 161): r 4, point 5 door $\Rightarrow$ tangent

$$3 \checkmark \text{ book ans (d)}$$

Q (EN 162): radii 4.5, 3.5, transverse tangent 6 $\Rightarrow$ d

$$\sqrt{36 + 64} = 10 \checkmark \text{ book ans (d)}$$

Q (EN 164): $\angle PAB = 55^\circ \Rightarrow \angle AOB$

$$\angle APB 70^\circ \Rightarrow 110^\circ \checkmark \text{ book ans (d)}$$

Q (EN 172): AB 9, BD 3 $\Rightarrow$ DE

$$\sqrt{3 \cdot 12} = 6 \checkmark \text{ book ans (c)}$$

Q (EN 178): tangents 110° par $\Rightarrow \angle POA$

$$\frac{180-110}{2} = 35^\circ \checkmark \text{ book ans (c)}$$

Q (EN 176): AB 7, DE 12 $\Rightarrow$ BD

$$BD(BD + 7) = 144 \Rightarrow 9 \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (33) EN 149, 156–180 · HI 164, 167–172

GE-8 — POLYGON MACHINE

PEHCHAN – Question mein ye dikhe

each interior / exterior angle of a regular polygon, number of sides from an angle, sum of interior angles, number of diagonals,
 interior – exterior = 36° , interior : exterior = 4 : 1; बहुभुज, विकर्णों की संख्या

$$\text{Exterior} = \frac{360^\circ}{n} \quad \text{Interior} = 180^\circ - \frac{360^\circ}{n} \quad \text{Sum} = (n - 2) 180^\circ \quad \text{Diagonals} = \frac{n(n - 3)}{2}$$

- n nikaalo: $n = \frac{360}{180 - \text{interior}}$.
- Interior – exterior = k $\Rightarrow$ exterior = $\frac{180-k}{2}$; interior : exterior = a : b $\Rightarrow$ exterior = $\frac{180b}{a+b}$.
- Exterior angles ka jod hamesha 360° .

TERMS – Har letter ka matlab

n = sides.

SOLVED – Asli book questions (verified)

Q (EN 181): regular octagon ka interior

$180 - 45 = 135^\circ$ ✓ **book ans (c)**

Q (EN 182): interior $168^\circ \Rightarrow n$

$\frac{360}{12} = 30$ ✓ **book ans (c)**

Q (EN 184): exterior $40^\circ \Rightarrow n$

9 ✓ **book ans (d)**

Q (EN 193): interior – exterior = $36^\circ \Rightarrow n$

exterior 72 $\Rightarrow$ 5 ✓ **book ans (d)**

Q (EN 201): interior sum $3240^\circ \Rightarrow n$

$18 + 2 = 20$ ✓ **book ans (b)**

Q (EN 204): interior $120^\circ \Rightarrow$ diagonals

n 6 $\Rightarrow$ 9 ✓ **book ans (a)**

Q (EN 213): interior sum $1440^\circ \Rightarrow$ diagonals

n 10 $\Rightarrow$ 35 ✓ **book ans (c)**

Q (EN 221): 27 sides $\Rightarrow$ diagonals

$\frac{27 \cdot 24}{2} = 324$ ✓ **book ans (b)**

Q (EN 194): interior : exterior = 4 : 1

exterior 36 $\Rightarrow$ 10 ✓ **book ans (b)**

BOOK KE SAWAAL (91) EN 181–235 · HI 173–207 | Ch 28 (Vividh): HI 29

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
GE-1	LINES & ANGLES (complementary, supplementary, parallel lines)	31	37
GE-2	TRIANGLE ANGLES (180° , exterior, bisector-meet)	26	28
GE-3	SIMILARITY + BPT + ANGLE BISECTOR + MIDPOINT	25	24
GE-4	TRIANGLE LENGTHS: Pythagoras, inradius, circumradius, centro	17	10
GE-5	QUADRILATERALS (parallelogram, rhombus, trapezium, cyclic)	62	61
GE-6	CIRCLE: CHORD, ANGLES, ARC	34	5
GE-7	TANGENTS	26	7
GE-8	POLYGON MACHINE	55	35
Total (book ke sawaal)		276/276 ✓	207/207 ✓

Chapter 26

Statistics & Probability (Sankhyiki aur Prayikta) सांख्यिकी और प्रायिकता

6 formulas · English book 265 sawaal · Hindi book 121 sawaal · 43 solved examples ✓

SP-1 — MEAN MACHINE (ungrouped, frequency, grouped)

PEHCHAN – Question mein ye dikhe

arithmetic mean of data, mean of frequency distribution ($\Sigma fx / \Sigma f$), grouped data with class intervals, missing value / frequency k when mean is given, wrongly read observation, each value increased by 10, combined mean, mean of first 6 primes, weighted mean; समांतर माध्य, बारंबारता

$$\bar{x} = \frac{\sum x}{n} \quad \bar{x} = \frac{\sum fx}{\sum f} \quad (\text{grouped: } x = \text{class ka mid-value})$$

- **Missing:** $\sum fx = \bar{x} \sum f$ likh ke k nikaalo.
- **Galti:** sahi mean = purana + $\frac{\text{sahi-galat}}{n}$.
- Har value $+k \Rightarrow \text{mean} +k$; $\times k \Rightarrow \text{mean} \times k$.
- Combined: $\frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$; ratio $n_1 : n_2 = (\bar{x}_2 - \bar{x}) : (\bar{x} - \bar{x}_1)$.
- 1, 2, ..., n with frequencies 1, 2, ..., n $\Rightarrow \text{mean} \frac{2n+1}{3}$.

TERMS – Har letter ka matlab

$x = \text{value} \cdot f = \text{frequency} \cdot n = \text{ginti}$.

SOLVED – Asli book questions (verified)

Q (EN 1): 10 values ka mean, har ek +10

$14 + 10 = 24$ ✓ **book ans (d)**

Q (EN 3): 33, x, 47, 83, 109 ka mean 67 $\Rightarrow$ 50, 64, 100, 126, x ka mean

$x = 63 \Rightarrow 403/5 = 80.6$ ✓ **book ans (c)**

Q (EN 5): marks 10...90, students 17, 28, 32, 24, 19

$\frac{6000}{120} = 50$ ✓ **book ans (b)**

Q (EN 7): 36 ka mean 42; 47 ko 41 padha

$42 + \frac{6}{36} = 42.17$ ✓ **book ans (a)**

Q (EN 12): mean 15: x 5...25, f 6, k, 6, 10, 5

$\frac{445+10k}{27+k} = 15 \Rightarrow 8$ ✓ **book ans (c)**

Q (EN 17): 100 ka mean 50; 50 ki jagah 150

$50 + \frac{100}{100} = 51$ ✓ **book ans (b)**

Q (EN 58): pehle 6 primes ka mean

$\frac{41}{6}$ ✓ **book ans (c)**

Q (HI 16): x 3, 2, 10, 4, 5; f 5, 3, 4, 2, 6

$\frac{99}{20} = 4.95$ ✓ **book ans (a)**

BOOK KE SAWAAL (94) EN 1-17, 19-39, 41-60, 218, 219, 226, 231, 236, 241 · **HI** 1-29, 95

SP-2 — MEDIAN MACHINE

PEHCHAN – Question mein ye dikhe

median of data (odd / even count), median of frequency / grouped distribution, median class, find x when median is given, effect of removing / replacing a value, median of factors of 48; माध्यिका

Sort karo: $n \text{ odd} \Rightarrow \left(\frac{n+1}{2}\right)^{th}$, $n \text{ even} \Rightarrow \frac{\left(\frac{n}{2}\right)^{th} + \left(\frac{n}{2} + 1\right)^{th}}{2}$

Grouped: $M = l + \frac{\frac{N}{2} - cf}{f} \times h$

- Frequency table: cumulative frequency banao, $\frac{N}{2}$ jis class / value mein pehli baar aaye wahi median.
- Sabse bade values badhe (order same) $\Rightarrow$ median wahi rahega.
- Ascending data mein x diya $\Rightarrow$ beech wale term(s) ko median ke barabar rakho.

TERMS – Har letter ka matlab

l = median class ki lower limit · cf = usse pehle ki cumulative f · f = class ki f · h = width · $N = \Sigma f$.

SOLVED – Asli book questions (verified)

Q (EN 63): 15, 2, 7, 8, 11, 5, 14

2, 5, 7, 8, 11, 14, 15 ✓ **book ans (a)**

Q (EN 61): 1, 1.1, 2.3, 0.1, 0.9, 5, 3.9, 2.5, 4.2, 4.6

$\frac{2.3+2.5}{2} = 2.4$ ✓ **book ans (b)**

Q (EN 67): CI 0-10 ... 60-70, f 2, 4, 7, 9, 10, 12, 6

$N/2 = 25$, class 40-50 $\Rightarrow 40 + \frac{25-22}{10} \cdot 10 = 43$ ✓ **book ans (b)**

Q (EN 69): ..., 50, x , $x+2$, 72, ... (10 terms), median 63

$x+1 = 63 \Rightarrow 62$ ✓ **book ans (b)**

Q (EN 65): 60, 68, 70, 72, 74, 76, 78, 80 se 70 hatao

71 $\Rightarrow$ 72: +1 ✓ **book ans (d)**

Q (EN 88): 48 ke factors ka median

1, 2, 3, 4, 6, 8, 12, 16, 24, 48 $\Rightarrow 7$ ✓ **book ans (d)**

Q (HI 42): x 1 ... 5, f 5, 7, 4, 8, 2

N 26: 13th, 14th dono 3 ✓ **book ans (b)**

BOOK KE SAWAAL (80) EN 40, 61–65, 67–70, 72–84, 86–110, 173, 192 · HI 30–42, 44–52, 54–60, 97

SP-3 — MODE MACHINE**PEHCHAN – Question mein ye dikhe**

mode of data (most frequent value), grouped mode, most demanded shirt size, mode is 12 $\Rightarrow$ k, bimodal data; बहुलक

Mode = sabse zyada baar aane wala Grouped: $Z = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$

- Do values barabar max $\Rightarrow$ dono mode (bimodal).
- Mode diya aur ek value k $\Rightarrow$ k ko wahi value banao jisse wo sabse zyada ho.

TERMS – Har letter ka matlab

f_1 = modal class ki f · f_0 = pehle wali · f_2 = baad wali · h = width.

SOLVED – Asli book questions (verified)**Q (EN 111):** 5, 4, 4, 6, 7, 6, 9, 7, 6, 56 teen baar ✓ **book ans (d)****Q (EN 112):** 15, 26, 15, 29, 19, 18, 19, 15, 24, 23, 15, 1915 chaar baar ✓ **book ans (d)****Q (EN 117):** mode 12 $\Rightarrow$ k (11, 15, 8, 9, k, 11, 12, 12, 15, 14)k = 12 ✓ **book ans (d)****Q (HI 70):** 23, 27, 38, 42, 27, 58, 69, 71, 4527 ✓ **book ans (b)****BOOK KE SAWAAL (53)** EN 111, 112, 114–133, 183, 195, 196, 203, 204, 214, 234 · **HI** 62–71, 73–81, 84–87, 117**SP-4 — EMPIRICAL RELATION (Mean – Median – Mode)****PEHCHAN – Question mein ye dikhe**

mean and mode given $\Rightarrow$ median, mode and median $\Rightarrow$ mean, mean – mode = 69 $\Rightarrow$ mean – median, which empirical formula is correct, mean / median / mode together of a data, sum or LCM of mean, median and mode; आनुभविक संबंध

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

$$\text{Mean} - \text{Mode} = 3(\text{Mean} - \text{Median})$$

- Median = $\frac{2 \text{ Mean} + \text{Mode}}{3}$, Mean = $\frac{3 \text{ Median} - \text{Mode}}{2}$.
- Mode – Median = $2 \times (\text{Median} - \text{Mean})$.
- Teeno ek saath poochhe $\Rightarrow$ SP-1, SP-2, SP-3 se alag-alag nikaalo.

TERMS – Har letter ka matlab

empirical = anubhav se bana (moderately skewed data).

SOLVED – Asli book questions (verified)**Q (EN 175):** mean 42, mode 60 $\Rightarrow$ median

$$\frac{84+60}{3} = 48 \text{ ✓ book ans (b)}$$

Q (EN 176): mode 24.6, mean 20.1 $\Rightarrow$ median

$$\frac{40.2+24.6}{3} = 21.6 \text{ ✓ book ans (c)}$$

Q (EN 177): mode 23.6, median 24 $\Rightarrow$ mean

$$\frac{72-23.6}{2} = 24.2 \text{ ✓ book ans (a)}$$

Q (EN 181): median 55, mean 58 $\Rightarrow$ mode

$$165 - 116 = 49 \text{ ✓ book ans (a)}$$

Q (EN 171): mean – mode = 69 $\Rightarrow$ mean – median

$$69 \div 3 = 23 \text{ ✓ book ans (c)}$$

Q (EN 174): mode – median = 2 $\Rightarrow$ median – mean

$$2 \div 2 = 1 \text{ ✓ book ans (a)}$$

Q (EN 180): mode 27, median 35 $\Rightarrow$ mean

$$\frac{105-27}{2} = 39 \text{ ✓ book ans (b)}$$

BOOK KE SAWAAL (58) EN 85, 113, 171, 172, 174–182, 187, 188, 191, 193, 198–202, 205, 206, 213, 215, 220–224, 233, 239 · **HI** 43, 61, 72, 82, 83, 93, 94, 96, 98–107, 109–113, 116, 118**SP-5 — SPREAD: RANGE, VARIANCE, SD, CV (+ class limits)****PEHCHAN – Question mein ye dikhe**

range of data, variance, standard deviation, coefficient of variation, each value doubled / increased by k $\Rightarrow$ new SD / variance, SD of first n natural numbers, mean of three numbers with range and differences, mid-value and width of a class; मानक विचलन, प्रसरण, परिसर

$$\text{Range} = \max - \min \quad \sigma^2 = \frac{\sum(x - \bar{x})^2}{n} = \frac{\sum x^2}{n} - \bar{x}^2 \quad \sigma = \sqrt{\sigma^2} \quad CV = \frac{\sigma}{\bar{x}} \times 100$$

- Har value $+k \Rightarrow$ SD / variance same; $\times k \Rightarrow$ SD $\times |k|$, variance $\times k^2$.
- Pehle n natural numbers: $\sigma^2 = \frac{n^2-1}{12}$.
- **Teen numbers** $a \leq b \leq c$: $a + b + c = 3 \cdot \text{mean}$, $c - a = \text{range}$, diya farak $\Rightarrow$ teeno.
- Class: mid-value = $\frac{\text{lower} + \text{upper}}{2}$, lower = mid - width/2.

TERMS – Har letter ka matlab

σ = standard deviation · σ^2 = variance · CV = coefficient of variation.

SOLVED – Asli book questions (verified)

Q (EN 134): mean 25, SD 5 $\Rightarrow$ CV

20% ✓ **book ans (c)**

Q (EN 137): {11, 7, 10, 13, 9} ka SD

mean 10, $\sigma^2 = \frac{20}{5} = 4 \Rightarrow 2$ ✓ **book ans (b)**

Q (EN 139): variance 36, har value double $\Rightarrow$ SD

$6 \times 2 = 12$ ✓ **book ans (a)**

Q (EN 152): 3, 6, 5, 2, 4 ka variance

$\frac{1+4+1+4+0}{5} = 2$ ✓ **book ans (c)**

Q (EN 157): 9, 5, 9, 3, 4, 7, 8, 4, 8, 9, 5, 9 ka range

$9 - 3 = 6$ ✓ **book ans (c)**

Q (EN 159): SD 4.5, har value $+K \Rightarrow$ variance

$4.5^2 = 20.25$ ✓ **book ans (b)**

Q (EN 186): SD 3, mean 6 $\Rightarrow$ CV

50% ✓ **book ans (d)**

Q (EN 155): mid-value 12, width 6 $\Rightarrow$ lower limit

$12 - 3 = 9$ ✓ **book ans (b)**

BOOK KE SAWAAL (72) EN 18, 66, 71, 134–170, 184–186, 189, 190, 194, 197, 207–211, 216, 217, 225, 227–230, 232, 235, 237, 238, 240 · **HI** 53, 88, 108, 114, 115, 119–121

SP-6 — PROBABILITY + COUNTING

PEHCHAN – Question mein ye dikhe

probability of getting a vowel / face card / sum 10 on two dice / at least one head / two balls of same colour / one red one black card, defective pens, independent shooters, handshakes, circular arrangement, selecting 3 of 5; प्रायिकता, पासा, सिक्का, ताश

$$P(E) = \frac{\text{Favourable}}{\text{Total}} \quad P(\text{at least one}) = 1 - P(\text{koi nahi}) \quad {}^nC_r = \frac{n!}{r!(n-r)!}$$

Experiment	Total
n coins	2^n
2 dice	36 (sum 7 $\Rightarrow$ 6 tarique, sum 10 $\Rightarrow$ 3)
Cards	52: red 26, har suit 13, face 12, ace / king 4

- Balls nikalna: $\frac{\text{favourable } C}{\text{total } C}$ (bina wapas).
- Independent events $\Rightarrow$ guna. Handshakes = nC_2 . Circle mein n log, do saath: $\frac{2}{n-1}$.

TERMS – Har letter ka matlab

P = probability · C = combination.

SOLVED – Asli book questions (verified)**Q (EN 254):** 3 coins, kam se kam ek head

$$1 - \frac{1}{8} = \frac{7}{8} \quad \checkmark \text{ book ans (d)}$$

Q (EN 248): alphabet mein vowel

$$\frac{5}{26} \quad \checkmark \text{ book ans (a)}$$

Q (EN 246): 2 black, 6 green, 4 yellow; 2 balls same (green)

$$\frac{{}^6C_2}{{}^{12}C_2} = \frac{15}{66} = \frac{5}{22} \quad \checkmark \text{ book ans (d)}$$

Q (EN 253): 2 cards: ek red, ek black

$$\frac{26 \cdot 26}{13 \cdot 26} = \frac{26}{51} \quad \checkmark \text{ book ans (a)}$$

Q (EN 260): dice do baar, sum 10

$$\frac{3}{36} = \frac{1}{12} \quad \checkmark \text{ book ans (c)}$$

Q (EN 261): face card

$$\frac{12}{52} = \frac{3}{13} \quad \checkmark \text{ book ans (c)}$$

Q (EN 263): 100 pens, 8 kharab $\Rightarrow$ achha

$$\frac{92}{100} = \frac{23}{25} \quad \checkmark \text{ book ans (a)}$$

Q (EN 243): 20 log handshake

$$20C2 = 190 \quad \checkmark \text{ book ans (b)}$$

Q (EN 258): A $\frac{1}{2}$, B $\frac{2}{3}$, C $\frac{3}{4}$ teeno hit

$$\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{4} \quad \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (27) EN 242–265 · HI 89–91**COVERAGE CHECK – Har sawaal ka formula**

Formula	Naam	EN Qs	HI Qs
SP-1	MEAN MACHINE (ungrouped, frequency, grouped)		64
SP-2	MEDIAN MACHINE		50
SP-3	MODE MACHINE		29
SP-4	EMPIRICAL RELATION (Mean – Median – Mode)		33
SP-5	SPREAD: RANGE, VARIANCE, SD, CV (+ class limits)		64
SP-6	PROBABILITY + COUNTING		24
$\rightarrow$ RP-5	PROPORTION MACHINE ($a : b :: c : d$) (Ch 9)		1
Total (book ke sawaal)		265/265 ✓	121/121 ✓

 $\rightarrow$ RP-5: EN 212 HI 92

Chapter 27

Data Interpretation (Aankdon ka Vishleshan) आंकड़ों का विश्लेषण

4 formulas · English book 217 sawaal · Hindi book 174 sawaal · 15 solved examples ✓

DI-1 — PIE CHART MACHINE ($360^\circ = 100\%$)

PEHCHAN – Question mein ye dikhe

pie chart / circle graph in % or degrees, central angle of a sector, “if total is ₹32,700, how much on decoration and DJ”, total from one part, ratio of two sectors, share after redistribution; पाई चार्ट, केंद्रीय कोण

$$360^\circ = 100\% \quad 1\% = 3.6^\circ \quad \text{Hissa} = \frac{\text{kon}}{360^\circ} \times \text{Total} = \frac{\%}{100} \times \text{Total}$$

- Central angle = $\% \times 3.6$.
- Ek hissa ki value di $\Rightarrow$ Total = value $\div$ (kon/360) (ya $\div$ %).
- Do hisson ka ratio = unke kon / % ka ratio (total ki zaroorat nahi).
- Sab kharche barabar % se ghatate $\Rightarrow$ share % wahi rahega.

TERMS – Har letter ka matlab

kon = sector ka angle · Total = poore chart ki value.

SOLVED – Asli book questions (verified)

Q (EN 3): Delhi 30% $\Rightarrow$ central angle

$$30 \times 3.6 = 108^\circ \quad \checkmark \text{ book ans (c)}$$

Q (EN 55): Rice 40° , total 9000 tonnes

$$\frac{40}{360} \times 9000 = 1000 \quad \checkmark \text{ book ans (b)}$$

Q (EN 86): families: Education 150, Food 400, Rent 40, Electricity 250, Misc 160 $\Rightarrow$ food ka kon

$$\frac{400}{1000} \times 360 = 144^\circ \quad \checkmark \text{ book ans (c)}$$

Q (EN 95): item 3 (10%) par ₹850 $\Rightarrow$ item 2 (20%) + item 5 (15%)

$$\text{total } 8500 \Rightarrow 35\% = 2975 \quad \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (113) EN 1–55, 86, 95, 200–203 · HI 1–47, 92 | Ch 28 (Vividh): EN 22, 56, 68 · HI 53

DI-2 — PERCENT + PERCENT-CHANGE MACHINE

PEHCHAN – Question mein ye dikhe

percentage increase / decrease from one year to another, “A is approximately what percent of B”, percentage share, maximum percentage growth, percent of literates, pass percentage; प्रतिशत वृद्धि, कितना प्रतिशत

$$\% \text{ change} = \frac{\text{Naya} - \text{Purana}}{\text{Purana}} \times 100 \quad A \text{ is } \left(\frac{A}{B} \times 100 \right) \% \text{ of } B \quad \text{Share} = \frac{\text{Hissa}}{\text{Total}} \times 100$$

- “Kis saal sabse zyada growth” $\Rightarrow$ har saal ka (farak $\div$ pichhla) compare karo — sirf farak nahi.
- Percentage difference poocha $\Rightarrow$ base = pehla (purana) saal.
- Approx options $\Rightarrow$ round figures se jaldi bhaag.

TERMS – Har letter ka matlab

Purana = base value · Naya = baad wali value.

SOLVED – Asli book questions (verified)**Q (EN 93):** CO₂: 2016 = 110, 2017 = 150

$$\frac{40}{110} \times 100 = 36.36\% \checkmark \text{ book ans (b)}$$

Q (EN 97): year 1 total 36, year 3 total 42

$$\frac{6}{36} \times 100 = 16.67\% \checkmark \text{ book ans (b)}$$

Q (EN 94): sales 2014: 7 + 3, 2016: 18 + 7

$$10 \rightarrow 25 \Rightarrow 150\% \text{ badhe } \checkmark \text{ book ans (c)}$$

Q (EN 56): loan 75, 85, 125, 145, 190 (2016–2020) $\Rightarrow$ max growth

$$13\%, 47\%, 16\%, 31\% \Rightarrow 2018 \checkmark \text{ book ans (d)}$$

BOOK KE SAWAAL (117) EN 56–58, 60, 63, 65, 66, 72, 74, 79, 80, 83, 84, 87–89, 93, 94, 96–98, 104, 109, 112, 116, 119, 120, 122–124, 126, 127, 134, 145, 147, 148, 150, 155, 168, 171, 172, 182–188, 191, 192, 194–197, 206, 213, 216 · **HI** 48–50, 53, 56, 57, 61, 62, 66–68, 71, 73, 79, 81–84, 87, 90, 91, 93, 94, 101, 102, 104–106, 110, 112–115, 117, 119, 121, 123–125, 129, 140, 143, 148, 151, 153, 155–157, 162–165, 168, 169, 172, 174 | Ch 28 (Vividh): **EN** 59, 60 · **HI** 48, 49

DI-3 — RATIO / DIFFERENCE / TOTAL**PEHCHAN – Question mein ye dikhe**

ratio of two values or two totals, difference between totals of two items, how many more, sum over years, which quarter had maximum difference, ratio of females to males; अनुपात, अंतर, कुल

$$\text{Ratio} = \frac{\sum (\text{pehla group})}{\sum (\text{doosra group})} \quad (\text{HCF se chhota karo}) \quad \text{Farak} = |A - B|$$

- Table / bar se pehle zaroori numbers **hi** uthao, poor table mat jodo.
- Common factor (lakh, thousand) ratio mein cancel ho jata hai.
- Har row ka farak likho, phir max / min chuno.

TERMS – Har letter ka matlab
 $\Sigma = \text{jod.}$
SOLVED – Asli book questions (verified)**Q (EN 82):** Physics 300, 425, 475, 515; Chemistry 450, 550, 590, 700 $\Rightarrow$ farak sabse kam

$$150, 125, 115, 185 \Rightarrow 2005 \checkmark \text{ book ans (a)}$$

Q (EN 100): quarter price: 36/40, 80/80, 65/40, 43/40

$$4, 0, 25, 3 \Rightarrow \text{quarter 3 } \checkmark \text{ book ans (d)}$$

Q (EN 68): females : males sabse zyada (C 414/950, B 414/650, ...)

$$\text{Group B } \checkmark \text{ book ans (b)}$$

Q (EN 101): 12 mahine, do saal ke daam; kitne mahine same

$$\text{Apr, May, Jun, Jul, Oct} \Rightarrow 5 \checkmark \text{ book ans (b)}$$

BOOK KE SAWAAL (89) EN 59, 61, 62, 68, 69, 73, 75, 78, 81, 82, 90–92, 100, 101, 105, 107, 108, 110, 111, 113, 115, 117, 125, 128–130, 132, 135, 136, 138, 141–144, 146, 151, 153, 154, 169, 173, 174, 181, 189, 190, 193, 199, 204, 205, 211, 212 · **HI** 52, 54, 58, 59, 63, 65, 69, 70, 72, 75–78, 80, 88, 96, 97, 103, 108, 109, 111, 116, 127, 133, 139, 141, 142, 144–147, 158–160, 166, 167, 170, 171

DI-4 — AVERAGE / MAX-MIN / SEEDHA PADHO**PEHCHAN – Question mein ye dikhe**

average of values, which year had maximum / minimum, value closest to the average, how many years above average, highest total, earnings = users $\times$ fee; औसत, अधिकतम, न्यूनतम

$$\text{Average} = \frac{\text{Jod}}{\text{Ginti}} \quad \text{Kamaai} = \text{sankhya} \times \text{rate}$$

- “Closest to average” $\Rightarrow$ average nikaal ke har value ka farak dekho.
- “Above average kitne” $\Rightarrow$ average se bade values gino.
- Max / min / “kitne the” wale sawal: graph ya table se value **seedhe padho** (calculation nahi).

TERMS – Har letter ka matlab

Ginti = kitni values.

SOLVED – Asli book questions (verified)

Q (EN 85): bulbs 225, 100, 150, 200, 75, 90 $\Rightarrow$ average ke paas

avg 140 $\Rightarrow$ Wednesday (150) ✓ **book ans (a)**

Q (EN 102): goals 2, 1, 0, 4, 3, 2 $\Rightarrow$ average

$12 \div 6 = 2$ ✓ **book ans (d)**

Q (HI 99): gym 300 users $\times$ ₹500

₹1,50,000 ✓ **book ans (a)**

BOOK KE SAWAAL (80) EN 64, 67, 70, 71, 76, 77, 85, 99, 102, 103, 106, 114, 118, 121, 131, 133, 137, 139, 140, 149, 152, 156–167, 170, 175–180, 198, 207–210, 214, 215, 217 · HI 51, 55, 60, 64, 74, 85, 86, 89, 95, 98–100, 107, 118, 120, 122, 126, 128, 130–132, 134–138, 149, 150, 152, 154, 161, 173

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
DI-1	PIE CHART MACHINE ($360^\circ = 100\%$)		61
DI-2	PERCENT + PERCENT-CHANGE MACHINE		57
DI-3	RATIO / DIFFERENCE / TOTAL		51
DI-4	AVERAGE / MAX-MIN / SEEDHA PADHO		48
Total (book ke sawaal)		217/217 ✓	174/174 ✓

Chapter 28

Miscellaneous (Vividh) विविध

4 formulas · English book 82 sawaal · Hindi book 59 sawaal · 29 solved examples ✓

MS-1 — WHOLE – PART – FRACTION CHAIN

PEHCHAN – Question mein ye dikhe

vessel $\frac{1}{4}$ full, 10 cups more $\Rightarrow \frac{3}{4}$ full; drum $\frac{3}{4} \rightarrow \frac{1}{2}$ after 9 litres; lost $\frac{1}{3}$, then $\frac{3}{5}$ of the rest, then $\frac{4}{7}$ of the rest; pillar parts $\frac{1}{4}$, $\frac{1}{2}$ of first, rest 10 m; girls = $\frac{3}{4}$ boys & fractions like mango juice; train passengers getting down in fractions; lottery share after tax; savings = income – expense

$$\text{Capacity} = \frac{\text{jo daala} / \text{nikaala}}{f_2 - f_1}$$

$$\text{Bacha fraction} = (1 - a)(1 - b)(1 - c) \dots$$

- Hisse (fraction) jodo $\Rightarrow$ bacha fraction = diya number $\Rightarrow$ Total = number $\div$ bacha fraction.
- “Baaki ka” (of the rest) $\Rightarrow$ guna chain; “total ka” $\Rightarrow$ jod-ghata.
- Ulta chalo (lottery, share): diya amount $\div$ (sab fractions ka guna).
- Bachat = aamdani – kharcha (mahine / saal units same karo).

TERMS – Har letter ka matlab

f_1, f_2 = pehle / baad ka bhara hua fraction.

SOLVED – Asli book questions (verified)

Q (EN 7): $\frac{1}{4}$ bhara, 10 cup daale $\Rightarrow \frac{3}{4}$
 $\frac{\frac{10}{\frac{3}{4}-\frac{1}{4}}}{1} = 20$ ✓ **book ans (b)**

Q (EN 36): drum $\frac{3}{4} \rightarrow 9$ litre nikaale $\rightarrow \frac{1}{2}$
 $\frac{9}{\frac{1}{4}} = 36$ ✓ **book ans (b)**

Q (EN 35): $\frac{1}{3}$ haara, phir baaki ka $\frac{3}{5}$, phir baaki ka $\frac{4}{7}$
 $\frac{2}{3} \cdot \frac{2}{5} \cdot \frac{3}{7} = \frac{4}{35}$ ✓ **book ans (d)**

Q (EN 11): pillar: $\frac{1}{4}$, phir pehle ka $\frac{1}{2}$, baaki 10 m
 $\frac{10}{1-\frac{1}{4}-\frac{1}{8}} = 16$ ✓ **book ans (b)**

Q (EN 1): girls = $\frac{3}{4}$ boys; $\frac{2}{3}$ girls, $\frac{1}{2}$ boys ko mango juice
 $\frac{\frac{1}{2}+\frac{1}{2}}{\frac{2}{3}} = \frac{4}{7}$ ✓ **book ans (b)**

Q (EN 45): 13/18 ke paas car; unmein 48/65 covered; 136 bahar
 $\frac{136}{17/65} \cdot \frac{18}{13} = 720$ ✓ **book ans (b)**

Q (EN 14): 450 yatri: $\frac{1}{9}$ utre, 20 chadhe; $\frac{1}{6}$ utre, 19 chadhe
 $400 + 20 = 420 \Rightarrow 350 + 19 = 369$ ✓ **book ans (b)**

Q (EN 63): 35% tax, baaki ka $\frac{1}{8}$ seller ko = ₹22,343.75
 $\frac{22343.75}{0.65/8} = 2,75,000$ ✓ **book ans (b)**

Q (EN 3): saal ka kharcha ₹4,20,000, salary ₹45,000/mahina
 $\frac{5,40,000-4,20,000}{12} = 10,000$ ✓ **book ans (d)**

BOOK KE SAWAAL (21) EN 1, 3, 5, 7, 9, 11, 14, 18, 33, 35, 36, 45, 50, 61, 63, 66, 69, 72 · **HI** 32, 51, 54

MS-2 — UNITS, SCALE & EK-KA-KITNA (unitary)

PEHCHAN – Question mein ye dikhe

map scale 1 : 50000, cm $\leftrightarrow$ km $\leftrightarrow$ mm, degrees $\leftrightarrow$ radians, earth rotates 360° in 24 h, cake / flour shared, glasses to fill a vessel, sweet with cashew & almonds, students $\times$ subjects, blood donated in 6 years; मानचित्र का पैमाना

$$\text{Scale } 1 : n \Rightarrow \text{asli doori} = \text{map doori} \times n \quad 1 \text{ km} = 10^5 \text{ cm} = 10^6 \text{ mm} \quad 1^\circ = \frac{\pi}{180} \text{ rad}$$

- **Unitary:** pehle “ek ka” nikaalo, phir guna — ya seedha proportion: $\frac{a_1}{b_1} = \frac{a_2}{b_2}$.
- Earth: $360^\circ / 24 \text{ h} = 15^\circ$ har ghanta.
- Same total, chhota glass $\Rightarrow$ zyada glass (ulta anupaat).

TERMS – Har letter ka matlab

n = scale ka bada number.

SOLVED – Asli book questions (verified)

Q (EN 25): $1 : 50000$, map par 12 cm
 $600000 \text{ cm} = 6 \text{ km}$ ✓ **book ans (b)**

Q (EN 30): $1 : 150000$, map par 6 cm
 9 km ✓ **book ans (c)**

Q (HI 7): $1 : 400000$, map par 5 cm
 20 km ✓ **book ans (b)**

Q (EN 27): earth 5 h 24 min mein
 $15^\circ \times 5.4 = 81^\circ$ ✓ **book ans (c)**

Q (EN 49): 54° in radians
 $\frac{54\pi}{180} = \frac{3\pi}{10}$ ✓ **book ans (d)**

Q (EN 15): $15 \text{ glass} \times 1.5 \text{ L} \Rightarrow 0.5 \text{ L}$ wale kitne
 $\frac{22.5}{0.5} = 45$ ✓ **book ans (d)**

Q (EN 74): $420 \text{ students} \times 5 \text{ subjects}$, har subject 60
 $\frac{2100}{60} = 35$ ✓ **book ans (a)**

Q (EN 62): 250 g sweet: 20 g cashew, 30 g almond $\Rightarrow$ 350 g sweet mein
 28 g aur 42 g ✓ **book ans (a)**

BOOK KE SAWAAL (17) EN 6, 13, 15, 17, 25, 27, 30, 34, 48, 49, 55, 62, 74, 77 · HI 7, 9, 58

MS-3 — PUZZLE PATTERNS (doubling, climb-slip, digits, pair-sum)

PEHCHAN – Question mein ye dikhe

basket doubles every minute, full in 30 min $\Rightarrow$ half?, monkey climbs 2 m & slips 1 m, typing numbers 2 to 222 (key presses), degree of a polynomial, place value, next pair in prime series, highest X in column addition, pair sums of money, $x\%$ of $y = y\%$ of z , max of abcd, data sufficiency

- **Doubling:** poora t par $\Rightarrow$ aadha $(t - 1)$ par, chauthai $(t - 2)$ par.
- **Climb-slip** (upar u , neeche d , height H , har cycle 2 sec): $\text{samay} = \frac{2(H-u)}{u-d} + 1 \text{ sec}$.
- **Key presses:** $1\text{-digit} \times 1 + 2\text{-digit} \times 2 + 3\text{-digit} \times 3 \dots$ (har range ki ginti).
- **Pair-sum** ($A+B$, $B+C$, $C+A$): sab ka jod = teeno ka jod $\div 2$; ek = total – doosri jodi.
- Degree = sab variables ki powers ka jod. $x\%$ of $y = y\%$ of $z \Rightarrow x = z$. Fixed sum of squares $\Rightarrow$ max product jab sab barabar.
- Column addition: carry ka hisaab rakh ke digit dhoondho.

TERMS – Har letter ka matlab

t = poora hone ka samay · H = unchai.

SOLVED – Asli book questions (verified)

Q (EN 16): har minute double, 30 min mein पूरी $\Rightarrow$ aadhi

29 min ✓ **book ans (a)**

Q (EN 80): 12 m ped, $+2 -1$

$\frac{2-10}{1} + 1 = 21$ ✓ **book ans (b)**

Q (EN 81): 24 m ped, $+4 -2$

$\frac{2-20}{2} + 1 = 21$ ✓ **book ans (b)**

Q (EN 8): $9x^5y^2z^7r$ ki degree 15

$15 - 7 = 8$ ✓ **book ans (c)**

Q (EN 44): 2 se 222 type karne mein key presses

$8 + 180 + 369 = 557$ ✓ **book ans (c)**

Q (EN 73): Rani + Malika 227, Rani + Tamali 273, Tamali + Malika 280

$\frac{227+280-273}{2} = 117$ ✓ **book ans (d)**

Q (EN 19): 1200 seb, har ladke ko ladkon ki sankhya ka teen guna

$3x^2 = 1200 \Rightarrow 20$ ✓ **book ans (b)**

Q (EN 47): $a^2 + b^2 + c^2 + d^2 = 1 \Rightarrow \max abcd$

$\text{sab } \frac{1}{2} \Rightarrow \frac{1}{16}$ ✓ **book ans (d)**

BOOK KE SAWAAL (15) EN 8, 16, 19, 32, 38, 41, 44, 47, 58, 73, 78, 80–82 · HI 57

MS-4 — DIGIT-RULE PUZZLES (ank badlo, phir poocha hua)**PEHCHAN – Question mein ye dikhe**

each odd digit $+2$ and each even digit -1 in 4723361, swap first and third digits, subtract each digit from 9, arrange numbers in descending order then product of digits, how many even numbers followed by a multiple of 3 in a series; प्रत्येक विषम अंक, सम अंक, अवरोही क्रम, श्रृंखला

- **Step 1:** har ank ke neeche uska naya ank likho — sam / vishham **purane** ank se tay hota hai.
- **Step 2:** jo poocha (sabse bada + chhota, 2nd from left, kitne sam) sirf wahi padho.
- Arrange wale: pehle पूरी list sort karo, phir “left / right se n-th” gino.
- Series wale: har jodi (ank, agla ank) par shart check karke gino.
- 0 sam hai; $9 - \text{ank se sam} \Leftrightarrow \text{purana ank vishham}$.

TERMS – Har letter ka matlab

sam = even · vishham = odd.

SOLVED – Asli book questions (verified)

Q (HI 1): 4723361: sam $\div 2$, vishham $+2 \Rightarrow$ bada + chhota

2 9 1 5 5 3 3 $\Rightarrow 9 + 1 = 10$ ✓ **book ans (b)**

Q (HI 8): 42514563: vishham $+3$, sam $-1 \Rightarrow$ bada – chhota

3 1 8 4 3 8 5 6 $\Rightarrow 8 - 1 = 7$ ✓ **book ans (b)**

Q (HI 46): 72514368 ka har ank 9 se ghatao $\Rightarrow$ kitne sam

2 7 4 8 5 6 3 1 $\Rightarrow 4$ ✓ **book ans (c)**

Q (HI 18): 8452, 6327, 7524, 4598, 5691 avrohi; right se 2nd ka 3rd ank $\times$ left se 3rd ka 2nd ank

5691 ka 9 $\times$ 6327 ka 3 = 27 ✓ **book ans (b)**

BOOK KE SAWAAL (42) HI 1–6, 8, 10–22, 24–28, 30, 31, 33–47

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
MS-1	WHOLE – PART – FRACTION CHAIN	18	3
MS-2	UNITS, SCALE & EK-KA-KITNA (unitary)	14	3
MS-3	PUZZLE PATTERNS (doubling, climb-slip, digits, pair-sum)	14	1
MS-4	DIGIT-RULE PUZZLES (ank badlo, phir poocha hua)	0	42
→ NS-10	SERIES + PLACE-VALUE + RATIONAL BOX (ratta facts (Ch 1)	1	0
→ IS-6	POWER LAWS (ghatank ke niyam) (Ch 3)	2	0
→ SIM-1	VBODMAS MACHINE (order of operations) (Ch 4)	1	0
→ SIM-4	IDENTITY CANCEL TABLE (bada calculation kabhi na (Ch 4)	1	0
→ LH-2	LCM + REMAINDER MACHINE (Ch 5)	1	1
→ PC-1	PERCENT BASICS (of · kitna % · poora number) (Ch 6)	2	1
→ PC-3	NET CHANGE MACHINE (do/teen change ek saath) (Ch 6)	1	0
→ PC-6	EXAM MARKS MACHINE (pass / fail) (Ch 6)	1	0
→ PL-1	BASIC P/L % (CP, SP, kharcha, fraction) (Ch 7)	1	0
→ DS-4	FREE-ITEM SCHEME (Buy x Get y) (Ch 8)	1	0
→ RP-3	BARABARI SE RATIO ($x\%$ of $A = y\%$ of B , $\frac{1}{2}A = \frac{1}{3}B$) (Ch 9)	2	0
→ RP-4	RATIO CHANGE MACHINE (jodna / ghatana / dena) (Ch 9)	1	1
→ WT-1	LCM / RATE MACHINE (saath mein, akele, kitna kaa (Ch 11)	1	0
→ WT-7	WAGES MACHINE (paisa $\propto$ kaam) (Ch 11)	1	1
→ MX-2	WEIGHTED MIX (mixtures jodna, price, profit) (Ch 12)	1	0
→ AV-2	TOTAL MACHINE (Total = Avg $\times$ n) (Ch 17)	1	0
→ ST-3	RATIO MACHINE (speed $\propto$ 1/samay) (Ch 18)	2	0
→ ST-4	AVERAGE SPEED (Kul doori $\div$ Kul samay) (Ch 18)	1	0
→ MN-4	CIRCLE PACK (circle, semicircle, sector, wheel, (Ch 21)	1	1
→ MN-5	SIZE-CHANGE / RATIO MACHINE (Ch 21)	1	1
→ MN-6	CUBE & CUBOID (ghan, ghanabh) (Ch 21)	3	0
→ AL-3	LINEAR PAIR (do seedhi equations) (Ch 22)	1	0
→ AL-7	EXPAND / SIMPLIFY / FACTORISE (Ch 22)	1	0
→ AL-9	VENN MACHINE (sets) (Ch 22)	1	0
→ TG-8	ELIMINATION + EQUATION SOLVE (Ch 23)	1	0
→ GE-8	POLYGON MACHINE (Ch 25)	0	1
→ DI-1	PIE CHART MACHINE ($360^\circ = 100\%$) (Ch 27)	3	1
→ DI-2	PERCENT + PERCENT-CHANGE MACHINE (Ch 27)	2	2
Total (book ke sawaal)		82/82 ✓	59/59 ✓

→ **AL-3:** EN 31 · → **AL-7:** EN 10 · → **AL-9:** EN 75 · → **AV-2:** EN 28 · → **DI-1:** EN 22, 56, 68 HI 53 · → **DI-2:** EN 59, 60 HI 48, 49 · → **DS-4:** EN 79 · → **GE-8:** HI 29 · → **IS-6:** EN 12, 57 · → **LH-2:** EN 76 HI 59 · → **MN-4:** EN 54 HI 23 · → **MN-5:** EN 67 HI 52 · → **MN-6:** EN 2, 24, 52 · → **MX-2:** EN 4 · → **NS-10:** EN 42 · → **PC-1:** EN 53, 64 HI 50 · → **PC-3:** EN 39 · → **PC-6:** EN 43 · → **PL-1:** EN 65 · → **RP-3:** EN 26, 51 · → **RP-4:** EN 70 HI 55 · → **SIM-1:** EN 20 · → **SIM-4:** EN 21 · → **ST-3:** EN 37, 46 · → **ST-4:** EN 40 · → **TG-8:** EN 23 · → **WT-1:** EN 29 · → **WT-7:** EN 71 HI 56

Appendix A – Book Errata & Recovered Questions

A.1 Book ki galtiyan (answer key / question text)

Ch	Chapter	Q	Kya galat hai – sahi kya hai
10	Partnership	EN 1	English book mein A ki capital ₹50,000 chhapi hai (galat); Hindi book (HI2) mein ₹4,50,000 hai $\Rightarrow 450000 \times 12 : 9B = 3 : 2 \Rightarrow B = ₹4,00,000 = \text{option (c)}$.
12	Alligation / Mixture	EN 42	1.5% aur 2.5% ko milakar 3.5% ban hi nahi sakta (mean hamesha dono ke beech). Question galat; book ka 1 : 2 cross rule ko ulta lagane se aaya.
12	Alligation / Mixture	EN 45	Book ne ₹160 aur ₹400 (SP) ko hi rates maan kar mean ₹240 se alligation lagaya $\Rightarrow 2 : 1$. Loss/profit se asli CP (₹200, ₹333.3) lein to 7 : 3 aata hai — option mein nahi.
13	Pipe & Cistern	EN 119	Book ne 2 ghante ka net $\frac{1}{4}$ maan kar $4 \times 2 = 8$ h likha. Par 5th ghante mein A akela $\frac{1}{2}$ se poora bhar deta hai $\Rightarrow$ sahi $4 + \frac{3}{4} = 4.75$ h = option (a).
13	Pipe & Cistern	HI 79	EN119 jaisa: sahi uttar 4.75 घंटे (a); book ka (d) 8 घंटे aakhri ghante ka trap bhool gaya.
13	Pipe & Cistern	EN 59	Wording ulti chhapi hai. Sahi arth: doosra pehle se 5 h KAM, teesre se 4 h ZYADA leta $\Rightarrow 15, 10, 6 (1/15 + 1/10 = 1/6) \Rightarrow 15 \checkmark$.
18	Speed, Time & Distance	EN 59	Question mein “reduced his speed to 2 km/h” chhapa hai, par book ka answer (12:30 pm) “by 2 km/h” (yaani 4 km/h) se aata hai: $18/6 + 12/4 = 6$ h. “to 2 km/h” lene par 3:30 pm aata jo options mein nahi.
19	Train	EN 39	Answer key mein (c) 700 m chhapa hai, par book ki apni solution bhi $L = 800$ m nikalti hai (55 m/s ki train). Sahi option (a) 800 m — Hindi book (HI38) mein bhi (a).

A.2 Wapas nikaale gaye / numbering-gap sawaal

Ch	Chapter	Q	Kya hua
6	Percentage	EN 189	Page se wapas nikala: question number printed as '18.'; stem garbled in print as '110 of 100% of 10%', book solution uses 110 of 110% of 10%
7	Profit & Loss	HI 81	Page se wapas nikala: book prints this as a second '80.'
9	Ratio & Proportion	EN 136	Page se wapas nikala: book prints this as '137.'
9	Ratio & Proportion	EN 137	Page se wapas nikala: extraction mein chhoot gaya tha
15	Compound Interest	EN 78	Book mein ye number chhapa hi nahi (numbering gap) — koi sawaal gayab nahi.
16	Problems on Age	HI 54	Book mein ye number chhapa hi nahi (numbering gap) — koi sawaal gayab nahi.
18	Speed, Time & Distance	EN 35	Page se wapas nikala: book prints the number as '35' without a dot
22	Algebra	HI 262	Page se wapas nikala: page text of Q262 and Q263 was interleaved; repaired
22	Algebra	HI 263	Page se wapas nikala: page text of Q262 and Q263 was interleaved; repaired
25	Geometry	EN 161	Page se wapas nikala: book prints the number as '161' without a dot
27	Data Interpretation	HI 97	Page se wapas nikala: book prints this number as a second '96.'
27	Data Interpretation	HI 98	Page se wapas nikala: extraction mein chhoot gaya tha
27	Data Interpretation	HI 99	Page se wapas nikala: extraction mein chhoot gaya tha

Master Summary – Ek Nazar Mein Sab

Chapter	Kitne	Formulas (yaad rakho)	EN / HI Qs
1. Number System संख्या पद्धति	10	NS-1 DIVISIBILITY TABLE (vibhajyata) NS-2 REMAINDER MACHINE (sheshfal) NS-3 NEAREST-MULTIPLE MACHINE (add / subtract / largest / smallest) NS-4 COUNT MACHINE (kitne numbers) NS-5 FACTOR MACHINE (gunankhand) NS-6 UNIT DIGIT MACHINE (ikai ank) NS-7 PRIME / CO-PRIME TOOLKIT NS-8 POWER-PATTERN DIVISIBILITY NS-9 TWO-NUMBER + EQUATION MACHINE NS-10 SERIES + PLACE-VALUE + RATIONAL BOX (ratta facts)	345 161
2. Decimal Fractions दशमलव भिन्न	9	DF-1 COMPARE + ORDER MACHINE DF-2 TERMINATING / RECURRING TEST DF-3 DECIMAL $\leftrightarrow$ FRACTION (9-0 RULE) DF-4 FRACTION OPERATION MACHINE (+ $- \times \div$, “kya jodein”) DF-5 RECIPROCAL MACHINE DF-6 DECIMAL SHIFT + ROUNDING MACHINE DF-7 TELESCOPING SERIES (cancel wali series) DF-8 FRACTION EQUATION MACHINE (ansh-har wale sawaal) DF-9 EQUIVALENT + CONVERSION BOX	232 133
3. Indices and Surds घातांक तथा करणी	7	IS-1 SQUARE ROOT IN 10 SECONDS (unit digit + range) IS-2 DECIMAL + FRACTION ROOT MACHINE IS-3 SURD SIMPLIFY + LIKE-SURD MACHINE IS-4 CONJUGATE + RATIONALISE + $\sqrt{A \pm 2\sqrt{B}}$ IS-5 NESTED ROOT + $(a^2 - b^2)$ ROOT IS-6 POWER LAWS (ghatank ke niyam) IS-7 ROOT EQUATION + WORD MACHINE	139 103
4. Simplification सरलीकरण	8	SIM-1 VBODMAS MACHINE (order of operations) SIM-2 DECIMAL COLUMN + “?” MACHINE SIM-3 APPROXIMATION MACHINE (“closest approximate value”) SIM-4 IDENTITY CANCEL TABLE (bada calculation kabhi nahi) SIM-5 NEAR-BASE MULTIPLY MACHINE SIM-6 FRACTION CHAIN + CONTINUED FRACTION + SERIES SIM-7 RATIO SUBSTITUTION MACHINE SIM-8 POWER + ROOT SHORTCUTS (infinite root bhi)	248 243
5. LCM & HCF लघुत्तम समापवर्त्य एवं महत्तम समापवर्तक	8	LH-1 POWER RULE (HCF/LCM nikaalna) LH-2 LCM + REMAINDER MACHINE LH-3 HCF + REMAINDER MACHINE LH-4 HCF APPLICATION (sabse bada naap / kam se kam hisse) LH-5 LCM APPLICATION (bells / lights / track — saath kab) LH-6 PRODUCT LINK ($HCF \times LCM = a \times b$) LH-7 RATIO LINK + PAIR COUNT LH-8 FRACTION + DECIMAL HCF/LCM	231 126
6. Percentage प्रतिशतता	9	PC-1 PERCENT BASICS (of $\cdot$ kitna % $\cdot$ poor number) PC-2 MULTIPLIER MACHINE (badha / ghata / ulta) PC-3 NET CHANGE MACHINE (do/teen change ek saath) PC-4 POPULATION / GROWTH MACHINE PC-5 REVERSE + CONSUMPTION MACHINE PC-6 EXAM MARKS MACHINE (pass / fail) PC-7 INCOME – EXPENDITURE – SAVINGS MACHINE PC-8 ELECTION MACHINE (votes) PC-9 GROUP % / MIXTURE % MACHINE	247 117
7. Profit & Loss लाभ और हानि	10	PL-1 BASIC P/L % (CP, SP, kharcha, fraction) PL-2 CP $\leftrightarrow$ SP MULTIPLIER + SP TRANSFER + CHAIN PL-3 DIFFERENCE MACHINE ($\text{₹ farak} \leftrightarrow \% \text{ farak}$) PL-4 ARTICLE-COUNT MACHINE ($x \text{ ka SP} = y \text{ ka CP}$) PL-5 RATE MACHINE ($a \text{ items ₹}x \text{ mein}$) PL-6 FALSE WEIGHT MACHINE (nakli baat) PL-7 SAME-SP MACHINE (do cheezein, ek hi SP) PL-8 WEIGHTED PARTS / TWO-ITEM MACHINE PL-9 MP – DISCOUNT – PROFIT CHAIN PL-10 “100 MAAN LO” CONDITION MACHINE	168 114

8. Discount बट्टा / छूट	4	DS-1 BASIC DISCOUNT (MP, SP, D%) + TAX / COMMISSION DS-2 SUCCESSIVE DISCOUNT MACHINE DS-3 MP – DISCOUNT – PROFIT CHAIN (= PL-9) DS-4 FREE-ITEM SCHEME (Buy x Get y)	76 79
9. Ratio & Proportion अनुपात और समानुपात	8	RP-1 RATIO KA BATWARA (share / ek hissa / farak) RP-2 RATIO JODNA ($A : B, B : C \Rightarrow A : B : C$) RP-3 BARABARI SE RATIO ($x\%$ of A = $y\%$ of B, $\frac{1}{2}A = \frac{1}{3}B$) RP-4 RATIO CHANGE MACHINE (jodna / ghatana / dena) RP-5 PROPORTION MACHINE ($a : b :: c : d$) RP-6 COINS / NOTES MACHINE RP-7 INCOME – EXPENDITURE – SAVINGS RATIO RP-8 RATIO ALGEBRA (componendo – dividendo)	178 162
10. Partnership साझेदारी	2	PT-1 CAPITAL $\times$ TIME MACHINE (profit ka batwara) PT-2 UNKNOWN CAPITAL / TIME (ulta PT-1)	54 25
11. Work & Time समय और कार्य	7	WT-1 LCM / RATE MACHINE (saath mein, akele, kitna kaam) WT-2 PAIR-SUM MACHINE (A+B, B+C, C+A) WT-3 EFFICIENCY RATIO MACHINE (twice as good, $x\%$ more efficient) WT-4 LEAVE / JOIN MACHINE (beech mein chhoda / juda) WT-5 ALTERNATE DAY / CYCLE MACHINE WT-6 M·D·H = WORK MACHINE (men, women, days, hours) WT-7 WAGES MACHINE (paisa $\propto$ kaam)	206 106
12. Alligation / Mixture मिश्रण / एलीगेशन	4	MX-1 ALLIGATION CROSS RULE (kis ratio mein milaayein) MX-2 WEIGHTED MIX (mixtures jodna, price, profit) MX-3 ADD / EVAPORATE MACHINE (ek cheez badli, doosri fix) MX-4 REPLACEMENT MACHINE (baar-baar nikaalo, paani daalo)	58 16
13. Pipe & Cistern नल और टंकी	6	PP-1 NET RATE MACHINE (bharna +, khaali karna –) PP-2 LEAK MACHINE (rasaav) PP-3 $\sqrt{ab}$ TRICK (saath T, ek a zyada, doosra b zyada) PP-4 OPEN / CLOSE MACHINE (beech mein band / chalu) PP-5 ALTERNATE HOUR MACHINE (ek ghanta bhara, ek ghanta khaali) PP-6 CAPACITY / LITRE MACHINE	155 95
14. Simple Interest साधारण ब्याज	6	SI-1 PRT MACHINE (SI, Amount, P, R, T) SI-2 DO AMOUNT MACHINE (T_1 pe A_1 , T_2 pe A_2) SI-3 RATE / TIME CHANGE MACHINE (farak se P) SI-4 n-GUNA / R = T MACHINE SI-5 SPLIT / MIX INVESTMENT MACHINE SI-6 INSTALMENT / PRESENT WORTH / MISC	200 76
15. Compound Interest चक्रवृद्धि ब्याज	6	CI-1 AMOUNT MACHINE (A, P, CI; half-yearly, quarterly, alag rates) CI-2 CI – SI DIFFERENCE / RELATION MACHINE CI-3 RATE MACHINE (dar nikaalo) CI-4 TIME / n-GUNA MACHINE CI-5 INSTALMENT / SPLIT / PRESENT WORTH CI-6 SI vs CI / YEARLY vs HALF-YEARLY MACHINE	206 70
16. Problems on Age आयु संबंधी प्रश्न	4	AG-1 DO-PAL RATIO MACHINE (t_1 pe a : b, t_2 pe c : d) AG-2 FARAK MACHINE (umar ka antar kabhi nahi badalta) AG-3 SUM / AVERAGE MACHINE (jod har saal badhta hai) AG-4 EQUATION / PRODUCT / FRACTION MACHINE	241 122
17. Average औसत	6	AV-1 SERIES AVERAGE (lagataar / AP numbers) AV-2 TOTAL MACHINE (Total = Avg $\times$ n) AV-3 COMBINED / WEIGHTED AVERAGE (do group) AV-4 JOIN / LEAVE / REPLACE MACHINE AV-5 OVERLAP / PART-SUM MACHINE AV-6 REQUIRED SCORE / INNINGS MACHINE	116 82
18. Speed, Time & Distance चाल, समय और दूरी	7	ST-1 BASIC D = S $\times$ T + UNITS + TUKDE ST-2 LATE-EARLY / TIME-GAP MACHINE ST-3 RATIO MACHINE (speed $\propto$ 1/samay) ST-4 AVERAGE SPEED (Kul doori $\div$ Kul samay) ST-5 RELATIVE SPEED (milna / peecha) ST-6 MEETING KE BAAD ($\sqrt{}$ rule) ST-7 DO SAWARI / DO SPEED KA BATWARA	149 78

19. Train रेलगाड़ी	6	TR-1 POLE / KHAMBHA / KHADA AADMI TR-2 PLATFORM / BRIDGE / TUNNEL TR-3 EK TRAIN, DO CROSSING (pole + platform / 2 platform) TR-4 DO TRAINS (aamne-saamne / ek disha) TR-5 TRAIN AUR CHALTA AADMI / GAADI TR-6 TRAINS KA MILNA / PEECHA (bina lambai)	159 80
20. Boat & Stream नाव और धारा	4	BS-1 D / U MACHINE (dhaara ke saath / khilaaf) BS-2 TIME-RATIO MACHINE (up = n × down) BS-3 AANA-JAANA (round trip) MACHINE BS-4 DO TRIP, DO EQUATION	47 13
21. Mensuration क्षेत्रमिति	10	MN-1 TRIANGLE (tribhuj) PACK MN-2 4-SIDED PACK (rectangle, square, rhombus, parallelogram, trapezium) MN-3 PATH / BORDER / RING + TILES, CARPET, POLES MN-4 CIRCLE PACK (circle, semicircle, sector, wheel, andar-bahar shapes) MN-5 SIZE-CHANGE / RATIO MACHINE MN-6 CUBE & CUBOID (ghan, ghanabh) MN-7 CYLINDER (belan) + PIPE MN-8 CONE (shanku) + FRUSTUM MN-9 SPHERE & HEMISPHERE (gola, ardh-gola) MN-10 MELT & RECAST + COMBINATION + PRISM / PYRAMID	354 279
22. Algebra बीजगणित	9	AL-1 SERIES MACHINE (AP / GP) AL-2 HCF / LCM OF EXPRESSIONS AL-3 LINEAR PAIR (do seedhi equations) AL-4 DO-VARIABLE + x + 1/x MACHINE AL-5 TEEN-VARIABLE IDENTITY PACK (a + b + c) AL-6 REMAINDER / FACTOR THEOREM AL-7 EXPAND / SIMPLIFY / FACTORISE AL-8 QUADRATIC MACHINE (Sum – Product – D) AL-9 VENN MACHINE (sets)	294 268
23. Trigonometry त्रिकोणमिति	8	TG-1 IDENTITY TRIANGLE + “ $\theta = 45^\circ$ DAALO” TRICK TG-2 PAIR TRICKS (sec + tan = k, tan + cot = k ...) TG-3 VALUE TABLE + EK RATIO SE SAB TG-4 COMPLEMENTARY + ALLIED ANGLES (90° , 180° , 360°) TG-5 COMPOUND + DOUBLE ANGLE TG-6 $a \sin^2 + b \cos^2$ MACHINE (min / max / angle) TG-7 HEIGHTS & DISTANCES TG-8 ELIMINATION + EQUATION SOLVE	221 124
24. Co-ordinate Geometry निर्देशांक ज्यामिति	5	CG-1 SECTION FORMULA + MIDPOINT + REFLECTION CG-2 DISTANCE + TRIANGLE TYPE CG-3 AREA OF TRIANGLE / QUADRILATERAL CG-4 STRAIGHT LINE (slope, equation, angle) CG-5 CIRCLE (centre, radius, tangent)	49 17
25. Geometry ज्यामिति	8	GE-1 LINES & ANGLES (complementary, supplementary, parallel lines) GE-2 TRIANGLE ANGLES (180° , exterior, bisector-meet) GE-3 SIMILARITY + BPT + ANGLE BISECTOR + MIDPOINT GE-4 TRIANGLE LENGTHS: Pythagoras, inradius, circumradius, centroid GE-5 QUADRILATERALS (parallelogram, rhombus, trapezium, cyclic) GE-6 CIRCLE: CHORD, ANGLES, ARC GE-7 TANGENTS GE-8 POLYGON MACHINE	276 207
26. Statistics & Probability सांख्यिकी और प्रायिकता	6	SP-1 MEAN MACHINE (ungrouped, frequency, grouped) SP-2 MEDIAN MACHINE SP-3 MODE MACHINE SP-4 EMPIRICAL RELATION (Mean – Median – Mode) SP-5 SPREAD: RANGE, VARIANCE, SD, CV (+ class limits) SP-6 PROBABILITY + COUNTING	265 121
27. Data Interpretation आंकड़ों का विश्लेषण	4	DI-1 PIE CHART MACHINE ($360^\circ = 100\%$) DI-2 PERCENT + PERCENT-CHANGE MACHINE DI-3 RATIO / DIFFERENCE / TOTAL DI-4 AVERAGE / MAX-MIN / SEEDHA PADHO	217 174
28. Miscellaneous विविध	4	MS-1 WHOLE – PART – FRACTION CHAIN MS-2 UNITS, SCALE & EK-KA-KITNA (unitary) MS-3 PUZZLE PATTERNS (doubling, climb-slip, digits, pair-sum) MS-4 DIGIT-RULE PUZZLES (ank badlo, phir poocha hua)	82 59
TOTAL	185	1,235 solved examples verified ✓ · har sawaal ek formula se	5,213 3,250